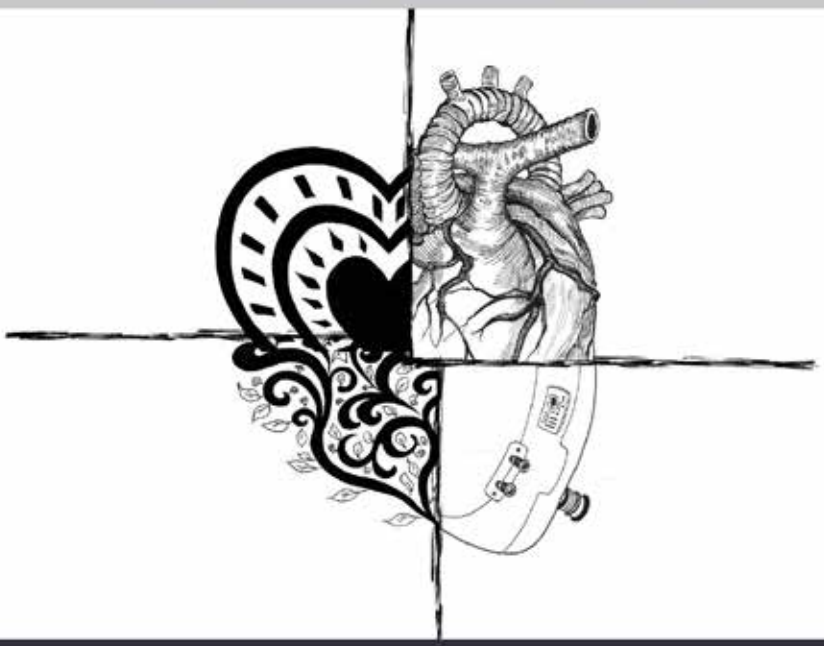
INTRODUCING MEDICAL ANTHROPOLOGY

A Discipline in Action

THIRD EDITION



Merrill Singer, Hans A. Baer,
Debbi Long, and Alex Pavlotski

Third Edition

Introducing Medical Anthropology

A Discipline in Action

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ROWMAN & LITTLEFIELD
Lanham • Boulder • New York • London

Executive Editor: Nancy Roberts
Editorial Assistant: Megan Manzano
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Published by Rowman & Littlefield
An imprint of The Rowman & Littlefield Publishing Group, Inc.
4501 Forbes Boulevard, Suite 200, Lanham, Maryland 20706
www.rowman.com

6 Tinworth Street, London SE11 5AL, United Kingdom

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First edition 2007. Second edition 2012.

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British Library Cataloguing in Publication Information Available

Library of Congress Cataloging-in-Publication Data

Names: Singer, Merrill, author.
Title: Introducing medical anthropology : a discipline in action / Merrill Singer, University of Connecticut, Hans A. Baer, University of Melbourne, Debbi Long, RMIT University, Alex Pavlotski, Aukland University.
Description: Third Edition. | Lanham : ROWMAN & LITTLEFIELD, [2019] | "Second edition 2012"—T.p. verso. | Includes bibliographical references and index. |
Identifiers: LCCN 2018048811 (print) | LCCN 2018051819 (ebook) | ISBN 9781538106471 (electronic) | ISBN 9781538106457 (cloth : alk. paper) | ISBN 9781538106464 (paper : alk. paper)
Subjects: LCSH: Medical anthropology.
Classification: LCC GN296 (ebook) | LCC GN296 .S57 2019 (print) | DDC 306.4/61—dc23
LC record available at https://lccn.loc.gov/2018048811

♾™ The paper used in this publication meets the minimum requirements of American National Standard for Information Sciences—Permanence of Paper for Printed Library Materials, ANSI/NISO Z39.48-1992.

Printed in the United States of America

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Preface

Students often do not bother with the prefaces to assigned textbooks, and for understandable reasons. The comments of authors about their book are not going to be on any test and it is the contents of the book, its ideas, concepts, theory, and examples, not the book as a product of author labor or the goals and intentions of authors, that is of most immediate concern to the student reader. We hope, in this instance, that these remarks capture some student attention because they deal with an important conceptual issue and provide a glimpse behind the curtain of book publishing. When the first (2007) and second (2012) editions of this book were published, the authors recognized that while medical anthropology was not the most fitting name for the field our book was introducing to students, it was the established and widely accepted term. Indeed, medical anthropology was a label that smuggled in a lot of problematic baggage. Since the last edition, however, our discomfort grew to the point that in 2016, along with our colleagues Debbi Long and Pamela Erickson, we published a paper in a leading anthropology journal entitled "Rebranding Our Field: Towards an Articulation of Health Anthropology." We feel strongly that the time has come to retitle medical anthropology using a more appropriate label that better reflects conceptual developments in the field. Our subfield, as shown clearly in the chapters of this book, has a wide range of concerns that includes a keen focus on biomedicine and other ethnomedical and health-care traditions. But the work of those we have come to call "medical anthropologists" addresses multiple other issues that are not specific to medicine or healing, including understanding health and illness within society and within the complex social and political-economic systems created by globalization. Consequently, we proposed the name "health anthropology"; this better describes the core issue that unites the subfield. We called for dropping the term medical anthropology because, as has long been recognized by many in the field, it is limiting and misleading, and reflects the hegemony of biomedicine at various levels. This renaming does not preclude health anthropologists working on or within biomedical, complementary, and alternative medical or indigenous medical settings, as reflected in the fieldwork of all of the coauthors of this book. It was our intention, however, that in its third edition this book be renamed *Introducing Health Anthropology: A Discipline in Action* to reflect the true breadth of the field.

This is where good intentions ran into marketing realities. Our book, and its title, were known, and *Introducing Medical Anthropology* was widely used as a textbook, the second-best-selling core text on the market in our topical niche. In our publisher's reasonable approach to their business it was not seen as a good idea to change the title of an established text. The book was seen as having instant name recognition for professors who might assign it in a class. Changing the title, the publisher felt, could cause confusion to teachers and to bookstores looking to order books for classes. As one reviewer of our proposed name change commented, "Why not continue to use it as a marketing tool to appeal to the

widest number of folks?" For these reasons, the title has not changed in the third edition, but the text has been updated to reflect our concerns on this matter. Throughout this text, if not on the cover, we use the terms "health anthropology," "health-oriented anthropology," and "health anthropologists."

One thing that has changed is the addition of Debbi Long and Alex Pavlotski as coauthors, which, based on their areas of expertise and fieldwork experience, allow us to further examine the wide range of issues investigated in a health-oriented anthropology.

One of the goals of the third edition of this book is to affirm that health-oriented anthropologists are very involved in the process of helping, to varying degrees, to change the world around them through their work in applied projects, policy initiatives, and advocacy. Not all anthropologists teach in a college or university—many are involved full time in directly applied work—and most who do teach medical anthropology courses are involved in application. Moreover, as the landscape of academia endures major transformations, with underpaid adjunct labor now increasingly replacing stable faculty positions, anthropologists must find creative ways to bring their skill-sets out of the ivory tower and into the world. Our subdiscipline both addresses specific health issues and analyzes them in their broader context. In other words, we seek to understand health-related issues and to use this knowledge in improving human health and social well-being.

A second goal of this book is a presentation of the fundamental importance of culture and social relationships in health and illness. Through a review of the key ideas, concepts, methods, and theoretical frameworks that guide research and application in health-oriented anthropology, the book makes the case that illness and disease involve complex biosocial processes and that resolving them requires attention to a range of factors beyond biology, including local systems of belief, structures of (often unequal) social relationship, the development and globalization of new technologies, and environmental and climatic conditions.

Finally, through an examination of the issue of health inequality, such as exposure to pesticides among farmworkers, unequal access to health care, the role of poverty in the spread of disease, or environmental degradation and environment-related illness, this book underlines the need for an analysis that moves beyond cultural or even ecological models of health toward a comprehensive biosocial approach. Such an approach integrates biological, cultural, and social factors in building unified theoretical understandings of the origin of ill health, while contributing to the building of effective and equitable national health-care systems. In this manner, health anthropologists have a broad vision of planetary health and seek to be part of a collective process aimed at creating a healthier world for both humanity and the biosphere. The ultimate goal is prevailing over the social causes of disease, the structures of social injustice that diminish well-being, and the social forces driving environmental destruction, including human-created or anthropogenic climate change. For us, ultimately, bringing these issues into the classroom, especially in a time when health has become a daily issue of deep concern, is of far greater importance than the specific way the field is labeled.

About the Authors

Merrill Singer is professor in the Departments of Anthropology and Community Medicine at the University of Connecticut. Dr. Singer has published 290 scholarly articles in peer-reviewed journals and book chapters, and has authored, co-authored or edited thirty-three books. His research and writing have addressed syndemics, HIV/AIDS and STDs in highly vulnerable and disadvantaged populations, illicit drug use and drinking behavior, infectious disease, community and structural violence, and the political ecology of health, including the health consequences of climate change. Dr. Singer has been awarded the Rudolph Virchow Professional Prize, the George Foster Memorial Award for Practicing Anthropology, both the AIDS and Anthropology Research Group's Distinguished Service Award and its Clark Taylor Professional Paper Prize, the Prize for Distinguished Achievement in the Critical Study of North America, and the Solon T. Kimball Award for Public and Applied Anthropology from the American Anthropological Association.

Hans A. Baer is Principal Honorary Research Fellow in the School of Social Political Sciences at the University of Melbourne. He has published twenty-one books and some 190 book chapters and articles on a diversity of research topics, including Mormonism, African American religion, sociopolitical life in East Germany, critical health anthropology, medical pluralism in the United States, United Kingdom, and Australia, the critical anthropology of climate change, and Australian climate politics. His most recent books are *Democratic Eco-Socialism as a Real Utopia* (2018), and *Urban Eco-Communities in Australia: Real Utopias or Market Niches?* (with Liam Cooper, 2018).

Debbi Long is senior lecturer in global studies at RMIT University (Melbourne, Australia). She is a critical health anthropologist and a pioneer of hospital ethnography in Australia. She is an experienced health ethnographer, having undertaken fieldwork in Turkey, Swaziland (eSwatini), and in a variety of contexts in Australian public hospitals, including maternity, spinal, intensive care, and dialysis units. She has worked as a consultant in clinical organization and management on projects including quality improvement, patient safety, behavior change, and in industrial relations contexts. Recent research has included family violence education and compensation industry analysis. She has taught at undergraduate and postgraduate levels in anthropology departments; international development programs; medical, nursing, and allied health programs; and in indigenous foundation and support programs.

Alex Pavlotski is teaching fellow at Auckland University, New Zealand, and an honorary research fellow at Latrobe University in Australia. He is an anthropologist and graphic artist. Alex has conducted fieldwork with comic artists in

Japan, and in Australia with an urban LGBTQI+ community; with CEOs and accountants across the country; and with recipients of government disability payments. His PhD thesis was on the global movement subculture of parkour, a multisited ethnography undertaken across twenty-four cities in eight nations. His research interests are psychological and neuroanthropology, visual representation in ethnography, cross-cultural communication, masculinity in leadership research, organizational ethnography, systems design, mind-body connectivity, and reactionary identities.

Introduction to the Anthropology of Health 1

Healing requires a legitimated, credible and culturally appropriate system.
—Mildred Blaxter (2004:43)

Introduction and Overview

In this chapter, we begin the process of defining and differentiating health anthropology in light of the range of disciplines concerned with health. We initiate this introduction to health anthropology by presenting three case studies. We then address both the practical and theoretical work and contributions of health anthropology, differentiate health and illness as technical terms in the discipline, engage the issue of health inequality, review the history of health anthropology, and compare and contrast health anthropology with other health-related fields including discussion of multidisciplinary collaboration.

Encountering Health Anthropology

People who encounter the term *medical anthropology*, or, as explained in the preface, "health anthropology," for the first time often are puzzled by what it means. Is it the study of how medicine is practiced, what doctors, nurses, or traditional healers from other health-care systems actually do? Or is it the study of what it means and feels like to be sick? Perhaps it is the study of folk illnesses in different societies? Might it be the application of cultural knowledge to the actual treatment of diseases? All these questions, in fact, can be answered in the affirmative. Health anthropology addresses each of these issues—and far more. A starting premise of health anthropology is that health-related issues, including disease and treatment, how and why one gets sick, and the nature of recovery, are far more than narrow biological phenomena. These processes are all heavily influenced by environmental, political-economic, social-structural, and sociocultural factors as well. Consequently, health anthropology has developed a *bio-sociocultural approach* in its effort to address health as an aspect of the human condition.

To take one example, health anthropologists ask questions such as the following: Could we really understand the AIDS pandemic and respond to it effectively simply by studying the human immunodeficiency virus, its impact on cells of the body, and medical interventions designed to stop the virus from destroying the immune system? Would we not also need to know how to reach and effectively engage those who are at greatest risk for infection, to figure out the structural and situational factors that contribute to their involvement in risky behaviors, to know how much they know and what they feel about AIDS and how these factors influence their behaviors, and to determine whether the ways

we go about interacting with them in the community and in the clinic draw them closer or push them away from our treatment programs? In other words, beyond biology clearly there are critically important areas of knowledge in the ongoing fight against the now about forty-year-old AIDS epidemic. Now, if we think about the AIDS epidemic as a global problem, a global pandemic, with different routes of infection, different populations at risk, different beliefs and behaviors associated with HIV/AIDS in diverse settings, and different health-care systems in different parts of the world or even different parts of a single country, we begin to get an initial sense of why a social science like anthropology might—as it certainly has—have a significant role to play in addressing the AIDS epidemic. This is of no small importance; we know from available research that HIV/AIDS is destined to take a greater toll on our species, proportionately and in terms of absolute numbers, than the bubonic plague, smallpox, and tuberculosis combined. Consider the epidemic in South Africa, the country with the highest burden of HIV/AIDS in the world. As Didier Fassin (2007a:261) points out, "In one decade, the rate of HIV infection went from less than 1 percent to over 25 percent of the adult population and AIDS became the main cause of death for men and women between 15 and 49 years of age," with an expected drop in average life expectancy in the country of as much as twenty years. Moreover, the global pandemic has helped to shape the social, cultural, and health worlds of people all over the planet, whether or not they are always aware of it. Within the broader story of the devastating impact of HIV/AIDS, however, there are many differing local narratives that together comprise the complex mosaic of the pandemic. The work of health anthropologists has been part of that story in many places and the same is true for a vast array of other health issues. Moreover, health anthropology, while sometimes contributing to cultural explanations of HIV/AIDS risk behavior, has struggled in recent years to counter explanations that fail to address the fact that the global economy or capitalism and social inequality are the primary driving forces in the epidemic (Hlabangane 2014).

■ ■ ■

Three Case Studies in Applied Health Anthropology

Coping with Cystic Fibrosis

The Reynolds family has two children. Carl is five and Stuart is seven. The younger of the two boys has cystic fibrosis (CF), the most common fatal genetic disease in the United States. Cystic fibrosis causes the body to produce a thickened form of mucus that clogs the lungs, leading to repeated bacterial infections and increasing lung damage. While the median age of survival among CF sufferers has been rising, most people with the disease do not live very far into adulthood before they succumb. Day-to-day care of a child with CF commonly falls on family members who must learn to cope with both a painful prognosis and the demands of responding to the patient's menacing symptoms, including pounding on the sufferer's chest and back for at least thirty-five to forty minutes

at a time, two to four times a day, to dislodge mucus. Some burdens fall particularly hard on the siblings of children with CF. Deana Reynold, Stuart's mother, notes one of these burdens that he must endure: "When Carl's sick, all the phone calls are, 'How's Carl?' Everybody who sees Stuart [says], 'How's your brother doing?' And all the presents. Carl gets all the presents. It has to have some kind of effect on him [Stuart]." How (and how well) do families with a child with CF cope? What toll does the disease take on family relations and on the emotional well-being of family members? How are siblings affected by growing up with a chronically ill brother or sister? How can health-care providers most effectively communicate with families at various stages in the natural history of CF progression? Having previously studied children with cancer, health anthropologist Myra Bluebond-Langner (1996) set out to answer these critically important questions. For nineteen months, in the clinic and in their homes, she repeatedly interviewed and observed families that were recruited from the patient rolls of the Cystic Fibrosis Center of St. Christopher's Hospital for Children in Philadelphia. She also interviewed attending physicians and reviewed patients' medical charts. Like most anthropologists, she immersed herself in the lifeworlds of the people she was studying. Her field notes and taped interviews filled thousands of pages and numerous three-ring binders. In the end, after many months of data collection and careful analysis, she was able to answer the key questions that motivated the study. Additionally—and tellingly—she was able to use her findings to develop a set of useful guidelines for physicians to use in clinical intervention with families with a CF sufferer. As a result, physicians now have a clearer idea of how best to communicate with families and to assist them in coping with the difficult challenges they face and the weighty burdens they must bear. Like the work of many other health anthropologists, this work by Bluebond-Langner has helped to make a positive impact in the tangled and often confusing world of health and illness. Addressing conflicts, miscommunications, and other problems in doctor-patient relationships as well as patient access to high-quality, culturally appropriate health care are central issues in health anthropology. But there are many other concerns as well.

The Bone Crusher

Dengue, the most prevalent mosquito-borne viral disease on earth, is found in more than one hundred countries and territories around the world, primarily in tropical and subtropical environments of Latin America, the Caribbean, and Southeast Asia, although a U.S. outbreak occurred in Hawaii in 2001. Since then, however, locally acquired cases of dengue have begun to appear in several geographic areas in the United States and researchers fear that the disease could be gaining a significant foothold on the U.S. mainland.

Current estimates are that each year fifty million to one hundred million people are infected with dengue when they are bitten by either the *Aedes aegypti*, the mosquito that also transmits yellow fever, or *Aedes albopictus* mosquitoes.

Mosquitoes become infected when they bite people who are infected, and, in turn, they subsequently transmit the infection to other people that they bite. In Southeast Asia and in most of Latin America and the Caribbean, the disease is pandemic, meaning that it is now firmly entrenched in the population and spreading. Malaysia has been particularly hard hit; thousands of people fall victim each year to this disease colloquially known—because of the fearsome joint pain it causes—as the "bone crusher." Other symptoms include stomach pain, headaches, nausea and vomiting, pain behind the eyes, and body flushes. In a more intense and even more frightening form, known as hemorrhagic fever, the sufferer's gums, nose, and internal organs bleed.

A number of health anthropologists have worked on preventing the spread of dengue. Karl Kendall (1998), for example, developed a strategy that involves studying and utilizing local health beliefs and practices in the development of community health campaigns about dengue in El Progreso, Honduras. In Kendall's approach, the first step in raising community awareness of effective prevention involved conducting in-depth interviews and surveys with community members to assess what they think and believe about dengue, its routes of infection, and the strategies they use to prevent becoming sick with the dreaded disease. This information was used to frame a locally meaningful education campaign designed to raise community awareness of the insects that transmit dengue, including effective pest control measures. This culturally sensitive project proved to be effective in reducing the populations of dengue-carrying mosquitoes, lowering rates of infection.

Sara Crabtree and colleagues (2001) built on this approach in the prevention of dengue in two communities in Malaysia. Like Kendall, Crabtree and coworkers began their work with a study of community knowledge, attitudes, and behaviors related to the disease. They also conducted focus groups with four different subgroups—women, youth, men who were heads of families, and village leaders—in an area that had not yet been hard hit by dengue. Through this research, it was found that the communities lacked much awareness of mosquito-borne disease transmission; they did not associate getting sick with being bitten by mosquitoes. Consequently, while they were available, people did not make much use of mosquito nets or spray repellents. The team then organized a set of three-day workshops designed to train volunteers to conduct a needs assessment on how to prevent dengue in their local communities. Under the guidance of the researchers, these individuals then carried out a door-to-door survey in their local communities. Researchers then worked with the needs assessment staff in translating findings into a strategic set of recommendations for practical, achievable activities to reduce mosquito populations. With the support of local leaders, actions based on these recommendations, such as burning accumulated rubbish, cleaning water containers, and identifying and eliminating breeding sites, were implemented to lower mosquito populations, a goal that was achieved in both participating communities. The health anthropologists involved in this project believed that this

success was due in large part to the initial assessment to ascertain community concerns, mobilize locally generated prevention ideas, and involve community members in all phases of the prevention initiative.

Despite the efforts described here, dengue continues to spread in the world, as do a range of old, new, and renewed diseases that were once controlled but are again spreading out of control (see chapter 6). From the fight against AIDS to the reduction of sexually transmitted diseases, health anthropologists, with their unique approach to understanding health and disease in terms of the interaction of human biology with social and cultural factors, are often on the front lines of infectious and other disease prevention as well as of the development of culturally and structurally appropriate and hence often more effective approaches to care. Although not all health anthropology projects are effective and success might be achieved at a much lower level of effect than would be desired, health anthropologists can point to a strong track record of making useful contributions to improving health, usually at the local level but sometimes even more broadly. In no small part, this is because a core realization of health anthropology is that disease vulnerability, including vulnerability to dengue, "is a relationship. It is always dependent upon cultural, social, and economic factors" (Nading 2017).

Pesticide Poisoning

The World Health Organization, a technical agency of the United Nations, estimates that there are more than a billion agricultural workers in the world, most in developing countries. Ethnographic research on the health of agricultural workers—specifically workers who work the fields by hands—are subject to "segregation . . . into a hierarchy of perceived ethnicity and citizenship" that produces significant social and economic inequality and leads to "displacement, migration, sickness, and suffering" (Holmes 2013:182). In the world inhabited by agricultural workers, their immediate supervisors, land owners, policy makers, and the wider society, the social and health inequalities suffered by agricultural workers are "considered normal, natural, and justified" (Holmes 2013:182) by ideologies of racialized biology. Rather than the consequences of unjust government policies that allow very low wages and limited occupational health protections, exploitation by employers who are dependent on very cheap labor for their profits, and a health-care system that, by blaming workers' culture and behavior rather than structural factors for their health problems, becomes inadvertently complicit in the functioning of a harmful social order.

Various studies have shown that one of the health problems commonly faced by agricultural workers is poisoning due to exposure to dangerous pesticides; indeed, they are the sector of society most likely to suffer health consequences from the powerful commercial poisons sprayed on food and ornamental crops to limit plant pests. Not only are those who work in agriculture at risk, but so are their spouses and children. Poisoning occurs because pesticide sprays are caught in the wind and drift into adjacent fields where people are working or into areas where

they and their families live, because workers are sent to work in fields in which pesticide has recently been applied, and because workers pick up pesticides on their clothing and other possessions, including their food containers, and bring them home unaware of potential risk. Even if exposures are limited, pesticides accumulate in the body, so that repeated contact increases risk for health-threatening outcomes. One of the most commonly used groups of pesticides, organophosphates (OPs), can be taken into the body through breathing, through ingestion, and through skin exposure. Organophosphates are known to damage nerves by reducing the availability of acetylcholinesterase, a necessary enzyme found at nerve endings. Organophosphate poisoning can produce rashes, nausea and vomiting, body fatigue, loss of consciousness, shock, and even death.

One OP pesticide, chlorpyrifos, has the distinction of being a neurotoxin that has been shown to be particularly dangerous to infants and young children. Approximately five to ten million pounds of the chemical, most of it produced by Dow AgroSciences, are applied to agricultural crops in the United States annually. A review of the pesticide by Environmental Protection Agency researchers during the Obama administration led to a recommendation that, given the threat it posed to children, chlorpyrifos be banned. The review found that residues of chlorpyrifos on food crops exceed existing safety standards and that drinking water exposure to the pesticide also exceeds safe levels. With the election of the Trump administration in 2016 and the subsequent appointment of an unabashedly probusiness and antiregulatory advocate, Scott Pruitt, to head the agency, the decision was made not to ban the pesticide. Opponents of the decision, health anthropologists among them, assert that the job of the agency, which increasingly has adopted a position of regulatory laxity, is to protect the health and safety of people, not the profits of corporations (Kolbert 2017).

Existing protections for farmworker health are limited. In 2002, for example, the Pesticide Action Network North America and a group of collaborating organizations issued a report called *Fields of Poison* based on data on pesticide poisoning collected by the California Department of Pesticide Regulation. The report found that farmworkers face a two-sided threat from pesticides: first, the existing set of regulations designed to protect them from harmful exposure to toxic chemicals is woefully inadequate to really provide safeguards against acute pesticide exposure, and, second, even the existing laws are weakly enforced. To address this issue, Thomas Arcury, Sara Quandt, and a team of colleagues (2005) recruited a group of nine farmworker households in North Carolina and Virginia for participation in an intervention study called ¡La Familia! Reducing Farmworker Pesticide Exposure, funded by the National Institute of Environmental Health Sciences. The research team conducted in-depth interviews with agricultural landowners and agricultural extension workers, completed interviews on beliefs about pesticide exposure and safety among primarily Latino farmworkers, collected samples in the homes of farmworkers to detect the presence of OP pesticides on household furnishings, and carried out urine

tests of farmworker adults and their children to assess body metabolite levels, which reveal whether OPs are present in the bodies of study participants.

These researchers found high levels of OP metabolites, which are the by-products of OP exposure, in the members of all of the households they studied, and all households had at least one member with especially high levels. Moreover, families that had carpeted homes but lacked a vacuum cleaner had higher-than-average OP metabolite levels. Bathing patterns also were linked to OP metabolite levels. As a result of their findings, this research team was able to identify specific policy changes that were needed to reduce farmworker exposure to OPs, including ensuring that all rented farmworker dwellings have shower facilities and working vacuum cleaners, that all farmworker dwellings are built at a safe distance from agricultural fields, and that all farmworkers receive training in pesticide risks and handling. Reflecting on the ultimate goals of their study, they conclude,

> Providing farmworker families (as well as all Americans) with safe and affordable housing will reduce their exposure to pesticides. This is not an instance of "blaming the victim" for exposure to pesticides, and attempting to address a systematic health disparity by educating those exposed to pesticides. Rather, it is an effort to build the capacity of farmworkers to defend themselves and to demand safe housing for their children. (Arcury et al. 2005:50)

Notably, most farmworkers live in countries with far fewer resources and weaker laws to protect workers than is the case in the United States. Pesticides produced in the United States, however, are shipped around the world, and anthropologists have observed them being applied by hand by workers who had received little or no information about how deadly they can be if not handled properly. For health anthropologists who work with agricultural populations, there is much work to do to help them protect themselves from occupational threats to their lives and well-being. In this instance, part of the problem is social inequality and the prevailing structure of power relations in society, such as the making and enforcing of laws that favor one social class, ethnic group, or gender over another. Indeed, health anthropologists have found that social relationships, such as those between ethnic groups, and social structures that determine access to resources and other things of value are a fundamental factor in health generally.

Although pesticide poisoning is a significant threat to the health of farmworkers, it is neither the only one nor the only one that has been effectively studied by health anthropologists. For example, Sarah Horton and Judith Barker (2010) have examined the issue of severe dental caries among the children of farmworkers in the Central Valley of California. They report that poor early oral health can have enduring effects both on children's physical development, including malformation of oral arches and crooked adult teeth, and on their emotional development, as a result of social stigmatization as young adults. This research examined the role played by inadequate diet as well as other anthropogenic environmental factors in the development of what they refer to as *stigmatized*

biologies. For example, these researchers present the case of Jorge, a young man who has "borne the marks of his lack of insurance as a child all his life. A star athlete and popular high school senior, Jorge feels his one social vulnerability is his stained and crooked smile" (Horton and Barker 2010:213).

Moreover, they reveal how market-based dental health insurance systems—and lack of access to insurance coverage among many farmworkers—contribute to enduring negative health effects. For this analysis, they use Margaret Lock's concept of *local biology* and recognition of the plasticity of biology to show how biology, rather than being a static or uniform phenomenon, in fact differs across groups as a result of factors like culture, diet, and the impress of a human-made environment and as consequence of differential social access to prevention care and treatment. Thus, they argue that market-based health-care systems create embodied differences between groups of people in society that both reflect and reproduce a structure of social inequality. Investigations of this sort affirm the value of a bio-sociocultural model in health anthropology.

■ ■ ■

Practical and Theoretical Contributions of Health Anthropology

The cases described here suggest an answer to the question, "why have a health anthropology?" The answer is this: because health anthropologists, using anthropology's traditional immersion methods for studying human life up close and in context, as well as the discipline's holistic picture of the human situation, a traditional disciplinary concern with understanding things from the insider's point of view and flow of experience, and an applied orientation to human problems, can make an important difference in the world. Revealing the nature of this difference is, as noted in this chapter, one of the main goals of this book.

While a primary emphasis of this book is on the practical contributions of health anthropology, the theoretical contributions of the discipline are equally important and guide the application of health anthropology in addressing particular health-related issues. Theory in health anthropology addresses questions such as the following: What determines health and illness? How and why do societies vary in their health-care systems, illness beliefs, and illness experiences? What role does culture play in treatment outcome? These questions are also addressed in this book.

Clarifying the Culture of Health and Illness

Beyond its initial goal, a second goal of this book is a presentation of the fundamental importance of culture and social relationships in health and illness. Through a review of the key ideas, concepts, methods, and theoretical frameworks that guide research and application in health anthropology, the book

makes the case that illness and disease involve complex biosocial processes and that resolving them requires attention to a range of factors beyond biology, including systems of belief, structures of (often unequal) social relationship, and environmental conditions.

Culture is a concept that has been central to anthropology throughout its 130-or-so-year history. In the past, anthropologists tended to think of culture as an established pattern of influential beliefs, practices, norms, and values that were shared in a social group and passed down and learned across generations to create unique configurations, the kind of differences among the ways of life and temperaments of people travelers report as they visit different destinations. These differences are found not only in behavior and ideas, but also in the ways people perceive and organize their worlds, their symbols, their sense of normal or proper behavior, and much more.

Over time, the somewhat fixed view of culture described above gave way to a more processual understanding of culture as an existing, discernable structure but one that is continually remade, rethought, and reinforced through social interaction. Moreover, anthropologists have come to see that a vital aspect of culture is its role in making life meaningful, purposeful, and understandable for people, although the meanings, purposes, and understandings people derive from their cultures differ across societies or even across sectors of the same society (e.g., the rich vs. the poor, men vs. women, dominant vs. subordinate ethnic or sexual identity groups, the disabled vs. the able-bodied, doctors vs. patients). Increasingly, in a globalizing world, where ideas, images, commercial products, technologies, corporations, people, and diseases move rapidly across national borders and local sites, cultures are changing. This is in turn leading to arenas of broadly shared behaviors and interconnectedness (e.g., global cell phone or internet use, global youth culture), sometimes referred to as the flattening of culture, as well as emergent forms of local difference that reflect the culturally specific ways global connections are mediated on the ground in different societies. Understanding these sociocultural changes and the forces driving them are issues high on the agenda of contemporary anthropology.

Health Inequality

Finally, through an examination of the issues of health inequality, such as the routine exposure to pesticides among farmworkers, on the one hand, and to environmental degradation that causes environment-related diseases, on the other, the book underlines the need for going beyond cultural or even ecological models of health toward a comprehensive, biocultural health anthropology. Such an approach integrates biological, cultural, and social factors in building unified theoretical understandings of the origin of ill health while contributing to the development of effective and equitable national health-care systems (e.g., Rylko-Bauer and Farmer 2002). In this manner, health anthropologists seek to be part of a collective process aimed at creating a healthier world for both humanity and the biosphere and thereby prevailing over widespread patterns of health and social injustice and environmental destruction.

Defining Health Anthropology

There is no simple definition of health anthropology because health anthropologists are involved in so many different issues and kinds of work. Stemming from this range, any easily crafted definition falters because it leaves out as much as it holds in. In effect, this whole book is designed to define health anthropology. Generally, however, health anthropologists are engaged in using and expanding many of anthropology's core concepts in an effort to understand what sickness is; how it is understood and directly experienced and acted on by sufferers, their social networks, and healers; and how health-related beliefs and practices fit within and are shaped by encompassing social and cultural systems and social and environmental contexts. In this multifaceted task, health anthropologists take a page out of Shakespeare's comedy *The Merry Wives of Windsor* in defining their domain of research: "Why, then the world's mine oyster, / Which I with sword will open." In other words, health anthropology is concerned not with a single society or with a particular health-care system but rather with health issues throughout the whole world and even through time. Our swords, so to speak, are strategies of research that are based on fieldwork and related methods that are close to the experience of those being studied. Further, our armory includes the application of knowledge gained through research in addressing pressing health issues, especially among populations marginalized by existing structures of power.

While recognizing the fundamental importance of biology in health and illness, health anthropologists generally go beyond seeing health as primarily a biological condition by seeking to understand the social origins of disease, the cultural construction of symptoms and treatments, and the nature of interactions between biology, society, and culture. Health anthropology "seeks to consider both the cultural and biological parameters of disease" as interacting biocultural processes (Joralemon 2017:11).

Also, health anthropologists tend not to accept any particular health-care system, including Western biomedicine, as holding a monopoly on useful health knowledge or effective treatment; rather, we see all health-care systems—from advanced nuclear medicine or laser surgery to trance-based shamanic healing or acupuncture—as cultural products, whatever their level of healing efficacy and however efficacy is defined within particular healing traditions.

Culture and Biology

Health anthropologists seek to understand and to help others see that health is rooted in (1) cultural conceptions, such as culturally constituted ways of experiencing pain or exhibiting disease symptoms; (2) social connections, such as the type of relations that exist within the family or within society and the encompassing world political and economic system generally; and (3) human biology, such as the threat of microscopic pathogens to bodily systems and the body's immune responses to such threats. In pursuing these lines of inquiry, health anthropologists are especially concerned with linking patterns of disease, configurations of

Pain and Bioculturalism

Bioculturalism refers to the significant interactions that take place between biology and culture in health and illness. Consider the issue of pain. In childbirth, a baby with a comparatively large head pushes its way through a small birth channel, a process that often produces "intense labor pains" among women giving birth in the United States. While these pains are expected, they are not accepted as tolerable, and the medical administration of painkillers is commonly demanded, sometimes vehemently so. In Poland, by contrast, labor pains are not only expected but also accepted, and painkillers are not normally requested. What accounts for these differences? While pain on one level is biological, part of a bodily communication system composed of nerves that ensures urgency in limiting bodily damage (e.g., pulling one's stray finger from the fire), health anthropologists have argued that pain expression and experience can be understood only in a cultural context. Culture teaches us how to think about, experience, and respond to the sensation of pain.

Health anthropology research on the intense and chronic pain associated with temporomandibular joint dysfunction (TMD)—diseases of the joint connecting the jawbone to the skull—is instructive (Eaves et al. 2015) While not debilitating, TMD pain is aggravated by routine and otherwise enjoyable activities of daily life, including smiling, eating, laughing, and kissing. Multiple interviews with a sample of TMD sufferers in Portland, Oregon, and Tucson, Arizona, showed that they tried to minimize pain through constant vigilance to avoid flare-ups, while seeking to remain stoic in the face of pain when it occurred. Further, participants attempted to avoid being perceived as someone who wants attention because of their pain. These goals often led to avoiding social activities, hiding pain in public places, and enduring some pain to diminish being perceived as unfriendly. Rather than being biologically determined, these TMD pain management behaviors appear to reflect core values of contemporary neoliberal market-based medicine, which emphasizes a "culture of self-help, privatization, and individualization of health" (Dutta 2016: 68).

In assessing a disease, it is important to consider how biology and culture interact. For example, it is likely that a devout Jew or Muslim could be made violently ill by being forced to eat pork, while consuming this meat is considered very satisfying among most people in New Guinea. The same could be said of monkey brains, worms, or hamburger, depending on the cultural traditions of the people involved. Cultural beliefs and practices are very involved as well in the spread and reaction to many infectious diseases, such as sexually transmitted diseases. Disease rarely acts as an independent biological force whose health impact is everywhere the same. Rather, disease expression is shaped by cultural values, beliefs, and expectations. Cultural practices may inhibit or promote disease spread, and, conversely, disease can significantly mold culture.

health-related beliefs and behaviors, and healing systems with cultural systems, social hierarchies, and biosocial relationships. Consequently, health anthropologists have tended to look at health as a "biocultural and biosocial phenomenon," based on an understanding that both physical and sociocultural environments in interaction determine the health of individuals and of whole populations. Central to our understanding is that while biology impacts social and cultural patterns, such as the ways recent emergent infections have affected our understanding and attitudes about the world around us, the reverse is also true. Culture and social organization impact biology. To cite but one example, research has shown that average adult heights consistently increase with improved economic conditions and decrease as economic conditions decline across time and location. This is tied to the impact of cultural and social forces on access to food and other resources. In other words, "economic conditions [are] indelibly written into and onto the body" (Goodman 2013:365). Human biology exhibits a high degree of flexibility and is as much a product of culture as it is of genetics. This insight has taught anthropology that the most accurate way of thinking about humans is conceiving of them as biocultural beings in which there is constant impactful interaction among our biology, our culture, and our surrounding environment.

Some health anthropologists, those who call themselves critical health anthropologists (including the authors of this book), stress what they call a critical biocultural model, one that is especially concerned with investigating the role of social inequality in shaping health, health-related experience and behavior, and healing, issues we will explore throughout this book. Whatever their theoretical perspective (and several alternative perspectives exist within the discipline), however, health anthropologists tend to have an applied orientation; they are concerned with putting their work to good use in addressing real and pressing health-related problems in diverse human communities and contexts (e.g., Rylko-Bauer et al. 2006).

History of Health Anthropology

The Straits Expedition

Interest in health-related issues within anthropology dates to the very origins of the discipline as a field-oriented social science. In 1898, three British researchers—W. H. R. Rivers, fellow physician C. G. Seligman, and Alfred Haddon—initiated the historic Cambridge University Torres Straits (Australia) expedition, one of the earliest anthropological research projects. Various kinds of data on indigenous Australian peoples were collected during this expedition, including information on traditional healing beliefs and practices. Rivers, who some see as the father of health anthropology, used data from the expedition to refute the popular notion among Western physicians and other worldly observers of the era that the ethnomedical practices of non-Western societies were "a medley of disconnected and meaningless customs" (Rivers 1927:51). Instead, he argued, ideas and practices around health and healing found in preliterate societies constitute internally coherent structures of cultural beliefs about the causes of disease. Now ninety years old, this perspective on healing systems around the

world has been abundantly supported by subsequent research in health anthropology and has guided numerous examinations of the nature of the relationship of health beliefs and practices to the encompassing cultural context in which they are found, such as faith healing in the context of folk Christianity, as in the case of El Santuario de Chimayó, a Catholic healing church in New Mexico, shown in figure 1.1. This pattern is not limited to folk traditions but includes the dominant healing system, biomedicine, within modern Western society and globally. For example, in the 1989 book *The Woman in the Body: A Cultural Analysis of Reproduction*, an influential text in health anthropology discussed in chapter 3, Emily Martin shows that the dominant metaphors in biomedicine for the delivery of babies come from the arena of industrial production. In her examination of medical textbooks, Martin found that reproduction is talked about and taught to students using analogies and concepts borrowed from factory production. In this biomedical cultural model of birth, (1) the doctor is portrayed as the manager of the laboring process, much like a factory foreman who oversees and regulates the production process; (2) the uterus is portrayed as the machinery of reproduction; (3) the mother is talked about as a kind of laborer, and hence she is said to be "in labor" during the birthing process; and (4) the baby is the product. This way



FIGURE 1.1 Crutches left behind at El Santuario de Chimayó Church in New Mexico, where pilgrims come to get "healing soil" from a hole in the church floor. Photo by Pamela Irene Erickson.

of viewing reproduction, Martin found, is often in conflict with the views held by women who have given birth or who are about to give birth, creating the potential for misunderstandings and conflicts in childbirth.

W. H. R. Rivers and Beyond

In Rivers's book on ethnomedicine, *Medicine, Magic, and Religion* (1927), a volume that has been called the "symbolic totem of medical anthropology" (Landy 1977:4) because it incorporates health-related issues into the agenda of anthropology, Rivers also maintained that non-Western ethnomedical traditions and biomedicine constitute completely separate entities. Indigenous healing, he asserted, is characterized by manipulation, using spells and other rituals, of assumed magical connections among objects and beings in the world and by beliefs about the actions of supernatural beings (e.g., spirits) in causing and curing illness. Biomedicine, by contrast, is grounded in natural laws and scientific principles. Ever since, health anthropologists have grappled with understanding similarities and differences in healing systems cross-culturally. The distinctiveness of biomedicine has been intensely debated within the field. While some health anthropologists have accepted it as a standard by which to assess the efficacy of other healing systems, other health anthropologists have sought to show the following:

- Biomedicine in its understandings and practices reflects its culture of origin no less than any other ethnomedicine.
- Folk healing systems around the world incorporate practices (e.g., bone setting) that are based on observation and practical reason as well as conceptions and protective behaviors that are believed to be derived from natural laws, such as the hot and cold properties of foods or other elements in nature, and not just on magical or religious belief.
- Many of the ideas and practices found in biomedicine are not, in fact, based solely on natural laws and scientific principles.

Pearl Katz, in her 1999 book *The Scalpel's Edge: The Culture of Surgeons* (which is described in greater detail in chapter 2), portrays modern surgery as an elaborate set of rituals that function to limit ambiguity, uncertainty, and error. The process of "scrubbing," for example, involves very strict behaviors, within precisely designated time periods, such as washing each hand in particular ways for a specified amount of time. While it is possible to significantly reduce the presence of pathogens on one's hands and arms through careful cleaning, the precisely defined rules of scrubbing appear to serve also to reduce anxiety about not being sufficiently germ free to avoid infecting the internal organs of the patient. By closely adhering to the rituals of surgery, the confidence doctors and nurses need to undertake a very delicate and potentially disastrous activity, such as cutting open and in some way changing a patient's internal environment, is heightened.

As a result of such studies, biomedicine is now analyzed within health anthropology as one among many ethnomedical systems around the world,

though a rather distinctive one because of its global reach, broadly recognized efficacy in handling many health problems, and ties to the international scientific community and dominant social classes worldwide. One consequence of this shift in perspective is that the medical anthropological perspective "helps us step back from medicocentrism by refusing to take [bio]medical categories and their various meanings as givens" (Saillant and Genest 2007:xxvii), including the biomedical tendency toward biological reductionism. As a result as well, diversity within biomedicine is more apparent, varying in its practice and perspectives across countries and within different medical specialties.

Rudolf Virchow

Returning to the origin of health anthropology, Otto von Mering (1970:272) contends that the emergence of the field dates to the late 1800s when Rudolf Virchow, a renowned German pathologist who is often regarded as the father of social medicine because of his interest in how the distribution of health and disease mirrors the distribution of wealth and power in society, helped establish the first anthropological professional society in Berlin. Virchow was an important early influence on Franz Boas, the father of American anthropology, while he was affiliated with the Berlin Ethnological Museum from 1883 to 1886. Virchow helped to lay the theoretical foundation for the approach known as critical health anthropology, which is described below.

Erwin Ackerknecht and William Caudill

During the 1940s and 1950s, Erwin Ackerknecht, a German émigré, emerged as an important figure in the evolution of health anthropology. Using field accounts from anthropologists working in various societies, Ackerknecht sought to develop a systematic understanding of healing beliefs and practices that emphasized that (1) healing behaviors and ideas tend to reflect the wider cultural traditions of the society in which they develop and that (2) whatever their ability to improve the health of patients, healing systems reinforce core cultural values and structures and contribute to maintaining the status quo by controlling social conflict and deviance.

The Postwar Period

After World War II, a growing number of anthropologists began to turn their attention to health-related issues, especially applied ones. Thus, the very first review of what we now call health anthropology was produced by William Caudill eight years after the end of the war and was titled "Applied Anthropology in Medicine" (Caudill 1953). Caudill's paper marked two important developments in the evolution of health anthropology: (1) the entry of a number of anthropologists after the war into international health development work and (2) the hiring of anthropologists to work in medical schools and clinical settings as teachers, researchers, administrators, and, in some cases, clinicians. Involvement in

the international health field actually began during the 1930s and 1940s within the context of British colonialism—an era during which the delivery of Western health services was seen as central to a larger effort to administer and control indigenous peoples.

Health Anthropology and National Development

Cora DuBois became the first anthropologist to hold a formal position with an international health organization when she was hired by the World Health Organization in 1950. Within a few years, Edward Wellin at the Rockefeller Foundation, Benjamin Paul at the Harvard School of Public Health, and George Foster and others at the Institute for Inter-American Affairs had joined the pool of anthropologists involved in seeking to address health-related aspects of technological development around the world, including the negative health consequences of ill-planned development projects. Another set of anthropologists became involved in efforts to facilitate the delivery of biomedical care to people in developed nations and underdeveloped sectors of technologically advanced nations. Alexander and Dorothea Leighton, for example, became involved in the Navajo-Cornell Field Health Project. This applied initiative created the social role of "health visitor," a Navajo paramedic and health educator who acted as a "cultural broker" or community liaison between the white-dominated health-care system and the Navajo people.

The Discipline Is Born

Whatever its diverse roots, as a distinct and labeled subdiscipline of anthropology, health anthropology has a relatively short history that can be traced to the period after World War II (Roney 1959). Organizationally, health anthropology began with the formation of the Group for Medical Anthropology in 1967, with Hazel Weidman, an applied anthropologist, as chair. Ultimately, this fledging organization became the now quite robust Society for Medical Anthropology, a formal section of the American Anthropological Association since 1972, with Dorothea Leighton, a psychiatrist-anthropologist, serving as its first president. As the field continued to grow and diffuse, health anthropology associations formed subsequently in other nations as well. Today, as described throughout the rest of this book as well as in several others that bring together the range of work done in the field (e.g., Brown 1998, Sargent and Johnson 1996), health anthropology continues to grow in size and diversity of work as well as in terms of the impact of the efforts of health anthropologists on a wide range of health-related issues internationally. As in any field, there have been disagreements and debates as well as frustration about structures that sometimes restrict the influence of health anthropology on health-related policy making, and there have been worries about how to have a bigger role in shaping public health discussions to include issues of culture, social organization, and the voices of socially marginalized populations. As reflected in this book, even the issue of what to call our field of study remains open to debate.

The Relationship of Health Anthropology to Anthropology and to Other Health-Related Disciplines

Health Research and the Subfields of Anthropology

Traditionally, anthropology in the United States, at least, has comprised four subfields: social and cultural anthropology, biological anthropology, archaeology, and linguistics. In other countries, such as the United Kingdom and Australia, the discipline has focused more narrowly on social and cultural anthropology. While some have proposed that health anthropology constitutes the fifth subdiscipline of anthropology, others see it as sitting on the cusp between cultural and biological anthropology and incorporating elements of each in its understandings of health, illness, and healing. An alternative perspective places health anthropology on the cusp between general or theoretical anthropology and applied anthropology. In this book, as we have stressed, a primary focus is on the applied side of health anthropology as a biocultural approach to addressing health issues.

Health anthropology has become one of the largest topical interest areas beyond the four primary subfields of the discipline. One of the current debates within health anthropology is the degree to which it has had an impact in shaping the ideas and orientation of the wider field of anthropology, including to what degree health anthropology has developed its own theories and to what degree it has merely borrowed and applied those found elsewhere in the discipline.

Health Anthropology and Paleopathology

Health anthropology has developed an important interface with archaeology. In that archaeology in the United States is often defined as part of sociocultural anthropology or at least as part of one of the four subfields of anthropology, the bridging nature of health anthropology is particularly apparent in the field called "paleopathology," which is the study of diseases in the past and, particularly, in prehistoric times. This research is accomplished through the study of human fossil remains recovered through archaeological excavations. Buikstra and Cook (1980) delineate four stages in the development of paleopathology: (1) the descriptive period during the nineteenth century, focusing on bone abnormalities; (2) the analytical period during the early twentieth century, when an attempt was made to interpret bone abnormalities; (3) the period between 1930 and 1970, when the field became more specialized, drawing on fields such as radiology, histology, and serology; and (4) the current phase beginning around 1970, when the field became considerably more interdisciplinary and incorporated genetic studies, including examination of microbial DNA in bone and soft tissue in order to diagnose disease in past humans.

As Roberts and Manchester (1995:9) point out, one of the important limitations of paleoanthropology is that the populations being studied are long dead, and consequently the number of cases being studied from any population are limited, a mere "sample of a sample of a sample." As a result, it is difficult to generalize to the whole population from which the cases are drawn.

Applied Anthropology in Action

Health anthropology straddles the line between theoretical and applied anthropology. Applied anthropology is the application of anthropological theories, concepts, and methods to solving problems in the world. Applied anthropologists work in many different areas, from A for aging, such as solving problems in the isolation aging people often feel in Western society, to Z for zoos, as seen in work done to determine how to use zoos to educate visitors about pressing environmental issues. While it is sometimes asserted that applied anthropology grew out of theoretical anthropology, the reverse is in fact the case. In the United States, for example, some of the individuals who first found employment as anthropologists worked for the Bureau of American Ethnology (BAE). The BAE was set up in 1879 as a policy research arm of the federal government to provide research needed to inform congressional decisions about the American Indian population. The bureau's first director, W. J. McGee, proposed that the organization focus on what he called "applied ethnology," which led to a series of descriptive reports that were prepared for policy makers. Despite this fact, there have been strong tensions at times between applied and theoretical anthropology. Some theoretical anthropologists identify the discipline's mission as understanding diverse pathways in human social life. They assert that to use anthropology to formulate planned social change violates two basic disciplinary principles: cultural relativism, which proscribes judging any given society by the values of any other, and avoidance of research bias, which includes, among other concerns, the scientific standard of minimizing opportunities for data contamination because of the value commitments of the researcher. Both of these principles suggest that intervention is, by definition, not anthropology. Applied anthropologists counter that science does not exist in a social vacuum and that its fundamental purpose is to apply its findings to solving human problems and improving the quality of human life. Because we live in a world of cultural contact and resulting social change that often leads to pressing problems and extensive human suffering, applied anthropologists feel obliged to apply their skills and understandings to solve real-world challenges. In recent years, these divisions have begun to break down, and older animosities about what is the proper role of anthropologists in society have begun to fade into history.

Many nurses have found anthropology and specifically health anthropology to be of great utility in their work. Consequently, numerous undergraduate nursing programs require their students to take a course in cultural anthropology, and many nursing schools now teach courses in health anthropology and related subjects. Indeed, many of the people teaching these courses are nurses who have obtained graduate degrees in health anthropology. Some anthropologists have argued that, in contrast to biomedical physicians, who tend to be "disease oriented," nurses are inclined to be more "person oriented." With this orientation, developing a health anthropology perspective appears as a useful and practical one for many nurses.

A specific example of one type of work that applied health anthropologists

do is seen in the efforts of Dan Small (2016) at Insite, a medically defensible but culturally controversial program located in Vancouver in western Canada. Insite is a legal, medically supervised facility for the injection of illicit drugs that is designed to reduce the potential harm (e.g., drug overdose) associated with drug use. Small's work began with helping to identify cultural barriers to implementing this controversial healthcare program. Once barriers were determined, a cultural initiative was developed to shift dominant social narratives about addiction that often create barriers to programs designed to address the health needs of a socially denigrated population. Clients of Insite are able to choose their injection equipment from on-hand sterile supplies and report which drug they are injecting. This information assists with implementing individualized emergency intervention if required. Drugs are not provided at the facility; users must bring their own. Participants are assigned to one of twelve injection booths, each of which contains a stainless-steel countertop and mirror to facilitate the user having a clear view of their chosen point of injection. There is also a nursing station for medical support staff, who provide crisis intervention, referrals to treatment, addiction counseling, detox, and supported housing referral. Rather than just study health risk among injection drug users, Small's work, as an advocate of harm reduction, focuses on lowering risk in a nonjudgmental, rapport-building manner.

Despite this shortcoming, paleopathology has much to teach us about diseases and related health problems of antiquity, including congenital defects, traumatic injuries, infectious diseases, metabolic and nutritional disorders, degenerative diseases, circulatory problems, caries, and even cancer. For example, bone spurs in the knees, toes, and spines of ancient Mesoamerican women strongly suggest that they spent long hours grinding maize to make flour.

Paleopathology can also contribute to the solving of medical mysteries. A case in point is the cause of death of Pharaoh Tutankhamun, more popularly known as King Tut, a relatively minor ruler historically but today the best-known member of ancient Egypt's dynastic line. On November 4, 1922, archaeologists discovered and began excavating King Tut's thirty-three-hundred-year-old subterranean royal burial tomb in Valley of the Kings, the royal necropolis of Egypt's eighteenth through twentieth dynasties, located on the west bank of the Nile River opposite the modern city of Luxor. The richness of the funerary artifacts in the tomb, and the discovery of the pharaoh's mummy, led to extensive press coverage and the rapid rise of King Tut in the popular imagination worldwide. A subsequent autopsy of the Tutankhamun mummy concluded that the pharaoh had died around the age of nineteen years. Speculation soon developed about the cause of death of the young ruler. Initial conjecture focused on the king's skull because of a small dense spot, suggesting the possibility of brain trauma inflicted by a murderous blow to the back of the head. According to Bob Brier, author of the popular book *The Murder of Tutankhamen* (1998), in light of the "violent, unstable times Tutankhamun ruled in, and [based on] forensic evidence . . . murder and intrigue emerged as the best explanation" (xix) of the pharaoh's untimely

death. Additionally, based on several peculiar features of statues, paintings, and other ancient depictions of King Tut and members of his family (e.g., suggesting androgynous characteristics among males as well as notably elongated heads and fingers), numerous speculative reports have been published about various diseases allegedly suffered by King Tut. Among these are Marfan syndrome, a genetic disorder that leads to above-average height, long slender limbs, and long fingers and toes; Wilson-Turner X-linked mental retardation syndrome, a congenital disease characterized by mental disabilities, childhood obesity, and enlarged male breasts; and Fröhlich syndrome, a disease caused by tumors of the hypothalamus that produces mental retardation and retarded sexual development. In part to test these various hypotheses, Zahi Hawass, an archaeologist and Egypt's secretary general of the Supreme Council of Antiquities, led an international team that carried out a set of now widely publicized anthropological, paleopathological, radiological, and genetic analyses of a group of mummies as part of the King Tutankhamun Family Project. While discounting various earlier hypotheses about King Tut's death, this study concluded that a likely cause of death was the occurrence of multiple health problems including malaria (based on findings of ancient malaria DNA in King Tut's mummy), juvenile aseptic bone necrosis or Köhler disease II (involving loss of blood to the king's left foot, causing bone death and collapse), and a fracture of his left femur that was unhealed at the time of death. King Tut's death, in other words, was a consequence neither of intrigue and ancient rivalry nor of a rare and disforming genetic disorder, but rather was caused by the adverse interaction of several diseases and trauma-induced infection (Hawass et al. 2010). Most important, this study suggested the possibility that malaria, alone and in interaction with other diseases and conditions, was an important factor in the health of all sectors of ancient Egyptian society.

Health Anthropology and Epidemiology

The AIDS Connection

Beyond its ties to subfields within anthropology, health anthropology also has an important connection with epidemiology, a discipline concerned with the patterns and spread of disease, including containing outbreaks of disease. At the U.S. Centers for Disease Control and Prevention (CDC), a branch of the national Public Health Service, scientists monitor the appearance and spread of disease all over the country and beyond. In the AIDS epidemic, for example, CDC researchers attempted to understand what was causing the disease, how it was spread, and how it could be stopped. There has been a long but in some ways not always deep collaboration between epidemiologists and health anthropologists. Well over three hundred health anthropologists, for example, have worked on some aspect of the global AIDS epidemic, often in close partnership with epidemiologists and other researchers and interventionists (Bolton and Orozco 1994), although far fewer work on the epidemic today. In this work, as Patricia Whelehan notes in her 2009 book *The Anthropology of AIDS: A Global Perspective*, health anthropologists have addressed various bio-sociocultural factors, ethical issues, cultural factors and social reactions, gender roles and relations,

sociopolitical and socioeconomic influences, and transnational and intergroup differences in the nature of the impact and range of responses that have developed to the pandemic. Health anthropologists have played various roles in the global AIDS pandemic. To cite one example, they have worked closely with outreach workers who locate and recruit hard-to-reach at-risk individuals, such as injection drug users, for interviews by the anthropologists on patterns of HIV risk and/or for participation in prevention interventions. Research of this sort has led to discoveries by health anthropologists and their colleagues of a range of behaviors beyond direct syringe sharing that can spread the virus that causes AIDS as well as sexual risk associated with injected and noninjected drug use (such as the use of crack cocaine). In addition, health anthropologists were involved in identifying social and behavior contexts in which risky behavior is most frequent, assessing the role of social networks in the spread of HIV infection and the importance of exposure to violence in risk behavior, and have played key roles in prevention research on syringe exchange.

Broader social and political issues have also been addressed by anthropologists in the AIDS pandemic. In her work in South Africa, for example, Ida Susser (2009) focused on women's avenues of hope, sources of resilience, and mounting community activism in response to the toll of AIDS. Central to Susser's objective was the development of a detailed account of women's on-the-ground responses to the entwined effects of disease, gender discrimination, and the world economic system through which wealthy countries promote deprivation, social suffering, and the spread of disease in poorer nations.

Notably, Melissa Parker (2003:179), who studied unsafe sex among gay-identified men in the back rooms of pubs, clubs, and saunas in London, has argued that anthropologists should "draw upon their ethnographic expertise and help to design interventions which target people and places with the explicit intention of promoting social change and saving lives."

Health anthropologists who continue to work in the HIV/AIDS epidemic have a strong focus on treatment and its impact on populations.

Cultural Epidemiology

In his book *Epidemiology and Culture*, James Trostle (2005), an anthropologist long concerned with building collaboration between epidemiology and anthropology, argued for the creation of a "cultural epidemiology" that would integrate the anthropological concept of culture into the set of explanatory variables used by epidemiologists to explain disease. Cultural beliefs and practices about condom use, for example, contributed to the spread of HIV infection. Because many people link condom use to casual intercourse with people other than their main partner, at times it has been difficult to convince them to use condoms in their primary relationships. Another example of how culture impacted HIV prevention was provided by Alexander Rödlach (2006) in his study of AIDS-related beliefs in Zimbabwe in sub-Saharan Africa. Rödlach found that many people in Zimbabwe wondered if the disease was spread either by sorcery or because of a conspiracy, such as a plot by the U.S. government to punish people of color. Importantly, he argued that studying such beliefs, which are widespread in the world, is not an

attempt to contribute to the construction of an *exotic other* (i.e., viewing people who are culturally different as bizarre or less intelligent). Rather, he emphasized that developing a better understanding of cultural beliefs about the nature and spread of AIDS "yields theoretical insights into how people explain and react to health problems, [which] in turn benefits health programs" (2006:13). Sorcery and conspiracy beliefs, he notes, are prevalent in social groups that have endured suffering, through poverty, discrimination, and disempowerment, at the hands of outsiders. Such beliefs, in other words, are culturally constituted *idioms of distress* that express the harmful experience of feeling helpless and unable to control much about one's life (and hence suspecting a nefarious outside agent as the source of one's painful lived experiences). A similar pattern has been described in accounting for the conspiracy beliefs that emerged during the 2014–2016 Ebola epidemic (Ayegbusi et al. 2016, Ngade et al. 2017). Health anthropologists use the term *idioms of distress* to describe the culturally specific and symbolically rich frameworks of understanding through which people both experience and articulate social discontent and suffering. In addition, they have come to recognize that even standard biomedical terms such as diabetes can come to be used popularly as idioms of distress when they refer to health conditions that reflect health and social disparities and express the lived experience of social suffering.

Health Transitions

Within epidemiology, the term *health transition* is used to label broad changes that have occurred within particular historic phases in the health profile of populations and the primary causes of mortality. The first health transition in human history began about ten thousand years ago when the development of agricultural modes of food production and more sedentary communities led to a marked increase in acute infectious diseases (often involving the adaptation of the disease-causing pathogens of domesticated animals such as cattle to human hosts and resulting in human epidemics). A second health transition occurred between the late eighteenth and early twentieth centuries with the rise of industrial economies. This economic and social transformation involved the restructuring of environments, industrial pollution of the air, water, and land, and changes in life conditions. These sweeping changes contributed to the growing importance of chronic diseases such as heart disease, stroke, cancer, and diabetes and what are termed *behavior problems*, such as substance abuse. Ultimately, especially as a result of public health measures supported eventually by biomedical interventions, a drop also occurred in many developed nations in rates of lethal infectious diseases as well. However, one effect of the second health transition is that poorer countries now suffer from what has been called the "triple burden" of acute disease, such as diarrheal disease; chronic disease, such as cancer; and behavioral pathology, such as the global spread of illicit drug injection. At the same time, health-care systems have been changing. One force pushing such changes has been the imposition of what are called *neoliberal reforms* by international lender institutions (i.e., banks that loan money for development to poorer countries), such as the World Bank. In poorer nations, such as Mongolia, as health anthropologist Craig Janes (Janes and Chuluundorj 2004) found during his research

there, lender-endorsed steep cuts in government investment in the health sector and the transformation of health services from a government-provided benefit to a purchasable commodity resulted in a significant drop in the quality and quantity of health services available in rural areas while restricting access to services that were still in operation. Janes discovered that women in particular were vulnerable to these changes, leading to increasing rates of both poor reproductive health and maternal mortality. Similar negative health effects of what has been called *structural adjustment* (i.e., changes in government policies and structures demanded by international development loan providers such as the World Bank and International Monetary Fund) have been described by health anthropologists working in various locations around the world. As Mark Nichter (2008) indicates in his book *Global Health: Why Cultural Perceptions, Social Representations, and Biopolitics Matter*, structural adjustment has led to a proliferation of service-providing nongovernment organizations (NGOs) in underdeveloped nations. One result is that in places such as Haiti there are multiple, uncoordinated, and competing health programs governed by the dictates of foreign funding institutions. This change has contributed "to the undoing of government services" (Nichter 2008:139) and the weakening of governments in developing nations, pushing them, as has occurred in places such as Haiti, Peru, El Salvador, and several countries in southern Africa, beyond the ability to meet the basic public health needs of their citizens (M. Singer 2010a, Smith-Nonini 2010). A study that reviewed a decade of the effects of structural adjustment policies in eight Latin American countries, for example, found that, in an effort by lender banks to make interest from the development loans they had made, health and social expenditures designed to meet human needs were cut drastically, leading to ever-widening disparities between the health and well-being of the rich and that of the poor (Petras and Vieux 1992). Many of the failures of neoliberal "development," in terms of human suffering and the promotion of poverty, in what amounts to a form of economic neocolonialism, are now acknowledged by some leaders of the World Bank.

In the last decades of the twentieth century, a third health transition began, and it continues into the twenty-first century. This transition involves the rapid appearance and global spread of new infectious diseases such as HIV/AIDS, SARS, Ebola, Zika, and avian influenza and the development of new levels of virulence and drug resistance in older infectious diseases such as tuberculosis. In the third health transition "poverty combines with urbanization and the dissolution of traditional family structures to intensify [health] challenges" from infectious agents, processes of globalization (e.g., rapid travel and the global shipment of commodities) enable the brisk movement of infectious diseases around the world, and microbial adaptation to medicines weakens the effectiveness of drugs like antibiotics to control infectious diseases (Barrett 2010). At the same time, adverse synergistic interaction among diseases (infectious, chronic, behavioral) has promoted the appearance and adverse health impact of disease "syndemics" that further increase the total health burden of affected populations (Bulled and Singer 2010, M. Singer 2010b, M. Singer et al. 2017). These factors are significantly shaping contemporary global health and appear likely to do so well into the future.

Illness and Help-Seeking Behavior

One of the important issues on which health anthropologists have focused is how local sociocultural factors come into play in "monitoring the body, recognizing and interpreting symptoms, and taking remedial action . . . to rectify the perceived abnormality" as well as "adherence to therapeutic advice, changes in treatment regimens [e.g., switching healers], and evaluation [and reevaluation over time] of therapeutic efficacy and outcome" (Christakis et al. 1994:277). In short, health anthropologists are interested in the ways that culture helps to shape the "illness behavior patterns," including help-seeking actions, found in a society. Of note, illness behavior is mediated by a sufferer's subjective interpretations of the meaning of experienced symptoms. However, these interpretations are not solely idiosyncratic but rather are influenced by wider cultural understandings of illness and the comments and actions of the sufferer's social network. For example, in "several areas of Africa where malaria is endemic, seizures [which in biomedicine are recognized as symptoms of malaria infection] are viewed as a 'folk illness' largely unrelated to malaria and are often attributed to supernatural causes" (Nichter 2008:74). Illness behavior is impacted by various factors, including gender (thus men and women often "get sick" differently) and socioeconomic status. Also, illness behavior in a society is not static; as a society changes, illness behaviors change as well, including patterns of use of health services. The global distribution of pharmaceutical drugs, for example, has contributed to the use of these commercial laboratory remedies in ways that go far beyond their intended purposes and patterns of use, such as crushing antibiotic capsules and applying them to wounds or the emergence of local healers who administer individual antibiotic injections. Documenting and assessing emergent patterns of illness behavior and their causes is a role that health anthropologists are playing in the realm of epidemiology. More broadly, health anthropologists seek to move beyond the examination of individual suffering to social suffering.

Health Anthropology and Public Health

Closely related to epidemiology is the discipline of public health, a field that is concerned with assessing and improving the quality of health of the general populations as well as that of especially vulnerable and at-risk subgroups therein. Health anthropologists contributed to public health by ethnographically examining disease-promoting behaviors in social context. As social scientists who attempt to "elucidate how and why people do what they do," health anthropologists have been able to contribute to disease prevention and control (Brown et al. 1996:198). Failure to understand the factors that shape the behavioral decisions people make, the frames of understanding they bring into play, and the influence of social relationships on their actions is a contributor to the disappointments of public health interventions. Another role for health anthropologists involves the issue of public trust in public health messages. While public health workers are motivated by a commitment to improving public well-being, the populations they seek to assist may not share their enthusiasm or trust their intentions. Resistance to public health measures, for example, has been described for populations at risk

for polio vaccination and Ebola infection. In the case of Ebola, people in some locales even came to question whether public health workers were bringing the potentially fatal disease rather than trying to curtail it (Cohn and Kutalek 2016). This pattern underlines the importance of ethnographic research on the drivers of fear and public health distrust in the development of grounded understandings of public responses to and following epidemic outbreaks. Health anthropology research on the drivers of public distrust informs efforts to improve public health communication, the nature of public health messages, the public presentation of public health interventions, and the role of community involvement in the context of infectious disease epidemics (Abramowitz 2014).

A number of methodological and conceptual developments in public health in recent years have created new opportunities for health anthropology to play a part in public health discussions and interventions. Among these new developments are (1) the growing public health and medical concern with health inequities, (2) the increasingly recognized need to enhance the cultural competence of health-care providers, (3) the emergence and growing influence of the community-based participatory research model, (4) the diffusion of and increasing funder emphasis on evidence-based interventions in public health, (5) the mounting demand for translational research that allows the findings and knowledge produced by health-related studies to shape health intervention efforts, and (6) the spiraling interest in what have come to be called complementary and alternative medicines. At the same time, a research method developed within anthropology—rapid ethnographic assessment—has spread into public health, creating additional opportunities for health anthropology to have an impact on health issues. Each of these increasingly important arenas of public concern in which health anthropology directly or indirectly has played a role is described next.

Health Inequity or Disparity

Health inequity or disparity refers to the significant differences in the health profiles (i.e., the distribution of diseases and wellness) across human populations, social strata such as social classes, or other segments of the population, such as rural areas compared to urban settings. Additionally, health disparities researchers are concerned with why disparities in health exist. In the United States, for example, as Grace Budrys (2003:39), observes in *Unequal Health: How Inequality Contributes to Health and Illness*, from a health standpoint "being at the top of the [social class] heap is a lot better than being on the bottom. . . . There has been an explosion of research indicating that social class is a powerful, and arguably the most powerful, predictor of health." Critical health anthropologists, for example, have examined the roles that economic structures (e.g., wage constraints) and health insurance companies (e.g., ability to pay for coverage vs. actual medical need) play in limiting people's ability to make healthy choices in their lives (Fletcher 2017).

The nature and importance of health inequities will be examined in greater detail in chapter 4. Suffice it to note here that it is an important health arena in which a number of health anthropologists have become quite active, such as Sabina Rashid, who works among the poor of Dhaka, Bangladesh (figure 1.2).



FIGURE 1.2 Children from poor families in an urban slum in Dhaka, Bangladesh. Photo by Sabina Faiz Rashid.

Community-Based Participatory Research

There are various approaches to research on health issues in specific populations and communities. In the "unilateral research model," researchers based at universities or research centers design a research project based on their understanding of the key issues and questions. The research agenda in this approach is almost completely if not completely determined by the researchers in terms of their conceptions and interests. After a project is largely designed, the researchers may contact and subcontract with a community organization to recruit participants from the community who will be interviewed in the planned study. Often strapped for funding, community organizations accept such subcontracts even though they may at the same time resent not having much voice in planning the study, including a say in what from the community's perspective are the key health issues in need of research. Another approach to research is called the "collaborative model." In this approach, while researchers initially conceptualize a study, they then contact one or more organizations based in the community of concern and invite them to participate in fleshing out the details of the study. While the community organization(s) participates at some level in the study, the bulk of project direction, decision making, and funding is still centered in the university or research institute. Both unilateral and collaborative research projects are very common and can be found on most university campuses, often funded by federal research grants or private health foundations.

As contrasted with both of these approaches to research, community-based participatory research (CBPR) is based on a full partnership between researchers and community representatives and organizations, from project conception to

completion, including publication of findings. Community-based participatory research grew out of recognition that traditional population-based biomedical research methods lack authentic community involvement and often result in community alienation from research and researchers. In some cases, researchers have come to be defined as exploiters of communities rather than as their natural allies. Consequently, in CBPR-guided projects, the community plays a key role in setting the research agenda. In this, communities are guided by their pressing need for specific health-related knowledge, which can be used in addressing community health problems and in making ongoing decisions about the direction of the research. As the concept of CBPR has developed and its value recognized, various efforts have been made to establish guidelines for successful and mutually satisfying participatory research projects.

Health anthropologists have contributed to the development of this alternative orientation to research. For example, the Institute for Community Research in Hartford, Connecticut, a community-based organization that has been led by anthropologists since its founding, established the Youth Action Research Institute to promote the use of a participatory action research model among youth. Central to the institute's work have been projects that involve minority youth in ethnographic research on issues of concern to the youth (e.g., AIDS and substance abuse), including the primary health and social problems faced by their communities. Anthropologists train youth in ethnographic methods and help them identify problems for action research. The goals of this project were personal growth among the youth participants, the development of positive peer norms, and the assessment of community health needs.

Diffused Evidence-Based Intervention

There has been a strong effort in recent years to accelerate the movement of scientifically proven (i.e., evidence-based) intervention models, such as training programs to assist people in avoiding communicable diseases. Rather than funding organizations to implement intervention models that have not been evaluated, funder institutions, such as the CDC, began emphasizing the use of models that have proven to be effective. The CDC's Diffusing Effective Behavioral Interventions project, for example, has helped implement eighteen different research-based infectious disease prevention models through community-based organizations, health departments, and other prevention providers across the United States. These were designed for specific populations, such as injection drug users, sexual and romantic partners of injection drug users, men who have sex with men, heterosexuals at high risk, people living with infection, and homeless and runaway adolescents. To prepare community organizations to successfully implement these intervention models in their respective communities, the researchers or those they have trained are called in to provide guidance and technical assistance to the frontline people who would use the models in day-to-day prevention work.

Translational Research

Translational research is a specialized type of research carried out with the intention of improving the flow of knowledge from research into action in public

health or some other arena of social intervention. Health practitioners and policy advocates, for example, have complained that the findings of behavioral research are slow in finding their way to public prevention efforts. For example, anthropologists and others who conduct ethnographic research with injection drug users have observed that they engage in a number of behaviors during the consumption of drugs that might lead to infection. One of these behaviors involves several drug users pooling their money to purchase a packet of an illicit drug, such as heroin or cocaine, and then mixing it with water, using one of the participant's syringes and the unit gauge on its barrel to measure the right amount of water to allow an equal distribution of the dissolved drug. If the syringe that is used for this purpose contains an infectious agent, then the virus may be flushed into the container (such as a bottle cap) that is being used for drug mixing and drawn up by all the individuals who are sharing the drugs from that container. In this way, all these individuals may be exposed to infection. An examination of the messages given by prevention programs to drug injectors to protect themselves, however, often urges them only not to "share needles." Because individuals who share drugs may never share a needle, they may falsely believe that they are protected from infection when in fact other behaviors in the drug preparation and use process are causing disease transmission.

As a result of examples like this, those involved in translation research have urged social scientists not only to publish their findings in professional journals, most of which are not widely read outside of academic and research settings, but also to take specific steps to ensure that their research is readily accessible to, relevant for, and understood by those who work in prevention programs or who make health policy decisions. A fundamental issue facing the health intervention is "how to translate the research findings for more widespread practice" (Sloboda 1998:203). The need for readily accessible and usable knowledge is particularly great in intervention efforts targeted to populations that are harder to reach and harder to retain in public health programs, a task that has been taken on by a number of health anthropologists. At the same time, interventions models proven effective at the local level, if they are to have a significant impact on disease, must be scaled up beyond the pilot level. For example, with regard to AIDS prevention programs in Africa, Binswanger (2000:2173) argues, "In most of Africa, there are examples of excellent HIV/AIDS prevention, mitigation, and care projects. These projects reach only a small fraction of the population however. Like expensive boutiques, they are only available to a lucky few." This too is a type of work that is well suited to the skills and interests of health anthropologists because "translating recent research advances into communities at risk . . . requires thoughtful adaptation to meet the needs of the community effectively with emphasis on social, economic, and cultural heterogeneities" (McGarvey 2009:242). It also requires creativity and a keen awareness of the cultural life and social context of target populations (figure 1.3).

Complementary and Alternative Medicine

There has been a fairly dramatic shift in recent years in the way policy makers, health-care providers, and the general public view healing systems beyond the dominant biomedical approach. One indication of this change is the remaking



FIGURE 1.3 AIDS educators in Havana, Cuba. Photo by G. Derrick Hodge.

of the American pharmacy. Today the shelves of the average major chain pharmacy in the United States and Australia (although not everywhere in the world; see figure 1.4) are filled with over-the-counter alternative medicines, such as St. John's wort, echinacea, and black cohosh, that in the past could have been found only in specialty health food stores or alternative markets. Moreover, in 1992 the U.S. Congress established the Office of Alternative Medicine, which in 1999 became the National Center for Complementary and Alternative Medicine, one of the twenty-seven research institutes and centers that make up the National Institutes of Health. The mission of the center is to promote scientific exploration of promising nonbiomedical healing practices and to disseminate research-based information on these practices to the public and health professionals. While some have argued that the new interest in complementary and alternative healing systems is driven, at least in part, by a desire to subordinate them to the dominant biomedical system (Baer 2004), there clearly has been a significant change in the way they are viewed and their place in Western societies if not globally. We will return to the issue of complementary and alternative medicine later in the book as we explore alternative health approaches. Here, suffice it to say that the rise in interest in complementary and alternative medicine rests on and grew out of a long history of anthropological study of nonbiomedical healers and their approaches to the treatment of illness.

Rapid Ethnographic Assessment

The rapid ethnographic assessment approach to research, which has now been adopted by health promotion institutions around the globe, was first fully



FIGURE 1.4 Modern pharmacy in Cuba lacks the many over-the-counter commercial "folk medicines" now found in U.S. pharmacies. Photo by G. Derrick Hodge.

described in the late 1980s and early 1990s by anthropologists Susan Scrimshaw and Elena Hurtado. Rapid ethnographic assessment is designed to bridge the gap between science and public health practice and policy by allowing the swift movement from community-based research on health or other pressing social issues to interventions based on research findings. While ethnographic research traditionally was both a labor- and time-intensive approach, Scrimshaw sought a methodology that would take advantage of close-up ethnographic insight in a community of concern without requiring the customary year or more of social immersion and extensive documentation of behaviors and events characteristic of traditional anthropological fieldwork. Rapid approaches do this by preparing researchers to build on existing knowledge of a target community and to implement several strategies for the rapid development of rapport, such as the use of local staff to collect information in their own communities and collaboration with local community organizations, in the assembly of highly focused data on very specific issues and problems. For example, in Project RARE (Rapid Assessment, Response, and Evaluation), anthropologists and other researchers used focus groups, quick-intercept interviewing, concentrated field observation, and social mapping to identify and describe gaps in the existing array of AIDS prevention programs and services in several dozen cities across the United States. In the Hartford RARE project, for example, the research team, which

consisted of people from the local community led by several community-based health anthropologists, found that late-night (midnight to 4:00 a.m.) sexual and drug use behavior was not being addressed by existing AIDS prevention efforts, resulting in a continued spread of HIV in the local population. As a result of the project, the city health department began to require late-night prevention efforts by some of the organizations that it funded to prevent the spread of AIDS in the city. The value of rapid research methods has been recognized in other fields as well, leading to the development of other types of accelerated assessment and evaluation models, such as rural rapid appraisal, rapid epidemiology, rapid disaster assessment, and rapid assessment of biomedical conditions. All these present emergent arenas of employment for health anthropologists.

Health Anthropology and Bioethics

Bioethics emerged as a new academic field during the 1970s and has quickly become an important force in science and medicine and, through the setting of health-related social policies and the rise of institutional review boards (IRBs), in health anthropology as well. The term *bioethics* can be defined as a branch of the field of ethics that is concerned with the establishment and application of standards and principles by which human actions within the arenas of health care, health-related decision making, and health research can be judged morally right or wrong. Many hospitals, for example, employ experts on bioethics to provide consultation on the treatment of terminally ill patients and to inform decision making regarding issues such as organ transplant, abortion, euthanasia, in vitro fertilization, and the allocation of scarce clinical resources. As Everett (2006:46–47) points out, while it is likely that, because of the kinds of research they do, health anthropologists would have something important to offer, "they have found it especially difficult to find a place within bioethics debates" that tend to be dominated by the fields of philosophy, law, and biomedicine. Health anthropologists, in fact, have sometimes been critical of bioethics because of a lack of sensitivity to cultural differences. Rayna Rapp (2000a:44), a health anthropologist who does ethnographic research on genetic counseling, for example, argues that bioethics is "self-confidently unaware of its own sociocultural context" and fails to consider whether the standards it develops reflect the values of non-Western populations. Thus, bioethics has emphasized the importance of respecting individual autonomy, free will, and self-determination and thus has opposed forcing patients or research participants to do things against their will or without their full consent. The problem, health anthropologists point out, is that the values emphasized in bioethics reflect the Western celebration of individualism, a moral stance that is not shared by cultural systems that emphasize collectivist models or rigid social hierarchies. The narrow application of Western ethical standards without sensitivity to alternative norms may be construed as ethical imperialism.

Moreover, bioethics has been used in the effort to set standards for ethical research in light of a past history of gross violations of the rights of human subjects in research. All universities and research centers now have IRBs that apply standards established for medical research to all forms of research involving

human subjects. According to these standards, all research that has potential risks for human participants, including research by health anthropologists, must be reviewed and approved by an IRB. A number of anthropologists have questioned the appropriateness of IRB review of ethnographic research on the grounds that it commonly involves the misapplication of standards that were established for biomedical and experimental research where life and death risks are not uncommon (Marshall and Koenig 1996). Further, there has been some concern that IRBs may require anthropological researchers to engage in behaviors that create rather than avoid ethical dilemmas, such as mandating that research participants sign informed consent forms that result in participants' names being part of a project's records. In the study of illegal behaviors, such as illicit drug use or prostitution, this may contradict the researcher's commitment to protect the confidentiality of study participants. Health anthropologists recognize that the kind of research they conduct commonly encounters perplexing ethical dilemmas and recognize the need for ethical principles to guide their research activities. Whether bioethics is the appropriate source for such principles is an issue of debate. Whatever the challenges, as discussed further in chapter 8, there is little doubt that health anthropology will continue to engage and develop in relationship with the field of bioethics (Lambeck et al. 2015).

Health Anthropology Theories

As is typical in science generally, health anthropologists understand the world in particular ways. One of the influences on how an anthropologist approaches issues of health or illness is the particular theoretical framework or school of understanding a health anthropologist uses. There are several such frameworks in health anthropology, although many individuals do not see themselves as adherents of any single perspective but rather take a more eclectic approach and allow the problem they are working on to shape the perspectives that they use. Other medical anthropologists consider themselves adherents or even advocates of particular points of view. Indisputably, however, the perspective they bring to their research will strongly influence the way a problem is approached, the questions that are asked, and the answers that are deemed sufficient and adequate. Among the primary perspectives found in health anthropology are medical ecology, the meaning-centered perspective, and critical health anthropology. These are introduced here and given fuller examination in chapter 5.

Medical Ecology

Rooted in both cultural ecology and evolutionary theory, this approach began with an emphasis on adaptation, defined as behavioral or biological changes at either the individual or the group level that support survival in a given environment, as the core concept in the field. From this perspective, health was seen as a measure of successful (or poor) environmental adaptation. In other words, a central premise of medical ecology initially was that a social group's level of health reflects the nature and quality of the relationships that exist at three levels:

within the group, with other groups, and with the physical environment. Beliefs and behaviors that improve health or protect societal members from disease or injury are adaptive. For example, there is the indigenous development of snow goggles that shield the eyes of Arctic dwellers from the damaging glare caused by the sun reflecting off ice and snow. This constitutes a small but important health-related cultural adaptation of the Inuit people, as do the fur clothing and ice houses of indigenous Arctic dwellers. Similarly, from the medical ecological perspective, behavioral complexes, such as medical systems, including everything from shamanistic healing of soul loss to biomedicine treatment of heart disease, can be viewed as adaptive sociocultural strategies for confronting illness and contributing to the ability of a society to survive or even thrive. Similarly, the ecological approach is concerned with understanding how, and how well, cultural and social formations address nutritional or other bodily needs. In recent years, as a result of dialogue with other perspectives, there has been movement toward the political ecology of health, indicating a recognition of the multiple ways the environment is shaped by the unequal structure of relations within societies.

Meaning-Centered Health Anthropology

The approach taken in medical ecology to the understanding of human biology and behavior, as an interactive set of adaptations to ecological and social challenges, makes a lot of sense to many health anthropologists. Yet others have questioned aspects of this approach on the basis that from the perspective of medical ecology, a disease, as is portrayed in science, is often treated as a natural object. Disease, in other words, is an object separate from human consciousness and human cultural interpretation of the world (B. Good 1994). In turn, medical systems come to be seen in medical ecology as utilitarian social responses to intrusive natural conditions. As a result of these premises, in medical ecology culture is absorbed into nature, and the work of health anthropology becomes analyzing the adaptive efficacy of health-related behaviors and beliefs. Lost in such understanding is a full appreciation of the human cultural/symbolic construction of the world we inhabit. Humans can experience the external material world only through their cultural frames and thus diseases, as they are known, through body sensations or observations and measurements, by sufferers and healers alike, are perceptions and sensations packed with cultural content. Even medical science and biomedicine do not offer culture-free accounts of the physical world; their understandings of disease or of the body also are cultural constructions. This is because both of these historically intertwined pragmatic ways of knowing the world emerged within particular cultural systems at particular points in the development of those systems, and they accept without thought or questioning many deep-seated cultural ideas and values derived from their encompassing cultural assemblage. For example, deeply embedded within and strongly supported by the day-to-day activities, theories, and organizational structures of biomedicine and medical science are Western cultural notions of (1) individualism, namely, that each individual is distinct and is responsible for his or her success or failure and self-improvement; (2) progress and the belief that history is a process of

steady social improvement; and (3) the responsibility of action or a belief in the appropriateness of changing the world to meet human needs and ourselves to meet cultural goals of maturity, wellness, and social success. Consequently, from the meaning-centered perspective, a goal of health anthropology is to "unpack" and analyze everything that makes up the health arena, from the experience of pain to the training and functioning of healers, as a set of systems for creating, experiencing, and communicating meaning in human life.

Some researchers have emphasized that a meaning-centered approach, in addition to its analytic utility, has practical value in addressing health issues. For example, a meaning-centered approach may have value in children's cancer treatment because it promotes the development of a way for caregivers to understand and respond to children's immediate experience of symptom distress, including the meanings and culturally influenced feelings children attach to particular symptoms throughout their illness trajectory (McClement 1998).

Critical Health Anthropology

During the early years of health anthropology's formation, explanations within the discipline tended to be narrowly focused on explaining health-related beliefs and behaviors at the local level in terms of specific ecological conditions, cultural configurations, or psychological factors. While providing insight about the nature and function of diverse medical models, the initial perspectives in the field tended to ignore the wider causes and determinants of human decision making and behavior. Explanations that are limited to accounting for health-related issues in terms of the influence of human personalities, culturally constituted motivations and understandings, or even local ecological relationships, some health anthropologists began to argue, are inadequate because they tend not to include examination of the structures of social relationship that unite (in some, often unequal fashion) and influence far-flung individuals, communities, and even nations. A critical understanding, by contrast, involves paying close attention to what has been called the "vertical links" that connect the social group of interest to the larger regional, national, and global human society and to the configuration of social relationships that contribute to the patterning of human behavior, belief, attitude, and emotion (Singer and Baer 1995). Consequently, what came to be called critical health anthropology (CHA) focused attention on understanding the origins of dominant cultural constructions in health, including which social class, gender, or ethnic group's interests particular health concepts express and under what set of historic conditions they arise. Further, CHA emphasizes structures of power and inequality in health-care systems and the contributions of health ideas and practices to reinforcing inequalities in the wider society. Moreover, CHA focuses on the social origins of illness, such as the ways in which poverty, unhealthy living and working conditions, discrimination, stigmatization, violence, and fear of violence contribute to poor health, as well as the political-economic and political-ecological sources of these causes of unequal health. The four authors of this book have been active throughout our respective careers in contributing to the critical health anthropology perspective.

From an applied perspective, critical health anthropologists seek to contribute to address questions such as: How are we to understand the nature of industries and governance structures that facilitate disease by polluting (or allowing the polluting of) the environment and furthering global warming (which constitutes a severe and multidimensional threat to human health) or other ecocrises? What drives an economic system that makes possible the manufacture and release of "killer commodities"—consumer products, such as those routinely advertised on children's television that are high in fat, sugar, or salt, that cause injury, illness, or death—onto the global market? What forces have pushed, and what kinds of social relationships support, the biocommodification of nature, including the production of genetically engineered food crops and invasive and exploitive biotechnologies, like the global systems that lead to the movement of transplantable organs and tissues from poorer countries to richer countries and poorer donors to richer recipients?

Contrary to what some have asserted, CHA does not take a "top-down approach" that denies people's ability to resist and fight back against oppressive relationships that harm their health. Rather, critical health anthropologists argue that lived experience and "agency," including individual and group decision making and action, are "constructed and reconstructed in the action arena between socially constituted categories of meaning and the political-economic forces that shape the context [and texture] of daily life" (Baer et al. 2003:44). In other words, people develop their own individual and collective understandings and responses to illness and to other threats to their well-being, but they do so in a world that is not of their own making, a world in which inequality of access to health care, the media, productive resources (e.g., land, water), and valued social statuses play a significant role in their daily options. At the same time, critical health anthropologists do not explain all experiences of body/self disorder as an expression of social critique or resistance. Often, in fact, illness experience and the way it is handled socially serves only to reinforce rather than throw open to question existing structures of power.

Additionally, while recognizing the fundamental importance of physical (including biological) reality in health, such as the nature of particular pathogens, CHA emphasizes the fact that it is not merely the idea of "nature"—the way external reality is conceived and related to by humans—but also the very physical shape of nature, including human biology, that has been deeply influenced by an evolutionary history of social inequality, overt and covert social conflict, and the operation of both physical power and the power to shape dominant ideas and conceptions in society and internationally through processes of globalization (Whiteford and Manderson 2000).

In the following chapters, we return to many of these ideas while emphasizing the ways in which health anthropology—guided by any of the perspectives described here—actively seeks to understand health, health beliefs and behaviors, and healing systems and practices across time and place and to use this information in addressing health problems, conflicts, and suffering in the world. We begin this process in the next chapter by examining what health anthropologists do, including the ways they approach problems, the kinds of problems they address, their collaboration with other health-interested disciplines, and their impact on the health arena.

Discussion Questions

1. A core question of this chapter is "why have a health anthropology?" What do you think?
2. Why is health far more than a biological or medical issue; what other factors need to be considered to understand both how and why people get sick and how and why they get better (or fail to do so)?
3. Why do health anthropologists tend not to accept any particular health-care system, including Western biomedicine, as holding a monopoly on useful health knowledge or effective treatment? Is this position justified?
4. What are some of the ways anthropologists have responded to dengue? With Zika and other spreading mosquito-borne diseases, what factors might contribute to a major mosquito-related U.S. epidemic?
5. Why are social relationships, such as those between ethnic groups, and social structures that determine access to resources and other things of value fundamental to health and disease?
6. Besides those mentioned in this chapter, can you think of any other cultural metaphors that express how we think and talk about health and disease?
7. How do anthropologists engage the issue of ethics? Why are ethics important in health anthropology?
8. What are the key distinguishing features of critical health anthropology? What do you think about this orientation?

What Health Anthropologists Do 2

The distinctive approach of anthropology to research is to go out and see what is actually occurring, and to talk to the people themselves.

—John Janzen (2001:18)

Introduction and Overview

In the previous chapter, a point we stressed is that health anthropologists are involved in a wide variety of health-related issues, places, and kinds of work. This chapter provides a closer examination of just what it is that health anthropologists actually "do" in these various settings. We begin by providing three additional cases of health anthropologists at work, including an exploration of the kinds of problems they address and how they go about applying health anthropology. Based on these cases, we then (1) discuss the range of issues health anthropologists study and the special focus they bring to their work because they are anthropologists; (2) describe the distinctive health anthropology approach to research within the broad, holistic vision of our field; (3) examine the methods used in health anthropology research; and (4) explore applied activities in health anthropology beyond research.

■ ■ ■

Three Settings, Three Case Studies, Three Health Anthropologists

Life and Death in Tanala

Studying Indigenous Healing

Sitting on the doorstep of Ranomafana National Park in the southern part of the island nation of Madagascar is the small Tanala village of Ranotsara (pseudonym). Located in the Indian Ocean off the southeastern shore of Africa, to the east of Mozambique, Madagascar is well known for its shimmering forests and seething mass of biologically unique and richly diverse plant and animal species, such as ancient and exotic baobab trees, exceedingly slow-moving chameleons with strangely rotating eyes, and loud packs of lemurs, including the very rare Golden Bamboo lemur. Sadly, the Republic of Madagascar also is known as a land of aggressive deforestation. Ranomafana National Park, in fact, was established in 1990 with private and public funding from the United States as well as from international entities such as the World Bank. It is administered primarily

by U.S. citizens with the express goal of preserving biological diversity and ecosystems by linking conservation with improved standards of living for the people who dwell in and near parklands.

When health anthropologist Janice Harper arrived in the village to begin her research in April 1995, about 180 people, about a third of them children and teens, were living there, divided among thirty thatch-and-tin-roofed homes, most of which were rusted and leaking. Surrounded by irrigated rice fields, the village constituted a manmade island nested on hard-packed reddish dirt. Harper had come to study the indigenous use of medicinal plants and nontraditional medicines, customary topics of health anthropology research, but right from the start events on the ground began to challenge and reshape her research plans.

The Cultural Construction of Hygiene

Harper initiated her research project with door-to-door introductions with all the villagers, followed by a set of "invasive but excessively polite demographic questions" (Harper 2002:166). Subsequently, she interviewed people about their health problems, understandings of health issues, and acquisition and use of forest and Western medicines, among other topics. She observed and participated in daily village activities and filled her notebooks with detailed accounts of what she saw and heard. As she lived among the villagers, witnessing their routines in the fields and around the village, and casually interviewed them about their use of healing plants and other mundane issues of day-to-day village life, Harper realized that the villagers' conceptions of health and cleanliness were different from her own, Western, middle-class ideas. Attempting to be ecologically sound, she tried to bury all her garbage, from crumpled papers (reflecting false starts in her writing efforts) to the tin cans and plastic bags from the used-up Western products she had carried with her to the field. The villagers, who produced little if any garbage, were astonished by the amount she produced—which, very likely, was far less per day than the average middle-class home in the United States—and they believed her to be very rich to be able to make so much garbage. Indeed, many of the things she discarded, such as used-up glass bottles, they coveted as useful containers. Everything else that she buried the pigs rooted up and scattered about. Washing was another issue. In a place where a bar of soap costs a full day's wage, water doesn't run from a tap, and the latrine consists of the spaces between the trees in the nearby coffee grove, keeping clean is a constant challenge that is complicated by the fact that daily labor—from sunrise to sunset—in the rice fields means standing in calf-deep mud that is thick with worms and feces, while work on the nearby hillside gardens involves toil in the blazing sun. Most people's clothes were tattered and their bodies scrawny but strong from daily manual labor. As for the children, Harper found that almost all had bloated bellies as a result of worm infestation, their skin was encrusted with lesions caused by scabies (a microscopic burrowing insect), their legs and arms were marked by large boils, their ears were leaking yellow pus from infections, and their bodies

were mostly underweight, and yet they were active and playful and filled the village with their laughter.

In the Company of Death

Harper also was struck by the frequency of sickness and death around her. Indeed, a woman who sought employment from Harper on the anthropologist's first night in the village had died by the following morning. During her fourteen months in the village, eighteen people died, a disturbingly high mortality rate of almost 100 per 1,000 population compared with just more than eight deaths per 1,000 population in the United States.

Her fieldwork ultimately ended as it began with the death of a villager. On the morning of her departure—under pressure from park officials not to express any criticism of the park or to imply that the park was a factor in village morbidity and mortality rates—Harper (2002:235) was awakened "not to the eerie cries of the lemurs," which had enchanted her mornings and nights in Ranotsara, but to "the ghostly cries of women wailing for the dead. Maily, a young woman in her twenties, had died during the night."

Writing Up

On her return to the United States, Harper wrote a book titled *Endangered Species: Health, Illness, and Death among Madagascar's People of the Forest* to document her findings and to show that while the people she studied use plants, barks, and roots from the forest to treat (but rarely cure) debilitating diseases that are readily cured with pharmaceutical medicines, the latter are usually inaccessible. Harper (2002:3) argued that "their continued reliance on the forest's botany for their health care is less conditioned by their 'culture' than it is by social inequalities that have rendered them cash poor." Even the forest at their doorstep is controlled by others, turning their social disparities into health disparities, an issue of keen interest to health anthropologists that is examined in chapter 4. Rather than existing in an exotic village with a traditional way of life sheltered by dense forest from the fast-paced and changing world system around them, the lives of the people of Ranotsara, Harper found, are shaped, although not narrowly so, by forces beyond their reach and certainly beyond their control. Harper seeks to reveal what can happen in contemporary development, including the ecological development movement, when people are not treated as part of the natural environment that is being protected, issues that are addressed more fully in chapter 5. To make her points, Harper relies not solely on her own research findings but also on an examination of historic records, economic reports, and population data.

Studying Surgeons

Problems at Meadowbrook

Many miles from Ranotsara and seemingly a world away from its humid green landscape lies Meadowbrook University Hospital (pseudonym) on the outskirts

of one of Canada's largest cities. With 800 patient beds, 120 of which are reserved for surgical patients, Meadowbrook doctors perform 15,000 operations each year. As a university hospital, in addition to providing health care to patients, Meadowbrook is a teaching institute and thus is home to a steady and changing flow of medical and other students seeking to become health-care professionals, and, further, it is a health-care research center oriented to increasing the fund of biomedical knowledge.

Despite its sanitized veneer of orderliness and control, Meadowbrook had a problem. The hospital began to receive urgent complaints from surgical residents in their final stage of training that the staff surgeons who were their teachers and supervisors were not spending enough time training them to be surgeons. In response, as noted in chapter 1, Pearl Katz, a health anthropologist, was invited by the chief of the Department of Surgery to carry out a study that would help resolve the problem. Katz realized that the study held the potential to be of even greater importance than the immediate needs of the Department of Surgery. Health anthropology, which views biomedicine as ethnomedicine—if of a special sort, namely, a medical system that has gained global importance and a dominant position in terms of other ethnomedical systems internationally—is keenly interested in understanding how biomedicine works, including its roles in society, its worldviews and underlying cultural models, social structures, and varieties across time and place.

Expect Rejection

When Katz (1999:16) began her study, she was "warned by friends and colleagues that [she] would have difficulty getting accurate information on what surgeons did." Katz's anthropological colleagues assumed that the surgeons, being privileged and, relative to patients and researchers at least, rather powerful, would limit her access to information, especially to unfettered backstage glimpses of their lives and behaviors. Certainly other researchers have had trouble gaining full access to biomedical practitioners. In her own study of surgeons, for example, health anthropologist Joan Cassell—a colleague who encouraged Katz to publish her findings—was once sternly asked by a surgeon after an operation to which a nurse had invited her to observe, "What's an anthropologist doing studying surgeons?"—to which she glibly responded, "Well, there were no other primitives left" (Cassell 1998:10). Cassell's experience notwithstanding, Katz found that the surgeons extended her an extraordinary level of trust and openness; they appeared to her to want to be understood.

Instead, Katz found herself challenged by her own attitudes toward surgeons. It was much easier for her to identify with patients and she was critical of the disparaging attitude surgeons expressed about the concerns of their patients. She was shocked to find that they referred to an especially frightened patient as the "beast" or the "colon." Only over time, after observing many operations, did Katz realize that she too was depersonalizing patients and why this

was a useful mechanism that enabled surgeons to "protect themselves against the [emotionally burdensome] experience of empathizing with the personal anguish of patients" (Katz 1999:8). Katz also found herself envying the power held by surgeons and came to realize that to some degree such envy fuels diatribes against the status of surgeons found both in the popular mass media and in some social science analyses.

Following Doctors

In her day-to-day work on the study, Katz focused intensely on six senior surgeons. With each of these key informants she spent approximately three continuous weeks, from the moment they arrived in the hospital early each morning until they left the hospital about twelve hours later, a method known as "shadowing." As one of the doctors she studied noted in introducing her to a colleague, "Look, I've got a girl following me around all the time, taking down everything I say" (Katz 1999:7).

Over time, Katz (1999:91) developed a number of keen insights about "surgeon culture," including the fact that while they act jovial and gregarious in the company of their colleagues, surgeons tend to carefully protect "information about themselves, their patients, their operating loads, operative techniques, referral sources, levels of specific knowledge, specific expertise (particularly deficits in knowledge and expertise), income, and doubts and concerns about medical decisions," in other words, anything that might allow another surgeon to have the upper hand in dealings with them. Feelings of competition with other surgeons, in short, were as strong as or stronger than any sense of collegiality and collaboration. This underlying feature of surgeon culture, Katz found, was a critical component of why residents felt that hospital staff surgeons were not spending adequate time training them for their jobs as surgeons.

Folk Illness in Haiti

Illness and Identity

Far to the south of Meadowbrook University Hospital, on the Caribbean island of Haiti, lies the rural village of Jeanty (pseudonym), tucked into the foothills above the Cayes plain, once one of the wealthiest sugar-growing areas of the world. The village itself is hilly and rocky and was probably first populated during the early nineteenth century by freed slaves fleeing plantation labor in the early years of Haitian independence from colonial France. Little more than a cluster of small tin-roofed houses strewn along a grid of unpaved streets, Jeanty is home to three thousand people with a total population of almost twelve thousand in nearby hamlets. With American funding, CARE, a relief agency (one of many in Haiti), had installed a potable water system in the village, and it was with the assistance of CARE that Paul Brodwin, a health anthropologist, had arrived there in 1987 to study folk conceptions of illness, healing, and mortality.

Specifically, Brodwin examined the crisis of illness as a fundamental social experience during which people commonly rework their social identities and confront the contradictions of their life experience, health-related processes addressed in chapter 3.

The Language of Research

Brodwin's research methods included participant observation of routine health-related activities among village members, semistructured interviews with village residents, attendance at community religious and healing rituals, the collection of life histories of village members, and interviews designed to record detailed narratives of actual illness experiences. He was aided in this effort by being a novelty for the villagers: only a handful of outsiders have ever lived in the village, and thus locals were interested in talking with him. Initially, Brodwin had an assistant, a young man from the village, but before long his ability to speak Creole—the language born of the merger of French and several languages from West Africa—was sufficient to conduct his own interviews.

Choosing a Healer

In his study, Brodwin addressed an issue that is central to health anthropology (see chapter 7): in contexts of medical pluralism (i.e., coexistent alternative healing systems), how do people make decisions about which ethnomedical system to go to—for what health issues and why—and what actually happens as patients and healers interact? When they get sick, people often must choose between multiple and competing healing traditions. This is the case both in the small-scale communities where anthropologists commonly have worked and in the urban centers where many anthropologists now do their research. To help answer these questions, Brodwin observed more than fifty consultations between patients and herbalist healers and about twenty interactions between patients and *houngans* (male healers in the Vodou folk health-care system). In each case, he carefully recorded his observations of what happened, including the nature of the problem brought by the patient, healer questions and actions, patient responses, the context and tone of the interaction, and the issue of payment. For example, about one consultation by a Haitian patient suffering from painful muscle aches and body fatigue, Brodwin (1996:134), who accompanied the patient, named Louis, to the healing session, recorded,

> André [a *houngan* known to the patient] greeted us at the door, and after the necessary introductions, led us into his small consulting room and sat at his table. It was covered with a heavy red cloth, and an assortment of small bottles filled with remedies lined the back. . . . As André settled in his chair, he lit an oil-wick lamp. . . . [After a while] André began to moan and gasp for air between coughs which wracked his entire upper body. . . . He soon began to rock back and forth and softly whistled, a serene smile on his lips. . . . From this moment on until the end of the consultation . . . we communicated only with his lwa [the spirit being named Byenzomal that had possessed André and began to speak

> through him]. . . . When he had finished the divinations [to determine the source of Louis's symptoms] and given to Louis the names of several [folk] remedies to purchase, Byenzomal rang the bell . . . and announced "We're through."

Studying Exotica

To his fieldwork project, Brodwin brought a strong curiosity about the exotic, one that was not disappointed by scenes such as the one just described. As Brodwin (1996:129) admits, "Years of graduate training in anthropology had not suppressed my typically American fascination with the exotic side of popular Haitian religion." To his initial disappointment, however, he was frustrated in this desire at every turn. Several months passed before he was finally able to make contact with and begin interviewing a *houngan*.

In his interviews with village members, Brodwin came to realize that while people may consult *houngans* for their pressing health problems, it is not without a degree of fear of the malign ability of these specialists to "send sickness." Fear—although of a different sort—plays a similar role in the hesitancy many people in the United States feel about seeing a dentist or having an operation. As Brodwin learned, ambivalence, uncertainty, and moral struggle are often critical features of illness and healing, lessons learned by health anthropologists working in diverse social and cultural settings.

■ ■ ■

A Case Study

Having Impact

As Erickson (2003:3), whose work in health anthropology addresses reproductive health issues, points out,

> Medical anthropology has a broad mandate—to understand and interpret humans, their diseases and illnesses, and their medical systems. Many medical anthropologists also take on the responsibility of making their research useful for clinical or health educational applications, for influencing public health policy, or for effecting social justice.

Through this brief examination of the fieldwork of three different health anthropologists in different parts of the world and with very different field sites and populations, as well as different research questions and methods (including, in addition, the three examples provided in chapter 1), we begin to grasp just what it is that health anthropologists do to fulfill the discipline's broad humanitarian and scientific mandate. Some of what they do, indicated by the six case studies presented thus far includes (1) at times, focusing their research and applied efforts directly on specific diseases (e.g., cystic fibrosis or dengue), and in other cases, concentrating on particular healing traditions (or multiple traditions in interaction), to reveal underlying factors missed by less intensive, less field-oriented

approaches to research; (2) addressing gaps in health-care services that often only are revealed by up-close, ethnographic research; (3) seeking to understand the provider-patient relationship and the nature of the healing process; (4) investigating local health beliefs, practices, and concerns to help shape public health efforts to match their target audiences and promote community involvement; (5) working "on the ground" among people as they live their everyday lives in order to spot and develop understandings of risk behaviors (and their causes) as well as risk contexts that promote unhealthy behaviors and exposures; (6) trying to understand the social origins of illness and use this information to make specific policy change recommendations to reduce health risk; (7) determining the contribution of social inequality and injustice to inequities in health outcomes; and (8) exploring biosocial and socioenvironmental interactions to better comprehend the making of wellness and disease relative to other species, environmental features, and the climate. Ultimately what health anthropologists do is seek to improve health, generally with groups that face the biggest health challenges and threats.

What Health Anthropologists Study

A Diverse Discipline

In the role of applied social scientists, health anthropologists commonly seek to answer practical questions about the nature of health, illness, healing systems, and related matters across cultural systems, populations, and social contexts. These questions derive from various sources. Some reflect ongoing debates and discussions generated at anthropology (and other health) conferences and in the published literature, primarily books and journals but also government or other reports, policy papers, and other documents of the discipline and related fields. Others are raised by institutions, government bodies, or communities that seek the help of a health anthropologist in addressing pressing issues. Still other questions arise because of changes in the world that impact health. For example, the rapid increase in electronic communication (especially cell phones and computers) has influenced health-related behavior in various ways (e.g., the rapid electronic accessing of health information on the internet). Increasingly, health anthropologists, like other researchers, have asked research questions about the intersection of health behavior and electronic communication including social media. During the 2014–2016 Ebola epidemic, for example, Olivia Marcus and Merrill Singer (2016) analyzed the appearance of an internet meme known as Ebola-chan that began to show up on various internet sites. This unexpected anime character—an anthropomorphic internet representation of the Ebola virus—combined the appearance of girlish innocence and an inversion of the meaning of caring (in the sense of loving someone to death) with morbid imagery to encapsulate conflicted social responses to a lethal and fast-spreading epidemic in the digital age. Drawn and posted anonymously, the meme appeared on various internet sites, garnered a range of posted reactions, triggered the creation of various other infectious disease memes, then faded away as the epidemic peaked and began to recede in the global social imagery. A similar internet response does not seem to have accompanied the 2015–2016 Zika epidemic despite the widespread fear

generated by the linkage of this mosquito-transmitted disease with microcephaly and fetal malformation. As these examples suggest, the internet is a domain of health-related behaviors and communication in critical need for enhanced health anthropology research and understanding.

Also of concern to health anthropologists is the role of governance in the framing of global health issues. Governance refers to how a governing body or other institution with sway in society views its mission and carries it out through policies and actions. Part of governance is what Michel Foucault (2008) called *governmentality* or the way governments and other social control institutions try to shape their citizens to be particular kinds of people or, in the case of the governmentality of hospitals and clinics, particular kinds of patients. The U.S. government, for example, stresses the making of responsibilized citizens who see issues like illness, health insurance, unemployment, and poverty as not being the responsibility of the state or society, but instead as the private burden of individuals and families. Consequently, there is a strong and often moralistic emphasis in American culture on self-care, often actualized as *care through consumption* (buying particular foods, joining gyms, taking yoga classes, purchasing the right gear for all manner of sports). In research on emergent popularized views on cancer, for example, adopting a healthy lifestyle is being defined as a way of preventing disease, based on the understanding that the epidemiological literature firmly demonstrates the role of factors such as weight, diet, and exercise on cancer risk. But, in fact, "scientific findings on the cancer/lifestyle link" show that they "are incomplete, ambiguous, and internally contradictory" (Bell and Ristovski-Slijepcevic 2015:219). At the global level, health anthropologists are examining global health governance and the ways global health issues are framed. This concern stems from recognition that how health problems are conceived will shape global efforts to respond to them.

Without doubt, health anthropologists have diverse points of view and do not by any means fully agree about what the key questions of the discipline are. Some health anthropologists, those interested in the experience of illness, focus much of their attention on humankind's phenomenological encounter with being sick, asking questions such as what does illness mean to people in particular social roles, sociocultural contexts, and social settings? What is the internal lived experience of illness like for sufferers and what shapes this experience? What impact does illness have on their identities, relationships, and values? Concern about answering such questions has led some anthropologists working on health issues to adopt the concept of subjectivity. Subjectivity refers to the internal thoughts, feelings, and worries of people as well as their intent, desires, and images of self—in short, the emotion-laden fabric of their inner lives in social context. Subjectivity, from the anthropological perspective, is not an individual or private state, but rather involves how people in common situations, facing common challenges, develop overlapping and meaningful ways of being, knowing, and experiencing in the world (Good 2012). Central to the notion of subjectivity is examination of behavior, including health-related behavior, as a dynamic and fluid process, an emergence, rather than one narrowly constrained by cultural conventions (Abadía-Barrero 2011). This means a study of behavior as conscious action relative to others and inner lives as products of interaction

and intersubjectivity. A study of gender and mental health treatment in different clinical and home settings in northern India (Pinto 2014), for example, focused on how women experience the vulnerabilities of love in their minds, through their bodies, and as part of their social world of interactions. In both the Hindi and Urdu languages, in fact, love and madness are both considered forms of *nasha* or intoxication. Loving the wrong person in an intense, driven way, crazy love, is constructed culturally as a destructive force that shattered the wellness of not only the women patients Pinto studied but their families as well. During her fieldwork, Pinto herself faced a jarring personal crisis, which she writes about as an intersubjective aspect of her research experience, a broken mirror that reflected the broken inner worlds of the women about whose lives her work poignantly examines.

Other health anthropologists are more interested in how people make decisions about what to do when they are ill, including their help-seeking strategies, how they select among alternative healing systems or combine different types of treatment, and the role of significant others in shaping these processes. Alternatively, there is keen interest in both the nature of the patient-healer relationship and the relative effectiveness of different approaches to treatment among various health anthropologists. Finally, some health anthropologists are especially concerned with the role of environmental, social, political, and economic factors in illness. In this book, we engage all of these issues of concern to the field by way of introducing the range found within the discipline.

Further, health has a global research mandate. As seen in the examples that have been provided, some health anthropologists work in rural villages in developing nations, others conduct their studies and applied work in modern hospitals, while others carry out their research on the street corners of cities large and small, or even on the internet. Many health anthropologists work in quite different settings over the course of their careers. A complete compilation of all the topics health anthropologists have addressed and all the places they have conducted research or implemented programs would be surprisingly long and varied. The discipline requires that its practitioners have a good sense of our changing world and the dynamic domain of health in human societies, and even beyond humans to interacting species that may be the source of zoonotic human diseases or cures.

Studying the Life Course

Another reflection of the broad mandate of health anthropology is the work done by members of the discipline across the various stages of the human life course, from birth (or even before, in terms of reproductive health) to death. In all societies, reproduction and reproductive health are heavily invested with cultural meanings and values. Women's menstruation, likewise, is commonly a focus of cultural elaboration; menstrual blood often is seen as being especially charged and possibly polluting. The process of giving birth also is culturally shaped and managed in all societies according to group understandings of appropriate birthing behavior. With her classic anthropological study *Birth in Four Cultures*, Brigette Jordan (1993) helped launch the anthropology of birth

and reproduction as a subfield of health anthropology. Work in this area has shown that cultures develop distinct beliefs, meanings, and associated practices around pregnancy, delivery, and the treatment of babies during the postpartum period. Further, researchers working on this topic have demonstrated that there is a need for an examination of birthing within structures of power in health care and in society generally, and in light of both women's culturally influenced priorities and factors such as their attitudes toward pain, fear, stress, and anxiety. In biomedicine, for example, the full medicalization of childbirth—characterized by (1) the use of high-tech machinery for monitoring the birthing process that begins early in the pregnancy and continues through delivery, (2) frequent use of surgery to both widen the birth portal and remove the baby, and (3) widespread use of medications to deaden pain or speed up the birthing process—has led to the use of the term "technocratic birth" to characterize having a baby in Western society (Davis-Floyd 1992). An informative study of the role of social class and educational influences on women's birthing behaviors in Chile focused on the period before delivery (Murray 2012). Rather than there being a single Chilean pattern of birthing preparation, there were different patterns structured by social class and access to differential material resources. However, as the time of birth approached and women's sense of vulnerability and risk increased, the class stratification in attitudes about the medicalization of birth collapsed into a general, cross-class acceptance of the choices made for women by the biomedical system. Fear and uncertainty came to override prior choice and led to acquiescence to socially enforced biomedical authority. By contrast, in many societies, traditional birth attendants (i.e., senior women with considerable personal and apprentice experience in birthing) provide the primary support to a woman in delivery. Low-technology, midwife-assisted birthing systems have resisted the spread of technocratic birth, often with anthropological support. An applied approach to the issue of birthing has led to anthropological work on issues such as teen pregnancy as a health and social problem, including showing why teen girls often view pregnancy as a sought-after social status (Erickson 1998). Similarly, health anthropologists have studied children's health and have been active in applied projects such as breastfeeding promotion. Some anthropologists, for example, became very involved in the effort to challenge the global promotion of commercial infant formula by Nestlé Corporation and other companies as a replacement for breast milk (e.g., Penny Van Esterik 1989). In many settings, manufactured formulas, they have argued, are detrimental to children's health and sometimes, if clean water for mixing the formula is not available, even life threatening. Breastfeeding, by contrast, not only provides a nutritious food for babies; it also passes important maternal immunities to diseases on to young children. Chapter 8 considers the human dimensions of the emergence of biotechnology, the role of biocapital and the emergence of markets for body tissues, and the growing array of bioethical issues in which health anthropologists are now engaged.

At the distal end of the life course, since the 1950s anthropologists have also paid considerable and increasing attention to aging. Research in this area has shown that old age commonly is marked by a shift in a person's role in society, such as a significant reduction in work responsibility, as well as changes in social status, including, depending on the local setting, coming to be viewed variously

as a valued storehouse of important cultural knowledge or as out of touch with the contemporary world. The nature of these transitions varies because the meanings attached to being "old" are culturally constructed in light of other features of a society. In a fast-paced, rapidly changing, and globalizing society, the knowledge and experiences of elders may be devalued; by contrast, in societies that look to tradition as a guide to the present, growing old may be accompanied by a gain in social stature. Increasingly, however, as the global market (e.g., labor migration) has had its impact, there have been significant but locally varying changes in intergenerational relationships. In China, for example, where the elder population has been growing rapidly and becoming a higher percentage of the total population, traditional values of elder respect, family support, and obligation to caregiving face new challenges and moral dilemmas (Sheng and Settles 2006). Broad economic and social changes have weakened the social foundations of family care for elderly parents and elderly expectations about being cared for later in life by their grown children. Parent-child relationships in China have become less hierarchical in recent decades, and, as contrasted with the past, elder parents exert shrinking influence on their grown children's life decisions. Elder parents, in fact, are contributing to their grown children's independence through their growing role in taking care of their grandchildren. While family support and caring certainly have not disappeared, remain expressed values among parents and children alike, and include both emotional and financial support, increasingly the government has played a role in elder care through the provision of nursing homes, day centers, and elder entertainment centers.

Another topic of concern in the anthropology of aging is medicalization. For example, since the introduction of Viagra (also see chapter 7 for discussion of this drug), there has been a biomedically driven redefinition of decreased erectile function accompanying aging vis-à-vis the disease diagnosis known as erectile dysfunction. The medicalization of older people's sexual health brings together Western social discomfort in response to aging with the valorization of penetrative sexual practice as "normal sex." In this context, drugs like Viagra provide biomedicine with the possibility of improving health, defined in this culturally influenced context as youthful sexuality. Such drugs are an enormous source of profit for the pharmaceutical industry (e.g., Viagra produced almost $8 billion in revenue between 2012 and 2016 [Statista 2017]). Moreover, they are marketed globally and represent the spread of Western cultural views of aging and the body, which reduce the aging process and the lived experience of the elderly to a form of fixable mechanical breakdown. However, not all men buy into this redefinition. Health anthropology research based in the urology department in a government hospital in Cuernavaca, Mexico (Wentzell 2013), for example, found that male patients understood their decreasing erections as both natural and a physical sign that it was time for them to move past a youthful focus on penetrative sex and live a more "mature" masculinity based on emotional interactions with their families. This decision, encouraged by their wives, enabled these men to embrace local cultural values pertaining to respectable older manhood. Such research demonstrates the continued impact of local cultural values on gendered life-course behaviors in a globalized world. It also represents the growing

concern that a consciously gendered focus on health not be limited, as it often has been in health anthropology, to women's health issues.

Applied work in the anthropology of aging has focused on many issues, including helping to give voice to elders whose needs and capacities have been overlooked in society. Applied health anthropologists have developed ties with an international movement working to promote elder-friendly communities and the promotion of well-being throughout the life course. Dying and death are also issues of concern in health anthropology; they constitute profound arenas of human experience, thought, and emotion. In contemporary health anthropology studies, dying and death are viewed as "subjects without clear boundaries, and any analytic exploration of those themes now problematizes their definition" (Kaufman 2004:245) rather than viewing them as clear-cut biological states. In other words, death and dying must be understood in sociocultural context because what it means to die in one society may be quite different from what it means to die in another.

Conducting Research: A Peculiarly Anthropological Approach

Holistic, Field-Based Understanding

In conducting research, health anthropologists tend to assume that the issues of immediate concern to them are embedded in wider sociocultural systems and are intricately interconnected with many other aspects of social life and with complex social environmental contexts. As a result, they tend to cast a wide research net and ask a broad set of research questions. To carry out their research, health anthropologists usually marshal a number of qualitative and quantitative methods for the collection of health-related data (e.g., see the seven-volume set *Ethnographer's Toolkit*, which fully describes the methods used by many health anthropologists in their research [Schensul and LeCompte 2010]). Moreover, health anthropologists generally seek to move out of the university or other institutional setting in which they are employed to gain direct and intimate access to the daily lives and activities of the people under study. Further, they often spend long periods of time in these field settings observing and participating in the flow of life. This is an approach to research known as ethnography.

Ethnography

Ethnography is the name of the core research strategy in health anthropology. It involves the immersion of the researcher or team of researchers into the social space and lifeways of the people under study. As part of this effort, health anthropologists seek to participate (to varying degrees) in the normal social life of the group under study. While not all research in health anthropology is ethnographic, there is a generally shared sense that ethnography offers keen insights about health and behavior that are not easily acquired through other means. Rather than a highly controlled and sharply focused approach to data collection, ethnography focuses on issues of concern within their natural social context and in terms of how they are seen and experienced by group members.

Choosing Risky Behavior

In studying health issues, health anthropologists often use ethnography to address difficult questions or seek to make sense of puzzling or misunderstood behaviors. Elisa Sobo, for example, was confronted with the challenge of understanding why people who know about AIDS and routes of infection nonetheless regularly participate in risky behaviors, such as not using condoms during sex. This is an important problem for public health because it is clear that simply teaching people about AIDS as an infectious disease does not eliminate risk; the disease continues to spread. In the United States, ethnic minorities have been disproportionately affected by AIDS since cases were first identified, and constitute the majority of new diagnoses in the ongoing epidemic (Kaiser Family Foundations (2017). In her study of risky behaviors among impoverished and socially disadvantaged women in Cleveland, Ohio, Sobo (1995) concluded that people who know about AIDS often act as if they are not at risk, a puzzling discovery that needed explanation. AIDS-risk denial, she found, appears to be rooted both in a cultural tradition that values monogamy and personal responsibility and in the actual position of women in society generally and relative to men. Thus, the women whom Sobo studied idealized and desired long-term monogamous romantic relationships and loyal, committed partners. The underlying cultural logic that supported this heartfelt desire is this: good women get good men, and unworthy women get bad men. If you are a good woman, then you should be able to trust your partner. This attitude reflects very basic cultural ideals in American society and leads people to interpret life course and outcomes in terms of how hard people work and how committed they are to achieving their goals. Simply put, those who succeed are seen as having worked hard, and those who do not are seen as slouches. Having an untrustworthy male partner, in short, implied that a woman did not deserve better, putting her very identity and sense of personal self-worth at risk.

Like others in the wider society, the women whom Sobo studied seek good relationships with good men, thereby proving to themselves and their social networks that they are, indeed, women of value. As a result, with their main partners, the women do not use condoms (because as good women they can trust their partners not to have sex with others). One consequence of poverty and social discrimination, however, is that the potential partners whom women like those Sobo studied are likely to meet tend to have comparatively high rates of unemployment and underemployment, lack of health insurance, and limited other resources. As a result of an inability to succeed economically in society and to reap the psychosocial benefits of being successful in one's own eyes and in the eyes of peers, it is a special challenge for poor men to feel good about themselves and about their own worth as human beings. As a psychosocial substitute for economic achievement and hence as a salve for the potential sense of failure in a success-driven society, street culture, in fact, tends to support "sexual achievement"—that is, having multiple partners over time or even at the same time—as a sign of success. Thus, one can still have a sense of achievement in life and the experience of having admirable qualities, even if it is acquired through a

nonmainstream method, although one that is not entirely beyond the mainstream; fantasies of multiple sexual exploits are hardly limited to the poor.

Consequently, poor men and women are pushed by social forces and cultural values to be at cross-purposes, a tension that finds expression in high rates of divorce, failure to marry, and intimate partner violence. In their effort to achieve monogamy and the psychological benefit of feeling that they merit a good man, Sobo argues, poor women are pushed to see their male partners as more loyal and more deeply committed than social circumstances allow them to be. Thus, they may begin prematurely to assume that their partners are not seeing other women. In this context, the decision not to use condoms affirms a woman's desire to feel worthy of a dependable man. Various behaviors reinforce this decision. Sobo (1995:99) notes, "A man who gives his wife or girlfriend gifts, services, or money lives up to—or at least begins to live up to and implies he intends to live up to—the cultural ideal of the male partner as breadwinner or provider and as a woman's protector." Under these conditions, stopping condom use is an act of commitment, an expression of trust, and an investment, through potentially having a child together, in a long-term partnership. The problem, however, is that most women who get AIDS are not infected during one-night stands with poorly known sexual partners because condoms tend to be used in such situations. Rather, they are infected with HIV by longer-term partners, people they trust with whom they are having sex without a condom.

Over time, ethnographic researchers are able to glimpse behind the public masks and front-stage performances of social actors to backstage and often hidden arenas of experience and social interaction. In this way, they are often able to develop understandings of behaviors that might otherwise appear irrational, meaningless, or inscrutable, or might otherwise go unknown.

For ten years, Andrew McDowell (2016) engaged in the ethnographic study in India of "Bengali doctors," unregulated popular healers who do not have medical degrees but make use of biomedical medicines (e.g., injections and intravenous transfusions of antibiotics, stimulants, painkillers, vitamins, and fever reducers, often in multidrug cocktails). These healers are said by their biomedical critics to treat immediate symptoms but not underlying disease, potentially causing longer-term harm to their patients. One of McDowell's study participants was a Bengali doctor named Mohit. He accompanied Mohit on fifty house calls to treat patients and spent countless hours with him at his clinic, observing and chatting about medicine, current events, and their respective lives. McDowell found that conversations about Mohit and health were commonplace among the people in southeast Rajasthan, where he carried out this part of his research. He heard such conversations in public places, like the village hand pump, bus stand, and squares. McDowell joined these conversations and carried out interviews with seventy local public health providers.

McDowell found through his deep and prolonged ethnographic immersion in everyday life and popular health treatment that while Mohit and fellow practitioners treat symptoms, they do so within a cultural frame that is meaningful to them and their patients. While their treatment practices may appear irrational in the eyes of

biomedical physicians, they are able to return patients to a satisfying feeling that they have a well-functioning body with steady breathing, cycles of hunger and satiety, and oscillations between tiredness and invigoration. In short, they reestablish a sense of organic and mechanical productivity from the perspective of their patients, until something else throws their bodies out of their normal cycling pattern.

Complex Sociocultural Tapestries

In explaining health-related behavior, health anthropologists pay attention to the interplay of a wide range of cultural, social, hierarchical, psychological, environmental, and even biological factors. Only by showing how all these weave together in complex tapestries, something that the ethnographic methods of health anthropology allow, can we really understand why people do what they do, believe what they believe, and get sick or stay well. In thinking about Sobo's AIDS study, one advantage of this perspective is the realization that sexual desire—which is often condemned in moralistic discussions of sexual risk—must be reframed "from an individual to a collective phenomenon" (R. Parker 2009:xiv). Culture provides a frame of reference "though which sexual meanings are organized—and in relationship to which conflicting and contrasting sexual scripts are produced and reproduced" (R. Parker 2009:xiv), and these, in turn, are shaped by other factors like social hierarchies and inequalities. As this example suggests, health anthropology does not provide simple answers to complex problems; rather it provides adequate understandings for the design of effective interventions.

This aspect of the health anthropology approach can also be seen in its work on a chronic disease like type 2 diabetes mellitus. The word diabetes comes from Greek and in its original form meant "to run through." The term reflects the ancient Greek view that diabetes causes food to rush through the body instead of feeding it, leading the body ultimately to melt down into urine. Today, biomedicine understands diabetes as a breakdown in the body's capacity to produce or respond to the hormone known as insulin, producing an abnormal metabolism of carbohydrates and elevated levels of glucose in the blood and urine. In recent decades, diabetes has become a global disease that causes suffering, disability, morbidity, and mortality worldwide, especially in disadvantaged populations. This development "reflects the evolving global reach of capitalism and colonialism, which precipitated increased access to commodity foods, decreased reliance on local food sources, and reduced physical activity" (Rock 2005:474).

Anthropologists, in particular, have been involved in the study of diabetes in indigenous populations such as American Indians, who often are said to be genetically vulnerable to this debilitating condition. Based on ethnographic research, health anthropologists have stressed that the high incidence of diabetes in indigenous populations cannot be explained by a case of bad genes. Diabetes appears to be a relatively new health threat to indigenous peoples, the symptoms of which do not show up in older accounts, historical records, or people's memories of prior generations. Rather, diabetes is a disease with social origins and reflects the biological consequences of intensely adverse structural factors such as poverty, discrimination, loss, trauma, and social stress, common threats to indigenous populations around the world (Ferreira and Lang 2006, Smith-Morris 2006).

The Health Risks and Benefits of Kissing

From a public health standpoint, is kissing risky? If so, what are the known health risks of kissing? Conversely, are there health benefits of kissing? These questions have received increasing attention in public health since the early 1950s when a bacteriologist reported that up to 250 colonies of bacteria can be transmitted during a single passionate kiss (fewer if one of the participants is wearing lipstick). To reach this conclusion, he recruited a sample of adults and adolescents to kiss a sterile glass slide or agar plate for various periods of time. Fortunately, this research found that the vast majority (95 percent) of kiss-transferred microbes are not pathogenic. Since then, of course, HIV/AIDS became a global pandemic and concerns about kissing as a route of lethal viral or other microbial transmission have become widespread. In their studies various social scientists have encountered popular uncertainty about kissing as a risk behavior. In one study, for example, researchers conducted semistructured qualitative interviews with thirty-three HIV-infected parents and their children to investigate fears about HIV transmission. They found that many of these families reported transmission fears, including specific trepidation related to blood contact, contact with bathroom items, food sharing, and kissing/hugging. Many of these fears, including those viewing kissing as an HIV risk, are based on misconceptions about modes of HIV transmission. While kissing is not a very likely route of HIV infection (because HIV cannot survive in saliva), kissing can be a health risk. In fact, an array of infectious diseases can be transmitted through this form of human intimacy, including strep throat, infectious mononucleosis (known colloquially as "kissing disease"), Herpes Simplex Virus-2, Hepatitis B, syphilis, scabies (a contagious skin disease caused by a mite), warts, and meningococcal disease (inflammation of the membranes that cover the brain and spinal cord). College freshmen, for example, especially those who live in dormitories, have been found to be at increased risk for bacterial meningococcal disease compared with age mates who are not attending college or living in a dorm setting. Moreover, research in Britain found that people involved in intimate kissing with multiple concurrent partners face a risk four times greater of developing meningitis than those not engaged in this behavior. Sexually transmitted diseases such as syphilis also can be transmitted through kissing if syphilis sores are present in the mouth or lips of one of the participants. In addition to various diseases, several studies have found that food allergens can be transmitted through kissing. For example, there is a case of a severe anaphylactic reaction in a young woman with shellfish allergies after kissing her boyfriend (who had just eaten several shrimp; Steensma 2003). On the plus side, research shows that there are several notable health benefits linked to kissing, including the fact that kissing, particularly passionate kissing, lowers cortisol levels, an objective sign of stress reduction. Moreover, kissing can improve self-esteem and feelings of being appreciated. Further, kissing helps stabilize cardiovascular activity and reduce blood pressure and cholesterol.

Further, health anthropology research in South Africa found that diabetes is linked to and adversely interacts with depression, and both are connected, in turn, to the role of gendered street violence and structural violence in creating everyday fear and chronic distress (Mendenhall 2015). Diabetes, in other words, does not develop into epidemic proportions independent of other diseases or taxing social and economic conditions associated often with social inequality.

As diabetes has gained global recognition as a disease under the global influence of biomedicine, researchers have found that it is being used at the popular level as an idiom of distress for talking about suffering and upsetting experience as a bodily condition in a culturally meaningful way (Mendenhall et al. 2010). This insight suggests the importance of investigating the relationship between experienced suffering and distress and the onset of chronic, noncommunicable diseases like diabetes.

Research Methods

Multimethod Research

When conducting a study, a health anthropologist or a team of collaborating researchers generally combines a number of specific methods. This "mixed-method approach" often includes direct observation and detailed recording of behaviors and events witnessed in the field, as seen earlier in this chapter in Brodwin's (1996) records of folk treatment observations in Haiti. In addition, health anthropologists generally conduct casual interviews with people as they are going about their everyday activities, which Pearl Katz (1999) did as she followed surgeons around the hospital she studied. Another method frequently used by health anthropologists is in-depth interviewing, such as the detailed interviews conducted by Bluebond-Langner (1996) in her study of cystic fibrosis. To gain a wider contextual framework for understanding what they find in the field, health anthropologists routinely review historic documents or other existing records, such as Harper's (2002) examination of the historical and economic literature on Madagascar.

Health anthropologists also use ethnosemantic elicitation, as well as a set of specialized systematic cultural assessment techniques, such as free lists, pile sorts, and Q-sorts, that allow researchers to glimpse how study participants think about and order the components of their world. Pamela Erickson and her colleagues, for example, used free lists and pile sorts, among other techniques, to study sexual health and decision making among inner-city African American and Puerto Rican young adults (M. Singer et al. 2006). In this study, called Project PHRESH, eighteen-to-twenty-four-year-olds were recruited through street outreach in Hartford, Connecticut, and Philadelphia, Pennsylvania. In small-group settings, they were first asked to list all the kinds of sexual practices that they knew.

Later, in one-on-one sessions, other participants from the same age and ethnic groups were asked to sort cards that were printed with descriptions of sexual behaviors that were listed by at least three of the small groups into piles ranging from least to most risky for HIV transmission. Participants were then requested to explain why they put cards together in the same pile. Using a special set of computer programs

called Anthropac, the team was able to prepare a scatter plot of the overall pattern in the sorted cards, based on an assessment of which cards people tended to put in the same pile or keep apart in different piles, across all the participants in the study. This approach allowed Erickson and coworkers to see underlying cultural associations that shape sexual and romantic relations in their study population. Other techniques commonly used by health anthropologists include life history interviews, focus groups, consensus analysis, and diary keeping.

One of the concerns of projects like PHRESH is the recruitment of a sample of individuals who represent a larger population when all of the features (e.g., overall size, makeup of subgroups) of the larger target population are not fully known (and hence the recruitment of a statistically representative sample is not possible). One approach that allows the researcher to combine location and time factors to obtain a large, diverse, and reasonably representative sample is known as venue-based sampling (VBS). The keys to the VBS approach are (1) rigorous exploration of possible recruitment sites (i.e., venues) where members of the target population can be found (e.g., a study of street commercial sex workers might attempt to identify "stroll" sites where commercial sex workers engage customers); (2) focused observation over a period of time at all identified sites to roughly determine the number of individuals from the target group that are found at each site; (3) random selection of venues at selected intervals from the list of identified venues; (4) recruitment of individuals at randomly selected venues combined with observation of the number of members of the target group present during the time/date of the recruitment; (5) calculation of the relationship of the number of individuals selected from those available at the site during the recruitment episode; and (6) data collection (e.g., through survey, structured interview, in-depth interview) of recruited individuals. As this description suggests, VBS combines qualitative and quantitative strategies to provide researchers with confidence that the individuals they collected data from do not represent a skewed sample that differs in significant ways from the larger population of interest.

Examining Lives

The life history interview is used to record the life story of an individual, with a strong emphasis on the meaning of life events for the person being interviewed. James Quesada (1998), for example, used this technique to explore the effects of war, endemic poverty, political instability, and social despair on Daniel, a gangly ten-year-old boy from the central highlands of Nicaragua. Quesada got to know Daniel and his mother while studying health and well-being in the aftermath of war in the Nicaraguan town of Matagalpa. Reflecting the growing interest of health anthropology in the health and social effects of war, aggression, and interpersonal violence, Quesada analyzed Daniel's life story as an embodiment of the pain and suffering shared by many Nicaraguan children as a result of the U.S.-supported Contra War that had been fought in the area. Daniel succinctly and coolly summarized his grim life experience one day when he told Quesada (1998:60) that sometimes he felt like dying: "Look at me, I'm all bones anyway. I'm already dying. I'm too small and I have stopped growing and I am another

mouth to feed. My mother can't keep taking care of my brothers and me, and I can't keep taking care of her. I can't do anything." Through this single life history, Quesada gained a far deeper understanding of the embodied experience of war, its profound health costs, the jagged social disruption it brings, and the abject poverty it often produces. Because of Quesada's intervention, Daniel did not die, but there are few Jim Quesadas and many Daniels in the world.

In another health anthropology study, Merrill Singer and Rebecca Allen (2017) used a life history approach to address a fundamental question in contemporary health care: how do we understand and respond to the pressing health problems of society? The conventional biomedical response to this question involves description of a type of care that is based on careful clinical assessment of the immediate health factors presented by each individual patient and the provision of science-based medical treatment drawn from an arsenal of pharmaceutical, surgical, and other established and approved biomedical strategies. In subtle tension with this dominant clinical model of medicine, the approach known as social medicine places an equal level of focus on the social conditions and structural inequalities that produce and worsen health problems, including social stigmatization, adverse physical factors in built social environments (e.g., overcrowding, impoverished neighborhoods, and inadequate diets), structural violence (like entrenched institutional racism), and social injustice. These social factors contribute to a biology of inequality, powered by the ways social disparity and deprivation get "under the skin" and are inscribed by disease on body systems. From a social medicine perspective, the fundamental underlying cause of disparities in health are social and economic inequalities that either produce illness directly or foster unhealthy behaviors that lead to poor health. While there are various personal accounts of the experience of becoming or working as a conventional physician, prospective medical students have few alternative role models of doctors who are not narrowly focused on traditional individual patient/disease-focused approaches to medical practice. Based on extensive life-history interviews over several months with a social medicine doctor, Singer and Allen present an alternative model of biomedical practice within the context of the existing structure of twenty-first-century medicine and in light of the consequential current unequal distribution of health, living and working conditions, and medical access. The type of social medicine practiced by this physician, which is informed by a keen awareness of the social origins of health and well-being, reflects many of the life experiences he had growing up and the identifiable influences of various friends, family members, and mentors as well as his often troubled experiences with powerful medical institutions.

Focus Group Interviews

Originally developed by social scientists, focus groups came to be a favored technique of market researchers concerned with assessing the appeal of new products among consumers. In more recent years, anthropologists and other social scientists have readopted focus groups and used them to study many issues, including health.

Focus groups entail bringing together a group of people and stimulating them to engage in conversations about issues of research and community interest. With this goal in mind, for example, researchers investigating a topic of concern among many Native Americans (Bletzer et al. 2011), alcohol consumption, organized focus groups with members of five tribes from southwest, northwest, and northern plains to discuss this issue. They then used the performance of humor as a stylistic feature of the conversations to explore aspects of adolescent and adult drinking with the insight that how social groups create and use humor is a serious issue for cultural analysis. The researchers found that the most common forms of humor in the conversations were parody, hyperbole, and word play and that all of these contributed to the ability of the group to discuss a highly sensitive topic. Interpretation of focus group data requires the systematic review of conversational transcripts to identify themes that can be developed and configured, in collaboration with community members, into effective program initiatives. In this instance, appropriate use of humor was found to be an important entrée into collaborative exchange, a lesson for culturally sensitive intervention.

Discovering behavioral practices like this allows health anthropologists to design health-related interventions that are grounded in actual beliefs and behaviors.

Considering Consensus

The term *consensus analysis* refers to a quantitative procedure used to determine the modal (i.e., most frequent) answers provided by a group to a set of questions about a particular topic (e.g., "what is a cold?"). Use of consensus analysis allows researchers to determine the degree of cultural agreement in a group or between subgroups of a larger population. For example, asthma beliefs and practices have been found to vary among ethnic groups. One important variation related to ethnicity is the different descriptors used to express an asthma attack. These descriptions can be both physical and psychological in nature. In a multicity cross-cultural comparison of four different Latino groups in the United States, Mexico, and Guatemala, health anthropology researchers found broad agreement about the major respiratory signs of asthma, including wheezing, cough, chest noise, and fast or difficult breathing (Pachter et al. 2002). Particular to Puerto Ricans in the Hartford, Connecticut, sample, however, was reference to symptoms such as chest pain; decreased activity; increased blood pressure; chest congestion; fast heartbeat; red, tired, or dark eyes; and difficulty breathing or talking to describe asthma attacks. Because these latter symptoms were peculiar to one group, they cannot be considered part of a pan-Latino cultural complex. Rather, they are core elements of only a Puerto Rican cultural conception of asthma. Failure to understand this kind of information can lead to public health efforts that are overly generalized and thus fail to be effective with specific populations, such as Spanish-language prevention materials that assume that the term "asthma" means the same thing to all Latinos.

Doing Diaries

Diaries commonly are thought of as very personal records of one's thoughts, feelings, experiences, and relationships, but they have been drafted into the study of

health as well. Researchers studying issues such as drinking behavior, food intake, and sexual risk have employed diaries as a means of getting insider descriptions of behaviors and experiences as soon after they occur as possible (when memories are at their best). In a study of access to sterile syringes, for example, a team of health anthropologists, epidemiologists, and other researchers recruited a small sample of active illicit-drug injectors in three cities in New England to record their acquisition, use, and discard of syringes in a daily diary. For example, one of the participants in the study recorded the following information in his diary:

> [Thursday] 4-13-00 Went to cop [acquire drugs] last night, nobody was around. So we went all over and found some [drugs] finally. Me and John used that one needle I have. It's really messed up. But after 20 minutes of trying he shot me with 3 bags [injected me with the drugs from three small plastic bags]. He then used my works [syringe]. He didn't clean them or anything. My works are bad. The tip is all bent.
>
> Fri. April 14th. The works wouldn't draw up the dope so we were going to put some cream on the black tip [the syringe plunger] so it wouldn't jam up and [would] slide easier. We continued to have the problem of the works not drawing up. They'd start sucking up the dope then would stop. Finally it sucked everything from the spoon and he hit [injected] me . . . the works are in bad shape. So it takes him longer to get it in the vain [vein]. But he finally did then we shared spoon and cotton [to filter the drug mixture] and works and he did himself next which took a while too because the shape of the needle. It has a burr in it. But we did it.

Through accounts of this sort, the study (Stopka et al. 2004) found that drug users' diaries elucidated useful information on (1) daily patterns of injection drug use, (2) the social contexts of high-risk events, (3) HIV and hepatitis risk related to the street life cycle of a syringe, and (4) emotional correlates of drug use. Furthermore, the study discovered an unexpected intervention effect that keeping a diary may have in the lives of drug users: a number of the individuals who kept diaries approached the researchers seeking help. Being pushed into paying closer attention to the impact of drugs and the frequency of risk in their lives, they decided to get into drug treatment. Assistance into treatment was readily provided. As this example shows, health anthropology research itself can offer a form of intervention or serve as a model for new intervention approaches in public health.

Quantitative Methods

In addition to these qualitative research strategies, health anthropologists use various quantitative techniques, such as the surveys Kendall (1998) used to assess people's beliefs about dengue in Honduras, described in chapter 1. Qualitative methods can be used, in fact, to improve the quality of quantitative data. In a project focused on the traumas endured by Cambodian refugees in the United States (Hinton et al. 2015), researchers began by developing an understanding of somatic symptoms and culturally meaningful syndromes. This was done using qualitative interviews with Cambodian patients at a clinic, as well as through conversations with Cambodian monks, traditional healers, and community leaders.

Based on the information that was collected, the ethnopsychological condition known as *kut caraeun*, which translates into "thinking a lot" (TAL) or "thinking too much," was identified as an important idiom of distress (a concept discussed in chapter 1) among Cambodian refugees. Suffering from TAL involves rumination about unpleasant topics, such as enduring a loss, suffering a setback or failure, or struggling with a current life concern, to the point of experienced distress. Symptoms of this condition include fear and anxiety, agitated irritability, depression, shortness of breath, insomnia, heart palpitations, neck soreness, and dizziness.

As part of a broader mental health assessment with two hundred treatment-seeking adult Cambodians in a psychiatric clinic in Massachusetts, researchers administered several TAL instruments, including a culturally salient checklist of experienced TAL symptoms and a questionnaire that measured the severity of TAL and of TAL-induced distress. Based on the findings, the researchers argue that exploration of this idiom of distress provides important clinical insight into local social concerns, the experience of trauma, and Cambodian ideas of ethnopsychology. This type of information, they maintain, is critical for the evaluation of psychological distress and the development of contextually sensitive interventions.

Broader Collaboration

Health anthropologists now collaborate with a broad range of other types of researchers. For example, a number of health anthropologists work with laboratory scientists and collect biological specimens, such as the urine samples described in chapter 1 that were collected by Thomas Arcury and coworkers (2005) among farmworkers to test for pesticide exposure. Multidisciplinary collaboration of this sort is now quite common in health anthropology.

As the global health impacts of climate change have become increasingly clear (Baer and Singer 2014), health anthropologists have begun to collaborate with climate scientists, health-care professionals, community activists, and other concerned individuals. Some of this collaboration has developed through anthropologists becoming involved in grassroots climate action groups and the larger climate social movement that promotes social, economic, and technological changes designed to mitigate climate change and its damaging impacts. Notably, one of the cofounders and cochairs of the 2017 March for Science—organized in protest of the Donald Trump administration's attacks on science, including climate science—was anthropologist Valorie Aquino. Interdisciplinary research on climate change and health is another way health anthropologists have participated in collaborative efforts in climate change mitigation.

Health Anthropology in Use

Mobilizing Research Findings

Beyond research, health anthropologists commonly are concerned with application, that is, with the practical use of their research-gained knowledge and experience-gained skills. Application in health anthropology takes many forms. Health anthropologists, for example, have carried out research on the lived experiences of women to help reduce the barriers they face in accessing reproductive health care.

This can lead to meaningful policy or programmatic changes that improve access to needed services (e.g., Ostrach and Matthews 2015). Ethnographic research has shown that in some settings access to health services is encumbered by street-level bureaucrats, such as security guards and nurses, or by strict bureaucratic adherence to official procedure. These barriers can override the clinical needs of patients in ways that actively harm their health. To overcome such barriers faced by indigenous Maya patients in rural Guatemala, Anita Chary and colleagues (2016) worked with Wuqu' Kawoqm, the Maya Health Alliance, a nongovernmental organization in Guatemala, to develop and implement a patient accompaniment program. This intervention helps Maya patients with limited Spanish language abilities and low literacy to successfully navigate hospital environments and gain access to needed care. In a very different type of applied project, researchers worked with local, small-scale gold mining communities in Peru to develop safe technologies for gold extraction (Baudin et al. 2016). This is of critical health importance because mercury has long been used by gold miners because it can be mixed with gold-containing rocks to form an amalgam, which, when heated (and the mercury vaporized) releases the desired gold. But mercury vapor is highly toxic to all body systems at even minute levels of exposure. To ensure that the new, safe technologies would be acceptable to the miners, the researchers worked closely with them at all stages of design. Miners tested and provided feedback on the various extraction prototypes they were shown and were pleased with the final models. In this instance, a participatory applied approach was used to prevent health problems and diminish environmental pollution.

Applied health anthropologists are also involved in the direct delivery of health care to communities in need. An example, from Haiti is the collaboration between HOPE (Health Opportunities for People Everywhere), an international health organization that provides humanitarian assistance, and Koodinasyon Gwoupman Peyisan Borgne, a Haitian peasant community development organization (Mazzeo et al. 2013). Applied anthropologists working with HOPE observed firsthand the positive transformations achieved through community-based peasant development activities. Their research helped to guide the work of HOPE with local partners to implement a range of projects, including health-care and sanitation interventions. These were of vital importance because a deadly cholera epidemic broke out in Haiti in 2010. The extensive, capillary-like social network developed by Koodinasyon Gwoupman Peyisan Borgne served as a model for the mobilization of volunteers in the fight against the cholera epidemic.

Recognizing the importance of social networks in addressing health issues has been an integral component of the approach of many applied health anthropologists. The value of a medical health anthropological focus on social networks in the AIDS pandemic (as opposed to one-on-one prevention education models that focus on individuals) was demonstrated in the work of Robert Thornton (2008) in sub-Saharan Africa. Thornton was particularly focused on sexual networks, which he noted are invisible even to the people who participate in them (in that people know who they have had sex with but not all of the people their partners have had sex with, and certainly not all of the people their partners' partners have had sex with). Thus, he referred to sexual networks as unimagined communities. Yet, in sub-Saharan Africa, where sex is the dominant route of HIV transmission, sexual networks are absolutely critical to whom, to where, and to how quickly HIV spreads.

Indeed, it became Thornton's view that it was precisely because they focused on individuals and not on sexual networks that many AIDS education campaigns had little effect in slowing the epidemic in many places. The AIDS prevention effort in Uganda, where Thornton did his research, was comparatively successful, by contrast, because the population is largely rural and dispersed, socially segmented by class, ethnicity, and religion, and not particularly mobile. Moreover, the country was hit early and hard by AIDS. Noted Thornton (2008:231): "All of these factors had the effect of eliminating links between clusters of previously sexually linked people, and the whole [sexual] network collapsed. HIV was no longer transmitted efficiently throughout the network, and HIV prevalence fell dramatically." Based on his analysis of the AIDS epidemic in Uganda and South Africa, and the behaviors that occur in sexual networks that promote the spread of HIV, Thornton recommended that effective HIV/AIDS prevention should (1) encourage people to limit themselves to one sexual partner at a time (as having concurrent sexual partners with people who also have multiple concurrent sexual partners facilitates the rapid spread of the epidemic), (2) promote the idea that at the end of a sexual relationship people should refrain from having new sexual partners for at least one month (because people are most likely to infect others in the first month or so after they become infected), and (3) uphold the idea that people should avoid sex when traveling away from home (which would slow the ability of the virus to jump to new areas).

As these examples show (and a very long list of additional examples could be cited), health anthropology is not isolated in the ivory tower of academia. It is being used daily in numerous places around the world, from the most remote villages to the largest and most technologically advanced megalopolises, to address health-related problems of many kinds, such as community nutrition education (figure 2.1). Very likely, there is a health anthropologist working in the city or town where you live (or at least nearby), attempting to use anthropological skills to address real-world health problems. Finally, not all of applied health anthropology is a post hoc product of a research project. Health anthropologists also design and oversee intervention programs, evaluate the effectiveness of programs, and are involved in helping to establish and use ethical standards in health-care decision making.

The Health Anthropology Crystal Ball

What is the future of health anthropology? Where is the field going? How might it change over the coming decades? As jocosely summarized by Danish physicist Niels Bohr: "Prediction is very difficult, especially if it's about the future." One thing, however, is clear: as Nancy Scheper-Hughes and Margaret Lock (1987) argued some time ago, it is a fallacy to believe we will discover a narrow biotechnological salvation or magical medical bullet to solve all of the many old as well as emergent health and social problems faced by humankind. Human health problems reflect human social life and social interactions with the environment, and thus addressing health issues means knowing about the details of the social structural, environmental, and cultural aspects of the human condition. Herein lies an ongoing role for health anthropology. Without question, health anthropology also will be shaped by the social and physical worlds of the future. That future, as Yogi Berra once quipped, "ain't what it used



FIGURE 2.1 Dietician speaking about healthy eating at a community health education program organized by medical anthropologists and colleagues. Photo by Merrill Singer.

to be." This is particularly true with reference to health. One of the reasons is the impact of *ecobiosocial* changes taking place on planet earth. A second reason has to do with the configuration of the social worlds we are constructing. As a result of these changes, in addition to many of the current areas of focus, health anthropologists of tomorrow will be progressively more concerned with issues such as the following factors.

First is the far-reaching and increasingly faster pace of globalism, which facilitates the global spread of diseases around the world (e.g., diabetes). Globalism also contributes to the restructuring of local social contexts and ways of life in ways that create or enhance health vulnerabilities. One critical factor in this regard is what have been termed "killer commodities," that is, global products distributed around the world for a profit that cause harm to consumers because they are physically dangerous or contain toxic substances. Linked to killer commodities is the growing rate of dumping in underdeveloped countries of damaged, used-up, and otherwise discarded consumer products by the developed world.

Second, anthropogenic climate change has multiple and diverse impacts on the environment. At the same time, the earth is beset by multiple other forms of environmental degradation, including air pollution, depletion of edible ocean

fish stocks, acid rain, loss of wetlands, pesticide and other chemical pollution, salinization of agricultural zones, ocean acidification, deforestation, and loss of biodiversity through extinctions. It is likely that over time a growing portion of the toll these changes take on human health will be due to *ecocrises interactions* that multiply adverse effects.

Third, the world now faces a global refugee problem, and human migration generally is reshaping the distributions of populations globally. With mass human movement, which often occurs under less-than-optimal conditions, comes significant risks for disease, injury, mistreatment, and death. Citizen resistance to the provision of refugee aid can produce tense border conditions and punitive policy changes. Yet wars, changing environmental factors, and economic conditions are likely to continue to promote human movement, creating the need for culturally sensitive understandings and responses to avoid catastrophes.

Fourth, growing barriers to access to food or ways to produce it and potable water are already significant sources of disease and death in many parts of the world, and the impact of these factors on human health is likely to grow worse as a result of climate change, other anthropogenic environmental degradations, and displacements and disruptions of human communities.

Fifth, the level of threat from infectious disease in the twenty-first century affirms that our world is very different from what health experts in the not-very-distant past imagined it would be. Globally, infectious diseases today are a leading cause of death, even in highly developed nations. Witness the SARS epidemic in 2003, the H1N1 influenza pandemic six years later, followed by the Ebola epidemic in 2014–2016, and the 2015–2016 Zika epidemic. The rapid emergence of new and renewed pathogenic sources of human disease, including growing drug resistance among pathogenic microorganisms, has become a major threat to the future of human health.

Sixth, the adverse syndemic interaction of diseases, both acute and chronic, and somatic, emotional, and behavioral in nature, is intensifying the impact of diverse threats to health. While syndemics increasingly have come to be recognized as a significant aspect of the contemporary health profile, especially of impoverished, disadvantaged, marginalized, and oppressed populations, much remains to be learned about the nature of the interactions that occur among comorbid diseases and between adverse social conditions and patterns of disease clustering in affected populations.

Seventh, global health increasingly will be shaped by the urbanization of the human population, including the rapid growth in size and number of megacities with more than ten million residents, including an ever-growing number of highly vulnerable, densely concentrated, underserved, and poorly housed marginalized peoples.

Eighth, the development of the "profitable body," associated biotechnologies, and tissue commodification (e.g., organ selling by the poor) are reshaping human health and the social distribution of suffering. At the same time, the global distribution of pharmaceutical drugs has led to the use of these substances in ways that go far beyond their intended or allowable purposes. In addition, the global pharmaceutical industry, in all of its activities, from bioprospecting and the commercialization and patenting of traditional medicines, to clinical trials

in developing nations, to the operation in local environments of manufacturing plants, is having increasingly dramatic global health impacts.

Finally, in coming years, mounting global tensions stemming from the growing demands of an expanding global population, unequal distribution of resources, and the dwindling availability of natural resources (e.g., peak oil extraction will occur during this century and then production will decline), as well as climate change, are expected to intensify the frequency of war and group violence. Consequently, the level of carnage to be produced by future wars—including both the number of battlefield casualties caused by ever-enhanced armaments and the fact that modern warfare increasingly targets civilians through sexual crimes, community destruction, torture, mass displacements, malnutrition, and ethnocide, as well as through the spread of infectious diseases—is likely to set new historic records.

In short, the tremendous need for socially, culturally, and environmentally grounded understandings of the construction of human health and illness, and health interventions informed by these understandings, will continue to grow in the future, and in fact, may become much more urgent. As expressed by João Biehl and Amy Moran-Thomas (2009:282), "Continually adjusting itself to the reality of contemporary lives and worlds, the anthropological venture has the potential of art: to invoke neglected human possibilities and to expand the limits of understanding and imagination." At the same time, health anthropology has demonstrated its ability as an applied field to address pressing problems of human suffering in a world in which most of our afflictions involve a complex interface of sociocultural and biological factors, and in which social, economic, and political inequality are the primary forces threatening our species and our planet.

Discussion Questions

1. Anthropologists often stress the importance of culture in understanding human behavior. In her study in Madagascar, Harper found that there was something more important than culture. What was it and what do you think about her argument?
2. What were some of the key findings of Pearl Katz about "surgeon culture," how was she able to make these findings, and what was their practical value?
3. The chapter identifies eight things that health anthropologists do. What do you think are the three most important things they do and why?
4. How have health anthropologists responded to the ever-increasing importance of electronic communication in people's lives?
5. Why might people choose risky behavior?
6. What has been the anthropological approach to addressing the high rate of diabetes in many indigenous and ethnic minority populations?
7. Why do health anthropologists emphasize the importance of social networks?
8. Health anthropology does not provide simple answers to complex problems; instead it provides adequate understandings for the design of effective interventions. Discuss this statement and its implications.

Understanding Health, Illness, and Disease 3

I am interested in physical medicine because my father was. I am interested in medical research because I believe in it. I am interested in arthritis because I have it.

—Bernard Baruch, *New York Post*, May 1, 1959

Introduction and Overview

This chapter addresses some of the core issues and challenging questions engaged by health anthropology. What does it mean to be healthy or sick and who determines a person's health status? What is the relationship of culture to *health*, to *illness*, and to patterns of mortality? Is there a difference between having an illness and having a *disease*? What does cure mean? What are the varieties of *health care* and *healers* cross-culturally? In this chapter, we define key concepts and attempt to answer questions like those listed here. In this way, we further clarify the way health anthropologists look at and respond to health problems in the communities where they study and work.

Conceptions of Health and Illness

Defining Terms

Health anthropologists have devoted considerable energy to examining conceptions of health and illness cross-culturally. While *health* and *illness* are everyday terms, what do they actually mean? Like many words in common use, a closer examination shows that they are often much more complex and much more loaded with cultural content than is at first evident.

In what is now a classic and widely referenced (and debated) definition, the World Health Organization (WHO; 1978) designated health as "not merely the absence of disease and infirmity but complete physical, mental and social wellbeing." This enduring characterization is appealing because it stresses that health is not simply a biological state; it reflects a unity of mind and body, and, moreover, it is deeply interconnected with the quality of social relations and social conditions. Nonetheless, it is evident that this definition is somewhat utopian. It is unlikely that many people in the world would be found healthy in terms of this definition.

An important distinction has been made between *experiential health* and *functional health* (Kelman 1975). The former refers to a sense of wellness, an experience of tranquility and fulfillment through social accomplishments.

The latter is what employers expect from their workers to ensure performance levels essential to capitalist production or perhaps what professors expect from their students in showing up for classes and examinations. Even though employees or students may not quite feel like going to work or showing up for a 9:00 a.m. lecture, they are deemed functionally healthy if they do so, even if it requires several cups of coffee, an "energy" drink, or other stimulants. As an alternative approach, critical health anthropologists define health in terms of access to and control over the basic material and nonmaterial resources that sustain and promote life at a high level of satisfaction.

In truth, health is difficult to precisely define because the term embodies cultural value judgments about what is a desired state of being, and because the notion of health is deeply rooted in cultural metaphors. Metaphors are pervasive in everyday language and thought. For example, in capitalism, an economy is viewed as *healthy* if it is growing while money is called the *lifeblood* of society. In the domain of a person's state of being, health is metaphorically equated variously with goodness, moral purity, divine favor, or living the "right way." In each of these cases, ideas about health are imbued with values that reflect underlying worldviews. Growing up in a particular society, we commonly internalize its key metaphors and may not realize that they are cultural constructs or that they have a lot of influence on us, intellectually, emotionally, and even physically. Yet, metaphors provide us with a mechanism for thinking and communicating simply about very complex and opaque issues of profound concern. Whatever one's perspective, it is evident that health is a hard-to-define elastic condition that is best understood within an encompassing sociocultural and historical context.

The terms *disease*, *illness*, *ill health*, and *sickness* are often used interchangeably in everyday speech, but much ink has been spilled trying to use them as technical concepts with specific meanings. Biomedicine has depicted *disease* as a maladaptive state, particularly in terms of the environment. For example, long ago, William White, in *The Meaning of Disease* (1926), emphasized that during a disease an organism, at least temporarily, is fighting a losing battle with temperature, microorganisms, disappointment, or other environmental foes.

George Engel (1962), who is often credited as having fathered the biopsychosocial approach in biomedicine, asserted that "When adaptation or adjustment fail and the preexisting dynamic steady state is disrupted, then a state of disease may be said to exist until a new balance is restored which may again permit the effective interaction with the environment" (quoted in Clouser et al. 2004:92).

As suggested by these comments, biomedicine understands disease as a suboptimal or abnormal condition, a disorder of a normal structure or function of the body or mind, that affects part or all of a person or other organism and is expressed through specific signs and symptoms.

Differentiating Disease and Illness

The meaning-centered perspective in health anthropology posits a distinction between disease and illness, with the former constituting a natural entity that can

be identified through various bodily signs, such as a high temperature, elevated blood pressure, or rapid heart rate (figure 3.1), and (possibly but not always) a sense of physical and emotional discomfort (Kleinman 1980). The presence of a disease, in other words, is established through a diagnosis by a professional healer, like a physician of biomedicine. A disease label (e.g., diabetes) is a unit for asserting why someone got sick and specifying what condition they are suffering from. The recognized diseases of biomedicine are codified in the International Classification of Diseases (ICD; World Health Organization 2017a). This catalog includes places for hundreds of distinct conditions. Carpal tunnel syndrome, for example, occupies code G560 while Ebola sits at code A984. The ICD is designed to be both "exhaustive—to include all conditions—and to ensure that no particular event of sickness will be classified under more than one code number" (Hahn 1995:18). In other words, as reflected in the ICD as well in the *Diagnostic and Statistical Manual of Mental Disorders* (American Psychiatric Association 2013), biomedicine conceptualizes diseases as (1) discrete entities (each disease, be it cholera or paranoia, is assumed to be an objective entity found in nature and not just a product of human attention) and (2) clinically identifiable and hence a boundable item in the world. Normal practice in biomedicine, whether in its diagnostic, research, or treatment capacities, is guided



FIGURE 3.1 Screening for high blood pressure. Photo by Merrill Singer.

by the conceptualization of diseases as distinct, unique, and disjunctive entities that exist (in theory) separate from other diseases and from the social groups and social contexts in which they are found at any point in time.

Illness, by contrast, as it is often used in health anthropology, constitutes a cultural construction that can be identified only through interpretive activities or narratives. Illness is rooted in a culturally constituted explanatory model that seeks to clarify the source of distress and to outline a course of treatment to be pursued by both the healer and the patient. Even an event as simple as a cough takes place in historic and cultural context:

> Illness depends on relatively stable biological features—a cough in every culture uses the same muscles and respiratory organs—but it is also deeply historical, no less changing than the microbes that surround and interpenetrate us. Indeed, the country of the ill assumes the distinctive features of whatever nation or social group inhabits it. (Morris 1998:22–23)

As discussed later in this chapter, healers and patients, such as the *curandera* (Hispanic folk healer) and her patient seen in figure 3.2, may reach different conclusions about the nature of a patient's health problem, with varying consequences for healer-patient communication and relationships and patient adherence to healer instructions.



FIGURE 3.2 Maria Ruiz Mendoza, a Zapotec healer (*curandera*) from Teotitlán del Valle, Oaxaca, Mexico, treating a child for "anginas" with a tomatillo coated with salt. Photo by Liz Cartwright.

Reconceptualizing Disease and Illness

There has been a tendency in health anthropology to use the term *disease* for the biomedical diagnosis of sickness and to reserve the term *illness* for both the lay, folk, or popular healer's definition and the patient's understanding of his/her sickness. However, we suggest that the term *disease* refers to the diagnosis of a sickness made by a biomedical physician or some other type of professionalized heterodox medical practitioner, such as an Ayurvedic physician in India or a chiropractor or naturopath in a Western society. Within this scheme, the term *folk disease* (or more accurately, folk illness) refers to the folk healers' label for a sickness, such as *susto*, a label used by *curanderos* in Latin American cultures for the distress that a person may experience following a frightful experience. Illness refers to the patient's experience of sickness and the understandings of the members of the patient's immediate social network.

The social stature of a healing tradition is a critical component of its ability to shape disease conception. Most diseases or illnesses are self-limiting for reasonably healthy individuals. Conversely, chronic diseases endure, often with great physical cost to the patient. Ultimately, of course, all bodies die under the weight of diseases, traumas, or other factors. Thus, death has been called the great equalizer, and this idea was even enshrined in a medieval European ritual known as the Dance of Death. It involved a processional through churchyards and cemeteries led by a figure macabrely dressed as a hooded skeleton. This ritual has been analyzed as an expression and enactment of deep collective anxiety under desperate social and economic conditions and recurrent deadly epidemics of the plague. In one very important sense, however, death is not the great equalizer: when you die, the conditions under which you die, what you die from, and the amount of suffering you endure before you die is very much a reflection of significant social disparities within and across societies. Death comes to us all, but the nature of its actual occurrence is the embodiment of inequality.

Understanding Cure

Biomedical practitioners often ascribe "cures" on the part of either professionalized heterodox practitioners or folk healers to the placebo effect. This has been defined as

> any therapeutic procedure (or a component of any therapeutic procedure) which is given (1) deliberately to have an effect, or (2) unknowingly and has an effect on a symptom, syndrome, disease, or patient but which is also used as an adequate control in research. The placebo effect is defined as the changes produced by placebos. (Shapiro 1964:713)

In fact, as numerous studies have indicated, the placebo effect occurs within all medical systems, including biomedicine, as randomized double-blind tests on drugs have repeatedly demonstrated. This is something that at least some biomedical physicians recognize as they seek to meet patient expectations. The term placebo is Latin for "I will please."

One schema developed by a health anthropologist delineates three human responses to sickness: (1) autonomous responses, or "all those processes which the organism can invoke to gain health or equilibrium, including the various immunological and related systems"; (2) specific responses of the body to treatment, be it an administration of a medicinal herb or penicillin; and (3) meaning responses, or the "psychological and physiological effects of meaning in the treatment of illness" (Moerman 2001:14). The meaning response tends to encompass most situations that have been generally associated with the placebo effect, whether it is the cure that a patient experiences after taking a sugar pill or the impact of a healing ritual conducted by indigenous practitioners of the kinds that we discuss in chapter 6. The placebo effect suggests that the subjective understandings and responses of patients to the treatments, rather than just the objective biological impact of treatments, play a role in patient recovery or deterioration. In biomedicine, for example, the conditions, procedures, and materials of treatment are meaningful to patients; they represent the healing power ascribed to this medical tradition:

> Clinicians often dress up in special uniforms that convey power and authority. They have very expensive machines that can look inside your heart or brain. All this, plus the magnificence of the hospital building . . . , the decor of the office, the lights of the operating room, the computers on every lap, the magical prescription pad, and the caring nurse, piles up meaning with increasing power regardless of what may be in the capsule or syringe. These meanings create expectations that can dramatically modify the effectiveness of even the most powerful proven treatments. (Moerman 2011:172)

Folk Understandings

Anthropologists have observed that peoples around the world operate with a wide array of definitions of health and illness, most of which tend to reflect core themes of their underlying culture or contact with other groups (Erickson 2008). The James Bay Cree, for example, a traditionally foraging society in subarctic Canada, view health (*miyupimaatisiun*) as a balanced relationship between individuals and their sociocultural and natural environments. Among the Cree, health "implies proper hunting practices and hence a respectful relationship between humans and the animals of the land: a successful hunt means healthy eating and appropriate social relations means that the hunted foods will be apportioned amongst oneself and one's kin" (Adelson 2003:617). The James Bay Cree operate with two categories of illness: those indigenous to their culture and those that they adopted (and adapted) as a result of their contact with Europeans. Half a world away from James Bay, the !Kung San of the Kalahari Desert in southwestern Africa interpret the ability to sweat as an indicator of good health since sweat is regarded to be a life-giving substance. The Matsigenka, a people in the upper Amazonian rain forest of southeastern Peru, a group with a mixed economy consisting of fishing, hunting, gathering, and horticulture, use the term *shinetagamtsi* to refer to a sense of contentment, productivity, being well fed, and freedom from illness.

To cite one additional example, in their traditional culture, the Han, by far the numerically and politically dominant ethnic group in China, defined health as a harmonious relationship between humans and the cosmos and among humans themselves. A healthy body is one in which Qi (a form of energy that permeates the universe), blood, vital essence, body fluid, and nutrients are in careful balance. An imbalance results in illness and may emanate from a wide array of factors, including wind, cold, heat, damp, dryness, fire, epidemics, emotional turmoil, improper foods, stress, physical injuries, parasites, and blood stagnation. As these examples suggest, popular understandings of health vary, but, as health anthropologists long have emphasized, to understand them they must be studied within social and cultural context; health, as people know it, is always entwined with other key themes and structures of society. This is true for college students as well. Abusive drinking on and near college campuses produces disturbing public health statistics about alcohol-related student deaths, injuries, sexual assaults, and negative academic performance, yet efforts to control campus drinking often are resisted by students who see it as an infringement on their rights and faculty who oppose policing students. This response indicates that alcohol-enhanced partying and even binge drinking on campus are enmeshed in strongly held cultural values.

Understandings of Disease Causation

As they vary in their understandings of health, societies also differ considerably in terms of what they regard to be the causes of illness. The Yanomamo, horticulturalists in the Amazon basin of Venezuela and Brazil, for example, believe that virtually all illnesses are caused by spirits or ghosts of their ancestors. Conversely, the Indians of the northwest coast of the United States traditionally identified several causes of disease, including soul loss, sorcery, taboo violation, disease-object intrusion into the body, and spirit intrusion. Table 3.1 summarizes many of the diverse theories of the causes of disease or illness (depending on whether a healer or a layperson makes the diagnosis) encountered by health anthropologists in their research.

Beyond the alternatives listed in the table, many societies employ a "humoral system" of health and illness of one sort or other. A humoral medical system is organized around the premise that the organs, body systems, and fluids inside the body parallel elements and structures of the natural world outside the body.

For the ancient Greeks, for example, the bodily "humors" of phlegm, yellow bile, black bile, and blood were believed to reflect the natural elements of air, water, earth, and fire, respectively. Maintaining good health, the ancient Greeks believed, depended on preserving a balance of the humors. When a person became ill, Greek healers would drain surplus humors by purging the digestive tract or through bloodletting. These therapies were believed to help restore a balance in the body's humors.

The three major humoral theories found in the world today are traditional Chinese healing, Ayurvedic medicine of South Asia, and the Latin American system, which has its historic roots in ancient Greek and other Mediterranean civilizations. Guatemalan Indians, for example, whose pre-Columbian beliefs

TABLE 3.1 Theories of Disease or Illness

Theories of Natural Causation

- Infection
- Stress
- Organic deterioration
- Accident
- Overt human aggression

Theories of Supernatural Causation

- *Theories of mystical causation*
  - Fate
  - Ominous sensations
  - Contagion
  - Mystical retribution
- *Theories of Animistic Causation*
  - Soul loss
  - Spirit aggression
- *Theories of Magical Causation*
  - Sorcery
  - Witchcraft

were reshaped following colonial contact with Europeans, generally acknowledge four causes of illness: natural injury such as snakebite, internal bodily imbalance, intrusion of pathogens, and witchcraft. Like other Latin American peoples, they believe that excessive consumption of "hot" foods (e.g., beans, beef, and chicken) or "cold" foods (e.g., corn, eggs, and oranges) can result in chills and coughing. The categories of "hot" and "cold" refer not necessarily to the actual physical temperature of a food or location but rather to its qualities and effects on the body, and the categories often vary from group to group in Latin America.

Humoral systems are not the exclusive domain of peasant populations but also are found in various indigenous societies, such as the Melpa, a horticultural village people in the Mount Hagan area of Papua New Guinea. The Melpa humoral system includes three primary elements: (1) blood, which represents kinship ties through women and is associated with red ocher and red-colored foods; (2) grease, which takes the form of breast milk, semen, fat, the nurturant component of vegetables, and the fertility capacity of the soil and which may become depleted during pregnancy; and (3) a complex hot/cool/cold distinction (Strathern and Stewart 1999). In the last case, blood that becomes "hot" may dry up as a result of sorcery, whereas high altitudes are believed by the Melpa to be too "cold" to maintain life on a permanent basis. Blood and grease exist in finite supply and may become scarce, resulting in illness. The Melpa view blood and grease as analogues to wealth as they know it, including pigs, shells, and money.

Another indigenous people, the Tuareg, a pastoral group that makes its home in the Sahara and Sahel of West and North Africa, also subscribe to a complex distinction between "hot" and "cold":

> Conditions of "hot/cold" and associated afflictions are sometimes literal, sometimes non-literal or metaphorical in connotations, for example, "hot" illnesses (tuksi) are believed to be caused by too much heat, from "hot" foods (dates, tomatoes), sunlight, or moon-beams (these latter may cause illness from direct [contact or exposure], as in sun-stroke, or, alternately, by sitting on warm mats or from their reflections inside doorways). Tuksi may also result from anger and other strong sentiments. . . . "Cold" illnesses (tessmat) are the counterpart of "hot" illnesses. These include urinary tract problems and STDS. (Rasmussen 2004:1005)

Humanizing Biomedicine

Health anthropologists examine conceptions of health, illness, and disease across societies. As part of this endeavor, various health anthropologists have sought to humanize biomedical care by recording patients' illness narratives (their stories of their illness experiences, concerns, and reactions) so as to assist physicians to better understand their patients, their patients' views of illness, and their patients' experiences of being sick. In his important book *The Illness Narratives*, Kleinman (1988), for example, a prominent physician-anthropologist, sought to persuade his colleagues within the corridors of biomedicine to listen closely to what their patients have to say, how they say it, and what they are trying to communicate. Patients, Kleinman stressed, order their experience of illness in light of what is significant to them and others. These personal narratives give coherence to the encounter with suffering. But patients do not tell completely private stories; rather they draw on plot lines and metaphors from their cultural heritage to create accounts that are meaningful to them and for telling their stories to others. Illness narratives express what matters most in patient lived experience. As this focus on patient narrative reveals, health anthropology often has as its applied goal the explication of sufferer experience, especially among those who suffer in silence because their voice is discounted by more powerful social groups or strata.

Sufferer Experience

Experience and Cultural Symbols

When Jerry, a gay man in Ohio, learned he was infected with HIV, he was twenty-six years old. When he subsequently was interviewed by a health anthropologist, he had been living with the infection for eleven years. He had remained relatively healthy during this period (diseased but not ill) until just about a year before the interview. By then, he had developed severe pain in his legs and a general decline in stamina, eventually forcing him to stop working. Still, he emphasized, "I'm a survivor, no matter what, y'know? I can make ends meet. I can get through anything. I guess maybe I get that from the streets, y'know? No matter what happens, you gotta keep going" (Bloom 2001:43).

A very different health-related experience was told to another health anthropologist by a thirty-five-year-old woman who was losing hope of ever having a child:

> My chances are getting smaller and smaller. It's like somebody has cancer and you say, "Well, you have a 20 percent chance that you will live"; and then go down to 10, and then to 5, and then you say, "You're not going to live, let's face it. You are going to die any day. Well, you just have to give up," and I guess that's where I am at now. . . . I really equate this with dying. (G. Becker 1994:396)

As these two contrastive insider accounts of "sufferer experience" suggest, tension between retaining hope, which is often expressed in a militant commitment to fight against disease, and losing hope, which, in this instance, is expressed by way of a progressing disease metaphor, is a common motif in the accounts of people facing illness in Western culture. In other societies, illness experience is shaped by quite different cultural symbols and metaphors. In northeastern Brazil, for example, the experience of having to suppress intense emotions, to avoid interpersonal conflict or social disruption, a behavioral event that is thought to be a common cause of illness, is expressed through the metaphor of swallowing frogs (*engolir sapos*). Thus, speaking of her relationship with a domineering and abusive man, one of the women interviewed in a study of folk illness in Brazil stated,

> I always had to swallow frogs, you know, because I was totally dependent on him. . . . Even when I knew I was right and he was unjust, I had to swallow it, I had to obey. I had to apologize, I had to humble myself and submit, and even thank him for mistreating me, in fear of my life. (Rebhun 1994:367)

Finally, a study with impoverished women in Nicaragua who reported suffering from "brainache" (*dolor de cerebro*), found that this symptom was reported in relation to women's persistent worries about the impact of death, abandonment, and outmigration of adult children on personal and family well-being (Yarris 2011). Brainache embodies the vulnerability and distress women experience as they confront conditions of extreme hardship in their local social worlds. In short, symptoms are culturally influenced although subjectively experienced and, as they are talked about, are "a necessary condition for us to articulate a relationship to the world and to others" (Biehl and Moran-Thomas 2009:273).

Cultural Emotions

Culture is not just a cognitive system, something we use to experience, understand, and talk about the world; it is, in addition, an emotional system, something that shapes our feelings and thus motivates our actions. By encapsulating frightening or disruptive illness experiences in familiar and meaningful cultural frameworks, culture binds "the past and future together . . . and give the appearance of coherence or 'return to the whole'" (G. Becker 1994:385). This is one of the ways in which pain, uncertainty, fear, and other physical and emotional challenges of illness are culturally managed and, to greater or lesser degree, tamed or even routed. Indeed, as health anthropology research on illicit-drug users living with HIV infection found, illness can even be experienced as salvation. As one of the participants in that study, a man who had lived with HIV disease for seven years, explained,

> The way I'm starting to look at it is that God got a plan to make you wake up and see things sometimes from this virus, to make you stop doing what you're doing [e.g., using drugs], killing yourself. . . . I guess He wanted me to stop suffering and start looking inside myself and He presented this virus to me and said, "Hey, this is it! You want to keep doing this and die, or do you want to live and look at life on life's terms?" And it worked. (Mosack et al. 2005:595)

Coming to view having HIV/AIDS as a "wake-up call" from God suggests the importance of cultural themes in individual sufferer experience. In this instance, the cultural leitmotifs mobilized by the sufferer are probably derived from Christian beliefs about individual agency and salvation through God's generosity. In this light, sufferer experience can be seen as a social product, even though individuals from the same complex society may experience the same disease in somewhat different ways. From the perspective of critical health anthropology, sufferer experience emerges from the action arena situated between socially constituted categories of meaning and social structures. This includes finding culturally meaningful reasons why a person gets sick, such as "I failed to exercise enough," "I was exposed to a virus when she sneezed on me," "I could not stay away from drugs" (figure 3.3), and the political-economic forces that shape daily life and thereby place individuals in harm's way, such as black lung disease among miners or malnutrition among children of the poor.

In her highly acclaimed book *Death without Weeping: The Violence of Everyday Life in Brazil*, Nancy Scheper-Hughes presents a vivid and moving portrayal of human suffering in Bom Jesus, a dismally poor shantytown in northeastern Brazil. She contends that the desperate and constant struggle for basic necessities in the community contributes to an indifference among mothers toward the weakest of their offspring. Scheper-Hughes recognizes that what mothers of deceased children have in common is intricately related to the collapse of the sugar plantation economy, which left many people without even a subsistence income. While there has been local development, most of the residents of Bom Jesus have not benefited from the growth of agribusiness and the industrialization sponsored by both transnational corporations and the Brazilian state, resulting in inadequate diet and a high level of infant death. Maternal indifference to infant death, rather than the mournful response we expect of mothers who have lost a small child—much like the laughter over tragic events reported among the shantytown dwellers elsewhere in Brazil (Goldstein 2003)—can be understood only in a social context. Goldstein (2003:39), in recounting the death of a poor man named Zeca, who died a prolonged, painful death and whose corpse initially was put in a refrigerator in a butcher shop while awaiting burial, notes,

> After Zeca died, the family had to pull itself together to dress him for the burial, but it was difficult to do so because his body was so stiff. The final, even more grotesque aspect of the story was that Zeca had a little "hard-on," a detail that during each telling of the story [of his death] brought laughter to those who listened and remembered one more time the details of this tragic story. For Soneca [his sister] and the others who witnessed his death, there was nothing left to do but laugh. The laughter was mad and absurd, similar to the conditions under which they lived.



FIGURE 3.3 Drug user preparing drugs for injection. Photo by Robert Booth.

Social Suffering

Suffering, in other words, like cognitions and emotions, is far from a narrow individual experience, and indeed much suffering in the world today is intimately connected to changes in the global capitalist economic system; these are played out and leave their effects on local physical and social settings and the people who inhabit them, including sparking rebellions against subjugation, some of which inflict considerable suffering of their own. Health anthropologists use the term *social suffering* (Kleinman et al. 1997) to link individual experience of pain and distress to the wider social events and structural conditions that often are the ultimate sources of human misery. Social suffering, in short, refers to the immediate personal experience of broad human problems caused by the cruel exercise of political and economic power, such as forced abortion, coerced sterilization, terror, political rape (often used as a form of ethnic oppression), exploitation, and abject poverty, topics that have gained attention in health anthropology in recent years. Examination of painful memory is now a major anthropological topic of research. Painful memories include *cultural trauma*, which are "wounds in the collective memory that contribute to the construction of identity in different social groups" (e.g., the trauma of slavery and the formation of African American identity or the trauma of the Holocaust and the construction of Jewish identity; Fassin and Rechtman 2009:25). In the analysis of the death of a Puerto Rican man in New York from the ravages of alcoholism (M. Singer et al. 1992),

for example, anthropology researchers sought to show the lines of connection between individual distress and disease and the colonial transformation of life and society on the island of Puerto Rico following the U.S. takeover in 1898. This included the subsequent dispossession of thousands of Puerto Rican farmers of their land as a result of federal taxation and the thirst of giant U.S. agricultural companies for choice farmlands on the island. This landgrab resulted in the migration of thousands of Puerto Ricans to the U.S. mainland, where they had citizenship imposed on them during World War I so that they could be drafted. After migration, the hopes of Puerto Ricans for a better life often were dashed as they encountered intense discrimination in housing, jobs, education, and everyday activities in their new home. Subsequent massive layoffs from factory jobs—one type of employment that, for a while, had been open to them—occurred when factory owners moved production south or out of the country in search of ever-cheaper, more exploitable labor. Many migrants consequently suffered hopelessness, worthlessness, and a sense of failure, including the Puerto Rican man whose life became the subject of a health anthropology study.

The scale and depth of social suffering in the world can at times overwhelm our ability to fully comprehend it:

> Knowledge of suffering cannot be conveyed in pure facts and figures, reportings that objectify the suffering of countless persons. The horror of suffering is not only in its immensity but in the faces of the anonymous victims who have little voice, let alone rights, in history. (Chopp 2007:2)

One objective of anthropology is to play a role in collecting the narratives of the survivors of the modern history of inequality and oppression as "testimony of suffering and yet surviving" (Chopp 2007:2).

Beyond Social Suffering

Even under the bleakest of conditions there are lessons to learn about the nature of our species, insights that go beyond recognition of our capacity for rampant brutality. Anthropologists in recent years have studied the lived experience of war and political violence, especially among noncombatants who nonetheless are among the most common victims of conflict and the unleashing of armies. Examination of the political shaping of war knowledge, including management of events, images, and experiences, often serves to make war appear inevitable, acceptable, and even desirable. Enemies and those being characterized as future adversaries "are never seen for what they are (i.e., people like us, with families and friends, fears and desires)" (Rylko-Bauer and Singer 2011:226). In the process, the full extent and enduring nature of the human costs of war and political violence are often rendered invisible. In this vein, Carolyn Nordstrom (1998) studied resistance to terror warfare and the crafting of identities based on an embrace of such resistance among the people of Mozambique during the war of 1976 to 1992. She captures the startling brutality of that war in the following description by Adriaan van Dis: "Thousands of children have seen their parents die before their eyes, hundreds of children have been boiled in the presence of their parents. Heads of old people have been used as stools by the bandits,

unwilling farmers are nailed to trees" (1998:108). Despite this inhumanity, Nordstrom's (1998:113) field notes show that most of the conversations she had with Mozambicans during the war "in one way or another reflected their preoccupation with both healing the wounds of war—physical, emotional, and social—and defusing the cultures of violence that war had wrought." These accounts and the heroic efforts they describe stand "as reminders that under the most extreme circumstances, the majority of people work to re-create a viable society, not to demolish it" (Nordstrom 1998:113). Telling this often-forgotten story too is a mission of health anthropology.

Disability and Chronic Illness

At the opening of this chapter, there is a quote from statesman and financier Bernard Baruch indicating that he became interested in arthritis when he became an arthritis patient. Similarly, the anthropologist Robert Murphy began his research on the issue of disability and being wheelchair-bound when cancer of the spine caused his legs to be paralyzed. On the basis of his experience and work in this area, he wrote a book (1987), discussed below, which has become an important piece in the disability and chronic disease literatures.

The Patient in the Body

Murphy focuses his book on the insider's experience: what it is like to be unable to walk, what it is like to be dependent on a wheelchair, and how this kind of loss is far more than a painful physical transition. In addition to the loss of the use of his legs, Murphy realized that he had lost an essential part of his sense of self. Despite strong social support from loved ones, he felt alone and isolated, a diminution of everything he used to be. At times, Murphy reported, he had a strong wish to withdraw from the world, to crawl into a hole and disappear, and to angrily mourn his loss of self.

Before he was paralyzed, Murphy realized that he really did not see disabled people; he, like others, had been practicing a kind of selective blindness when in the presence of a disabled person. This radically changed when he was the disabled one, as he became hypersensitive to the social position and social treatment of the disabled, including the common tension and awkwardness of personal interactions between the disabled and the able-bodied. Social situations of this sort often lack clear-cut cultural guidelines about how to act, leaving people quite uncomfortable or in open distress. Murphy realized that people were giving him a wide berth and avoiding directly looking at him. He had begun to vanish socially. He also retreated from his own body; like many paralytics, he felt less and less attached to his own body, less that he was his body. He suffered, in short, not just a disease of the body but a disease of selfhood and a disease of social relationships.

As Murphy's poignant account makes vividly clear, disability is far more than a physical problem; rather, examination of disability, an issue that has engaged a number of health anthropologists, raises important questions about the basis of selfhood as a culturally constructed yet often taken-for-granted attribute.

Additionally, it raises difficult questions about the ways in which cultural expectations shape social interactions and social relationships while challenging conventional ideas about what being normal means.

The Cultural Construction of Disability

An especially important figure in the anthropology of disability is Joan Ablon. In her various studies of disabled groups, Ablon (1992:10) has focused on "the daily insults which endanger their personal identity and self-image, their social life, their economic opportunities." She became especially concerned with social exclusion associated with being disabled and how to overcome these barriers, including both those internalized by disabled people and those imposed by the able-bodied. In her well-known research with dwarfs, for example, Ablon (1984) emphasized that the dwarf body is different but not disabled and that dwarfs experience themselves and their bodies as normal. In interactions with the wider society, however, they are subject to discrimination and frequent messages about abnormality. Increasingly in recent years, health anthropologists have studied a growing range of disabilities and chronic health problems, including blindness (Ainlay 1989), deafness (G. Becker 1980), post-polio syndrome (Scheer and Luborsky 1991), impairments of mobility (Luborsky 1995), and psychiatric disability (Duncan 2017).

Regarding the last of these topics, increasingly anthropologists have been putting ethnographic inquiry to productive use in assessing the strengths and weaknesses of existing treatment modalities for psychiatric patients. An important contribution to this body of work is Neely Myers's 2015 book *Recovery's Edge: An Ethnography of Mental Health Care and Moral Agency*, an area of study influenced by Myers's challenging experiences with her brother, who was diagnosed with schizophrenia at a young age. The book is based on a three-year ethnographic study of a U.S. mental health treatment organization that she calls Horizons (pseudonym). This program uses a treatment model known as recovery-oriented care. This involves treatment designed to assist and support an individual suffering from a serious mental health condition to "return to the life one might have expected to lead if one had never become ill—or possibly even a better life" (Myers 2015:4). Invited by the directors of Horizons to study its approach to implementing patient-centered, recovery care, Myers's goal was to understand how the organization defined recovery, what this approach required of staff and care recipients, and the nature of facilitators and hindrances of the recovery process.

As used at Horizons, and elsewhere in the contemporary U.S. mental health treatment and support circuit, the recovery model asks "people to raise . . . out of [their] situation by becoming rational, autonomous, hard-working adults" (Myers 2015:19). In other words, the model prioritizes workforce reentry over the building of meaningful personal relationships, or, as Myers phrases it, "work in place of wellness" (Myers 2015:52). Recovery is conceptualized within the program as a kind of existential journey that leads an individual to steady employment. On the one hand, this journey reflects American cultural images about the hero's transformative journey of self-discovery and personal growth,

as expressed in successful movies like *Harry Potter and the Philosopher's Stone*, the Star Wars series, *The Matrix*, *Spider-Man*, *The Lion King*, and the Lord of the Rings trilogy. On the other hand, the goal of the Horizon recovery journey aligns with American cultural values about taking active personal responsibility for one's life, maintaining a record of accomplishment, and gaining economic success. In Myers's view, it is only once care receivers are employed that they prove themselves worthy of social recognition and reintegration with society. Myers refers to this as achieving "moral agency" in the eyes of program staff. Recognition of having a right to moral agency, Myers contends, is only granted to care recipients after they evidence that they are rational, autonomous, and productive, hardworking people. Stemming from the fact that the emphasis is on gaining outside employment rather than on building supportive relationships outside of the mental health-care system, care recipients find they must travel their journeys alone. Based on her experience with people diagnosed with serious mental health conditions such as schizophrenia, Myers argues that establishing dependable intimate relationships should be the first goal of treatment, not something delayed until after they are able to work, if people are to succeed in their recovery journeys. Also, she is critical of psychiatric treatment programs that heavily medicate their patients because it dulls their experience and blocks the spiritual, emotional, and psychological work that is critical to the recovery process.

In a related study, an intimate examination in Japan of the residents of a treatment community for people primarily suffering with schizophrenia, a health anthropologist lived for eight months with patient residents gaining their trust and getting to know about their lives and experiences (Nakamura 2013). As a result, through the presentation of detailed stories of patients' lives, it became possible to see beneath the usual stereotypes of people who historically have been stigmatized by Japanese psychiatry and society, and hidden at home or warehoused in impersonal, sterile institutions. Treatment, which includes full patient involvement in decision making, involves a limited use of psychiatric medications because they are seen as dampening patients' emotional lives and social experiences. Emphasis is on self-reliance, social skills training, social support, and helping patients to manage the disturbing inner voices they hear. Patient hallucinations (referred to as *Gencho-sans*, "honorable voices") are externalized from patients and modeled by dolls or by members of the therapeutic group. The treatment approach is based on the idea that sufferers do not recover from schizophrenia; rather they learn to live with it, similar to the conception of alcoholism promoted by Alcoholics Anonymous. The study also addressed social factors in the emergence of mental health problems through the telling of patient stories, such as a young man who prior to hospitalization had been working in the Japanese motor industry. When he saw that shoddy work was being passed off as acceptable, he openly complained. In response, his supervisors subjected him to disturbing "reeducation" sessions that provoked the onset of his symptoms.

In their work at the Philadelphia Geriatric Center, health anthropologists and colleagues (Scheer and Luborsky 1991) found that the decisions patients make about disability-related issues are infused with broader concerns about their identity and the fulfillment of personal and cultural expectations. People

who survived polio, for example, often learned from their health-care providers that they should forget about having had a bout with polio, put the past behind them, and move on with their lives. This attitude reflected a wider cultural orientation during the era that polio was prevalent in the United States in the 1950s. For polio survivors who suffered polio-related disabilities later in life, the early polio-recovery cultural expectations they learned and embraced came back to haunt them as they encountered a new wave of functional losses many years after their initial bout with the disease. It became clear to them that in health the past may not always be left behind easily.

Trauma and PTSD

Posttraumatic Stress Disorder (PTSD) has been a consequence, no doubt, of every war in history as well as all other violence and traumatizing events (e.g., earthquakes). Yet it was not until 1980 that the American Psychiatric Association accepted PTSD as a legitimate diagnosis. The change was not fully welcomed at first but met with controversy. Emphasized in the new diagnostic category was exposure to an external catastrophic trauma. In fact, PTSD is unique among psychiatric diagnoses because of the need for there to be an etiological agent, the traumatic stressor. What was not realized when it was first introduced as a health condition is just how common PTSD is among both veterans and nonveterans. Also realized over time have been the range of symptoms of this painful condition, including negative cognitions, flashbacks, and mood states such as grief and despair as well as disruptive behavioral expressions such as outbursts of anger, violence, impulsiveness, recklessness, hyper-vigilance, and self-destructive actions including abusive drinking and other illicit and prescription drug abuse. Comparatively high rates of domestic violence among PTSD sufferers create the potential for "secondhand PTSD" among their traumatized children.

In their approach to the study of PTSD, health anthropologists have emphasized the interview-based stories of sufferers, including their perceptions of traumatizing events and their lived experience of the way their symptoms have impacted their lives and social relationships. In a thorough account of such stories told by combat veterans presented by Finley (2011), for example, we hear about the disturbing things witnessed by men and women deployed as soldiers in Iraq and Afghanistan. After they return to the states, often with little time to adjust to the transition between two very different experiential worlds, they report struggling to cope with confusing internal conflicts, intense grief, and jarring images. Finley stresses that not everyone who has fought in these wars suffers from PTSD and not to the same degree. How returning soldiers are treated and how PTSD is constructed by friends, family members, doctors, and other caregivers is an important component in the development or severity of the condition. In Haiti, the frequent experience of trauma is mediated by the cultural concept of *ensekirite* (Creole for insecurity), which labels the collective experience of the costs of living at the crossroads of multiple threats produced by political instability, economic subordination, environmental precarity, and the social strains of poverty (James 2016).

A health anthropology lesson is that "Culture matters in PTSD, from resources that foster social and emotional support to interpretations of symptoms

and how to make sense of trauma" (Collura and Lende 2012:135). A second lesson is that there is a need for a shift in the cultural dialogue from portraying PTSD as a chronic condition to recognizing that it need not be a lifelong illness.

Stigmatization

Another and related topic of concern to health anthropologists is the process of health-related social stigmatization and the creation of what have been called socially "discredited selves" in which disease sufferers are blamed for their health problems. People with AIDS, for example, often have been accused of being responsible for their illness through immoral or illegal behaviors. What health anthropologists have found is that the experience of illness is far more than the exposure to physical symptoms. Fear of discrimination and stigmatization may prevent people from seeking health care. Fear of being diagnosed with a stigmatizing disease is a primary factor in why Mexican laborers in California waited an average of eight and a half months before going to see a doctor for symptoms that turned out to be tuberculosis (Rubel and Garro 1992). Similarly, a study of breast cancer beliefs among Latina women in California found that about one-third of women who had immigrated from Mexico (but only about 10 percent of those from El Salvador) believed that God gives people breast cancer because they lived a "bad" life, such as using illegal drugs (Chavez et al.1999). Notably, various studies have shown that Latinas have comparatively lower rates of self-screening or mammogram screening for breast cancer and as a result tend to be diagnosed at a later and more life-threatening stage of disease development.

A major contribution of health anthropology to the study of stigma has been in the examination of its local social construction cross-culturally and over time. As part of this work, researchers have examined the interface between local cultural beliefs and practices and stigmatization for a variety of diseases. Additionally, they have challenged approaches that limit the analysis of stigma to narrow cognitive explanations without regard to larger structural forces, such as the exercise of power, the existence of social conflicts, and the use of structural violence.

Additionally, by seeking to know sufferers directly, by striving to understand their experience and point of view, and by bridging the barriers that divide the well from the sick, the able-bodied from the disabled, and credible from discredited selfhoods, health anthropologists are able to use this otherwise unobtainable information in developing needed recommendations or programs designed to minimize the many social harms attached to physical illness, including the damage done by stigmatization. In the case of the study of beliefs about breast cancer among Latinas, researchers used their findings to design a culturally appropriate intervention aimed at improving Latinas' knowledge about and attitudes toward breast cancer as a means of increasing their use of breast self-examination and mammography. In developing their program, these applied researchers "felt that it was imperative to incorporate the Latinas' beliefs into the intervention rather than dismissing them as silly or folkloric" (Chavez et al. 1999:133). Outcomes with a sample of eighty-eight Latina women, in which fifty-one were assigned to the culturally appropriate intervention and thirty-seven to a control group, found that the culturally based intervention was associated with greater increases in

Stigma as a Social Driver of Obesity

In recent decades, a global biological trend has been of the spread of overweight and obesity. Scientific evidence points to the effects of interrelated structural factors including poverty, import-dominated food systems, changing work patterns, the development of more sedentary life patterns, and the nature of the built environment to explain why so many people around the world are becoming obese. Despite the widespread nature of this trend, and the existence of local deviations from the norm, the dominant cultural model in both biomedicine and public health around the world and beyond these in the media and in popular discourse emphasizes individual responsibility and personal failure for becoming overweight (Brewis and Wutich 2015). The core social message stigmatizes significant weight gain, which is portrayed as irresponsible, shows a lack of willpower, reflects laziness, and indicates a lack of self-respect. In short, fat is stigmatized as bad. These negative conceptions shape national and local policy and public health strategies, while undergirding the billion-dollar weight-loss industry.

Paradoxically, fat-stigma, rather than contributing to individual weight loss, undermines such efforts. People who experience weight discrimination are more likely to remain obese; "weight stigma begets weight gain" (Tomiyama 2014:8). This is because the feeling of being "stigmatized and being explicitly aware of exclusions or mistreatments because of one's 'fatness' predicts greater stress, fewer health-enhancing behaviors, and worsened health outcomes" (Brewis et al. 2017:258). In short, stigma is a social driver of obesity and, in fact, many of the health problems commonly attributed to being overweight are products of discrimination and stress resulting from the stigma.

In everyday social life as they move about the built environment, people with high body weight encounter punishing architectures that block them from fitting in socially. These "fat stigmatizing environmental cues," include too-small chairs, seat belts on airplanes that do not fit, narrow booth spaces in restaurants, and doors on low-slung cars (Brewis et al. 2017:259). Although interpersonal interaction (from staring or unwanted comments) are an important source of stigmatization, the reminder of not fitting in, not fully part of society, is often built into the environment showing that "space, architecture, clothing, and so on can also render bodies pathological through systematic exclusion" (Brewis et al. 2017:272).

knowledge about breast cancer and a significant lowering of stigmatizing beliefs about the disease compared to the control group.

Human Rights and Health

The issue of suffering—and medical anthropological approaches to it—raises an important issue of urgent concern within contemporary anthropology, namely,

the discipline's approach to human rights, including its stance toward the notion of health as a human right. Since just after World War II, anthropology has struggled with this issue. This chapter in the history of the discipline opened with the "Statement on Human Rights" written by Melville Herskovits and thereafter adopted by the American Anthropological Association's Executive Board (American Anthropological Association 1947) as a reflection of the discipline's initial perspective on this issue of global concern. The statement was drafted as part of an international effort led by the United Nations to craft what would eventually be called the Universal Declaration of Human Rights. The latter was intended to serve, in the grisly shadow of Nazi atrocities, as a collective affirmation of global respect for human dignity and inalienable liberties. Herskovits, as an internationally known anthropologist, was asked to bring cross-cultural anthropological insight to the task of establishing an agreed-on bedrock of universal moral imperatives concerning human rights and justice. Instead, reflecting an important theme within the discipline, he rebuffed the idea that anthropology could contribute to this effort on three grounds: (1) the conception of rights and morality varies considerably cross-culturally; (2) as a science of culture, anthropology does not make normative judgments about particular cultural practices, an approach the discipline deems ethnocentric; and (3) in use, a universal declaration can be used to justify sanctions against a society acting within the boundaries of its own culturally constructed moral system.

While this approach was formally adopted by the American Anthropological Association, it never won universal agreement even among American anthropologists. Anthropologists long have been active in various social justice and human rights movements, including the fight against McCarthyism, the civil rights movement, the debunking of racism, the environmental movement, the women's movement, and the struggle to gain acceptance of health as a human right (di Leonardo 1998, Rylko-Bauer et al. 2006). These efforts ultimately challenged the finality of Herskovits's stance while, for the most part, recognizing the importance of being alert to his cautions. The foundation of this challenge is that (rigorous scientific) anthropology can make important contributions to lessening human suffering and that moral detachment in the face of suffering is not neutral; rather, it helps sustain suffering in the world. Silence, as they say, can speak louder than words.

In recent years, the moral compass has swung away from Herskovits's position to the point that at least some anthropologists now believe that in a world of exploitation and suffering, "only anthropological knowledge that is politically instrumental is worth pursuing" (Goodale 2006:2). More broadly, at the turn of the twenty-first century, the American Anthropological Association (1999), in accepting that anthropology in use involves the application of scientific skills and knowledge to the solution of immediate human problems, adopted its Declaration on Anthropology and Human Rights, which unambiguously rejected Herskovits's position. According to this document,

> Anthropology as a profession is committed to the promotion and protection of the right of people and peoples everywhere to the full realization of their humanity, which is to say their capacity for culture. When any culture or society denies

> or permits the denial of such opportunity to any of its own members or others, the American Anthropological Association has an ethical responsibility to protest and oppose such deprivation.

This statement well reflects a broadly held view within contemporary applied health anthropology that it is possible to "create knowledge that is at once empirically grounded, theoretically valuable, and contributes to the ongoing struggle for greater justice" (Speed 2006:75).

The issue of social suffering stemming from structural inequalities or human conflict and the problem of hindered and unequal access to quality health care are natural entry points for health anthropology into the issue of human rights. Today, health and human rights activists are waging a struggle to gain global acceptance of health/health care as a human right. Health anthropologists have a wealth of data corroborating the suffering associated with lack of health and health care internationally. In the view of many people within the discipline, health anthropologists have a moral responsibility to use their knowledge to promote health as a human right in order to reduce suffering in the world and improve access to and availability of quality health care for everyone.

Some health anthropologists promote the study of human rights as one of many idioms of social justice mobilization and call for the field to pay attention to the power and limits that human rights discourse has in any given circumstance. From this perspective, health anthropologists should not simply rally behind health as a right. Rather, the right to health should be treated as an object of ethnographic exploration allowing anthropologists to assess how it is called on, debated, negotiated, and actively resisted in specific contexts (Willen 2011). Among the infamous "human wrongs" for which the twentieth century will forever be marked is that of *genocide/ethnocide*. The link that some have described between globalism and genocide raises it as a critical human rights issue for the twenty-first century. In response to such calamity, health anthropologists are in a position to contribute to the establishment of universal health care, safe and sustainable living environments, freedom from poverty, access to adequate diets and clean water, and protection from physical and structural violence as inalienable human rights with significant health implications. As Paul Farmer (2004:245–46) asserts,

> Utopian ideals are the bedrock of human rights. By arguing that we must set standards high, we must also argue for redistribution of some of the world's vast wealth. . . . Claims that we live in a world of limited resources fail to mention that these resources happen to be less limited now than ever before in human history. . . . [I]t is time to take health rights as seriously as other human rights, and . . . stand side by the side of those who suffer most from an increasingly harsh "new world order."

Illness Narratives

Personal accounts of illness experience, like the "swallowing the frog" example cited previously, are called illness narratives, and they have become an important subject of research in health anthropology. Telling stories about life is a central

human social activity around the world, with good storytellers being socially rewarded. Narrative has been defined as "a discourse featuring human adventures and sufferings connecting motives, acts, and consequences in causal chains" (Mattingly 1998:275). As contrasted with other kinds of talking, a narrative is a story about a specific event or a set of linked events in which the actors perform specific behaviors with socially meaningful outcomes. Narratives need not be full, elaborate tales with a moral, and they may be embedded in the course of other kinds of talk. In the case of illness narratives, expressing feelings and recalling actions around a particular illness form the core elements.

The Social Uses of Narration

Illness narratives have been interpreted by health anthropologists in several different ways, including as a common way "by which people organize, display and work through experiences" (Rubinstein 1995:259). A primary way we "make sense of . . . experience is by casting it in narrative form" (Gee 1985:11). A narrative often "affixes blame and assigns responsibility" (Mattingly 1998:286), thereby setting moral boundaries around what is acceptable and what is unacceptable while pointing to a course of action. In this, narratives may be future oriented. Hearing a narrative augments a listener's "fund of cultural knowledge" with which to confront future life challenges, including key cultural information about symptoms of various illnesses, their course and causes, and beneficial responses (Price 1987:315). At the rise of the AIDS epidemic in Haiti, for example, narratives about individual cases helped Haitians come to grips with understanding this new threat by "providing the matrix within which nascent representations were anchored" (Farmer 1994:801). Moreover, narratives often point to the future to imagined ends our lives are intended to fulfill (B. Good 1994). In addition, narratives are a cultural mechanism people use to acquire or assert particular social identities and self-understandings (Cain 1991).

Analyzing Narrative

People who are suffering from significant illness tend to tell three kinds of stories (Frank 1995). The first of these are "restitution narratives," a term that refers to linear stories that begin with a serious illness and end with the restoration of health, often through medical intervention. Such stories affirm to listeners that illness need not be an end point: people recover. Such stories often are told to people when they are ill to help them maintain hope and good spirits. The second type of illness narrative are called "chaos stories." These are filled with uncertainty, confusion, and a sense of powerlessness in the face of overwhelming forces. For example, in a study of drug users living with HIV, one of the participants stated,

> I can [tell] . . . my body is getting smaller, like something sucking me at like the neck, and everything is getting stringier and weaker, and I know this. And everything's showing, like my ribs and stuff, and the flab outside my face is gonna sink and I know this. (Mosack et al. 2005:597)

The final narrative type is called the "quest narrative." Rather than recover, the person advances to a higher or superior state of being. While the person may not always return to physical health, spiritually or morally illness delivers them to an elevated plane. New religions, in fact, often are started by individuals who have been sick but during their illness undergo a transformation, believe they have communicated with the supernatural, and never resume their prior identity but instead become missionaries for their new faith. Typical is Handsome Lake, a Seneca Indian and alcoholic who lived at the turn of the eighteenth century. Lying in bed, deathly ill, Handsome Lake had a series of three transformative visions involving the receipt of messages from the Creator. These events led to the development of the Code of Handsome Lake, a religious doctrine for healing the Seneca people of the social and individual devastation wrought by colonial contact.

From an applied standpoint, M. Singer and coworkers (2001) use the narratives of drug users to gain lessons for culturally appropriate AIDS prevention. They argue that analysis of the corpus of narratives they collected from street drug users holds three important lessons for confronting the AIDS epidemic in this highly vulnerable group. First, it is evident in the stories that acts of generosity and caring on the street are unexpected and that when they occur they are seen as especially welcome surprises. As a result, these applied researchers maintain, programs that work with drug users should go out of their way to treat them with dignity, respect, and authentic concern. Second, drug user stories emphasize the survival value of always remaining wary of betrayal. Avoiding actions that can be construed (rightly or wrongly) as duplicity should be avoided to the degree possible. Finally, the narratives stress the skills and abilities of drug users, attributes that fly in the face of the usual way they are depicted in society, namely, as "losers," unproductive burdens, and threats to social tranquility and safety. Reversing the social message of denigration and worthlessness, by treating drug users as socially resourceful, knowledgeable, and goal-oriented individuals, offers promise of engaging them in health programs.

One of the ways that narratives currently are used in health intervention is through what is called "graphic medicine." This term was coined by a physician and cartoonist who defined it as "the role that comics can play in the study and delivery of health-care" (Williams 2012). The originator of the term subsequently launched a graphic medicine website to draw attention to the growing number of illness graphic narratives that were available. Graphic narratives, or graphic novels, are book-length stories in the style of comics. Graphic narratives of health experiences include stories about mental illness, sexually transmitted diseases, obsessive compulsive disorders, epilepsy, and cancer. As the second-leading cause of death in the world, cancer has been the topic of a number of graphic narratives, including graphic memoirs and biographical comics. Anthropological analysis of cancer graphic narratives suggests that they tend to legitimize biomedicine and biotechnology, upon which cancer patients come to be dependent for life, while diminishing issues of social inequality (McMullin 2016).

Embodied Health Experience

Over the course of the past three decades or so, anthropologists and other scholars have turned their attention to historical and cross-cultural studies of conceptions of the human body and of embodied human experience (i.e., experiencing the world from within the confines of our bodies). The anthropology of the body, however, harkens back to French anthropologist Marcel Mauss's assertion that bodily sensations are mediated by culture and Maurice Merleau-Ponty's concept of "embodiment," which draws attention to how the body internalizes cultural meanings. Today, the body is a central focus of work being done under the banner of health anthropology, including a focus on embodied experience, such as the folk illnesses in which people "feel" invasive forces or beings in their body, representations of the body, and the actual cultural shaping of the body through decoration or tattoo or as a result of unintentional factors such as culturally constituted or structurally imposed diets. Although the body is generally viewed as a biological entity, particularly within biomedical circles, it also in large part is a sociocultural construction. Further, the body functions as a vehicle for learning about our environment. In the view of many societies, perhaps particularly indigenous ones or peasant communities, it is an open system that links the self, society, nature, and spiritual world.

Why Bodies?

In part, interest in the body is driven by recognition of its increasing fragmentation and commodification within the contexts of biomedicine and global economics. With reference to the former, some health anthropologists argue that in diagnosing and treating sickness, biomedicine focuses on human biology, more than on human pathophysiology (Hahn and Kleinman 1983). Hospital physicians often refer to patients as body parts based on the primary site of the patient's disease (e.g., the liver in room 327 or the kidney in room 783). Biomedicine tends to radically separate the body from the nonbody, although critiques of biomedicine in recent decades have prompted some physicians to turn to a biopsychosocial orientation and/or to embrace complementary and alternative therapies as well as to improve their bedside manner by more actively touching patients.

Various social factors also have contributed to a growing interest in the body in the social sciences. Modern art, particularly as exemplified in the classic work of Picasso with his Cubist paintings depicting rectangular-shaped body parts, stimulated an awareness of the way the body has become fragmented in modern societies. The growth of a consumer culture with its emphasis on the cult of the body exemplified by cosmetics, fashion, and advertisements (as well as tattooing and piercing) also played an important role in enhanced focus on the body. At the same time, the feminist movement, particularly with respect to its emphasis on reproduction and sexual violence and the harmful effects of the objectification and manipulation of the female body, technological innovations associated with reproduction, and the replacement of bodily organs, as well as cosmetic surgeries like face-lifts and breast implants, all contributed increased focus on the body as the site of immediate human experience. Last but not least, diseases that can

leave marks on the body (e.g., skin lesions and discoloration, pocks, rashes, severe loss of weight), have contributed to a scholarly interest in the body.

Body Theory

Michel Foucault, Pierre Bourdieu, and Bryan Turner are perhaps the three most important social theorists in recent times whose work has contributed to an enhanced appreciation of the role of the body in societal processes and the role of society in bodily attitudes and practices. Foucault argued that the transition from premodern to modern society entailed a shift from sovereign power to disciplinary power. Whereas in premodern society power, at least in the European context, resided in the body of the monarch, in modern society it resides in the bodies of diverse populations. Disciplinary power, which refers to the ways in which bodies are regulated, trained, maintained, and understood, is exerted particularly in schools, prisons, hospitals, and asylums. According to Foucault, disciplinary power is expressed by various instruments. One of these is an elaborate system of hierarchical observation and tracking designed in the form of a Panopticon in which officials can observe all "prisoners," normalizing judgments made particularly by teachers, physicians, social workers, psychologists, and health promoters and allowing a detailed comparison of the actions or attributes of individuals. An instrument of particular interest to the health anthropologist is the physical examination or "clinical gaze" of biomedicine. Within the context of the clinic, the biomedical examination facilitates the categorization of diseases in terms of various symptoms, such as the patient's expressed experience of pain, and signs, such as pulse rate. Knowledge of the body translates into power over the body, or what Foucault called biopower, on the part of not only the physician but also the state he or she serves. For Foucault, biomedical knowledge is one of the several discourses by which modern populations are disciplined and controlled.

Pierre Bourdieu argued that the body as manifested through its shape, speech, and gait translates into physical capital, which, along with economic capital (e.g., money, wealth, and property), cultural capital (e.g., education and knowledge of arts and high culture), and symbolic capital (e.g., presentation of self), serves to influence an individual's position in the social hierarchy of a particular society. In other words, people experience the bodies of others and, in turn, may attempt to have their own bodies experienced by others in terms of various qualities, such as (1) being trim and fit as signs of having high self-esteem; (2) speaking "properly," such as use of what is termed "the King's English"; and (3) having good upright posture as an expression of confidence that conveys meaning about the character and social standing of the person being observed. This kind of body language, in fact, can speak louder than words in establishing a person's rights and entitlements.

Bryan Turner (2008) argues that modern communities have been moving toward a new "somatic society"—a social system in which the body constitutes a central field of political and cultural activity as exemplified by an emphasis on safe sex, sex education, free condoms, clean needles, controlling pollution, healthy eating, proper amounts of exercise, and other physical concerns expressed by

various social movements, particularly those focusing on feminist, abortion, fertility, and environmental issues. Turner's work focuses on the vulnerability of the body, and has addressed issues of aging, disease, and injury. These provide metaphors for conceptualizing the dominant institutions of society that increasingly are thought of as vulnerable to disease (e.g., computer viruses), at risk because of the aging of the infrastructural technology on which they depend and threatened with injury from cyber-attacks.

Ultimately, even dead bodies matter, culturally. This is seen, for example, in the social response to the emergence of body donor decisions in which individuals voluntarily donate their entire body, after death, to anatomical science for education and research. One impact of this kind of decision is that the people who feel close to the deceased are denied access to a mourning space such as a cemetery. This has led to the construction of donor monuments that provide the bereaved a symbolic resting place for their lost loved ones. Dead bodies also matter to the students who learn anatomy through the dissection of corpses. Anthropological research indicates that "students find themselves shifting back and forth between considering the corpses either as objects of study or a human beings who had real relationships" (S. Bolt 2012:619). To avoid this ambiguity, medical students try—although not completely successfully—to distance themselves from the bodies they work on by sticking to scientific terminologies, avoiding the bodies unnecessarily, focusing their attention on abnormalities, and employing humor to lighten the otherwise somber mood of the dissecting room.

Bodies in the Age of Immunology

In a pioneering book, Napier (2003) examines one of the important ways in which culture shapes the sufferer conception and experience of illness in bodily terms. In his thesis (2003:3), "immunology as a cultural paradigm . . . has found its way into our lives, and the power of that way of seeing may be evidenced in various personal and professional domains, including—because they focus on 'well-being and its absence'—the medical sciences." Immunology is a biomedical field concerned with the structures, systems, and processes within the body that protect it from pathogens and infection. Napier argues that immunology offers a paradigm (i.e., a particular way of understanding, ordering, and experiencing "reality") that shapes how patients, doctors, and many others in and far beyond the health/healing domain think about their bodies but also about the world as well. Central to this paradigm is the notion that the world exists as two opposed categories. The first of these is *self* (at the individual level: one's identity, body, internal organs, and at the social level: us, members of one's family or social group, friends). The second is *not-self* (at the individual level: things that may intrude into my body and cause illness, such as infections, toxins, bad influences, drugs, and at the social level: them, outsiders, foreigners, enemies, terrorists). In what has been called the immunological age, the era in which we now live, "the self/not-self opposition is fundamental. In this confrontation, selves become immune—safe—if and only if the dangerous not-self—bacteria, viruses, tumors, or radically different others—is either destroyed or neutralized" (Stoller 2008:30). As a result, to take the example of cancer, doctors and patients

(depicted as allies in the language of modern biomedicine) fight an internal war on disease (which is portrayed as an invading not-self entity), complete with reconnaissance missions (X-rays and other technological weapons for internal imagining), incendiary bombing (radiation), search and destroy missions (surgical removal of tumors), and use of biochemical warfare (chemotherapy), all of which are expected to produce a certain amount of damage from friendly fire (i.e., killing cells that are part of self, not other) but, if the battle is fought well, lead to victory with the patient becoming a cancer survivor. As a result of exposure to this culturally constructed metaphoric world, "it is not uncommon for . . . people . . . to think of malignant cells as alien invaders that are completely separate from . . . [their] bodies" (Stoller 2008:34). Notably, however, it is in fact the case that malignant cells are self; they are something the patient's own body produces (although possibly upon exposure to pollutants or other toxins). But cancer sufferers tend to deny this fact because of the pervasiveness of the cultural division of self and not-self, a theme that extends far beyond the health of the individual body to debates about the health of the social body (e.g., debates about immigration).

Despite the prominence of the immunological paradigm, it is good to remember that biology tends not to be ordered by simple self/other relationships. In autoimmune diseases, it is the body's own immune cells that attack other healthy body cells. Similarly, while it is technically a foreign body (with antigens different from those of the mother), the mother's immune system usually does not attack a developing fetus. Moreover, even differences we take for granted, like the existence of biological distinct male and female bodies within our species, is subject to rethinking. The frequency of deviation from standardly defined male or female bodies may be as high as 2 percent of live births (Blackless et al. 2000), although this is a disputed statistic.

Cyborg Bodies

The boundaries between bodies and machines have become increasingly blurred, as anyone knows who has watched people engaged in nonstop video gaming, internet chatting, texting, cell phone calling, use of social media sites, or listening to music through a headset. This is particularly exemplified in the contemporary incarnation of cyborgs, mythical creatures that were half human and half animal or machine. Modern-day cyborgs include "test-tube babies" (assisted conception), people with organ transplants, people using artificial body parts (prosthetic limbs), as well as those who have undergone plastic surgery. Moreover, various social processes, including sports, dance, yoga, military training, construction work, body building, and computer work, as well as social class, ethnicity, nationality, and culturally defined gender shape bodies. Within the context of late capitalism, the commodified body functions as a site of production, a source of labor power, a "baby machine," an object of sexual pleasure in the form of prostitution, a supplier of transplanted body parts, a source of beautification in the form of cosmetic surgery, and a source of profit. These are important health-related developments addressed in chapter 8.

Mindful Bodies

Within health anthropology, Nancy Scheper-Hughes, Margaret Lock, and Emily Martin in particular have made major contributions to the scholarly study of the body. Lock and Scheper-Hughes point out that in addition to its array of anatomical features, such as a set of interrelated biological systems, the human body is mindful in that it is the site of individual experience, social expression, and the inscription of structural relations. On this basis, these researchers delineate what they refer to as the three bodies of concern, a delineation that has impacted thinking within the discipline for several decades.

The first of these is the "individual body," which comprises the "lived experience of the body-self" (Lock and Scheper-Hughes 1996:45). While the notion of the individual body, conceived of as a separate and unique entity, driven by intention and will, is quite compatible with Western ideas about individualism, this is not always the case. The Gahuku-Gama of Papua New Guinea historically conflated individual and social identity (Read 1955:276). In other words, they did not conceive of themselves as fully distinct individuals; rather, their identities were embedded in their roles and responsibilities as members of a social group. In such cases, the good of the group overwhelms personal interest or intention, and thoughts of oneself separate from the group are unsettling. Traditionally, a version of this way of conceptualizing the self has been found in Chinese culture. A core aspect of Chinese culture is its emphasis on relationalism (*guānxì xué*), which involves viewing human relatedness, rather than separateness, as the essence of human existence. The Chinese "self" traditionally was not an independent entity; one's life could be meaningful only through coexistence and interaction with others. The Chinese term *guānxì* refers to both a relationship type and to the idea of closeness.

Under the influence of globalism, the individualization of self has become more common everywhere. The world of the Gahuku-Gama, for example, has changed dramatically, with urban encroachment, involvement in a global economy, and significant cultural change (Read 1986). In the contemporary world, there are whole new dynamics that must be considered in understanding the body at the individual level. One such area of growing interest among health anthropologists is the topic of organ transplantation. This operation, which often is seen as a pinnacle of biomedical skill and achievement, a "medical miracle," is of keen interest to health anthropologists in part because it involves the crossing of traditional cultural and physical boundaries (e.g., the notion that body and self are intertwined and integral to each other) and the corporal mixing of "self" and "other" in ways that challenge cultural notions of personhood, personal property, the distinction between life and death, and what Joralemon has called "the intuitively unambiguous connection we feel to our bodies" (1995:347). Undergoing such an operation, as a recipient, can be a personally transformative experience. Afterward, the individual's sense of self and self-worth often is changed (Sharp 1995). At the same time, cases have been ethnographically documented of family members of a deceased organ donor seeking out the recipients of their relative's transplanted organs, people they had come to think of as new kinsmen, a phenomenon that has begun to occur, as well, among the children

of diverse mothers with a common sperm donor biological father. On the other side of the transplant operation, in the arena of organ procurement, especially in low-resource countries, the poor often are subject to exploitation and may be pressed to literally sell their body parts to survive, while large numbers of citizens in wealthy countries refuse to donate organs even at death. In an anthropological study in India, for example, researchers found that kidney donors were paid approximately $1,500 and that most used the money "to pay off debts incurred for wedding dowries and other types of loans," among other expenses (Marshall and Daar 2000:213). In addition, at times, criminal elements in the procurement business have kidnapped individuals to steal transplantable organs such as kidneys for sale on the organ market. Indeed, a worldwide system that has been called "organ tourism" has emerged in which individuals from wealthier countries gain life by acquiring organs from the people of poor countries, a practice that some have interpreted as a veiled form of structural violence.

In a grisly turn of events in the world of organ transplant, over the past several decades more than fifteen thousand families in the United States have brought suit against funeral home employees, crematorium operators, and morgue workers claiming that body parts, from knees to skin, were harvested without consent from their deceased relatives and sold for profit. Police have found that entire bodies have been sold for tens of thousands of dollars, with elbows and hands fetching $850 each (Amour 2006). While the sale of body parts is illegal in the United States, the sheer demand has motivated the creation of a black market in transplantable organs and other body parts. From an anthropological perspective, this turn of events represents a further commoditization of human life and experience, namely, the transformation of objects, nature, emotions, relationships, indeed, ultimately everything, into salable products in the capitalist marketplace.

One health anthropologist who has long studied living organ donation and is the founder of Organs Watch, an organization focused on exposing the egregious moral abuses of international organ trafficking, has likened it to "sacrificial violence" (Scheper-Hughes 2007). This organization asserts that the poor often must sell a kidney to a transplant harvesting company (known by their critics as "kidney hunters") in order to raise the money to flee as refugees from conflict or the damaging environmental effects of climate change. But having two kidneys does not make one of them a spare organ, and the long-term medical, psychological, and social consequences of selling a kidney are not widely reported. These include chronic pain, depression, self-loathing, distorted body image, a sense of bodily emptiness, anger, isolation, and even suicide (Scheper-Hughes 2016).

The "social body," the second component of Lock and Scheper-Hughes's tripartite model, refers to representations of the body as a "natural symbol with which to think about nature, society, and culture" (Lock and Scheper-Hughes 1996:45). Indeed, one often hears expressions such as a "healthy society" or a "sick society" to refer to either societies with a high degree of social cohesion or societies with a high degree of fragmentation, as if societies could get illnesses the way bodies do. To a large degree, Lock and Scheper-Hughes note, talk about the body and sexuality tends to be talk about the nature of society and about socially acceptable standards of moral behavior. This assertion reflects a broader point

made in anthropology that "the human body is always treated as an image of society [and therefore] . . . there can be no natural way of considering the body that does not involve at the same time a social dimension" (Douglas 1973:98). This point is well captured in a statement made by Richard Carmona, former surgeon general of the United States, about the problem of obesity in the United States. He stated, "Obesity is the terror within. Unless we do something about it, the magnitude of the dilemma will dwarf 9-11 or any other terrorist attempt" (quoted in J. Harris 2006). Terrorism, by intent, is targeted at societies, not individuals per se. In likening obesity, a problem of individual bodies, to terrorism, a social threat, the surgeon general affirmed our tendency to use the body as a symbol of society and vice versa, a pattern found around the world.

Finally, there is the "body politic," a concept that draws from the work of Foucault and that refers to the "regulation, surveillance, and control of bodies (individual and collective) in reproduction and sexuality, work, leisure, and sickness" (Lock and Scheper-Hughes 1996:45). With respect to biomedicine, a medicalized body is a "manifestation of potent, never settled, partially disguised political contests about how aging and rebellious bodies should be managed" (Lock and Scheper-Hughes 1996:68). We are reminded of Rudolf Virchow's often-quoted historic statement that "politics is medicine on a grand scale." In looking at the body from the standpoint of political relations in society, health is a reflection of the unequal hierarchy of relations in society, and health anthropologists are mindful of the ways in which inequality, exploitation, and structural violence are recorded "under the skin" in the social distribution of health and in the patterning of biology across social classes and other lines of social difference. A nutrient-deficient diet, for example, is more likely to be found among the poor than among the wealthy in a society, and this difference is reflected in the actual formation of the teeth and bones of societal members. Society, quite literally, is inscribed on our bodies and one need go no further than a tattoo or piercing parlor to see contemporary expressions of this. Thus modern tattooing has been interpreted as "a symptom of a culture that has commodified the body" (Fisher 2002:91).

Gender is another important arena of bodily social control. As contrasted with societies characterized by limitations on food availability, in which a plump figure often is seen as a sign of beauty, in modern America a slender, androgynous body has become a valued female physique. Expectations of thinness are reinforced heavily by the advertisement industry. Studies of magazine content have found high numbers of ads and articles about food, including recipes and accounts of the newest chic dishes. At the same time, magazines frequently give voice to social angst about obesity while exhibiting images of perfectly sculpted bodies attained through the use of exercise machines, workout routines, and an endless array of miracle diets advertised nonstop in media commercials. The cultural celebration of thinness and stigmatization of obesity are critical factors in the emergence of epidemic levels of food-related disorders, such as anorexia and bulimia, in U.S. society. Although eating pathologies have been a peculiarly Western and middle-class phenomenon, in recent years they also have begun appearing in non-Western countries, as well as across socioeconomic classes, ethnicities, age groups (including among the elderly), and even genders. Recent

analysis has focused on the role of globalization and the diffusion of American or other Western images of the "ideal" body type as critical elements in the changing international face of eating disorders.

Engendered Bodies

Much of the work on the anthropology of the body focuses on gender issues, including the cultural construction of gendered bodies, that is, bodies that are not just different biologically (e.g., sexual organs, roles in reproduction, capacity to nurse infants) but are constructed as different within cultural frames and conceptions. In *The Woman in the Body*, for example, Martin (1989) asserts that the body in the United States is viewed through the cultural lens of a machine metaphor that emerged during the industrial era. Biomedicine, she maintains, treats women's reproduction of the social group by giving birth as a cultural form of factory production in which "uterus machines" produce babies with the physician managing this process. It also regards menstruation as a failed production process. Biomedicine also has treated the body as a passive object over which particularly women came to lose control, although the women's movement eventually challenged this trend.

In another case, in contrast to the anorexic-like female body that has become idealized in Western societies, Fijians, a Melanesian people in the South Pacific, value a well-proportioned, robust body (A. Becker 1994). They believe that such a body indicates that a person is capable of arduous labor and that he or she has been well nurtured.

A third example comes from the Mediterranean area, where it has been observed that women "represent the code of *honor* of the family and the code of shame via the blood revenge for nonfamily member's transgressions. . . . The honor/shame dichotomy is evident in the highly guarded aspects of women's virginity, chastity, marital virtue, and especially fertility" (Olujic 1998:34).

As these examples indicate, cultures ensnare and diversely elaborate biological differences, with significant but different social and health implications for women and men.

Understanding Medicalization

The term "medicalization," mentioned in chapter 2, was introduced into anthropology from sociology and refers to the role of medicine in social control. At the simplest level, medicalization means giving a bodily condition or a behavior a medical label, defining the problem in medical terms, and using a medical intervention to treat it. While many health anthropologists now use the term, medicalization was coined by the influential sociologist Irving Zola. He pointed out that when a problem, such as overeating, is medicalized, there tends to be a reduction in the degree to which the disorder is stigmatized because other causes besides individual moral failings are ascribed, be they the consequence of pathogens or the effects of genetics. This process has been called the transformation from "badness to sickness" (Conrad and Schneider 1980). At the same time, as

conditions are medicalized, there is an ever-growing expansion of the jurisdiction of medicine over arenas of social life and experience. In that medicine is now highly commodified, medicalization also involves the medical marketing of profit-making commodities to address body issues, such as menopause treatments, clinical weight-loss programs, sleep deprivation interventions, and smoking cessation initiatives. Critics have warned about the ways physicians are expanding their power over the public through what has been called "the medicalization of life" (Illich 1975). In recent times, there have emerged "demedicalization movements," such as the largely successful struggle to have homosexuality removed from the list of medically recognized psychopathologies.

Within health anthropology, an issue of concern is the ever-creeping boundaries of problems defined as appropriate for medical intervention and hence medical social control. Lock (2003:117), with reference to the medical management of everyday life, notes:

> First, . . . emotions and sexuality become targets of medical technology, with the result that reproduction of populations and even of the species are medicalized. Similarly, other activities, including breastfeeding, hygiene, exercise, deportment, and numerous other aspects of daily life are medicalized—largely by means of public health initiatives and with the assistance of the popular media.

With some issues, such as substance abuse, medicalization may be less repressive than the alternatives, such as criminalization. Ironically, in the United States, during the very period that the National Institute on Drug Abuse, an arm of the National Institutes of Health, began declaring chemical addiction and dependence to be diseases of brain chemistry, involving a restructuring of neural pathways in response to exposure to drugs, the country continued to build many new prisons each year to house the large number of individuals arrested for the possession of these substances, including, increasingly, illicitly sold and used prescription drugs. Alcoholism and alcohol dependence, by contrast, have been much more fully medicalized, and street alcoholics, for example, are no longer arrested and incarcerated for public inebriation.

At the same time, as Lock (2003:123) points out, with "molecularization, geneticization, digitization, computerization, and globalization . . . the potential exists to make the body increasingly the site of control and modification, much of it carried out, in the West at least, in the name of individual rights or desires."

Healer versus Sufferer Conception of Disease

The Two Sides of Compliance

As documented in many studies, patients do not always follow doctors' orders, a behavior that often is labeled "noncompliance" in biomedicine. But this finding is not peculiar to patients in biomedicine and has been documented for alternative medical systems as well. In fact, studies have shown that doctors themselves often are out of compliance with existing standards of care. As Gawande (2002:236), a physician, comments, "Physician compliance with various evidence-based

guidelines ranges from over 80 percent of [cases] in some parts of the country to less than 20 percent in others." Supporting Gawande's assertion, a twelve-city U.S. study of medical care found that patients received recommended care for their diagnosed health problem only 54.9 percent of the time, ranging from a high of 78.7 percent for senile cataract treatment to a low of 10.5 percent for the treatment of alcohol dependence, findings that led the researchers to conclude that the failures of clinical adherence pose significant threats to the health of the American public (McGlynn et al. 2003). Other research shows that the way doctors handle the same diagnosis varies considerably; the likelihood, for example, that a patient with a gallbladder problem will be sent for surgical removal varies by 270 percent by city, and for a hip replacement the variance is 450 percent (Wennberg 1999). Not only are physician decisions highly variable, but in a series of articles in the prestigious *Journal of the American Medical Association*, a physician who studies clinical decision making described them as often being arbitrary (Eddy 1990).

While health anthropologists have not focused much of their attention on physician compliance with established medical procedures, they have been quite interested in patient behavior. In this, they have been quick to point out that an important component of illness behavior stems from the fact that patients are not blank slates waiting to be inscribed by their healers; rather, they have beliefs and ideas of their own. Illness, especially one with familiar symptoms, does not elicit an empty response; rather, sufferers and their significant others, including those in one's social network who are thought to have greater experience in handling health problems (e.g., grandmothers), commonly engage in folk diagnosis, and it is on the basis of such reflection that specific kinds of professional, folk, or popular treatment are sought. As one health anthropologist observed, "Health programs that fail to recognize and work with indigenous beliefs and practices also fail to reach their goals" (Scrimshaw 2001:53). A case in point is that of Tamil refugees living in Norway but originally from Sri Lanka, where they were used to being treated in the Ayurvedic medical tradition. Suffering from diffuse aches and pains in their bodies, the Tamil seek therapy from the local biomedical health-care system. The experience, however, is often disappointing. As one patient complained to a health anthropologist, "The problem is that Norwegian doctors only see my body and organs. They don't see my person. Very often doctors do nothing. We feel very uncertain" (Grønseth 2006:152). The problem in this instance is that Tamil think of their health problems in terms of a humoral system of hot/cold balance. The failure of Norwegian physicians to address the imbalance, Tamil patients feel, makes them appear unconcerned and useless. Another patient stated,

> I don't know how to explain. I only know that sometimes a sickness makes you too hot or too cold. So, you need to make food and use mixtures that will bring you back in balance. . . . Here, the doctor doesn't want to know our problems and give no advice and no pills. They can't cure our difficulties. (Grønseth 2006:153)

Insider and Outsider Assessments of Health Status

What do sufferers know, subjectively, about their health status, and how does sufferer knowledge compare to that of the healers who are treating them? Several studies on this topic have produced interesting results. In a study in Denmark, researchers (Geest et al. 2004) asked 456 middle-aged individuals who subsequently had a health screening performed by their general practitioners to rate their health status on a five-item scale ranging from "very poor" to "excellent." In 68 percent of cases, doctors and their patients agreed about the patients' health status. In a U.S. study of noninstitutionalized elderly individuals, it was found that self-assessments of health were a better predictor of mortality than objective biomedical assessments (Idler et al.1990). Similarly, self-assessment has been shown to be a superior predictor of return to work among individuals suffering from chronic pain than objective clinical or radiological test results (Yelin et al. 1980), while, based on a study among cancer patients, researchers concluded that self-assessment "is valid, reliable, and responsive to change as a predictor of survival of advanced cancer" (Shadbolt et al. 2002:2514). These findings are striking because traditionally, biomedical physicians have tended to discount patient self-assessment as a source of information about patient health.

One Word, Two Meanings

Another aspect of the sufferer versus healer conception of disease involves the existence of various disease names that mean one thing to sufferers and another to healers. For example, health anthropologist McCombie has studied a phenomenon she refers to as "folk flu." In her study in the American Southwest, she found that the word "flu" "does not mean the same thing to the epidemiologist that it does to the public, who include illnesses with a number of clinical presentations" (McCombie 1999:31). Importantly, McCombie found that the discrepancy between the health-care system and lay sufferers in the meaning of the term "flu" is more than a matter of semantics because people think they have the flu when, from a biomedical perspective, they are suffering from a different disease, including giardiasis, viral hepatitis, and salmonella infection. Similarly, in a study in Washington, DC, Merrill Singer and coworkers (1984) found that the folk illness "hypoglycemia" (low blood sugar) was fairly common, especially among middle-class women, who reported a range of symptoms. However, many physicians would agree and would find that the majority of those who believe they are sufferers of this affliction do not meet diagnostic requirements for the disease. While some physicians have suggested that underlying emotional problems are the real cause of widespread folk beliefs about hypoglycemia, those who believe themselves to suffer from the disease are critical of biomedicine's failure to understand this affliction and its victims. In this sense, hypoglycemia functions as an idiom of distress among middle-class women who think that their lived experience of suffering is not taken seriously by biomedicine.

As these examples show, even when patients and healers use the same disease labels, they may be, in effect, speaking different languages or, more precisely, marshaling different bodies of information and different understandings.

The potential for provider-patient miscommunication in such cases is enormous. By exploring diagnostic labels and their meanings to the parties involved in a clinical encounter, health anthropologists are in a position to help minimize this problem.

Diseased but Not Ill

Given the definitions that have been offered for illness (as subjective experience) and disease (as objectively measured phenomena), it is possible to have a disease but not feel ill. This disjuncture is quite common with diseases such as hypertension, which has been called "the silent killer"; the early stages of various cancers; and infectious diseases such as HIV/AIDS that generally do not produce many symptoms, although very early symptoms are often misinterpreted as the flu until sufferers have been infected for a number of years. In such cases, adherence to medical instructions is complicated by the fact that the patient feels fine and has difficulty believing they are suffering from a life-threatening disease. Conversely, it is possible for sufferers to feel quite ill but be told by their physicians that nothing is wrong (e.g., folk hypoglycemia) or that their problem is psychosomatic. In some cases, however, the real problem is that disease exists but is unknown to biomedicine. Chronic fatigue syndrome, for example, was producing misery among sufferers long before it was recognized as a bona fide disease by biomedicine. It was not until 1988 that the disease was recognized by the Centers for Disease Control and Prevention, although the term "myalgic encephalomyelitis" began appearing in the medical literature in 1938 and is still used today in some parts of the world, particularly in the United Kingdom, to label this disease. Even so, controversies about the disease, which was once derided as the "yuppie flu" and its sufferers mocked as "rich, whiny white women," continued until 1994 when an international panel of researchers was convened and a clearer definition of the disease developed to assist clinicians in making a diagnosis. Accepted signs of the disease, which afflicts about a million people in the United States, most of whom are neither rich nor white, include having a significant level of fatigue that has lasted for over six months and having four or more of the following motley array of symptoms: substantial impairment in short-term memory or concentration; sore throat; tender lymph nodes; muscle pain; multijoint pain without swelling or redness; headaches of a new type, pattern, or severity; unrefreshing sleep; and malaise lasting more than twenty-four hours following exercise. Like the definition for AIDS, there is no clear-cut indicator but rather a set of factors occurring together that definitively signals the presence of the disease. Complex diagnostic criteria such as these often are the source of communication problems between patients and physicians.

Mismessaging

Another type of miscommunication that can occur is exemplified by findings in a study of HIV testing among drug users carried out by several health anthropologists and in collaboration with researchers from other disciplines. In this project, the researchers offered oral HIV testing because it was far easier for participants and less risky for research staff in terms of accidental "sticks" with a syringe that

had been used to draw blood from a participant who was HIV positive. In oral HIV testing, a small antibody collector that looks like a short-handled toothbrush is inserted for a short time between the inside cheek and gum of the participant. The collector does not capture the HIV virus (which does not tend to survive in the mouth) but rather draws out antibodies (if they are present) that the body produces to fight off infection. While HIV is not frequently transmitted orally (e.g., through kissing), it occurred to the researchers that being tested orally might convince participants that HIV could, in fact, be transmitted through oral contact. Subsequent interviews with individuals who had and had not been tested orally affirmed the researchers' suspicion. Those who had been tested orally were significantly more likely to believe that HIV could be transmitted through kissing, sharing a cup, or other types of oral contact (Clair et al. 2003). Why is this finding important? Does it matter if people think HIV can be transmitted in ways that are not in fact common routes of infection? If people believe that no matter what they do they are likely to get HIV, then they are less likely to follow any prevention strategy (e.g., using condoms). Consequently, it is critical that false beliefs about transmission be understood and corrected to increase the likelihood of adopting effective behaviors for preventing disease. Here again, health anthropology, which generally sees understanding the insider's point of view as a topic of concern, can make a contribution to community health.

Analyzing Health Discourse

This discussion of communication in the health domain suggests the growing importance of discourse analysis in health anthropology. This type of study focuses on the communication (be it verbal or involving gestures, written texts, or electronic media) that takes place between people in the health domain, such as clinical conversations between doctors or other healers and their patients. In part, this work has focused on the ways health-care providers and patients construct shared cultural worlds through conversation or other communication. Additionally, this kind of analysis has emphasized the ways that inequalities in access to knowledge shape health-related communication and help to undergird medical authority (Briggs 2005). Similarly, a keen interest to health anthropologists involved in medical discourse analysis is the social origin and structural functions of the messages exchanged in clinical encounters. Thus, medical discourse may have as its effect "the creation and maintenance of the interests of certain hegemonic groups" (MacDonald 2002:464). Silences are also of importance. Consider the implications in a biomedical encounter in Australia as a provider attempts to record a patient's medical history and the patient happens to be a member of an aboriginal community that culturally bans talking about one's past serious illnesses. Exploration of change is also essential, especially in the kind of fast-paced globalizing world that we all now inhabit. Based on detailed analysis of dyadic clinical encounters between providers and patients and other forms of medical discourse, a body of recent studies traces interactions between discourse forms that circulate globally and local traditions of practice to assess patterns of change over time and place (Wilce 2009).

Discussion Questions

1. What is health? What are the ways it has been defined? Why is it so hard to develop a widely shared definition? What does anthropology add to the definitional process?
2. Is death the great social equalizer or not? What are the implications of your answer for the human condition?
3. How and why do health anthropologists differentiate illness and disease? Is this a useful differentiation?
4. Why do beliefs across disease causation and approaches to cure vary across societies?
5. Why is it important to understand sufferer experience? How do health anthropologists study sufferer experience?
6. What have health anthropologists contributed to the study of disability and mental health (including PTSD) issues? Do you see these as important contributions?
7. How have health anthropologists' views of human rights changed over time? Why did this happen?
8. What are illness narratives? Why do health anthropologists study them?
9. Why has the conceptualization of "the body" become so important in health anthropology? What insights has studying the body provided? What is meant by the notion of the "mindful body"?

4 Health Disparity, Health Inequality

The causes of many diseases are a complex interplay of multiple factors, many of which are due to social injustice.

—Barry Levy and Victor Sidel (2006:11)

Introduction and Overview

Health and disease are unequally distributed across populations. Malaria, for example, is found in some places but not in others, while some segments of a population are much more likely to get tuberculosis than others. Moreover, access to health care and treatment within health-care settings is not uniform; many studies of health care in the United States, for instance, show that African Americans get different—generally poorer—health care than white Americans. These differences have been variously labeled health disparities or health inequalities. As will be seen in this chapter, these terms are not fully synonymous. Addressing inequalities in health and health care has become an important arena of work in health anthropology research and application. In this chapter, we (1) define health disparity and health inequality; (2) examine the nature of disparities in health within the context of U.S. society; (3) review the debates about the causes of health disparities; (4) closely scrutinize three key issues in U.S. health disparities: asthma, access to health insurance, and the development of culturally competent health care; (5) move from the United States to a cross-cultural examination of health disparities; (6) discuss methods and approaches in ameliorating disparities and inequalities in health; (7) discuss the roles that health anthropologists are playing in addressing health disparities; and (8) conclude with a discussion of the issue of the biological concept of "race" as a factor in health disparities.

What Is Health Disparity?

It is noteworthy that at no period of time "in the history of the United States has the health status of minority populations—African Americans, Native Americans, and, more recently, Hispanics and several Asian subgroups—equaled or even approximated that of white Americans" (Geiger 2003:417). Even though important scientific and technical advances in biomedicine have occurred over the past several decades, for many people of color these have not contributed to an equalization of health and well-being with other groups in society. In other words, certain groups have not benefited fully or equitably from the growth of biomedical knowledge and medicine's expanded capacity to heal. As noted by

the Institute of Medicine (2012:16–17), "health disparities exist from the cradle to the grave, . . . the data show elevated rates of mortality for African Americans and American Indians at all ages compared with those for whites. . . not only are mortality rates higher for racial and ethnic minorities, but these groups must also contend with earlier disease onset and a greater severity of disease."

The term "health disparities" refers to disproportionate or excess morbidity, mortality, and decreased life expectancy as well as unequal access to health care and other health-supportive resources (e.g., insurance or access to good nutrition) especially but not solely in disadvantaged groups in society or in the world at large. This has emerged as a major public health issue worldwide. While the term *health disparities* references differences in health across groups, the term *health inequalities* points also to underlying structural causes in disease distribution, namely, that social inequalities produce health inequalities. The term *health inequality*, in short, focuses on the role of unequal social relationships embedded in the economy, political system, educational system, medical system, and elsewhere in society in creating worse health in some groups compared to others.

Health Disparity in the United States

Although the United States is the richest country in the world and the most technologically advanced, this has not translated into having the best health or the greatest life expectancy for either men or women. Overall life expectancy in the United States for the total population, however, masks significant differences by subgroups. For example, while the life expectancy on average is 76.3 years for white males, among African American males it is 67.8 years, almost eight years less than white men. Among white females, the average life expectancy is 79.9 years, while for African American females it is 74.7 years, about five years less than white women. African Americans have been found to be at significantly greater risk of death, including premature death, than whites, with most existing studies showing a 30 percent higher age-adjusted risk of mortality among African Americans. Higher prevalence of several diseases, including hypertension, HIV infection, ischemic heart disease, stroke, and cancer, accounts for the higher death rate of African Americans. Much of the difference in this burden of disease—but not all of it—is explained by differences in the socioeconomic status (SES) of African Americans compared to whites. In a meta-analysis of several national studies, it was found, for example, that

> African-Americans experience about 67,000 more deaths than they would have had their mortality rates been similar to whites. This translates into 2.2 million more YLL [annual years of life lost]. After adjusting for SES, these numbers drop to about 38,000 lives and 1.1 million YLL. Thus, roughly 29,000 of the lives lost and 1.1 million years lost annually may be attributable to differences in income and education between the groups. (Franks et al. 2006:2472)

The additional burden of morbidity and mortality among African Americans after statistical adjustment for SES is believed to be a consequence of the health damage done by racism and discrimination throughout the lifecourse leading to the

direct and indirect bodily effects of stress, including adverse chemical changes. African Americans who perceive that they are being subjected to racism have been found to have higher blood pressure than those who do not (Krieger and Sidney 1996). Moreover, SES and being the object of racism have been found to interact, increasing the adverse health effects of each. As these data suggest, ethnicity, and how it is handled socially, appears to be an important factor in health in the United States.

In fact, researchers have found that there are considerable and consequential inequalities in the distribution of health, disease, and access to health care across populations, subgroups, and regions of the United States. While the issue of health disparity has received a growing level of public health, policy, media, researcher, and public attention in recent years, it remains a topic of considerable debate, especially with regard to the causes of disparity and its social solutions. Studies have found multiple indicators of health disparity; for example, with reference to the impact of ethnicity on access to health care, the Institute of Medicine (2002) reported the following findings in a review of multiple studies:

- health-care providers appear to give less intensive care to African American patients compared to white patients.
- African Americans are significantly less likely than whites to receive a major therapeutic procedure.
- African Americans receive fewer health care services than whites.
- Mexican Americans with myocardial infarction are 40 percent less likely than whites to receive thrombolytic therapy.
- African Americans are less likely to receive even low discretionary care (e.g., for clinically urgent health problems like appendicitis and abdominal aortic aneurysm).
- Latinos were found to be less likely than non-Latinos to undergo major medical procedures.
- African Americans with diabetes were found to be significantly more likely to undergo amputations and less likely to receive lower-limb arterial revascularization.
- Latinos with long-bone fractures are twice as likely as non-Latinos to receive no medication for pain.
- Minority cancer patients are less likely to receive adequate analgesic medication for pain.

In short, the care you receive in biomedicine is determined by many factors, including who you are in terms of the social construction of ethnicity.

Gasping for Breath

The various studies of the health-care system noted above—and many more that could be mentioned—affirm considerable inequity based on ethnicity and income in the United States. Similarly, there are major inequalities in the distribution of diseases. Consider, for example, asthma, a disease that has become more prevalent in the twenty-first century. More than twenty-five million people in the United States suffer from asthma, about a third of whom are children under eighteen years

of age. Asthma has been identified as one of the most common chronic illnesses of childhood in the nation; however, this breath-robbing disease is not equally distributed among children of different ethnicities or socioeconomic statuses. National surveys report an overall lifetime asthma prevalence of 12.2 percent for children (under eighteen years of age), although there are dramatic differences across ethnic groups and subgroups (Dey et al. 2004). Rates of asthma are especially high among African Americans and some Latino groups. Compared with white children, African American and Puerto Rican children have 1.6 and 2.4 times higher asthma prevalence, respectively. Lower socioeconomic status also is linked to increased asthma morbidity, and minority children are disproportionately impacted by lower socioeconomic status (Carroll 2013).

Houston, Texas, the fourth-largest city in the United States and the home of large numbers of Latinos and African Americans, for example, is one of the fifteen most polluted cities in the country and exhibits high rates of respiratory ailments, including asthma, in large part because of air pollution from motor vehicles and petrochemical plants (Harper 2004). In the Houston metropolitan area, people of color and working-class people in general tend to live in closer proximity to freeways and petrochemical plants than do affluent whites. As a result, different communities in Houston report different rates of asthma diagnosis. Only 12 percent of white households with children ages five to seventeen have a house member with an asthma diagnosis. This compares with 24 percent of African American families in the city. African American families report being much more likely to have gone to the emergency room or urgent care for an asthma-related incident than other families (Binkovitz 2017).

While distressing, reported rates of asthma among Puerto Ricans probably underestimate the actual prevalence of the disease in this population because many symptomatic children are not diagnosed. Research by Whitman and coworkers (2004) in Chicago, for example, found that when cases of "possible asthma" based on a report of patient symptoms rather than actual medical diagnosis were added to cases of physician-diagnosed asthma, the prevalence level for Puerto Rican children jumped to an alarming 34 percent. This higher rate suggests the potential overall significance of asthma among Puerto Ricans. In a study of asthma among children in East Harlem, New York, Puerto Rican children with asthma were found to be more likely to miss school because of asthma than children of other ethnic groups from the same neighborhoods (Findley et al. 2003). This study also found that low-income Latino families that have young children with asthma lack the necessary information, training in asthma management, and medical resources for good asthma control. The problem, however, is not simply one of the health-care system failing to adequately prepare Latino parents in asthma management. Notably, the Childhood Asthma Severity Study (Ortega et al. 2002), which surveyed over a thousand children and their families, found that Latino and African American children were given fewer beta2-agonists, which is a standard medicine in asthma management, and, further, that Latino children received fewer inhaled steroids than white children.

One hazard of asthma for children is stigma, a factor that is associated with limiting needed asthma medication in public or in front of peers to avoid the embarrassment and exclusion of being stereotyped as a sick child. While asthma

is often portrayed in the media as a minor or manageable disease, in the life experience of children it can be terrifying. Anthropological content analysis of Hollywood movies, including those viewed by children, indicates that they have systematically stigmatized asthma and its sufferers. Interviews with children with asthma about representative scenes from a sample of child-oriented movies showed that they felt those with asthma were portrayed as "wimpy," "a geek," "a scaredy cat," "freaked out" or "handicapped." The children objected to these portrayals as unfair and said they make them feel uneasy. Based on viewing such movies, the children expressed worry about what might be in store for them as asthma sufferers during their lives (Clark 2012).

Causes of Health Disparity: Lifestyle versus Social Inequality

The public debate over health disparities in the United States encompasses many different factors, but of central importance is a heated discussion of lifestyle versus social structural causation of health disparities.

Living Right

The lifestyle view asserts that differences in health are a consequence of eating right, exercising, avoiding smoking, not using illicit drugs, drinking moderately, learning to overcome stress, getting proper rest, and securing and adhering to good medical care—that is, taking personal responsibility for your health. It emphasizes individual-level decision making and behavior as the primary factor in wellness and poor health. This perspective flows from very core American values concerning individual hard work and achievement and a tendency to believe that we are personally responsible for our own life situation, social standing, and success or failure.

Exemplary, in the book *Real Age: Are You as Young as You Can Be?* the author, a physician (Roizen 2001), provided a formula for people to calculate how many months of life they were losing because of their own behaviors (e.g., eating fatty foods or not flossing) based on the questionable assumption that individuals are the primary determinant of their own longevity. On the basis of the book, which sold millions of copies, the author founded RealAge, which administers a test to establish a respondent's lifestyle-related health status. The company was subsequently sold for more than $60 million and findings from its popular online test (found at https://www.sharecare.com/group/realage) are used by pharmaceutical companies to target specific marketing messages for their drug products to test takers, which is contrary to the whole lifestyle perspective (Clifford 2009).

Structural Explanation

By contrast with the lifestyle perspective, the social structural view holds that health disparities are more than the consequences of individual decisions and

Social Status Inequality and Health

In an interesting study of the effects of inequality on health, researchers studied the Tsimane, a forager-horticultural group living along the banks of the Maniqui River in lowland Amazonian Bolivia (von Rueden et al. 2014). While it is a comparatively egalitarian society, the researchers did find degrees of status inequality among Tsimane men; higher-status men have more influence in group decision making. These same men are called on by the group to represent the community in its dealings with outsiders. The researchers measured status by asking men to evaluate one another in terms of their influence in community affairs. The men also were asked to provide urine samples. Analysis of these two datasets showed that men with less political influence had higher levels of the stress hormone cortisol. Chronically elevated cortisol can impair cardiovascular and immune function. Accordingly, the less influential men had a higher risk of respiratory infection, a common cause of sickness and death among the Tsimane. The researchers concluded that higher-status men experienced a greater sense of control and greater social support, factors known to have positive impacts on health. If even in a small-scale, largely egalitarian society, status impacts health, how much more so does it affect health in societies structured by unequal social class, gender, or other social divisions?

actions; they are the products of social inequality in many spheres of life, from opportunities in education and housing to treatment in the criminal justice system and access to health care. From this perspective, the key issue is not health disparity per se but health inequity and social injustice. From this understanding, health inequalities are unnecessary, avoidable, and unfair and they reflect the ability of dominant groups to control needed resources in society.

As Krieger (2005:15) observes,

> Social inequality kills. It deprives individuals and communities of a healthy start in life, increases their burden of disability and disease, and brings early death. Poverty and discrimination, inadequate medical care, and violation of human rights all act as powerful social determinants of who lives and who dies, at what age, and with what degree of suffering.

Health adversities due to structural factors are often lifelong despite improved socioeconomic conditions. As the organization Voices for Children (Canny et al. 2002:29) notes with reference to women's health,

> Women from lower income conditions as children [are] 58 per cent more likely to show high insulin resistance than those who lived under higher income conditions as children. Women living in better social and economic circumstances as adults still had a 29 per cent greater chance of being insulin resistant if they grew up in low-income families. Women who grew up poor [also are] more likely to

> . . . have higher levels of bad cholesterol and obesity than those who lived under better socioeconomic circumstances as children. These relationships remain . . . after taking into account adult social and economic conditions.

Consequently, there is a need to study the expressions and interconnections of health inequalities across the life span (Adler and Stewart 2010).

While poverty appears to be a critical factor in health disparities, household income is not the only issue involved. While health disparities are strongly influenced by poverty, disparity is greatest among poor people who live in areas with a high percentage of poor people in the local population. Thus, poor people in cities with smaller impoverished populations are at lower risk of dying than those in cities with large impoverished populations. In other words, the density of poverty in an area appears to be a crucial determinant of the health status of the poor.

Moreover, studies of differences by location among the poor show that the sociophysical environment in which people live—that is, their experience of their surrounding community, including issues of danger, stress, comfort, and appeal—is also an important determinant of their health. Feelings of hopelessness and powerlessness in a community have been found to be good predictors of health risk and health status. One direction of the study of health among the poor developed by anthropologists involves the investigation of human biological responses to impoverishment, or what has been called the biology of poverty.

Recalling the disparities in health care reported earlier, it is also evident that ethnic discrimination by health providers, whether conscious or unconscious and whether driven more by individual or institutional bias, is another important structural cause of disparities in health. An analysis of existing research led to the conclusion that the key factor is "institutional racism," a systemic tendency to view ethnic minorities as less worthy patients, an ideology that is grounded in power imbalances between minorities and professional elites in medicine (King 1996). In their contribution to the Institute of Medicine review of health disparities, anthropologist Mary-Jo DelVecchio Good and coworkers (2003) emphasized the importance of the "culture of medicine," including how medical students and residents are trained in the perpetuation of institutional racism in medicine.

Biology of Poverty

Poverty, quite literally, is inscribed on and in their bodies. One of the consequences of the biology of poverty is that, to the degree that poverty-induced bodily defects and diseases are increased in a population, the challenge for that population of overcoming conditions of poverty without outside resources is greater. As a result, "the very conditions that give rise to disparities are likely to persist and perpetuate a dialectical interaction of inequalities and health" (Leatherman and Jernigan 2015:176). Under these conditions, the goal for health anthropology is to demonstrate the links between structures of inequality, like social class, and local biologies in local contexts.

Insuring Disease

Health insurance coverage has become an important measure of access to health care in the United States. Yet tens of millions of Americans, especially ethnic minorities, have never had access to insurance coverage. As Figueroa (2004:i) aptly points out,

> Not having health insurance is deadly and costly. Even those with health insurance suffer from the insecurity of inadequate coverage and care. . . . [P]roposals to reform the nation's health care system over the last several decades have succumbed to partisan politics and intensive lobbying campaigns by insurance companies. Meanwhile, more than 44 million Americans are without health insurance.

Nationally, Latinos consistently have been found to be the group most likely to lack health insurance across all age groups. Thus, Latino children are the most likely not to be covered by health insurance, with only 8.8 percent of non-Latino white children lacking coverage in 1999 compared to 27.1 percent of Latinos under eighteen years of age. In the same year, while 68.4 percent of non-Latino whites had employer-based health insurance, among Latinos this figure was only 43.4 percent. Despite some improvements, by 2003 only 79 percent of Latino children had health insurance compared with 93 percent of white children and 86 percent of African American children (Holahan et al. 2003). Poverty, discrimination, parental lack of awareness of eligibility, language barriers, enrollment barriers, and fear (especially among undocumented immigrants) of repercussions for using publicly funded insurance have been cited as factors to explain why so many Latino children do not have health insurance. Thus, for example, the fact that in 2003 at least 30 percent of Latino children nationally lived in households with incomes below the federal poverty line, compared to 10 percent of white children, is likely an important factor in health insurance disparities faced by Latino youth (Interagency Forum on Child and Family Statistics 2005).

Significantly, the Commonwealth Fund (Doty 2003) found that the number of uninsured Latinos has steadily gone up, doubling between 1990 and 2002. Lack of health insurance coverage is a significant determinant of both being able to be seen by a doctor when in need and having a primary care provider. Moreover, the uninsured are much more likely to report that they are in poor or only fair health than are the insured. The Robert Wood Johnson Foundation (2005) reported that nationally 37.7 percent of uninsured Latinos are unable to see a physician at a time of need compared to about 10 percent of insured Latinos. Notably, among the insured, Latinos were the most likely to report inability to be seen by a physician at a time of need. Additionally, uninsured Latinos are the ethnic group most likely to lack a personal physician, and, whether they are insured or not (but especially among the uninsured), Latinos are the most likely to report being in poor or only fair health. While 69.6 percent of uninsured Latinos nationally lack a personal physician (compared to 48.8 percent of uninsured whites), almost one-fourth (24.5 percent) of insured Latinos also do not have a personal provider. Lacking personal physicians results in Latinos being the group most likely to rely on a community or public clinic for whatever health care they

do receive (Doty 2003), with Spanish-speaking Latinos being almost five times more likely than whites to get care in a community or public clinic. While 85 percent of whites receive care in a doctor's office, this is true of only 43 percent of Spanish-speaking Latinos. In short, many Latinos have little or no choice of where to go for medical care.

Additionally, lack of health insurance coverage is a significant barrier to preventive health care for Latinos in the United States. Thus, the Commonwealth Fund (Doty 2003) found that comparatively low rates of insurance among Latinos limits their access both to routine preventive care, including visits to the dentist, routine Pap tests among women, routine mammograms, routine prostate examinations, and help with management of diabetes, and to acute care at the same levels as the insured. As a result, the health of Latinos is further compromised, with the resulting development of a number of significant chronic health problems.

Culturally Competent Care

Health disparity research suggests that ethnic minority groups such as African Americans, Latinos, and Native Americans suffer a triple burden in seeking health care: (1) they are significantly less likely to have health insurance than whites, so accessing care is a major challenge, and while adequate acute care is hard enough to come by, preventive care is all but impossible for those who are not insured; (2) in the health-care system, ethnic minority patients face individual and institutional discrimination at every level; and (3) culturally competent health care is in especially short supply.

What is culturally competent health care? The Commonwealth Fund has defined it as the "ability of systems to provide care to patients with diverse values, beliefs and behaviors, including tailoring delivery of care to meet patients' social, cultural, and linguistic needs" (Betancourt 2006:1). In other words, from a health anthropology perspective, culturally competent care begins with the perspective of the patients, their culturally conditioned understandings, attitudes, expectations, and feelings; more precisely, its starting point is patient experience and the often emotionally charged cultural meanings attached to that experience (G. Becker 2004). At present, there are major gaps between the health-care systems in place around the country and what would be considered culturally competent care in light of local ethnic diversity. Yet the need for health-care providers to meet the needs of ethnic minority patients is inescapable, as seen in the changing ethnic composition of the U.S. population.

While individuals of white European and related heritage accounted for about 75 percent of the U.S. population at the time of the 1990 national census, current trends suggest that by 2030 whites will make up only 60 percent of the population. At that point, almost one in five *norteamericanos* will be Latino. Since culturally competent health care is seen as being better health care, unless health-care systems adapt, the quality of care will continue to drop as the American population diversifies.

Recognizing these realities, there has been a push in health-care systems to (1) develop principles and standards of culturally competent care, (2) implement curricula to train health-care providers in culturally competent care and to evaluate their adherence to learned approaches, (3) conduct research on the contribution of cultural competency to reducing disparities in health and health-care provision, (4) publish and disseminate reviews of best and promising culturally competent practices, (5) identify strategies to overcome linguistic and cultural barriers, and (6) develop organizational structures to support culturally competent care. Reflecting these trends, the Office of Minority Health of the U.S. Department of Health and Human Services has established national standards for health-care institutions such as hospitals and clinics to follow in responding to the medical needs of their culturally diverse clients. These standards include the following components that indicate the nature of culturally appropriate care: (1) ensuring that patients receive effective, understandable, and respectful care that is provided in a manner compatible with their cultural health beliefs and practices and preferred language; (2) implementing strategies to recruit, retain, and promote a diverse staff and leadership that are representative of the social and demographic characteristics of people living in the area served by the health institution; and (3) ensuring that staff at all levels and across all disciplines receive ongoing education and training in culturally and linguistically appropriate service delivery.

Given their focus on the role of culture in health, illness, and healing, health anthropologists recognize the value and support the adoption of such standards, although—often citing the costs they might entail—health-care providers are not always immediately receptive to changing their practices. Health anthropologists have found that it is important to demonstrate that culturally and linguistically appropriate services, or what is called culturally competent health care, is, in fact, more effective in terms of producing better health outcomes, which is generally an accepted and valued goal among health-care providers.

An alternative to the cultural competence approach, and a critique of it, is supported by physician and health anthropologist Helena Hansen and her sociology colleague Jonathan Metzl. Rather than training in what often becomes simplistic understandings of culture in medical school, they propose that medical students receive education in what they call *structural competency*. In this approach, training is focused on the social determinants of health, an area that traditionally has not received adequate attention in biomedicine. For example, Hansen and Metzl argue that medical students should be trained in holding providers (and insurance companies) responsible for the outcomes of their patients. Clinical training, they maintain, should teach medical students the ability to "discern how a host of issues defined clinically as symptoms, attitudes, or diseases (e.g., depression, hypertension, obesity, smoking, medication 'non-compliance,' trauma, psychosis) also represent the downstream implications of a number of upstream decisions about such matters as health care and food delivery systems, zoning laws, urban and rural infrastructures, medicalization, or even about the very definitions of illness and health" (Metzl and Hansen 2014:128). Training in structural competency includes five components: (1) recognizing the social

Collating Cultures in Biomedical Care

One expression of the new medical sensitivity to culture is the notable popularity of Anne Fadiman's (1997) compelling and very touching book *The Spirit Catches You and You Fall Down: A Hmong Child, Her American Doctors, and the Collision of Two Cultures*, which has become the most widely used text in cultural competence training for health-care practitioners, including many first-year medical students, in the United States. The book recounts the story of Lia Lee, a Hmong child who was born in Merced, California, in 1982. At three months of age, Lia began having intense seizures, which were interpreted as signs of "soul loss" by her parents and a Hmong shaman but diagnosed by her biomedical doctors as epilepsy. Convinced that Lia's parents were not administering the medicines they prescribed because of their faith in Hmong ethnomedicine, Lia's doctors launched a successful effort to remove her to foster care. Only after a year of battling the system were Lia's parents able to have their daughter returned home. Ultimately, Lia suffered a massive seizure that left her permanently brain damaged. By poignantly describing a fatal clash of cultures in medicine, Fadiman's book has made clear to many why culture is so important in the practice of medicine (though some have criticized the book for presenting an overly static "fixed-in-time" view of culture rather than seeing it as a moving process that is created and changed as it is used).

structures that shape clinical interactions; (2) developing an extra-clinical language for clinicians to talk about structure; (3) rearticulating "cultural" formulations in structural terms; (4) observing and imagining the nature of effective structural interventions; and (5) developing structural humility as physicians, which entails that physicians must be listeners and collaborators or are not the final authority on patient lives.

As the importance of culturally competent care caught on in biomedicine, a number of organizations concerned with the education of physicians and other health-care providers have developed medical and nursing school curricula that reflect an awareness of the importance of paying close and sensitive attention to the patient's cultural and linguistic heritage and its role in their health. For example, the Society of Teachers of Family Medicine developed and has continued to update the "Core Curriculum Guidelines on Culturally Sensitive and Competent Health Care." These guidelines were motivated by the view that "understanding . . . sociocultural variables in health care settings will facilitate the clinical encounter toward more favorable outcomes and enhance the potential for a more rewarding interpersonal experience" (Like et al. 2017). Various medical and other health education institutions in the United States and elsewhere have adopted elements of these kinds of tools into their educational curricula. In the view of one physician advocate of culturally competent training of health-care

providers, "Cultural competence means understanding beliefs of your patient, and reaching into those beliefs to better understand their perception of their illness or treatment, to give them comfort or to give them what they need, so that they can either make it through treatment or perhaps let go" (quoted in Meldrum 2010:94).

These changes, at least in part, reflect the impact of health anthropologists who work in clinical settings within the subfield of health anthropology called clinical anthropology or clinically applied anthropology. Exemplary of the anthropological contribution to cultural competence in health care is a two-volume set produced by the California Endowment. Edited by anthropologist M. Jean Gilbert (2003a, 2003b) of California State University–Long Beach, these manuals include a set of principles and recommended standards for culturally competent health-care education, a resource guide, and a manager's guide for running a culturally competent health-care institution. The manuals illustrate once again an important way in which health anthropology is a discipline in action actively seeking to address pressing issues in health. Another example, from Australia, demonstrates the practical collaboration of anthropology with people from the health professions and public agencies. The Intercultural Interaction Project in the School of Occupation and Leisure Sciences at the University of Sydney involves health science students, and organizations such as the Transcultural Mental Health Centre in Sydney have produced classroom and workshop training materials and teaching models to enhance the cultural competence of health professionals. From this work has developed the idea that cultural competency can be considered as being of three types: culture specific, intercultural, and culture general (Fitzgerald 2000). The emphasis in this work on culture general competence highlights issues particularly in relation to multicultural societies, which with global migration is now most societies, where there is a need to recognize that cultural issues in the provision of health care and social services arise and must be addressed for all people, not just minority ethnocultural communities.

Health and Social Disparities Cross-Culturally

Health disparities are not peculiar to the United States or to the Western world but are a global problem. Internationally, there are two expressions: disparities between wealthier and less wealthy nations and disparities within the populations of all nations around the world. Regarding the first of these, the Health Evidence Network of the World Health Organization (2006b) notes that over the past several decades the number of people in the world living on less than $2 per day has grown to almost three billion. Further, income disparities between the rich and the poor around the world are growing. While the income of the richest 20 percent of the population in the world was thirty times that of the poorest 20 percent of the population in 1960, today it is eighty-two times greater. Within this context of an ever-wider gap between the rich and the poor, health disparities are driven by factors such as social exclusion and discrimination, inadequate diet and malnutrition, a rapid reemergence of both waterborne and blood-borne

infectious diseases, poor housing, low-status employment, unhealthy early childhood living conditions, degradation of the physical environment, disinvestment in the public health and health-care infrastructure serving the poor, escalating violence, and lack of health insurance (figure 4.1).

One way to get a better understanding of health disparities across nations is to compare the top causes of death between a wealthy nation such as the United States and worldwide, as seen in table 4.1. It is evident from reviewing this table that, aside from the high ranking of heart disease and cerebrovascular disease (stroke) as well as unintended injuries, the primary causes of death in the United States and the world as a whole are quite different. Of even greater significance from the standpoint of health disparities is the fact that while chronic conditions dominate the United States, a number of acute conditions and diseases that could be treated if appropriate health care were available (i.e., diseases marked by an asterisk in the table) dominate the global list. In other words, if health care were equitably distributed in the world, millions of people around the planet would not be dying of diseases for which cures and treatments already exist.

A critical factor contributing to health disparities around the world is the emergence of new infectious diseases and new vector-borne diseases. Infectious diseases—diseases whose immediate cause is a pathogen, such as viruses, bacteria, protozoa, or helminthes—are on the rise and kill millions of people each year, particularly in resource-poor nations. The era of "new diseases" began with the



FIGURE 4.1 Backyard pond in East Medinipur District, West Bengal, India, in which this family must bathe, fish, clean fish, and wash dishes and clothes. Photo by Anna Marie Nicolaysen.

TABLE 4.1 Top Ten Causes of Death in the United States and Worldwide

United States	Worldwide
1. Heart disease	1. Heart disease
2. Cancer	2. Cerebrovascular disease
3. Stroke (cerebrovascular disease)	3. Acute lower respiratory infections*
4. Chronic lower respiratory diseases	4. Chronic obstructive pulmonary disease
5. Unintentional injuries	5. Diarrheal diseases
6. Diabetes mellitus disease	6. HIV/AIDS*
7. Pneumonia and influenza	7. Tuberculosis*
8. Alzheimer's disease	8. Lung/bronchial/tracheal cancer
9. Nephritis	9. Unintentional injuries
10. Septicemia	10. Premature birth and low birth weight

Sources: Centers for Disease Control and Prevention and World Health Organization.
*Acute diseases that are treatable if health care is available.

appearance of AIDS in the early 1980s, followed quickly by the emergence of SARS, Ebola, hantavirus, Rift Valley fever, Zika, and a rising tide of other new or renewed infections (i.e., diseases that had been brought under some control but have since reemerged, often in more virulent form, and are resistant to the drugs that had once controlled them). One such reemergent disease, tuberculosis, often associated with AIDS (a combination most frequently seen in the poorest populations), has become one of the world's greatest causes of mortality linked to an infectious agent. Vector-borne diseases (those spread by a blood-sucking arthropod such as a mosquito), including malaria, dengue, yellow fever, plague, filariasis, louse-borne typhus, trypanosomiasis, and leishmaniasis, also have become an ever more important source of health disparities in the world. In 2015, the global number of new malaria cases was 212 million including 429,000 deaths. Across Africa, in particular, millions of people still lack access to the resources that are needed to prevent and treat this disease. Inadequate funding and fragile health-care systems block people's access to life-saving interventions like bednets. According to the 2016 World Malaria Report (World Health Organization 2017b), less than half of the ninety-one malaria-affected countries and territories in the world are on track to achieve the 2020 global health goal of a 40 percent reduction in case incidence and mortality.

Various factors have contributed to the changing profile of infectious diseases in the world, including the breakdown of public health prevention systems in poorer countries; the social disinvestment in health in debt-ridden nations under pressure from development banks and wealthy lender nations; the increased concentration of people living in megacities with huge impoverished population sectors living in overcrowded, unhygienic conditions; the frequent movement of populations fleeing low- and high-intensity wars and other military threats; the

rapidity of global travel made possible by jet aircraft; the global movement of consumer products that can harbor infectious agents; the environmental discard of nonbiodegradable products such as worn-out tires that become mosquito breeding sites in the rainy season; deforestation; and, increasingly, environmental change associated with global warming.

Child and Maternal Health Disparities

An important measure of health internationally, one that is used commonly to assess the health status of a nation, is the general health and well-being of young children and their mothers. The WHO reports that more than ten million children and more than five hundred thousand mothers die each year, mostly from preventable causes. These numbers suggest the painful significance of health problems in the world today, usually rooted in social marginality, poverty, and powerlessness, despite tremendous national and international efforts to make the world a healthier place (figure 4.2).

In addition to reducing diarrhea-related diseases through the distribution of a simple oral solution for rehydrating children with diarrhea, international health efforts focused on childhood immunization against major infectious diseases such as tuberculosis, diphtheria, pertussis, tetanus, measles, and poliomyelitis. As Justice (2000:25) observes, these "two technological interventions—oral rehydration and immunization—emerged as the twin-engine approach to child health and reducing infant mortality" (figure 4.3).



FIGURE 4.2 Family awaiting the demolition of their home in slum clearance in Dhaka, Bangladesh. Photo by Sabina Faiz Rashid.



FIGURE 4.3 Two mothers with sick children from the Dani people of Papua New Guinea. Photo by Leslie Butt.

As a result of these efforts and the investment of resources in child survival, the rate of mortality of children under five years of age fell throughout the world during the latter part of the twentieth century, going from 146 per 1,000 population in 1970 to 79 per 1,000 population in 2003. Between 1990 and the present, the rate of childhood death dropped by about 15 percent, resulting in the lives of two million children being sheltered from death in 2003 alone. As the millennium was ending, however, the general global downward trend in childhood mortality began to level off. In the years between 1970 and 1990, the mortality rate of children under five years of age dropped by 20 percent every decade; by contrast, between 1990 and 2000, it dropped by only 12 percent. This slowdown signaled shifts in relations among richer and poorer nations, a fact that is revealed by moving from global averages to an assessment of regional differences in child mortality. In two regions of the world, Africa below the Sahara Desert and the western Pacific, the slowdown in the tempo of deaths among children under five years of age began during the 1980s. Sub-Saharan countries of Africa began the last thirty years of the twentieth century with the highest levels of under-five child mortality, witnessed the smallest reductions (about 5 percent per decade between 1980 and 2000), and displayed the most marked slowdown in the process of improving children's health by the end of the century. By contrast, progress in reducing childhood mortality continued or even accelerated in the Americas, Europe, and Southeast Asia throughout the period of concern.

As reflected in these statistics, disparities in the health of children in richer and poorer countries around the world have been growing in recent years.

The mortality rate for children under five years old is now seven times higher in sub-Saharan Africa than it is in Europe. In 1980, it was just over four times higher. Indeed, child death in the world increasingly is concentrated in sub-Saharan Africa, where 43 percent of the global total of child mortality is found, reflecting an increase of 13 percent between 1990 and 2003. Another 28 percent of child deaths occur in Southeast Asia. Indeed, more than half of all child deaths in the world occur in just six countries: China, the Democratic Republic of the Congo, Ethiopia, India, Nigeria, and Pakistan.

Even short of actual mortality, disparities in children's health cross-culturally are telling. For example, 6 percent of children around the globe are born with serious birth defects. More than 90 percent of births with serious defects are found in middle- to low-income countries (March of Dimes 2006). Poverty has been found to be one of the most important factors in birth defects internationally; other factors include the mother's age, marriage among close blood relatives, and living in the "malaria belt" of countries with significant levels of malaria in which inherited diseases such as sickle cell disease, thalassemia, and glucose-6-phosphate dehydrogenase deficiency are most common. Similar sociogeographic disparities are found in childhood nutritional status and nutrition-related health conditions, according to the WHO. From 1990 to 2000, the global prevalence of both growth stunting (due to poor diet) and being underweight dropped by 20 and 18 percent, respectively. But high rates of malnutrition among children and other nutrition-related health and physical symptoms continue in southern and central Asia and sub-Saharan Africa. For children in particular, the planet increasingly is a world divided between comparatively wealthy and comparatively poor places, and where you are born is a major determinant of what will happen in your life, including the likelihood that you will live beyond your fifth birthday.

One of the key forces impacting the health of children around the world is the condition of the environment in which they live, including the quality of the air they breathe, the water they drink, the food they eat, and the hidden threats that lurk in the areas in which they play, issues that will be addressed more fully in chapter 7. Children are at particular risk from environmental toxins because their brains, nervous systems, and immune capacities are still developing and because exposure to environmental poisons can cause lifelong damage and chronic disadvantage. Air pollution, for example, both outdoors and within the home, is a primary cause of acute lower-respiratory infections in children, especially pneumonia (figure 4.4).

Such infections are among the major sources of mortality among young children, causing more than two million deaths each year, the vast majority in resource-poor nations. Similarly, lack of access to clean water is common in poorer countries. Each year, diarrhea disease, a consequence in part of unclean drinking water, takes the lives of 1.5 million children. Moreover, it is estimated by the WHO (2006a) that 2.5 billion people, most of them in poor countries, lack access to adequate sanitary facilities.

War and internal conflict are also forces impacting children's health internationally (figure 4.5). Countries that have been ravaged by internal wars, such as Sierra Leone, Angola, Afghanistan, Liberia, and Somalia, tend to have the highest child mortality rates in the world. In Sierra Leone, for example, 250 of



FIGURE 4.4 Factory air pollution is a source of respiratory disease. Photo by Merrill Singer.

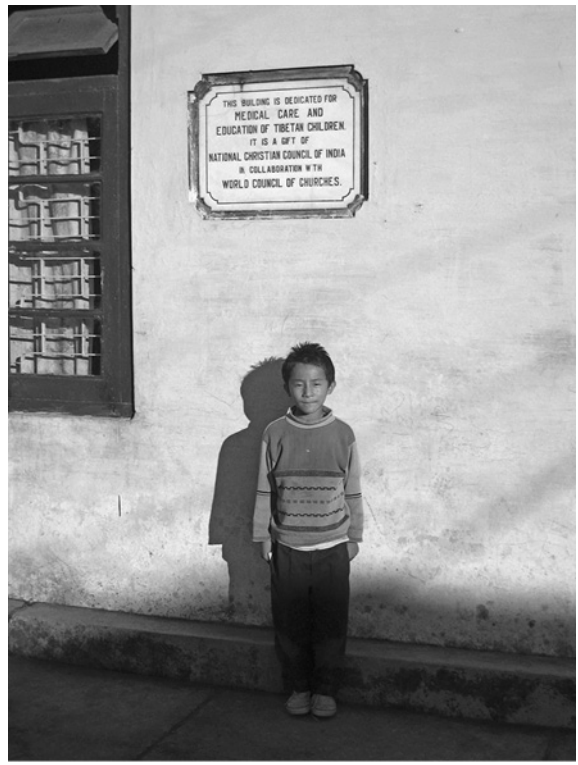


FIGURE 4.5 Tibetan orphan in front of the Medical Care Unit at the Tibetan Homes Foundation in Uttarakh, India. Photo by Anne Marie Nicolaysen.

every 1,000 children die before they reach their fifth birthday, compared to 8 per 1,000 in the United States, according to UNICEF. Since 1990, 90 percent of conflict-related deaths have been civilians, and, of these, 80 percent have been women and children. UNICEF reports that in a typical war, the death rate among children under five years of age goes up by 13 percent. Wars often take a significant toll on health facilities and other societal infrastructure, loss of which is especially damaging to the most vulnerable sectors of a population. Moreover, many children are left orphaned, and others are abducted to be used as soldiers or are raped or otherwise injured. The emotional scars from exposure to violence, loss of family members, dislocation from their homes, and resulting impoverishment can be lifelong. Typical is Sumaya, a fifteen-year-old girl from Sudan whose village was attacked by an Arab tribal militia force called the Janjaweed. Sumaya was interviewed by UNICEF workers in the Kalma refugee camp in South Darfur:

> "I was at school when they attacked us," says Sumaya. "My sisters ran back to the village, and I ran with some friends. My cousin Mona was running ahead of me when she was shot. I stopped and held her hand. When she died, her hand slipped out of mine. Some boys came and told me that I had to run, so I did." Along the way Sumaya found her grandmother and her 4-year-old brother, Mozamel (whom everyone calls Baba). She took the little boy in her arms and started running. "We ran and ran until I felt that I couldn't go on any longer," remembers Sumaya. "I thought about throwing my brother in the grass because he was so heavy, but my grandmother took my hand and told me that we should all stay together." Two agonizing weeks went by before Sumaya and Baba were reunited with the rest of the family. Together, they walked 147 kilometres to Kalma Camp [along with 70,000 other refugees]. (UNICEF 2006)

Despite major international efforts and significant initial improvements in child health, continued improvements in rates of success have not been sustained, nor have they been equitably distributed across the nations of the world. In Justice's (2000) assessment, several factors are important. While international funds were made available to cover initial costs of improving maternal and child health, individual countries were required to use their own resources to implement programs at the local level. Poorer countries with limited human and financial resources were unable to meet all the local challenges of program implementation. The international community, moreover, failed to recognize the fragile nature of the health systems and scarcity of resources (e.g., availability of trained personnel in local settings) in poorer countries. Additionally, immunization was tested in a number of small-scale pilot projects without full consideration of how to scale up immunization in resource-poor nations. Finally, the international health priorities and programs shift from year to year, leaving earlier programs, such as childhood immunization, on their own without international support, leading to drops in immunization rates in many countries.

In other words, from Justice's perspective as a health anthropologist trained to shift her analytic lens between the macrolevels and microlevels in human social systems, overemphasis on a top-down approach, inadequate attention to

disparities in national health resources, and failure to recognize local diversity within nations became important barriers to sustained improvements in child health. In the following section on globalization (see chapter 5), we will consider some other factors as well.

One final indicator of health disparities across nations is access to health insurance. Lack of health insurance among the poor globally is an especially damaging problem. According to the World Health Organization,

> Insurance protects people against the catastrophic effects of poor health. What we are seeing is that in many countries, the poor pay a higher percentage of their income on health care than the rich. . . . In many countries without a health insurance safety net, many families have to pay more than 100 percent of their income for health care when hit with sudden emergencies. In other words, illness forces them into debt. (World Health Organization 2000)

Addressing Health Disparities

In a keynote address at the 2004 International Conference on Overcoming Health Disparities: Global Experiences of Partnerships between Communities, Health Services and Health Professional Schools, David Satcher (2004), the former surgeon general of the United States, noted that to eliminate health disparities globally, we must "know enough" (i.e., carry out much-needed research on existing health disparities and their causes), "do enough" (i.e., deliver desired outcomes in terms of removing barriers to health-care access, eliminating the social causes of poor health among disparity populations, and implementing culturally competent care), "care enough" (i.e., have the commitment needed to accomplish desired goals), and "persevere enough" (i.e., recognize that the elimination of health disparities is a long-term objective and avoid discouragement). Each of these agenda items for achieving global health equity represents a major challenge for policy makers, health-care providers, consumers, and the field of health anthropology.

Health disparities have become an area of concern to the work of a growing number of health anthropologists. In particular, health anthropologists who are active in this arena have been attempting to understand biocultural and biosocial interactions as causes of health inequality and marshaling research findings on these connections in the development of targeted intervention programs intended to reduce or eliminate disparities.

Addressing Health Disparities in the Community

One issue that continues to come up in applied health work is the issue of access to health insurance among people who are at or below the official federal poverty line. Is the federal implementation of universal health-care coverage the answer to this problem as it has been in countries such as Canada, the United Kingdom, and Australia? A related question that applied health anthropologists must ask is whether it is possible to achieve universal health care in the United States. It would seem so, given the vast resources of the nation and the fact that other

countries have been able to implement programs designed to cover the health-care costs of all residents. Yet, there has been tremendous resistance to this idea historically in the United States, especially from the powerful insurance industry and other wealthy sectors of society. The nature of this opposition was played out during the Barack Obama administration in efforts to implement some form of accessible insurance coverage through the Affordable Care Act. While this limited bill did bring coverage to millions of people, it was firmly resisted by political conservatives, who scrambled to eliminate it—despite repeated national polls indicating its popularity—and roll back all federal assistance for essential health coverage for the poor, seniors, women, and others with the election of Donald Trump.

As this example suggests, the elimination of health-care disparities is not easy because it would involve significant social changes that are not supported by sectors of the society that benefit from the existing way society is organized. As Townsend (1986) argued in what is now a classic article in the literature on health disparities, it is not simply the case that there are rich and poor in the world; instead, there are rich precisely because there are poor. Consequently, to eliminate health disparities he argues that it is necessary "to restrict the power and wealth of the rich, to dismantle the present structures of social privilege, and to build social institutions based on fair allocation of wealth and on social equality" (Townsend 1986:29). More recently, Chernomas and Hudson (2010) noted that historical broad health and social inequality has been justified by the claim that the poor are intellectually and behaviorally inferior to the rich. While this assertion is no longer made, at least publicly, the wealthy tend to maintain that redistribution of wealth is unnecessary because a free market will efficiently and justly distribute income among members of society. This neoliberal argument, which Chernomas and Hudson refer to as "the myth of conservative economic policy," has not proven to eliminate either health or social disparities, suggesting the accuracy of Townsend's proposition.

Upstream Analyses of Health Disparities

A critical role that health anthropologists can play in addressing the issue of health disparities emerges from the broad, holistic perspective of anthropology. To date, much of the work that has been done on health disparities that analyzes the "social determinants of health" tends to frame its discussion of causal factors in terms of intermediate-level issues, such as poverty, local or neighborhood conditions, malnutrition, and SES, without embedding these variables in a more expansive macroscopic context.

However, critical health anthropology posits the ultimate determinants of health in the capitalist world system, one that started out in Europe some five hundred years ago but has diffused to every human-inhabited region of the globe (Baer et al. 2013). Global capitalism has progressively impacted and shaped and reshaped social life, including health and access to health care. Given this macroscopic reality, CHA seeks to examine health-related issues within the context of class, racial/ethnic, and gender and imperialist relations inherent in the capitalist

world system and seeks strategies to transcend them, which we discuss in the concluding chapter.

Two very important books have moved beyond an intermediate-level focus. Additionally, they address a key question about disease causation: "Just because most diseases have a social connection of some type, does not necessarily mean that such links can actually cause a disease to occur—or does it?" (Cockerham 2013:2). Both books, like critical health anthropology generally, answer this query about the dominant forces shaping global health in the affirmative. Central to these forces is "the growth of inequalities between the rich and the poor in countries around the world" (Parker and Sommer 2011:1), inequalities between countries, and other factors like gender or minority status that are often entwined with wealth inequality. Inequality contributes to the contemporary significant disease burden of lesser developed countries consisting of communicable diseases (such as Ebola, tuberculosis, or HIV), noncommunicable diseases (such as diabetes, cardiovascular problems, and most cancers), and behavior disorders linked to alcohol and drug abuse.

The first of these books, *Dying for Growth*, emanated from the collaborative work of an interdisciplinary team based at the Institute of Health and Social Justice in Cambridge, Massachusetts (Kim et al. 2000). One of the editors of this volume is Jim Yong Kim, a physician, health anthropologist, and head of the World Health Organization, and another editor, Joyce Millen, also a health anthropologist, is on the faculty of Willamette University. This important anthology explores the linkages between neoliberalism or late global capitalism and health problems among the poor in various countries. In its various essays, *Dying for Growth* argues that the World Bank's policy of "structural adjustment," which has fostered privatization of social and health services in developing countries, has in turn adversely affected the poor around the globe. Further, the book explores the adverse impact of the practices of transnational corporations (those that have operations in multiple countries) on the health of the poor, such as in the case of the 1984 Bhopal gas disaster in India and the harsh working conditions experienced by maquiladora workers who are employed in factories owned by American companies that were built just over the border in Mexico.

The second book that provides a macroscopic perspective on health disparities is *Global Inequality and Human Needs* by Laurie Wermuth (2003), a medical sociologist who adopts a political economic perspective in her examination of regional trends in world health; the impact of gender, race, and ethnicity on both physical and mental health; and the relationship between economic development and health. She argues,

> Among the social factors that influence health and life chances is location in the global economy. Economic projects designed by capitalists in affluent parts of the world have ripple effects in the countries where the production occurs. . . . A person's location in this global economic web of production and consumption is likely to influence how well he or she is fed, clothed, and housed, and what kinds of educational and occupational opportunities he or she has. (45–46)

Obviously, social factors such as food, clothing, shelter, and educational and occupational background play an important role in people's health and access

to health care. By "looking upstream" to the factors (e.g., social policies) and unequal social relationships that ultimately determine health disparities, health anthropologists can help seek effective responses to health inequality.

Gender and Global Health

Gender, a socially constructed distinction between the social position, authority, rights, and access to resources between men and women in society, is an important dimension of global health. Much of the focus on gender and health has been on women's health. Globally, women experience a disproportionate burden of both disease and death as a consequence of inequities in access to basic health care, nutrition, and education. Gender influences exposure to health risks, ability to learn about health information, access to treatment, and health outcomes. As the World Health Organization reports (2013), noncommunicable diseases, including cardiovascular disease, cancer, diabetes, and chronic respiratory disease, are the leading causes of death among women globally and cause two-thirds of deaths among women annually. While these are also diseases of men, diseases often impact women and men differently. Women who die suddenly of heart disease, for example, are less likely than men to experience previous symptoms.

Additionally, girls are far more likely than boys to suffer sexual abuse. Adolescent girls and young women (fifteen to twenty-four years) face double the risk of HIV infection compared to boys and young men in the same age group. This higher risk level is linked with unsafe and often unwanted and forced sexual activity. HIV and other sexually transmitted diseases pose a heightened threat to women and put them at special risk because of an interrelated set of biopolitical factors, hormonal and immunological factors, on the one hand, and poverty, gendered power relationships, and violence, on the other (Ostrach and Singer 2012). It is estimated that 35 percent of women worldwide have experienced either intimate partner violence or nonpartner sexual violence in their lifetime. Further, work assigned to women, such as getting water, may place them at special threat for certain diseases and injuries.

Anemia, especially iron-deficiency anemia, is common among adolescent girls in poorer countries and is known to increase the likelihood that hemorrhaging and sepsis will occur during childbirth. In poorer nations, death during pregnancy and delivery is a major threat. Preterm birth complications are ranked as the tenth leading causes of disability of women, while maternal death is the second biggest killer of women of reproductive age. Indeed, among women of childbearing age, the health risks associated with pregnancy and childbirth are the leading causes of death, disease, and disability internationally. More than three hundred million women in poorer and less-developed countries suffer from short-term or long-term illness brought about by pregnancy and childbirth, according to the WHO, and there are few indicators that improvements are occurring in these areas. Indeed, in some parts of the world, the health of mothers is falling. Not surprisingly, maternal death is especially common in poorer countries, especially in sub-Saharan Africa, with less than 1 percent of maternal deaths in the world occurring in wealthy countries. While the lifetime risk of maternal death is 1 in 16 in the world's poorest nations, it is 1 in 2,800 in

the world's wealthiest nations. In some settings, such as parts of Latin America, women experience neglect and even abuse in health-care institutions, a phenomenon that has been called "obstetric violence" (Dixon 2015).

In sum, there are multiple ways in which girls and women face special health risks, yet neglect of women's health is common in many societies.

Focusing on Disparity in Diseases

One health anthropologist who has been quite active in addressing gender-related health disparity issues over time is Deborah Erwin, the director of health disparities at the Roswell Park Cancer Institute in Buffalo, New York. Much of Erwin's career has focused on reducing and preventing health disparities in cancer. She has worked for many years developing culturally appropriate programs to enhance access and screening for cancer among minority and underserved populations. As part of this work, Erwin helped develop the Witness Project, a breast cancer education and screening intervention specifically targeted to African American women. The Witness Project model has been replicated in over thirty sites in twenty-two states. At the heart of the project are African American women living with cancer who share their personal narratives with patients with a more recent diagnosis of cancer. These narratives affirm that "African American women are not helpless as patients in the biomedical system [and stress] the fact that other sisters can help navigate the process, much like shaman traversing unknown worlds." Moreover, they call attention to the need "to recognize and address the social justice issues and social determinants integrated into cancer detection, treatment, and causation" (Erwin 2008:144–45).

In another applied project, Erwin and colleagues (Jandorf et al. 2014) focus on reducing disparities in breast and cervical cancer for Latinas in Arkansas, Buffalo, New York, and New York City. Participants were randomly assigned to group educational programs on either cancer or diabetes screening. They found that involvement in educational programs significantly increased getting mammography, clinical breast exams, and Pap tests, with no significant differences by message type. These researchers concluded that the specific health messaging may not be as important to increasing breast and cervical cancer screening as cultural- and language-specific outreach in the community.

Another health anthropologist who works on cancer as a health disparity issue is Christine Makosky Daley of the University of Kansas Medical School. Her primary areas of research focus are the reduction of health disparities in Native North America through primarily cancer control and prevention and cross-cultural communication in the health-care setting, and by way of communication technology (Choi et al. 2011, Daley et al. 2012). In addition to working on other applied health anthropology projects, Daley has worked as the project director on an American Lung Association–funded assessment of a smoking cessation program culturally tailored for American Indians and Alaska Natives. Although American Indians and Alaska Natives have the highest smoking rates in the country, few smoking cessation interventions are specifically targeted to these populations. She subsequently began working on a culturally targeted smoking cessation program with the Haskell Health Center and the urban Native

American population in Kansas City. Also of concern in her work are barriers to cancer screening faced by Native Americans. Based on this concern, she worked with the Fort Apache Indian community in Arizona, where she helped to implement the Well Woman Healthcheck Program, a program of mobile breast and cervical cancer screening. In addition to this work, Daley helped to develop a touch-screen computer system to increase patients' open and relaxed discussion of colorectal cancer screening with their physicians.

Multidisciplinary Approaches

In his work at Northwestern University, Thomas McDade focuses on understanding health and human development in the context of social and cultural processes, with a particular emphasis on the role of stress in the development of disparities in health. For example, McDade (2003) has studied the impact of stressful life events on the function of the immune system among Samoan adolescents. Additionally, McDade helped develop Cells to Society (C2S), a center for innovative, multidisciplinary approaches for studying health disparities. Specifically, C2S was developed to unite the social, life, and biomedical sciences in expanding the understanding of the origins and consequences of health inequalities in the United States and using this information to generate policy solutions. Of special interest to McDade and other researchers at the center is how broad social, ethnic, and economic disparities "get under the skin," such as through the impact on the immune system, and affect human development, psychological well-being, and longevity.

One potential pathway through which psychosocial stressors increase the risk for cardiovascular disease is chronic inflammation. However, McDade and colleagues (2013) from several disciplines found that prior research on stress and inflammation was conducted almost exclusively in high-income, industrialized populations that have low levels of infectious disease. Consequently, they carried out their research among young adults in the Philippines who have grown up in an ecological and epidemiological setting substantially different from that of high-income Western countries. Their results suggest that early environments influence the development of inflammatory phenotypes in ways that moderate sensitivity to psychosocial stressors in adulthood, and they support the importance of carrying out comparative, developmental research on social environments, inflammation, and disease.

Studying Local Mediation of Global Health

A final example of the roles health anthropologists play in health disparities is seen in the work of Lee Pachter, whose research has focused on accounting for patterns of health disparities among children, among other related topics (Pachter 2006). In his research with inner-city populations, he has argued that purely economic indicators, such as household income, do not fully account for the unequal distribution of health and illness among children. In addition, he has shown that residential segregation and the number of

urban poor families living in concentrated poor neighborhoods is of critical importance. Pachter identified three mechanisms through which residential segregation magnifies child health disparities: (1) having to pay higher prices in smaller neighborhood stores while having limited employment opportunities; (2) the breakdown of interpersonal cooperation for mutual benefit with resulting loss of interpersonal trust and community participation; and (3) enduring exposure to structural violence (e.g., racism and discrimination) giving rise to psychological deficits, such as low self-esteem, a low sense of self-efficacy, and disempowerment. The interconnectivity and mutual reinforcement of these factors produces heightened rates of poor health among the children of households in highly concentrated poor neighborhoods. As a pediatrician/anthropologist, Pachter is able to translate his findings into hospital-based programs to improve the health of poor children.

In more recent work, Wade and colleagues (2017) examined the lifetime health effects of adverse childhood experiences (ACEs). Traditionally, such experiences have been limited in research to the enduring negative health effects of various aspects of family dysfunction. In their work, these researchers emphasize the importance of adding community-level variables such as community violence, urban crowding, and perceived racism. In research using this expanded definition, it was found that elevated family dysfunction ACE scores were significantly associated with adult health risk behaviors, and physical and mental illness (Bethell et al. 2017). High community-level ACE scores, in turn, were associated with substance abuse history and suffering from a sexually transmitted infection. These results support an understanding of the primacy of family relationships on a child's health life course and the need for specifically targeted interventions designed to support families living under adverse conditions in socially and economically marginalized neighborhoods.

As this discussion suggests, with reference to public health, "place matters." Indeed, this is the title of one of the seven episodes in the widely viewed public health documentary video *Unnatural Causes: Is Inequality Making Us Sick?*, produced by California Newsreel (see www.unnaturalcauses.org/episode_descriptions.php?page=5). As this educational program emphasizes, people's street address is a good predictor of their health status. Exemplary is the economically depressed city of Richmond, California. During World War II, Richmond had been a boomtown, with the local shipyards turning out ships for the war effort twenty-four hours a day. But at the end of the war the shipyards shut down, throwing workers into unemployment. White families were able to take advantage of federally backed home loans and move to the suburbs, but because of overt discrimination this was not the case for ethnic minorities. The result was the emergence of a legacy of poverty and poor health in places like Richmond. Today Richmond has high rates of asthma hospitalization and diabetes and a lower life expectancy than suburban areas. Research indicates that in places like Richmond, there is a 50 to 80 percent increase in risk for heart disease related to chronic stress. Additionally, while fresh fruits and vegetables are hard to find locally, tobacco, liquor, and fast foods are readily available.

Pushing Back on Health Disparities

Communities do not often passively accept significant health disparities. Rather, they find ways, large and small, visible and more hidden, and perhaps not always articulated in the language of health, to resist and fight back against health inequality and injustice. One such strategy was identified in a study of inner-city community perceptions of being stigmatized due to suffering from food insecurity and living in a *food desert*, a term used to refer to marginalized sections of cities with few if any supermarkets. Residents who were interviewed about the difficulties they faced in providing healthy food for their families, and the stigma they endured because of it, reported resistance in the form of taking practical if difficult steps such as riding a bus to distant areas of the city to shop at full-service grocery stores. While this was very time-consuming for them and somewhat of a burden in light of the planning involved, participants "expressed a determination to seek the best possible nutrition for their families and to not let themselves fall victim to nutritional exclusion" (Xu et al. 2016:1306).

At a broader scale, a review of published studies (Anderson et al. 2015) found that interventions targeted at addressing the unequal health status of ethnic minority populations that are led by community coalitions can build alliances among health and human service providers and ethnic racial minority communities in ways that contribute to improved health outcomes and behaviors, as well as to enhanced quality of the care delivery systems. Community coalitions are created when a lead community agency or other convening group takes action on a community health issue (e.g., disproportionate rates of diabetes). Other organizations and residents in the community are attracted by the initiative, which sometimes leads to or is the result of a funding opportunity (e.g., government funding for community-based obesity prevention) or sometimes from a mandatory requirement by the funding source (like the Robert Wood Johnson Foundation's funding of the Fighting Back Initiative to reduce the use and abuse of alcohol and illicit drugs). Governments and foundations have come to see community coalitions as a means of reducing costs and duplication of effort through merging of resources at the local level where interventions are intended to have an impact. Communities may see coalitions as challenging but effective ways to address the health disparity issues they perceive as most pressing.

"Race" and Health Disparity

One of the debates over health disparities concerns the role of genetics and race as determinants of health status. For example, it is known that throughout the Americas, "people of African ancestry have higher blood pressures and higher rates of hypertension than do others in the same societies" (Gravlee et al. 2005:2191). Hypertension plays a major role in lowering the life expectancy of African Americans in the United States—more so than cancer, diabetes, stroke, or AIDS—and this is true elsewhere in the Americas among people of African origin. Moreover, among African Americans, those with darker skin color tend to have higher blood pressure than those with lighter

skin color. One hypothesis about why this might be so is that dark skin color is a visible marker of African ancestry and reflects a genetic, race-based predisposition for higher blood pressure. To test this hypothesis, Gravlee and coworkers (2005) conducted a study in Guayama in southeastern Puerto Rico in which they first used cultural consensus analysis to determine local study participants' assessment of the color of seventy-two standardized facial portraits that varied by skin tone, hair texture, nose shape, and lip form. This led to the identification of five different color groups (e.g., blanco, or white; trigueño, or wheat colored; and negro, or black) that represented meaningfully different groups based on color in the local community. One hundred individuals, one-fourth drawn from each of four neighborhoods in Guayama, were recruited to participate in the study. Based on their observation of each participant's appearance, interviewers assigned them to one of the five color groups. Additionally, a narrowband reflectometer was used to objectively determine each participant's color. Then an automatic blood pressure monitor was used to measure each participant's blood pressure three times during an hour-long interview. These researchers found that objectively measured skin color was associated neither with blood pressure nor with which culturally defined color group a person was assigned to. However, the culturally defined color group was associated with blood pressure, with those labeled blanco or trigueño having statistically lower blood pressure than those labeled negro. In other words, the key factor associated with blood pressure was not one's actual skin pigmentation (which would reflect genetic influence) but rather how people in the community culturally defined someone's "race." Rather than finding support for a racial determination of health disparity, this study provides support to the primary competing hypothesis, namely, that having perceived darker skin subjects individuals "to racial discrimination, poverty, and other stressors related to blood pressure" (Gravlee et al. 2005:2191).

Race and Racism

In recent years, anthropologists increasingly have dropped the term *race*, which traditionally was used to refer to biologically distinct groups, from their lexicon of scientific terms. As the American Anthropological Association affirmed in its 1998 "Statement on 'Race,'" most physical variation, about 94 percent, lies within so-called racial groups. Conventional geographic "racial" groupings differ from one another only in about 6 percent of their genes. This means that there is greater variation within "racial" groups than between them. In neighboring populations, there is much overlapping of genes and their phenotypic (physical) expressions. Throughout history, whenever different groups have come into contact, they have interbred. The continued sharing of genetic materials has maintained all of humankind as a single species. Importantly, while anthropologists no longer believe it is empirically valid to use the term *race*, they nonetheless recognize that racism—bias based on assumed biological differences between populations—is quite real and does significant damage to the health of those who are subject to it.

In sum, in the arena of health inequalities, social factors (including social inequality) rather than genetic factors tend to be most important. From an applied perspective, this means that health inequalities will be limited or eliminated to the degree that social disparities are limited or eliminated. As succinctly summarized by Krieger (2007:662): "social justice + human rights = health equity."

Discussion Questions

1. There has been eye-catching growth in biomedical knowledge and medicine's expanded capacity to heal in recent decades and yet certain groups have not benefited fully or equitably from these developments; why not? What is the triple burden in seeking health care suffered by ethnic minorities?
2. What is the difference between health disparity and health inequality? Is this a useful differentiation?
3. What did the Institute of Medicine review of existing studies find with regard to differences in health-care access in the United States? How can we account for these notable differences?
4. What is culturally competent health care? Why is it important?
5. Why is there a need to study the expressions and interconnections of health inequalities across the life span? What are the special risks of children and mothers in today's world?
6. What did Krieger mean in arguing that social inequality kills? Do you agree?
7. Why have disparities in the health of children in richer and poorer countries around the world been growing in recent years?

Health and the Environment 5

Toward a Healthier World

It is no measure of health to be well adjusted to a profoundly sick society.
—Krishnamurti

Introduction and Overview

This chapter focuses on the impact of both the natural environment and the socially constructed environment on human health, placing special emphasis on the influence of climate change on the environment, human settlements, and health. Further, the chapter discusses the ways in which corporate globalization is playing a crucial role not only in creating health disparities, climate change, and an ecological crisis, but also in contributing to diseases such as cancer and even AIDS.

The health of a population related to the environment in which the population lives has been central to health anthropology for many years. Many health anthropologists have attempted to answer this question for specific populations, and some have attempted to develop general ecological theories of human health that apply across local situations. From an epidemiological perspective, population health is rooted in "a stable and productive natural environment that: yields assured supplies of food and fresh water; has a relatively constant climate in which climate-sensitive physical and biological systems do not change for the worse; and retains biodiversity (a fundamental source of both present and future value)" (McMichael and Beaglehole (2003:4–5). This is an ideal that no human populations experience in full. The reasons they do not, the nature and determinants of alternative human-environment relations, variations in environment-related health status within and between ecological zones, and the causes and health implications of environmental change are the concerns of this chapter.

The *natural environment* refers to the earth's surface and atmosphere, including biological organisms (of which humans are one species), along with air, water, soil, and numerous other resources required to sustain life. Due to having developed cultures that include technoeconomic, social structural, ideological, and attitudinal components, humans are able to survive in a wide diversity of habitats, indeed, in most habitats on land surfaces around the planet. They do not do so passively by merely adapting to the contingencies in the physical conditions around them. Rather, humans are capable of having a profound impact on the natural world, and in this sense the environments humans inhabit are, to varying degrees, not really natural in the sense of being pristine and free of human influence. At the same time, human health is significantly affected by the environment

produced by the daily interaction of natural and sociocultural forces. To take one vitally important and increasingly pressing example—one that, because of its significance, we will return to in greater detail later in the chapter—the World Health Organization estimates that the earth's warming climate currently causes 150,000 deaths and five million illnesses each year. WHO (2014) estimates that approximately 250,000 deaths will be caused by climate change per year between 2030 and 2050, although it recognizes that this may be an underestimation. Climate change is contributing to rising rates of malaria, diarrhea (a common cause of death among children in developing nations), and malnutrition. For instance, higher temperatures in South Asia have caused elevated rainfall levels and contributed to improved breeding opportunities for mosquitoes. As a result, diseases spread by mosquitoes, such as dengue fever and malaria, have skyrocketed. Notably, rising temperatures on the planet have a disproportionate impact on poorer nations, those that have through their limited production of industrial pollution or other environmental effects of human technology contributed the least to causing global warming. As many scholars have observed, those people who are the most vulnerable to the devastating impact of climate change have not been and in the next several decades will not be the ones most responsible for causing it. This responsibility falls upon the shoulders of the affluent, particularly in developed countries but more and more so in developing countries. In short, humans both make and are made (and, in the case of disease and death, are unmade) by their environment. As this discussion suggests, the relationship between human health and the environment is complex, and it has certainly been an arena of debate within health anthropology.

Medical Ecology and Critical Health Anthropology on the Environment

Two perspectives within health anthropology—medical ecology and critical health anthropology, referred to as critical anthropology until recently—have especially concerned themselves with human-environment interactions. As noted in chapter 1, medical ecology, a biocultural perspective in health anthropology, views health and disease within an elaborate ecosystem, one that includes physical, biological, and cultural components (McElroy and Townsend 2014). The physical, or abiotic, environment includes climate, energy sources, and material resources, such as those that can be converted into tools for procuring food, building shelters, and making clothes to protect people from the natural elements. The biotic environment includes plants and animals that can be converted into food, clothing, and even shelter as well as predators, vectors, and pathogens that can endanger the life of the individual or the group. Finally, the cultural environment includes technology, social organization, and ideology, all of which serve to sustain health and counteract disease, which is not to say that culture does not play a large role as well in producing illness and death.

According to McElroy and Townsend, changes in any of the components of their model can cause imbalances in the other components (e.g., new subsistence strategies can lead to exposure to new risks), and a very severe imbalance

can generate stress and disease. For example, "a change in climate may lead to a decrease in food supplies. Erosion of soil may undermine agricultural productivity" (McElroy and Townsend 2014:20).

Built into the medical ecology model are the following assumptions: (1) there are no single causes of disease (e.g., a virus, a vitamin deficiency); rather, disease ultimately flows from human-ecosystem imbalances; (2) health and disease develop within the interface of physical, biological, and cultural subsystems; (3) the environment is not only physical habitat but the human-made environment of villages, towns, and cities; and (4) perceptions of the environment are shaped by culture. McElroy and Townsend stress that their ecological-based systems approach precludes easy explanations of health problems. This is a reasonable assessment; health is a complex phenomenon that is influenced by many factors.

While there are points of overlap between medical ecology and critical health anthropology in their respective approaches to health and the environment, there are some notable differences in emphasis as well. Both of these perspectives in health anthropology adopted a version of a biocultural approach that sees health and disease as products of the continuous interaction of human biology and human ways of life and social organization. Both perspectives also overlap with a multidisciplinary approach that first emerged within anthropology called *political ecology*, which incorporates political economy but goes beyond examining the relationship of social inequality to health to also consider the impacts of social inequality on the environment and, through human interactions with the environment, on human health.

Contributors to political ecology come from many fields, including sociology, anthropology, political economy, and geography (Foster et al. 2010; Robbins 2012). One scholar provides a succinct overview of the basic premises of political ecology in his assertion that "economy is a matter of ecology: it has to do with the production and distribution of goods and services in the context of human society and nature" (Parsons 1977: xii).

Scholars interested in the political economy and political ecology of health have considered a wide array of politically and ecologically induced health problems, including malaria, tuberculosis, occupational hazards, and cancer. To fully understand the political ecology of health in the contemporary world, it is necessary to recognize that "the analysis of illness, work, and the environment must consider the connections between these issues and the contradictions of capitalism" (Waitzkin 2000:8); this is the world system in which, in varying ways, almost all people on the planet participate.

In assessing the role of the environment in human health, the critical perspective in health anthropology draws on the notion that nature has been affected by human events. In other words, the political economy of specific sociocultural systems and the world system generally has profoundly shaped the actual physical reality of nature. This approach differs notably from the common view of nature as an autonomous reality that operates in terms of its own principles separate from human society, a perspective that is reflected in the separation of "natural history" and "human history" into two different and largely separate fields of study but also is seen, more mundanely, in Western ideas about getting

back to nature or escaping to nature as refreshment from life in a human-made environment. The critical perspective, in contrast, attempts to consider "the relation of people to their environment in all its complexity" (Turshen 1989:48) and to treat political economy and political ecology as inseparable.

Crosby (1986), who uses the term "ecological imperialism" to label the biological expansion of European life forms (including European peoples, their plants, animals, parasites, and pathogens) into all corners of the globe, has contributed an important concept for the exploration of the impact of capitalist political economy on "nature." He asserts that this impact involves "a condition of continual disruption: of plowed fields, razed forests, overgrazed pastures, and burned prairies, of deserted villages and expanding cities, of humans, animals, plants, and microlife that have evolved separately suddenly coming into intimate contact" (Crosby 1986:291–92). Crosby argues that the ability of earlier civilizations—such as the Sumerians and, much later, Europeans—to conquer indigenous peoples often had more to do with the fact that they had exposed the latter to a multiplicity of alien diseases than with technological and military superiority. For example, shortly after the First English Fleet arrived in the harbor of Sydney, Australia, in 1788, smallpox spread like wildfire both up and down the coast and inland among the aboriginal peoples, eventually affecting even residents in the remote outback of New South Wales, Victoria, and South Australia.

Health and the Environment in the Past

Ancient populations developed fitness to resist environmental threats, including climatic ones, periods of food shortage, microbial parasites, and predators, particularly during the Paleolithic era. Most skeletal remains from the Mesolithic (10,000 to 8,000 years BP) and the Upper Paleolithic (40,000 to 10,000 years BP) eras depict vigorous adults who were relatively free from organic diseases. People in indigenous societies have reportedly enjoyed cleaner environments and better health than the majority of people in agrarian civilizations (Cohen 1989). Even today, indigenous people exhibit a scarcity of cancers, cardiovascular diseases, and other degenerative diseases, although researchers often have observed that few people live long enough to contract them, in large part because they are vulnerable to parasites, predators, poisonous snakes, accidents, warfare, and unexpected harsh climatic conditions that may cause periods of food and water shortage or otherwise lead to illness and death.

All human societies encroach on and modify the natural environment and have created environmental crises of greater or lesser degree throughout their long evolution (Bodley 2012). Foragers contributed to the creation of grasslands, pastoralists overgrazed their lands, and peasants caused deforestation. Each of these levels of social and technological complexity in human social evolution probably contributed to the extinction of certain species. While there is a tendency in Western society to romanticize indigenous peoples, such as pre-Columbian Native Americans, various species, such as indigenous small horses, camels, and large game animals in North America disappeared after the arrival of the first humans in the Americas, perhaps in small or large part because

of overhunting. Environmental degradation was one of the principal factors contributing to the demise of the Anasazi ("the ancient ones," as they are referred to by contemporary southwestern Native Americans) horticultural villages of Chaco Canyon in western New Mexico, the chiefdom societies of Easter Island, and the Norse settlements in Greenland (Diamond 2005).

The emergence of social mechanisms for harnessing large amounts of energy from the environment contributed to the emergence of what Ruyle (1977:623) has called "predatory ruling classes." In addition to the role of class conflict in contributing to the demise of the earliest state societies (e.g., ancient Sumer, Egypt, and Rome), as Hughes (1996:29) observes, "their failure to maintain a harmonious balance with nature" was a significant factor in their demise. In the New World, environmental devastation created by the classic lowland Mayan civilization contributed to a yellow fever epidemic that may have played a significant role in the collapse of the Mayan state structure and unifying institutions.

The dangers of ecological self-destruction that plagued ancient state societies became even more pronounced with the advent of industrial capitalism beginning around 1800, and it has been growing ever greater with the passage of time. In contrast to the smaller-scale environmental modifications resulting from the intervention of indigenous societies, global capitalism, with its treadmill of production and consumption, introduces a new array of industrial and chemical pollutants that disrupt the natural environment and have devastating effects on human health. The Industrial Revolution allowed for the harnessing of nonrenewable fossil fuels, namely, coal, petroleum, and natural gas, and humans came to rely increasingly on machine power rather than the energy derived from humans and animals.

In order to meet the needs of the capitalist system for functionally healthy workers, the development of water supplies, sanitation, and agricultural practices dramatically improved health conditions and longevity worldwide, but these benefits were unevenly distributed: some societies benefited far more than others, and within societies some social classes profited disproportionately. Conversely, capitalists often ignored the health of their workers if they had access to a surplus labor pool and could increase profits by ignoring the physical environment of the workplace, the pace of the work itself, and the housing and living conditions of the workers. One consequence is the significantly substandard housing found among the poor and working people of a highly marginalized country such as Haiti, as seen in figure 5.1.

Within the context of the global economy or the capitalist world system, any effort to create a healthy global community must come to grips with what has been called "the productivist ethic," a belief that continual economic expansion is necessary, socially beneficial, and natural, as summarized in folk sayings such as "you can't stop progress." Closely tied to the productivist ethic is a "culture of consumption" that assumes that humans have insatiable material needs and live in an environment with endless resources. The productivist ethic, which is part of the ideology that rationalizes the capitalist mode of production, historically helped drive many of the key social changes that despoiled and continue to despoil the ecologically fragile biosphere while channeling an increasingly large



FIGURE 5.1 Poor-quality housing in Cité Soleil, Haiti, reflects the overall poverty of the country. Photo by Merrill Singer.

share of the planet's limited resources to the privileged few at the expense of the masses of people. The global economy has been transforming more and more of the natural environment—including water, forests, beaches, and plants—into private commodities to be purchased and sold on the market.

Health and the Environment Today

The fossil fuel–based technologies that make up industrialization brought about sweeping environment-reshaping changes, including the growth and pollution of heavily populated urban centers. These changes have had a profound impact on human health. During the twentieth century and continuing in the present century, the world has witnessed the emergence of teeming metropolitan areas such as Mexico City, São Paulo, Mumbai (Bombay), Beijing, and Los Angeles, places that environmental scientists call unhealthy "heat islands" because of the elevated temperatures their technology and population densities produce. Rapid industrialization and technological innovation, including changes in transportation and telecommunications, are part of a process that has come to increasingly be referred to as "globalization" or, perhaps more appropriately, "corporate globalization." A global energy-intensive food production system, the ever-increasing adoption of motor vehicles and air-conditioning systems, oversized houses, a myriad of consumer items for the affluent, the increasing use of air transport for business and pleasure, and other "developments" act to strain a fragile ecosystem and ultimately threaten human health. Environmental degradation, as a result, has become a primary public health issue internationally in the contemporary world.

In their drive for profits, private multinational corporations in the West and state corporations in state capitalist societies such as China have created not only a global factory but also a new global ecosystem that has resulted in the burning of fossil fuels, motor vehicle pollution, acid rain, toxic chemical and nuclear radioactive waste, desertification, defoliation, ozone depletion, and greenhouse gases. The last of these, which include chlorofluorocarbons, methane (largely from vast cattle breeding facilities), and carbon dioxide (a direct product of fossil fuels) emitted by motor vehicles, airplanes, ships, and coal- or oil-fired electric generators, have been contributing to ozone depletion and global warming. Carbon dioxide acts like the glass covering of a greenhouse in that it allows heat from the sun to pass through to the earth's surface while preventing much of it from escaping from the planet.

Depletion of Natural Resources and Environmental Degradation

While to a greater or lesser extent all human societies have historically encroached upon the natural environment, global capitalism has done so at a much more pronounced level. In essence, capitalism lacks a concept of sufficiency, of individuals having enough, and demands perpetual economic growth. It operates with the assumption that natural resources are infinite, when in reality many of them are finite. Capitalism contributes to the depletion of natural resources and environmental degradation, including climate change. While in their thorough and critical analysis of capitalism Marx and Engels did not focus on ecological issues, they were certainly cognizant of the dialectical relationship between sociocultural systems and the natural environment. In *The Condition of the Working Class in England*, Engels (1969; original 1845) described the devastating impact of industrialization on the natural environment. Marx notes numerous examples of capitalist pollution: industrial chemical waste-product; machine tool manufacturing iron filings; flax, silk, wool, and cotton washes in the textile industry; rags and discarded clothing; and human excreta in the Thames River in London (Merchant 2005:146). In essence, global capitalism has created an ecological disruption or a "metabolic rift" between society and nature that is "alienating us from the material-natural conditions of our existence and from succeeding generations" (Foster et al. 2010:7). Capitalism has been reliant on what Moore (2015:155) terms the "Four Cheaps," namely cheapened raw materials, cheap coal, cheap iron, and cheap steel, all of which exist in limited supply.

Researchers at the Stockholm Resilience Center identify nine "planetary boundaries" that are essential to maintaining a global ecosystem in which humanity can survive and thrive safely: (1) climate change, which is on the verge of reaching a tipping point; (2) ocean acidification, also on the verge of reaching a tipping point; (3) stratospheric ozone depletion, also close to a tipping point; (4) disruption of natural nitrogen and phosphorus cycles; (5) human pressure on global freshwater availability; (6) conversion of forests, wetlands, and other vegetation types to agricultural lands; (7) biodiversity loss; (8) atmospheric aerosol loading; and (9) toxic chemical pollution (Rockström et al. 2009). These

boundaries constitute the "safe operating space for humanity with respect to the Earth system" but exceeding them would "push the Earth outside the stable environmental state of the Holocene, with consequences that are detrimental or even catastrophic for large parts of the world" (Rockström et al. 2009:472). For climate change, the boundary is 350 ppm of CO_2 in the atmosphere, but this boundary has been surpassed at over 406 ppm of CO_2 and continues to rise because serious mitigation efforts have not been put in place, despite the fact that the Kyoto Protocol was passed in 1997. Despite the ongoing *Limits to Growth* discussion since the 1970s, governments, corporations, the World Bank, and even the United Nations, as evidenced by its 2012 Conference on Sustainable Development in Rio de Janeiro, are committed to ongoing economic growth.

The development of capitalism expanded the metabolic rift from Europe to a global scale of colonialism, imperialism, and corporate globalization (Foster et al. 2010:78). Foster provides a long list of "warning signs" signaling the environmental unsustainability of the capitalist world system, which we summarize in condensed form:

- There is virtual certainty that the conventionally critical threshold of 2°C will be crossed in the near future.
- Various studies suggest that with each 1°C increase in global temperature, rice, wheat, and corn yields could drop 10 percent.
- With the world approaching peak oil production, a global crisis and mounting resource wars appear to be likely (J. Foster 2009:254–255).
- Global freshwater supplies are diminishing due to the drawing down of aquifers in many locations.
- Two-thirds of the global fish stocks are being depleted.
- Species extinction is at the highest level that it has been in sixty-five million years.
- The National Academy of Sciences in the United States released a study in 2002 that indicated that humanity had exceeded the earth's regenerative capacity in 1980 and by 1999 exceeded it by as much as 20 percent.
- Just as various earlier civilizations and societies have experienced ecological collapse, the capitalist world system faces a similar fate.

Capitalist production has played a central role in a dramatic increase in greenhouse gas emissions as well the creation of a global water crisis. As for the first of these developments, the Working Group I component of the 2013–2014 Intergovernmental Panel on Climate Change Fifth Assessment reports the following grim scenario regarding the increase of greenhouse gas emissions:

> The atmospheric concentrations of carbon dioxide, methane, and nitrous oxide have increased to levels unprecedented in at least the last 800,000 years. Carbon dioxide concentrations have increased by 40% since pre-industrial times, primarily from fossil fuel emissions and secondarily from net land use change emissions. The ocean has absorbed about 30% of the emitted anthropogenic carbon dioxide, causing ocean acidification. (Alexander et al. 2013:11)

Whereas only 8 percent of freshwater supplies are utilized for domestic use, industry consumes 22 percent of the world's water and irrigation agriculture uses 70 percent (Parr 2013:65). In contrast to a global average water footprint of 1,243 cubic meters per capita per year, the per capita annual footprint of China is 702 cubic meters, with only 7 percent of Chinese water consumption coming from offshore. The water footprint of somebody living in the United States is 2,483 cubic meters per year, with 19 percent of that consumption coming from outside the country. Meanwhile, in Kenya the average per capita water footprint per year is just 714 cubic meters, with 10 percent coming from outside the country (Parr 2013:65–86). In addition to wide discrepancies of water use varying from country to country, there is tremendous class variation in the access to freshwater, varying from households that have swimming pools and large lawns to people who have limited access to water, particularly clean water.

Capitalism and Climate Change

Marx's metabolic analysis can be extended to an examination of global climate change involving the linkage of three dimensions: (1) the impact of capitalism on the global carbon cycle; (2) the Jevons Paradox, in which technological improvements in energy efficiency lead not to decreased but to increased utilization of natural resources, and thereby contribute to environmental degradation; and (3) the role of capitalism in the destruction of carbon sinks (Foster et al. 2010:122). As is evident from the imperilled list of planetary boundaries, climate change is not happening in a vacuum but is growing at a time of multiple and other anthropogenic ecocrises, some of which, like ocean acidification and stratospheric ozone depletion, are verging on their own respective tipping points.

The 2013 Intergovernmental Panel on Climate Change Working Group 1 report projects a rise in global temperature anywhere from 0.4°C to 2.6°C by 2046–2065 and from 0.3°C to 4.8°C by 2081–2100, depending on what strategies humanity takes to mitigate anthropogenic climate change. The World Bank (2014) and Price Waterhouse Coopers (2012), hardly radical organizations, reports indicate that the world is presently heading to 4°C or more of warming this century. This global average could mean around 6°C of warming over land, and perhaps 7–8°C in far northerly latitudes, mind-boggling possibilities to say the least.

The National Aeronautical and Space Administration (NASA) reports that most of the warming of earth since the 1880s has occurred since the 1970s, with the twenty warmest years having occurred since 1981. Sixteen of the seventeen warmest years on record have occurred since 2001, with 2016 being the warmest year on record (NASA 2017).

Global warming and associated climatic events are a direct effect of an increase of greenhouse gas emissions since the Industrial Revolution, which kicked off reliance on fossil fuels, initially coal but later petroleum and natural gas. Reportedly 556 billion metric tons of CO_2 emissions have been added to the atmosphere since 1750 as a result of fossil fuel consumption and land cover change due to increased agricultural production and deforestation. While

climate scientists have debated for a long time whether recent climate change is primarily a natural phenomenon rather than an anthropogenic one, the vast majority of them now agree that it is has been largely created by the emission of various greenhouse gases, particularly carbon dioxide, nitrous oxide, and methane. In one survey, research found that 97.4 percent of the climatologists and 82 percent of the earth scientists in their sample maintain that human-related activities are a significant factor in increasing global temperatures (Doran and Zimmerman 2009). Despite the existence of many climate denialists or skeptics, particularly in the United States and Australia, they argue that the "debate on the authenticity of global warming and role played by human activity is largely non-existent among those who understand the nuances and scientific bases of long-term climate processes" (Doran and Zimmerman 2009:23).

Atmospheric alone CO_2 has increased from 280 ppm at the time of the Industrial Revolution to 406 ppm in early 2017 (NASA 2017). Whereas during the 1990s acceleration of CO_2 emissions stood about 1.1 percent per annum, since 2000 the acceleration of CO_2 emissions stood at about 3 percent per annum (more than 2 ppm per year; Rogner et al. 2007).

China in particular has become the factory of the world, a place where labor costs are much cheaper than developed countries. According to Malm (2016:329), "between 2000 and 2006, 55 percent of the global growth of CO_2 emissions happened there: in 2007, the figure stood at *two-thirds*." An increasing percentage of China's emissions are "export emissions" generated in the process of largely exporting manufactured products to developed countries in North America, Europe, and Australia, which do not generally count "import emissions" into their totals of emissions for which they are responsible. Since China joined the World Trade Organization in 2001, thus ending remaining barriers to foreign investment and ownership, China's export emissions increased from a third during 1990–2002 to nearly half during 2002–2008 (Malm 2016:331).

Multinational corporations and state and joint-venture companies in post-revolutionary societies, particularly China, have created a global factory and a new global ecosystem that includes industrial and motor vehicle pollution, toxic and radioactive wastes, deforestation, desertification, and climate change. As a result of its emphasis on ever-expanding production, global capitalism has been having adverse impacts upon large sections of much of humanity and a fragile ecosystem. It has been heavily reliant on fossil fuels, which prompted Lewis Mumford (1934) to refer to the world system as a form of "carboniferous" capitalism. The "industrial-fossil revolution" or "fossil capitalism" relied first upon coal, then oil, and finally natural gas, including coal seam gas.

Infectious Diseases in a Globalizing World

One of the consequences of corporate globalization has been the emergence of new infectious diseases and the rapid spread of infectious diseases (new and old) to previously uninfected populations. The discovery of new pathogens, some of which are already widespread by the time they are recognized as a threat to human health, is now occurring at a swift pace. Approximately three new

human infectious diseases are identified every two years, and a new pathogen is reported in the health literature weekly. New diseases in human communities commonly have their origin in nonhuman animal species (and hence are called zoonotic diseases). This process of cross-species infection involving humans has been going on at least since the earliest domestication of animals, and it continues to occur today. HIV/AIDS, for example, transferred to humans from chimpanzee populations (perhaps as long as a century ago), but the disease did not come to human attention until it spread around the world as a result of the global movement of people as tourists, on business, as refugees, as armies on the move, and so forth. Severe acute respiratory syndrome (SARS) is another global pandemic that had its origin in nonhuman animals and later spread to and adapted to living in human bodies (as a result of mutation and natural selection).

The possible role of corporate globalization in the spread of infectious disease is illustrated by the influenza pandemic that began in April 2009. This zoonotic disease developed in pigs, and the initial epicenter of the outbreak among humans was in a part of Mexico that is home to the massive Smithfield Foods hog production center, which supplies approximately one million animals annually to the world food market. Smithfield, a global corporation, is the world's largest pork processor and hog producer, with revenues that exceeded $11 billion in fiscal 2010. Industrial farm animal production (known as IFAP) centers, like Smithfield's operation in Veracruz, Mexico, have been found to produce more than livestock. They also generate vast quantities of animal waste that are too large, concentrated, and toxic to be converted into reusable manure. The Pew Charitable Trusts and Johns Hopkins Bloomberg School of Public Health have reported that livestock centers like Smithfield are a grave potential source of land and water pollution with toxic and potentially infectious animal waste products. Among the prominent concerns explicitly raised by these institutions is whether large-scale corporate livestock production creates a risk environment that fosters the emergence of novel viruses such as H1N1 that could lead to human infections, sickness, and death.

The Impact of Climate Change on Health

Although the global burden of disease due to climate change remains relatively small compared to the impacts on health resulting from poverty, poor sanitation, and tobacco consumption, climate change over the course of the present century, particularly under a more or less business-as-usual scenario, will result in increasing adverse impacts upon health, particularly among the poor and people living in regions relatively close to the equator. The 2014 United Nations Intergovernmental Panel on Climate Change (IPCC) synthesis report states that there was a 0.85°C (1.35°F) rise in global temperature over the period 1880–2012 and a 0.19 meter rise in sea levels over the period 1901–2010. While this may not sound particularly serious, ongoing economic development, coupled with the increasing demands of an ever-growing population—which, in part, is a response to the conditions of poverty that push people in particularly developing countries

to have more children in order to have enough laborers to meet their subsistence needs—further aggravates the problem.

In addition to the general warming of the planet as discussed earlier, climate change has resulted in the melting of icecaps and glaciers around the globe. Forty percent of the surface of the Arctic icecap has retreated during the past several decades; glaciers in the Himalayas, Andes, high African mountains, the North American Rockies, and sub-Antarctic island mountains have been rapidly retreating; much of the tundra of Arctic areas has been melting and releasing a growing level of methane gas into the atmosphere; and portions of the Antarctic icecap also have been breaking up. Because of climate change, plants and animals are moving into regions closer to the poles because these regions are becoming warmer. Animals, such as polar bears, are in danger of extinction due to the contraction of the Arctic ice pack and a reduction of the animals, such as seals, upon which they feed. Coral reef damage is occurring all over the world, including in the Greater Barrier Reef off the coast of Australia.

Climate change is already having serious impacts on human, animal, and plant life around the world. South Pacific islanders in particular face a threat to their traditional horticultural lifestyle because of rising sea levels that inundate their fields and water supplies and threaten to submerge their islands. The Inuit of Arctic Canada and Alaska are experiencing a contraction of the polar bear populations that they have traditionally hunted, and they are unable to access seals because the icepack is freezing later than it once did. Andean peoples face the possibility of the eradication of their way of life as glaciers from which they have drawn water for themselves, their fields, and their animals retreat. Peoples living in coastal areas around the world face the possibility of increased flooding and hurricanes because of the rise and warming of the oceans. A growing number of prominent climate scientists now believe that global warming contributed to the devastating effects of hurricanes Katrina, Rita, and Wilma in 2005. Among these is Judith Curry of Georgia Tech University, whose research has shown that an ever-greater percentage of hurricanes are of the most intense kind, in no small part, she believes, because global warming is increasing the temperature of the world's oceans. According to Curry, Hurricane Katrina was "the 9/11 of global warming" (quoted in Lang 2006:4).

Further, many climate scientists contend that the droughts that people in parts of sub-Saharan Africa and Australia have been facing are related to climate change warming as well. Moreover, climate change appears to be the primary impetus behind the spread of infectious diseases to environments north and south of the equator and heat waves that threaten the lives and health of vulnerable populations, such as the elderly and the sick.

Warmer temperatures have contributed to the spread of mosquito-borne diseases, such as malaria, in South Africa, the highlands of Kenya and Tanzania where it was previously nonexistent, to the higher elevations in the Andes of Colombia; disease-carrying mosquitoes that once lived at altitudes no higher than 3,200 feet have now appeared at the 7,200-foot level (Berger 2000). While climate change is not the only factor involved, today it is estimated that there are 300 million to 500 million cases of malaria in Africa each year, resulting in

between 1.5 million and 2.7 million deaths, more than 90 percent occurring among children under five years of age. Global warming also has been implicated in the resurgence of various other epidemics, including cholera in Latin America in 1991, pneumonic plague in India in 1994, and the outbreak of a hanta virus epidemic in the U.S. Southwest also in 1994. Even in temperate zones, climate change may contribute to greater frequency of diseases such as tick-borne viral encephalitis, which occurs in parts of Western Europe, Scandinavia, and Russia, and Lyme disease (McMichael 2001:302).

Additionally, climate change may endanger people's health and even their lives through heat exhaustion, as was the case during the summer of 2003 in Western Europe and Russia in 2010. The estimated mortality of some 70,000 people in the case of the heat waves in 2003 in Western Europe and 55,000 people in Russia was associated not only with the high temperatures but also with the fact that nighttime low temperatures have been rising nearly twice as fast as daytime temperatures (McMichael 2017). The lingering nighttime warmth deprived people of normal relief from blistering daytime temperatures and of the opportunity to recuperate from heat stress. Air pollution linked to longer, warmer summers particularly affects those suffering from respiratory problems, such as asthma. The heat waves in Western Europe in 2003 and Russia in 2010 contributed to 30 percent declines in grain harvests, accompanied by increases in food prices, food shortages, and nutritional deficiencies among the poor in grain-importing countries (McMichael 2017). Elderly people with higher blood pressure and/or heart disease along with possible obesity are particularly likely to die or require medical attention during heat waves.

Given the health consequences of climate change, we can speak of diseases of climate change. These would not necessarily be new diseases, although they might be, but would include any "tropical" disease (e.g., malaria and dengue fever) that spreads to new places and peoples because of global warming as well as diseases linked to poor nutrition due to desertification of pastoral areas or flooding of agricultural areas.

Merrill Singer (2009, 2010b) developed the term *ecosyndemic* to label the role of anthropogenic changes in the environment that increase the frequency of adverse disease interactions and, as a result, increase the health burden of affected populations. Ecosyndemics are likely to concentrate among the poor and other marginal communities, who are least able to obtain adequate nutrition, sanitation, shelter, low-stress life experiences, and preventive and curative health care. Climate change is disrupting natural ecosystems and contributing to the geographic diffusion of disease-causing viruses, bacteria, fungi, and other pathogens. Diseases that have increased in range because of global warming include malaria, dengue, Rift Valley fever, West Nile disease, Chikungunya fever, yellow fever, hanta virus, Seoul virus, and Zika. Increased planetary warming is also a factor in the spread of disease-bearing rodents (as well as other disease vectors) like those responsible for the outbreak of hanta virus in the U.S. Southwest in 1994 and pneumatic plague in India the following year. Furthermore, water-borne diseases such as cholera are spreading because of ever-warmer oceans.

There is growing recognition that the time is overdue for anthropologists, including health anthropologists, and other social scientists to begin examining the impact of global warming on human populations, including their health. As McMichael (2003:15) aptly observes, "This topic is likely to become a major theme in population health research, social policy development and advocacy during this first decade of the twenty-first century." While archaeologist Brian Fagan (1999:76) is correct in his assertion that "global warming is nothing new for humanity," the magnitude of the global warming or climate change that the planet has been experiencing over the past several decades and that is predicted will continue for decades to come, even if it is checked by monumental preemptive measures, is on a magnitude never experienced before by humanity.

Climate change constitutes one of the most important issues of the twenty-first century, along with the growing gap between the affluent and the poor within most nation-states and between nation-states, thanks to corporate globalization and ongoing conflicts in many parts of the world. The latter in part can be attributed to various states, led by the United States but also including the United Kingdom and Australia, doing the bidding of multinational corporations in what Klare (2002) calls the "resource wars." We have called for the development of a critical health anthropology of climate change warming that recognizes its relationship to the treadmill of capitalist production (Baer and Singer 2009). The four authors of this textbook maintain that, ultimately, climate change or global warming is a problem that is part and parcel of global capitalism with its drive for profits that results in a pattern of ever-increasing production and consumption.

There is much debate on how to adapt to and mitigate climate change, ranging from turning to alternative energy sources (such as wind, solar, biomass, and even nuclear, which many argue would be an utter ecological and health disaster) to planting trees and developing environmentally friendly technology and energy-saving devices. Many of these proposals would be steps in the right direction, but others, particularly schemes such as carbon trading, carbon offsetting, geoengineering, growing biofuels on land that could produce food, or eliminating forested areas to clear land for biofuel crops, are dubious. Conversely, there would be a significant drop in climate change if the majority of people in the developed world and the affluent in the developing world were to opt to live simpler lifestyles by restricting the use of motor vehicles; relying more on mass transit, bikes, and their feet for transportation; and consuming less. Ultimately, as we state in our concluding chapter, an authentic program of mitigation to reverse the trend toward climate change will require transcendence of global capitalism.

Global climate change is only the exposed "tip of [a] broader global-sustainability iceberg" that includes a litany of environmental degradations that are now "converging rapidly in a manner not previously experienced" (Spratt and Sutton 2008:xi). All of these threats (e.g., acid rain, air pollution), which are connected to the transformation of the earth's biomass into an ever-growing human population, have vital implications for a host of human diseases and other threats to health (e.g., exposure to toxins). Even more significant than the number of health-related degradations that are occurring is the issue of *eco-crises interaction*. Rather than the various environmental calamities we face and

their respective configurations of health indicators being stand-alone threats to human well-being—which is the conventional but limited outlook that leads to fragmented and even competitive mitigation efforts—adverse human impacts on the environment intersect and the resulting interaction significantly exacerbates the overall human (as well as other animal and plant) health consequences; these interacting ecocrises create the potential for catastrophic outcomes (Singer 2009).

Although all countries and regions of the world are suffering the ill effects of climate change as well as other forms of pollution and environmental degradation, poor countries and poor people in all countries, namely those who are generally least responsible for pushing the earth system toward even more drastic changes, are the victims disproportionately dealing with the consequences. While the injustice of the existing global economy is recognized by mainstream global organizations such as the United Nations in the form of the Sustainable Development Goals and the World Bank in its call to eradicate extreme poverty, such bodies generally do not call for a drastic redistribution of the wealth and resources of humanity, a stance that is imperative for addressing the health ravages of climate change and other environmental problems for particularly the poor. As we argue in the final chapter, achieving a healthier world population will require an alternative world system based upon social justice, environmental sustainability, and a safe climate.

Other Environmental Impacts on Health

Chlorofluorocarbons also play an important role in ozone depletion, which has allowed a significant increase in the amount of ultraviolet light striking the earth's surface, especially in portions of the Southern Hemisphere nearest the pole, namely, southern Australia, New Zealand, Argentina, and Chile. While skin cancers are a well-known health consequence of ozone depletion—and rates of these diseases have been rising steadily (e.g., skin cancers account for around 80 percent of all new cancers diagnosed in Australia each year)—increased ultraviolet radiation at ground level also can lead to mutations, especially among viruses, whose only protection from the air is a thin layer of protein.

As a reflection of the underlying productivist ethic that led to their creation, national indices of growth and development, such as the gross domestic product, count only the production of goods and services that are sold in the marketplace (i.e., commodities) but do not take into account the impacts of their production, such as environmental degradation. While the emergence of global capitalism has ensured plentiful (if far from equitably distributed) food resources (including nutritionally dubious ones), modern sanitation, and biomedicine for some, these material benefits have often been obtained at the expense of the masses of people in developing nations and the working class in the developed world. Even the privileged few in the developed world face the danger of ecocide.

Two categories of health risks are posed by corporate globalization: (1) primary health risks stemming from changes in the social and natural environment and (2) more specific forms of health risks (McMichael and Beaglehole 2003). The former includes the following: (1) growing socioeconomic disparities both

within and between nation-states, a topic that we discussed in the previous chapter; (2) the fragmentation and weakening of labor unions as multinational corporations attain greater influence over governments around the world and thereby endanger occupational health in their insatiable drive for profits; (3) the depletion of the biodiversity of plants and animals, as reflected in vast acres devoted to a single plant, such as wheat or corn, and the destruction of the habitats that support many species; and (4) the spread of "invasive" species, most notably humans themselves, resulting from imbalances in biodiversity and the dispersal of persistent organic pollutants. The latter include the following: (1) the spread of tobacco-related diseases and dietary diseases resulting from food consumption patterns shaped by marketing campaigns intended to convince people to eat nutritionally dubious foods that are high in carbohydrates, starches, and fat but low in protein and vitamin content; (2) the proliferation of motor vehicles, including more recently in developing societies, such as China; (3) the rise of obesity in urban populations where most people follow a relatively sedentary lifestyle; (4) the expansion of international drug trafficking related in part to demoralization among socioeconomically disadvantaged populations, including youth, searching for an escape from the alienation of modern life; (5) the spread of infectious diseases due to increased global travel; and (6) the increasing prevalence of depression and other mental ailments in aging and socially isolated people, particularly in urban areas.

Although public health measures and secondarily biomedicine have done much, although by no means enough, to counter the threat of infectious diseases, there has been growing concern in recent years about what have been termed "emerging diseases." There exist four types of emerging infectious diseases: (1) newly described conditions that have been discovered or recognized only within the past few decades (e.g., HIV and SARS, or severe acute respiratory syndrome); (2) expansion of a familiar disease into a new region or habitat (what have been called "reemerging diseases," such as malaria or Zika); (3) marked increase in the local incidence of a disease, as seen with Ebola; and (4) increased severity or duration of an older disease or its increased resistance to previously effective treatment (e.g., tuberculosis). The spread of emerging diseases appears to be a consequence of several factors. These include major environmental changes such as deforestation and reforestation, intensification of agriculture, dam construction and irrigation, mining, and housing and road construction; the increasing concentration of people in overcrowded and densely packed cities (particularly in the developing world); the global movement of people due to migration, business, and tourism; and the development of pathogen resistance to overused and misused antibiotics.

Water and Globalization

In terms of health and illness, particularly with reference to infectious diseases, whether of the older variety or the new emerging variety, access to clean freshwater is crucial, perhaps more than most of us in the developed world think, because, as it is available at the turn of a faucet handle in so many places, we often take it

for granted. Although water covers more than 70 percent of the world's surface, particularly in the form of oceans, less than 1 percent of the water on the planet is available for human consumption, thus making it a precious natural resource that has increasingly become commodified within the context of corporate globalization. A growing number of anthropologists have addressed this issue, as reflected in the book *Globalization, Water, and Health.* In this volume, Linda and Scott Whiteford (2005:9–10) delineate four categories of diseases related to water: (1) waterborne diseases, such as cholera, typhoid, and hepatitis, emanating from the ingestion of water containing pathogenic organisms; (2) water-washed diseases resulting from a lack of water for washing or personal hygiene; (3) water-based diseases, such as schistosomiasis, resulting from exposure to nonhuman parasitic hosts in activities such as bathing, swimming, or washing clothes; and (4) water-related diseases, such as dengue, West Nile fever, malaria, and yellow fever in which humans come into contact with water that an insect vector uses as a breeding ground. Many of these diseases could be prevented if people had access to clean water for drinking, cooking, washing, and recreational purposes as well as access to sanitary living conditions and adequate nutrition. Whereas for eons access to water, particularly in indigenous societies, was viewed as a human right, with the spread of global capitalism, water is being redefined as a commodity. In today's world, increasing competition for clean water has been contributing to a tremendous increase in the number of deaths and diseases that could be easily avoided if world resources were not so inequitably distributed. Thus, we can speak of a *political ecology of water* that "can be and should be more explicitly linked to human health and well-being" (Whiteford et al. 2016:233). The political ecology of water concerns itself with the utilization, meanings, distribution, cost, and ultimately of water as a crucial resource for sustenance, but also one that is highly contested on the part of a multiplicity of actors, ranging from powerful multinational corporations and governments to ordinary people who find themselves on the margins of a capitalist world system.

Clean water supplies increasingly are being threatened by climate change, mining, agriculture (which uses about 69 percent of the world's freshwater supply), deforestation of mountainous woodlands, and urban sprawl. With respect to the last issue, in his contribution to the volume mentioned previously, Carl Kendall (2005) discusses the impact of Lima, Peru's rapid growth on the availability of fresh clean water in an arid coastal plain. The Rimac River, Lima's primary water source, has become heavily polluted with heavy metals from nearby mines, high levels of nitrates and other organic compounds, and untreated sewage. Furthermore, the Lima aquifer has become polluted with fecal matter and other pollutants. Lima residents experience high rates of diarrhea that are intimately related to a shortage of water and contamination of their water supply.

Water insecurity also constitutes a serious challenge in remote Alaska communities, where many households lack running water and sewer services (Eichelberger 2016). High water tables, permafrost, seasonal flooding, limited economic resources, and poverty all compromise reliable access to water. In some villages more than 10 percent of households lack access to piped water, a grim

reality that contributes to high rates of invasive pneumococcal infection. As the Whitefords so aptly observe,

> Protecting the physical environment and human health must be part of a deeper change in societal values. For anthropology and other social sciences to participate in this transformation through methodologically rigorous, theoretically informed, and socially relevant research is the biggest challenge. (S. Whiteford and L. Whiteford 2005:265)

Fortunately, ordinary people in various parts of the world, such as women in South Africa—a country where theoretically access to water is enshrined in the country's constitution as a human right—have been campaigning for "water citizenship" that will ensure access to a resource essential for sustenance and health (Bulled 2016).

The Political Ecology of Cancer

Cancer is an ancient disease that was little understood in the past, and its prevalence prior to the nineteenth century is not known. Whereas many infectious diseases have been on the decline, although AIDS and some other emerging diseases are notable exceptions, cancer rates rose dramatically during the twentieth century and have continued to do so into the present century. Cancer is largely a disease of development rather than underdevelopment, as indicated by the World Health Organization's estimate that about one-half of cancers occur in the most industrialized fifth of the world's population (Proctor 1995:2). Within specific nation-states, cancer rates vary in terms of the type of cancer, regionally, between the sexes, and among ethnic and religious groups.

Cancer etiology is a frequently debated topic, one about which scientists find themselves adopting diametrically opposing interpretations. Some argue that cancer is caused primarily by an array of environmental factors, others that it is related to lifestyle, and still others that it is related to stress, emotions, hereditary factors (referred to as "genetic susceptibility"), aging, and hormonal disturbances. Very likely, each of these factors plays some role in cancer etiology, and which ones are more significant varies greatly from person to person and by type of cancer. Nevertheless, each of these theories ultimately has political implications that have contributed to the "cancer wars," which include disagreements about the most effective way to treat it, ranging from the conventional approaches of radiation, chemotherapy, and surgery to a wide array of alternative therapies.

Several epidemiologists and public health scholars have sought to address the narrow reductionism they find inherent in prominent approaches to disease etiology generally and cancer etiology specifically by adopting a multicausal model that shows that a huge number of phenomena go together to produce illnesses. Within medical health anthropology, some proponents of the biocultural approach identify six factors contributing to disease: genetics, nutrition, the environment, psychogenesis, iatrogenic forces, and infection (Brown et al. 1996). However, the multicausal model tends to lead to political inaction because "prevention policy has to attack all possible causes at once—a strategy

that could stretch available resources far beyond their capacity and end up by devoting only pittance to each" (Tesh 1988:62). Furthermore, the multicausal model fails to come to grips with the issue of primacy—that is, whether certain factors, particularly political economic and political ecological ones, play a more important role in disease etiology (Turshen 1989).

There exists considerable evidence that a disproportionate incidence of cancer occurs among members of lower-income groups and workers exposed to industrial carcinogens, although there are some exceptions. For instance, higher breast cancer rates among professional women in affluent communities, such as San Francisco and the Upper West Side of Manhattan, may be related in part to the fact that many of these women postpone bearing children until later in their lives. Theories that emphasize political economic and environmental factors in cancer etiology have tended to receive short shrift in health policy and cancer prevention circles, such as in the National Cancer Institute and the American Cancer Society, largely because addressing them would require major social structural changes that ultimately would require transcending the existing global economy.

Samuel Epstein in *The Politics of Cancer* (1978) documents links between industrial pollutants and rising cancer rates. He estimates that 70 to 90 percent of cancers are induced by exposure to environmental and occupational exposure to carcinogens. More recently, Epstein (2005), on the basis of data from the Northeast, argued that breast cancer, as well as other cancers, has been associated with living near hazardous waste sites.

In light of these findings, Epstein (1979, 2005) argues that industry has adopted an elaborate strategy that seeks to exonerate itself from its contribution to cancer etiology, including blaming the victims by postulating the existence of "hypersusceptible workers," who allegedly are more prone to contract cancer, and moving hazardous chemical facilities to developing nations when labor unions and environmental groups have managed to convince politicians to pass environmental legislation.

Drawing upon the pioneering research of Epstein on environmental factors contributing to cancer etiology, various critical health anthropologists have come to view cancer as an "environmental and economic disorder rooted in material practices" around the globe (Burke and Mathews 2017:179). Unfortunately, while biomedicine has made tremendous strides in comprehending the biology of cancer cell development, it along with scientific, corporate, and governmental bodies continues to downplay the environmental causes of cancer. The prevailing approach in these circles is to place personal responsibility for cancer on the patients themselves, an approach consistent with neoliberalism (Karakasidou 2015).

Cancer in the Community

Anthropologist Martha Balshem (1993) was intimately immersed in the debate between the lifestyle and environmental theories of cancer etiology when she worked as a health educator at the Fox Chase Cancer Center in Philadelphia, a role in which she found herself feeling increasingly uncomfortable. Her highly

engaging and readable book *Cancer in the Community* explores "cancer control" in a European American working-class inner-city neighborhood in Philadelphia that Balshem calls Tannertown (pseudonym). The Fox Chase Cancer Center focuses on the prevention of suffering, namely, lowering cancer morbidity and mortality, by encouraging its clients to change their lifestyles, including ceasing smoking, following certain dietary regimens, and having regular cancer screening examinations. Expressive of this approach, with funding from the National Cancer Institute, the Fox Chase Cancer Center initiated Project CAN-DO, a five-year cancer education campaign between 1983 and 1988 in Tannertown. According to Balshem, Tannertowners tend to reject the assertion that their high cancer rates were a result of their lifestyles but instead viewed air pollution emanating in large part, but not entirely, from nearby chemical plants, occupational exposures in the plants, motor vehicle pollution, and the chemical adulteration of their food and water as the primary sources of cancer. Balshem (1993:75) found that respondents expressed frustration and fear in response to their perceived arrogance of the scientists and a discourse that while acknowledging environmental factors in disease etiology, asserts that people can prevent cancer through lifestyle changes.

In twenty-five one-on-one interviews and eight focus groups, Balshem found that while her respondents tended to view cancer as resulting from both lifestyle and environmental factors, they believed that making lifestyle changes would be a useless preventive measure in lowering cancer risk. The health educators at the center followed the official health prevention paradigm that stresses individual responsibility for good health. Balshem (1993) maintains that the skepticism that Tannertowners expressed about the views of scientific experts and health educators on cancer constitutes a manifestation of the loss of control that they feel over their daily lives. Interestingly, she goes on to argue "that the community critique of medical science mirrors that of critical studies in medical anthropology" (Balshem 1993:89). This is only partly true in that Tannertowners often assume that God or fate plays a role in whether they develop cancer. In coping with their sense of powerlessness, they—particularly the men—had come to view themselves as being tough and determined to beat the odds. Perhaps if scientists, who undoubtedly Tannertowners view as privileged members of society, were more willing to join forces with them in challenging corporate power, of which the former had some sense, they might not feel so powerless as a collectivity and have to resort to individualistic coping mechanisms rooted in U.S. culture.

Cancer and Industry

In contrast to Balshem's cultural interpretive account of Tannertowners' view on cancer etiology, David Michaels (1988) conducted a provocative political economic analysis of corporate policies toward bladder cancer in the U.S. dye industry—one that appears in a special issue of *Medical Anthropology Quarterly* on "Health and Industry" that was guest edited by Ida Susser, a well-known critical health anthropologist. Despite the fact that European scientists demonstrated in the early twentieth century that beta-naphthylamine and benzidine, two

chemicals commonly utilized in dye production, are deadly bladder carcinogens, there is reason to believe that the U.S. dye industry has systematically ignored and suppressed these and later findings concerning the link between these two chemicals and bladder cancer (Michaels 1988).

China's Cancer Villages

China's massive industrial expansion, with annual economic growth rates of 9–10 percent at its zenith, has been supported by its heavy export of a wide array of products, ranging from automobiles to electronic goods to clothing. Coal, of which China has massive supplies, provides for 68.7 percent of China's total energy consumption, but also results in carbon dioxide emissions and sulfur dioxide emissions, the latter which causes acid rain. Modernization has contributed to China having sixteen of the twenty most polluted cities in the world (Gerth 2010:185). Under Deng Xiaoping's policies, the costs of economic modernization and massive urbanization were tremendous, including environmental degradation, industrial and motor vehicle pollution, increasing greenhouse gas emissions, lack of adequate safety standards, growing socioeconomic inequality, and social conflicts. Increasing inequality in China contradicts revolutionary egalitarian ideals and has contributed to an increase of social struggles in both urban and rural areas.

In early 2013 the Chinese government finally acknowledged the existence of "cancer villages," pockets of high incidence associated with pollution. Unfortunately, serious epidemiological studies of these cancer villages remain spotty. Nevertheless, cancer villages have been linked primarily with the chemical industry and secondarily with paper factories, resource extraction and processing, and heavy metals emanating from these activities. Given the lack of conventional epidemiological studies, villagers have taken matters into their own hands by engaging in *popular epidemiology*, in which they seek to find an association between industrial pollution and deleterious health consequences. Conventional scientists, the Environmental Protection Bureau at the county level, and the Chinese Center for Disease Control and Protection at the national level generally dismiss these studies.

Anna Lora-Wainwright (2016), a health anthropologist, conducted ethnic research on Baoma (a pseudonym), a "cancer village" in 2004–2005 and for shorter periods thereafter. Baoma village is situated in a hilly area six kilometers from Langzhong city in the northeast of the Sichuan Basin. Under modernization, chemicals have been introduced into village farming practices, such as the use of pesticides and fertilizers, and very likely have contributed to rising rates of cancer, along with polluted water streams that carry heavy metals and toxic chemicals. Indeed, cancer communities tend to be situated along China's major rivers. Nevertheless, many villagers attribute cancer to repressed anger and anxiety induced by close relatives who may blame a woman who does not produce a son, for instance. In a local epidemiological study of fifty families residing in Baoma during 2003–2007, eleven of them reported cancer, six among males and five among females. Cancer treatment may be unavailable to many households

due to its high costs. According to the village secretary in 2004, payments for cancer treatment may come to as high as 20,000 yuan, an extremely high cost given that the typical annual income was estimated to be about 1,500 yuan.

Furthermore, the CDC has difficulty assessing disease rates at the local level, meaning that "villagers are trapped between their own village-level evidence and the scale at which the government operates" (Lora-Wainwright and Chen 2016:403). As a consequence of serious government response to pollution-related cancer cases, residents in some cancer villages have resorted to violent protests, petitions, and lawsuits. In the view of Lora-Wainwright and Chen (2016:411–12), while the "official recognition of the existence of cancer villages may serve as an opening for civil society forces concerned with the environment to raise their complaints or to legitimize their activities," it may also constitute an attempt by the government to "reclaim the terms of debate on cancer villages and to silence further protests by suggesting" that it is addressing the problem.

Anthropological Examinations of Cancer Treatment

Numerous health anthropologists, often in collaboration with research from other disciplines, particularly ones in the health sciences, have conducted ethnographic and survey research on cancer treatment. Instead of focusing on environmental factors in cancer etiology, cancer conferences in North America often pinpoint lifestyle factors, such as smoking and diet, as being the primary factors contributing to cancer (Bell and Ristovski-Slijepcevic 2015). While indeed lifestyle may constitute a contributing factor in cancer etiology, cancer conferences tend to neglect how alienation in the workplace or social life may lead individuals, particularly from working- and lower-class backgrounds, to adopt certain lifestyle habits as means for coping with their marginal status in the larger society. Other studies have pinpointed the differential access to cancer treatment in various developed societies, ranging from a welfare capitalist society such as Denmark where cancer treatment falls under the umbrella of a national health place with extensive services (Høybye and Tjørnhøj-Thomsen 2014) and other Scandinavian countries (Andersen et al. 2014), to the United States, a laggard in terms of cancer treatment for working- and low-income people, particularly ones from various ethnic minorities (Luque et al. 2010, 2011; Joseph and Dohan 2012). Some cancer patients, particularly women who have or have had breast cancer, have engaged in activism in the United States and served as a key force in the passage of the Breast Cancer and Environmental Research Act of 2008 (Anglin 2016).

Nuclear Reactors and Health

Radiation is highly dangerous to living tissue. The uranium split in a nuclear reactor or in spent fuel can damage a cell, including the DNA that carries the cell's genetic information. Damaged DNA in an egg or sperm can be passed onto successive generations. The nuclear industry mines, mills, and distributes uranium and leaves radioactive waste products resulting from nuclear fission in the process of creating electricity. Despite nuclear plant mishaps at Three Mile Island

in Pennsylvania in the 1970s and Chernobyl in the Ukraine, the nuclear energy industry managed to reposition itself as constituting a green energy source that does not emit greenhouse gas emissions and thus contribute to climate change, in contrast to fossil fuels. The nuclear industry defines itself as a key component in the economic growth of many developed societies and growing economies, particularly China, India, and Russia. Furthermore, some notable natural scientists, such as James Lovelock and Tim Flannery, and James Hansen, one of the most prominent climate scientists in the world and a high-profile climate activist, have come out in favor of using nuclear energy as a method for mitigating greenhouse gas emissions.

However, nuclear energy is not a renewable resource. Aside from the issues of the dangers posed by nuclear waste materials, the construction of nuclear plants creates significant greenhouse gas emissions and pollution (Caldicott 2006).

Fears of the risks of nuclear power reactors came to fruition with the Fukushima nuclear disaster in the wake of the earthquake and the tsunami on March 11, 2011. The mainstream and alternative media asked various anthropologists to comment on this event. Despite the fact that the nuclear disaster would have serious impacts on health, as Johnston (2011:3) observes, "what is the most disturbing for Japanese citizens, residents, and to a much lesser degree for those downwind, is that—despite the acknowledgments that the worst has happened, that emissions continue to persist, that there is no viable plan for the safe control and cleanup—the average citizen still struggles to access current and meaningful information that might inform proactive action that reduces exposure and minimizes risk." The Japanese government restricted the shipping of foods, farm animals, and fish from many prefectures (Eddy and Sase 2015). Highly radioactive waters discharged into the Pacific Ocean. Unlike the nuclear disasters at Three Mile Island and Chernobyl, the Fukushima disaster was caused by a natural event and tells us that nuclear disasters can have multiple origins, including possibly terrorism.

Unconventional Natural Gas Extraction and Health

Natural gas extraction, known by the term *fracking* or *hydraulic fracturing*, is an unconventional natural gas extraction process surrounded by much controversy for both environmental and health reasons (Kaden and Rosie 2015). In reality, fracking is one step in unconventional gas extraction (UGE), which includes "pad construction, drill setup, drilling, hydraulic fracturing (or fracking), gas processing, well decommissioning, and land restoration" (Paulson and Tinney 2015:1–2). Unconventional gas extraction results in air pollution during well production, numerous truck trips to create a pad, setup of the drilling equipment, transporting workers and supplies to the pad, drilling and fracking the well, removing waste from the pad, and transporting methane from the site either by trucks or in pipelines. Methane and volatile organic compounds (VOCs) releases can occur during exploration and upon storage of methane in various storage sites in the form of leakage. Sand used to open the fractures in the shale allowing the gas to escape can aerolize, thus presenting a serious health hazard

to miners and workers at the pad. Other forms of pollution resulting from the extraction process are benzene and ozone; water may become contaminated in various ways. In Pennsylvania and Texas there have been "clear, well-documented episodes of migration of natural gas into the drinking water supply" (Paulson and Tinney 2015:12). Ultimately, the UGE process releases methane, a very powerful greenhouse gas, into the atmosphere, and natural gas derived from the process contributes CO_2 to the atmosphere.

The Impact of Private Motor Vehicles on Health

Motor vehicles, with their internal combustion engines, embody the social structural contradictions of the world system. The world now has over a billion automobiles along with other private motor vehicles, particularly recreational or camping vehicles and motorcycles, and this number continues to rise rapidly as many developing countries such as China, India, Indonesia, and Brazil but certainly others, imitate developed societies in their adoption of the car culture. Motor vehicles impact patterns of production and consumption, settlement, traffic congestion, mass transportation, the environment, and environmental health. They constitute a major contributor to air and noise pollutions, greenhouse gas emissions, and ultimately climate change as well as fatalities, injuries, stress, social isolation, and obesity.

The private automobile constitutes the ultimate commodity for an increasing number of people around the world, indeed a *killer commodity* or *death machine*. The automobile industry and a massive road network are pivotal to the history of global capitalism (Flink 1988). Along with industrial pollution, motor vehicles have transformed many cities around the world into ecological disaster areas accompanied by a wide array of health problems. Car-related injuries, deaths, and disease are ultimately proportionally inflicted by the relatively affluent upon the poor, many of whom cannot afford to purchase an automobile. Due to weaker environmental protection laws in developing societies, their cities in particular experience high levels of pollution from a wide array of motor vehicles, including not only private automobiles and motorcycles but also buses and trucks (Bohren 2009).

Motor vehicle emissions consist of sulfur oxides, nitrogen oxides, carbon monoxide, hydrocarbon, volatile organic compounds, toxic metals, lead particles and particulate matter, and carbon dioxide, which contribute to acid rain, climate change, and cardiovascular, pulmonary, and respiratory diseases. Benzopyrene, a by-product of gasoline exhaust, is a carcinogenic chemical that becomes suspended in urban air. Aside from the amount of CO_2 emissions that motor vehicles produce, their manufacture and the construction of highways also result in CO_2 emissions (Robbins et al. 2010:136).

Automobile accidents reportedly result in more than 1.24 million deaths per annum worldwide and an estimated twenty million to fifty million injuries, of which 91 percent occur in low- and middle-income or developing societies where safety standards are not as developed as in high-income countries (WHO 2013). One study concludes that "motorized violence" takes the forms of obesity, diabetes, and cardiovascular disease, ailments that are in large part a consequence

of diminished active travel that has contributed to a global obesity epidemic (Woodcock and Alfred 2008). In essence, heavy reliance on cars as a mode of transportation results in less exercise through walking and cycling and thus to obesity and various related diseases.

Despite the existence of massive corporate support for the ongoing use of private motor vehicles, there have been counterhegemonic efforts, particularly in developed societies, to resist the automobilization of society by emphasizing the need for people to rely on other forms of mobility, including public transportation, cycling, and walking. Even walking to a train station or a bus stop has health benefits in the form of modest exercise. Some urban planners and public transportation advocates are calling for *pedestrian or carless cities.*

The Political Ecology of AIDS: Assessing a Contemporary Syndemic

Acquired immune deficiency syndrome (AIDS) has emerged as one of the most devastating diseases in human history. While there has been a decline in the level of AIDS-related deaths in recent years, the World Health Organization (2017c) estimated that 1.1 million people died of the disease in 2015, indicating that the epidemic is far from over. Caused by a contagious virus known as the human immunodeficiency virus (HIV), the disease has spread rapidly to all human populations in all corners of the planet. The actual origins of HIV are not precisely known, but it appears to have evolved from similar viruses, called simian immunodeficiency viruses (SIVs), that infect monkey and ape species. Genetically, HIV is most similar to the SIV found among chimpanzees, our closest relative in the animal kingdom. Current thinking is that SIV underwent a mutation that allowed it to infect human hosts sometime in the late nineteenth or early twentieth century. This kind of "species jump" is a not-uncommon occurrence with infectious pathogens in nature. The political ecological context for this transition was the expansion of human settlements into previously forested areas in Africa, associated with the population explosion, deforestation, expanded agriculture, and more rapid transportation produced by colonialism and neocolonial economies. Expansion into previously unsettled areas in tropical zones—where the concentration of microscopic life is highest on the planet—appears to have been the source of a number of the emerging infections of recent decades. Whatever its point of origin, HIV found the human body, especially components of the blood system, a habitable environment; human sexual contact and pregnancy initially and other routes of transmission eventually (e.g., injection drug use and blood transfusion) all proved to be effective mechanisms for moving between hosts and led to its rapid spread within and across populations.

The global count of people living with HIV/AIDS infection reached forty million by the end of 2004; millions more had already succumbed to the disease. Although HIV/AIDS is now found everywhere, it is unequally distributed among the populations and subpopulations of the world. One way of understanding the unequal spread of this disease is by examining the number of people living with HIV/AIDS disease along a geographic continuum. At one end of the continuum is sub-Saharan Africa, which remains the region hardest hit by the disease. Near

the opposite end of the continuum falls North America. While life expectancy is increasing in the United States and Europe despite the presence of HIV/AIDS, in many Third World countries the disease is significantly shortening how long people live on average. How can we understand the significant disparities found in HIV/AIDS prevalence and impact among countries of the world?

A starting point in answering this question is to examine disparities in HIV/AIDS in the United States because of the great differences in the social conditions that exist for different populations in the country. Wealth is highly concentrated in the United States, with the wealthiest 2 or 3 percent of the population controlling 40 percent of the nation's wealth and the poorest 20 percent of the population controlling hardly any wealth at all. These disparities in wealth are matched, as discussed in chapter 4, by great disparities in health. Moreover, these inequalities of wealth and health reflect important contrasts across ethnic populations.

HIV/AIDS is more prevalent—and significantly so—among African Americans than any other ethnic population in the country, a pattern that holds across age and gender subgroups. Importantly, while the number of deaths among whites living with HIV/AIDS has steadily fallen, among African Americans the picture is more complicated. In 2014, approximately 3,500 African Americans died of AIDS-related conditions, accounting for 53 percent of total deaths attributed to the disease that year, although African Americans make up about 12.3 percent of the total population of the United States. Notably, HIV diagnoses among African American gay and bisexual men increased by 22 percent between 2005 and 2014, although this rate has since declined (Centers for Disease Control and Prevention 2017). In other words, while compared to some other parts of the world the HIV/AIDS prevalence is low in the United States, HIV-related sickness and death are heavily concentrated in the African American sector of the U.S. population. Moreover, there is a distinct parallel between disparities in the distribution of HIV/AIDS between the wealthier and poorer nations of the world and between the wealthier and poorer populations within the United States.

This parallel suggests that HIV/AIDS does not exist in isolation from other diseases or from the social and political economic environment that shapes a population's health, access to food and shelter, and availability of medical treatment. To help frame an understanding of these interrelationships, health anthropologists introduced the concept of *syndemic* in the early 1990s (Singer and Clair 2003). While biomedical understanding and practice traditionally have been characterized by the tendency to isolate, study, and treat diseases as if they were distinct entities that existed separately from other diseases and independently of the social contexts in which they emerge, a syndemic model directs attention to social and biological interconnections in health as they are shaped and influenced by inequalities within society. At its simplest level, the term *syndemic* refers to two or more epidemics (i.e., notable increases in the rate of specific diseases in a population) interacting synergistically with each other inside human bodies and contributing, as a result of their interaction, to an excess burden of disease in a population beyond what would otherwise be expected. It refers not only to the temporal or locational co-occurrence of two or more diseases or health problems but also to the health consequences of the biological interactions among

copresent diseases, such as between HIV and tuberculosis (TB). HIV-positive individuals infected with TB, for example, are one hundred times more likely to develop active disease than those who are HIV negative with TB. Similarly, research has shown that individuals coinfected with hepatitis C (HCV) and HIV have higher HCV viral loads (i.e., levels of infectious agents in their bodies) than those infected with only HCV.

Beyond the notion of disease clustering in a social location or population and the biological processes of interaction among diseases, the term "syndemic" also points to the critical importance of social conditions in disease interactions and consequences. As Farmer (1999:51–52) has emphasized, "The most well demonstrated co-factors [for HIV] are social inequalities, which structure not only the contours of the AIDS pandemic but also the nature of outcomes once an individual is sick with complications of HIV infection." Living in poverty, for example, increases the likelihood of exposure to a range of diseases, including HIV. Given the impacts of (1) stress that is associated with poverty, unemployment, population density, and exposure to unsafe environments; (2) poor diet (especially early in life); and (3) heightened exposure to human-made toxins in local environments—which tend to be disproportionately common in lower-income areas where populations are more dense and opportunities to escape exposure to toxins are lower (e.g., high concentrations of lead in older homes and poor heating systems leading to sealed windows and poor ventilation), diseases tend to concentrate among impoverished populations. The concentration of multiple diseases in one population increases the biological likelihood of disease interaction of various kinds (e.g., gene swapping or "assortment" among co-occurring pathogens or opportunistic infection by one pathogen because of damage done by another) as well as the adoption of behaviors to cope with and medicate the emotional injuries of oppression and relative deprivation, such as abusive drinking and illicit drug use, which damage the human immune system. Additionally, poverty and discrimination place the poor at a disadvantage in terms of access to diagnosis and treatment for HIV as well as ability to adhere to treatment plans because of structurally imposed residential instability and the frequency of disruptive economic and social crises in poor families.

Racism, for example, is a critical social condition that appears to contribute to higher levels of HIV risk and infection. In Brazil, for example, while ethnic-based oppression is denied officially and at the popular level, studies show that "the structures of racism are present in everyday experience" (Goldstein 2003:105). Consequently, writing of internalized racism in Brazil, Neusa Santos Souza (1983) notes that dark-skinned Brazilians commonly feel inferior and ugly because of all of the subtle reminders to which they are subjected each day that whiteness equals beauty. Internalized racism, no less than open color-based discrimination, is linked with heightened levels of HIV risk and infection (Baer et al. 2013). Ultimately, social factors, such as poverty, racism, sexism, and marginalization may be of far greater importance in HIV morbidity and mortality between and within countries than the nature of the virus. Poor populations and populations of color encounter HIV/AIDS not as a single life-threatening disease but as part of a set of interacting diseases and toxic social and biological conditions with a resulting significant toll on their health and well-being.

In summary, this chapter discusses the complex relationship between health and environment, delineated in large part by two theoretical perspectives in health anthropology, namely medical ecology and critical health anthropology. In terms of the present era, it focuses on how the depletion of natural resources and environmental degradation, particularly anthropogenic climate change, affect the spread of infectious diseases. Finally, the chapter also examines the political ecology of cancer and HIV/AIDS, two pandemics that have spread throughout much of the modern world.

Discussion Questions

1. How do medical ecology and critical health anthropology compare and contrast in their respective approaches to examining human-environment interactions? Discuss in particular how critical health anthropology draws upon political ecology of health.
2. What does Alfred Crosby mean in his use of the term "ecological imperialism"? Discuss how indigenous people in the Americas and Australia were impacted by ecological imperialism.
3. Compare and contrast the overall health status of people in early foraging, horticultural, and agrarian state societies. What was the impact of the emergence of capitalism on workers, particularly in urban settings?
4. Discuss how transcending the nine "planetary boundaries" identified by Johan Rockström and his colleagues at the Stockholm Resilience Center endanger both the earth's ecosystem and ultimately human health. What do John Bellamy Foster and his colleagues mean when they argue that capitalism is in a "metabolic rift" with nature?
5. In what ways do you agree with Foster in his listing of "warning signs" that signal the long-term unsustainability of the capitalist world system? Discuss McMichael and Beaglehole's two categories of health risks posed by corporate globalization.
6. How has global capitalism contributed to climate change? In what ways might capitalism be reformed to mitigate its present deleterious impact upon climate change, or ultimately may it be necessary to conceptualize an alternative world system that leads to a safer climate for all human beings?
7. What impacts do epidemiologists and other health scientists envision resulting from climate change in terms of the spread of infectious diseases? Delineate and discuss the four types of "emerging infectious diseases" in the modern world. Discuss Merrill Singer's concept of *ecosyndemic* in elucidating the impact of anthropogenic changes in the environment that contributes to negative health outcomes.
8. In what ways is access to clean water crucial in maintaining global health?
9. Discuss various theories on the etiology of cancer. What has been the impact of capitalist industrial and agrarian practices on the spread of cancer, not only in developed societies but also in developing societies such as China?

Ethnomedicine 6

The Worlds of Treatment and Healing

Formerly, when religion was strong and science weak, men mistook magic for medicine; now, when science is strong and religion weak, men mistake medicine for magic.

—Thomas Szasz, *The Second Sin* (1973)

Introduction and Overview

In chapter 3, we examined conceptions and explanations of health, disease, and illness with which people around the world operate. In this chapter, we introduce the notion of ethnomedicine in order to demonstrate that all medical systems, be they national guises of biomedicine or indigenous medical systems or folk medical systems in complex societies, either preindustrial or postindustrial, are part and parcel of culture and society. We also discuss how various anthropologists have attempted to categorize medical systems. Finally, we introduce an evolutionary model of disease and healing systems drawing on the work of others that examines how various types of societies ranging from nomadic foraging to postmodern societies seek to explain the causes of illness as well as the categories healers use to identify and treat sickness.

Approaching Ethnomedicine

Historically, health anthropologists have devoted considerable attention to the study of medical systems in indigenous or tribal societies, in peasant communities, and among subgroups of urbanites in developing nations. Even before the emergence of medical anthropology or health anthropology as a named area of specialization within anthropology, ethnographers recorded data about health beliefs and practices, including healing techniques, in indigenous societies. Medical systems in indigenous societies and peasant communities and among other ordinary peoples have often been defined within anthropology as "ethnomedicines." Charles Hughes (1978:151), a renowned health anthropologist, defined ethnomedicine as "those beliefs and practices relating to disease which are the products of indigenous cultural development and are not explicitly derived from the conceptual system of modern medicine." However, many medical anthropologists since Hughes did his seminal work have noted that biomedicine also constitutes an ethnomedical system, one that has diffused from Western societies to many other societies around the world. An elaborate overview of the purview of ethnomedicine is summarized in table 6.1.

TABLE 6.1 The Purview of Ethnomedicine

Domain	Concerns
Ethnomedicine of the body	
Body image	Relationship between culturally specific images of the ideal body and their relationship to health beliefs and practices
Ethnophysiology	Cultural conceptions about body structure and function, including internal organs
Ethnomedicine of the mind	
Ethnopsychiatry	Culture and recognition of mental illness Culture and expression of mental illness Culture and occurrence of mental illness
Ethnopharmacology	Focuses on drugs or medicine utilized within specific cultures

Source: Adapted from Quinlan (2011).

Ethnopsychiatry or culture psychiatry as a sub-genre of ethnomedicine concerns itself with *culture-bound syndromes*, such as *latah* in Indonesia and Malaysia, which refers to highly exaggerated reactions to various stimuli, *susto* in Latin American cultures, which is a response to a frightful experience, and *pibloktoq* among the Inuit of the Arctic region, which entails tiredness or confusion accompanied by an outburst of culturally inappropriate behavior, such as tearing off one's clothes, frantic running, or rolling in the snow. Anthropologists view these conditions as cultural idioms of distress. They serve as cultural vehicles through which to express emotional upset in the form of physical symptoms that draw the attention and support of significant others and the broader community. Within different cultural systems, different body organs may be the focus of these syndromes. In Iran, for example, the heart is a common focus, while in China the liver, kidneys, or spleen tend to be emphasized. As a result of globalization and enhanced global communication on the internet and in the media, questions have been raised about the disappearance of culture-bound syndromes, although it is increasingly clear that various syndromes that once were regarded as culture-bound are more widely dispersed than was originally realized and global communication may be a factor in the spread of these conditions.

Ethnographers, including those working in the anthropology of religion, have long been fascinated by the shaman, a part-time religious healer who contacts the supernatural realm, often by going into a trance, in order to alleviate the distresses and diseases of his or her patients. Others have been struck by the role of the modern biomedical physician, a healer with unparalleled technological resources for measuring, visualizing, and treating disease who still must deal face-to-face with a nervous patient in need of as much care as cure, such as the patient interacting with a podiatrist in figure 6.1.



FIGURE 6.1 Podiatrist examining a diabetic patient's feet at health program organized by medical anthropologists and public health activists. Photo by Merrill Singer.

In reality, all medical systems constitute ethnomedicines in that they developed from and are embedded in particular sociocultural systems, regardless of whether they are small-scale or state societies. In this sense, all healers, from shamans to cardiovascular surgeons, are ethnohealers, even though the latter have far transcended the boundaries of any single ethnic group or population because their orientation to healing is rooted in Western culture (Erickson 2008).

Moreover, as Lynn Payer (1988) notes in her highly entertaining and readable book *Medicine and Culture*, biomedicine is shaped by its specific cultural or national setting, as she found to be the case in France, Britain, Germany, and the United States. American biomedicine relies much more on invasive forms of therapy, including cesarean sections, hysterectomies, breast cancer screenings, bypass heart operations, and high dosages of psychotropic drugs, than does biomedicine in the other three countries. British physicians are less concerned about moderately elevated blood pressure and cholesterol counts than are their U.S. counterparts, and German physicians are much less likely to prescribe antibiotics than U.S. physicians. In large part, the latter pattern is because German biomedicine emphasizes the self-recuperative aspects of the body and has long been open to natural healing techniques. German biomedicine, in fact, consists of two varieties: (1) *Schulmedizin*, or "school medicine," which is taught in major medical institutions, such as Humboldt University and the Charité, and (2) *Naturheilkunde* (nature cure), in which biomedical physicians may acquire

training through special workshops and apprenticeships. In contrast to Americans and the British, French biomedical practitioners generally take rectal rather than oral temperatures because of the greater accuracy of the former although this method can result in rectal bleeding. Finally, whereas American women tend to prefer breast enlargements, French women tend to prefer breast reductions.

While this chapter focuses on the ethnomedical practices of indigenous peoples around the world, it also discusses biomedicine as both a "global medical system" (i.e., an ethnomedicine that has transcended multiple cultural boundaries) and a structured array of somewhat interconnected ethnomedical subsystems. Ethnomedical research involves "the study of how well-being and suffering are experienced bodily as well as socially; the multivocality [i.e., having many voices or, more precisely in this context, having multiple means of expressing different kinds of information] of somatic [bodily] communication; and processes of healing (their relative success and limitations) as they are contextualized and directed toward the person, household, community and state, land and cosmos" (Nichter 1991:138).

Indigenous and Folk Medicine Systems

While the term "primitive medicine" was in use in the early days of anthropological interest in health and healing, the term is problematic for several reasons. In addition to implying a priori (before it is empirically tested) that indigenous or folk medicine is not effective, it also lumps all nonbiomedical systems into one box, but in fact there is considerable diversity in such healing systems.

Ways of Healing

Indigenous societies often incorporate members of the community who function as a *therapeutic management group* into the healing process. All of these dimensions are exemplified among the Navajo where the singer (as healers are known in the Navajo or Dine tradition) constructs an elaborate sand painting depicting the Holy People and chants over the patient in order to reestablish harmony, the loss of which is believed to cause illness. As the singer destroys sections of the sand painting, it is believed that various aspects of the patient's sickness are eradicated. Traditional medicine, like Navajo healing, often stresses emotional catharsis for the patient and draws the supportive attention of fellow members of the community to the patient's illness. At the same time, Navajo healing has a naturalistic orientation, as is evidenced by its broad pharmacopoeia consisting of herbs, powders, teas, and animal substances (e.g., powdered insects, cow dung, and so on). The Incas of the Andean region practiced trephination, which entailed the removal of part of the skull in treating fractures or removing brain tumors, purportedly in order to release a malevolent spirit.

An estimated 25 to 50 percent of the substances in the traditional nonbiomedical pharmacopoeia of indigenous healers have been shown empirically to be effective as measured by conventional scientific and biomedical standards. Other naturalistic techniques found in traditional medicine include herbs, powders, bone setting, massages, sweat baths, and mineral baths, as well as the suturing

of wounds with various items, such as sinews among Native American groups, thorns among the Masai, and biting mandibles from the heads of termites among both Somalis and Brazilian Indians. The Aleut and Inuit people are adept surgeons as a result of a pool of anatomical knowledge that they have developed from their hunting of sea mammals. The Masai, in turn, operate on abscesses of the liver and spleen. Mayan healers in Belize, in a manner similar to that of traditional Chinese medicine and Ayurvedic medicine in India, rely upon pulsing as a diagnostic technique, accompanied by various indigenous healing prayers (Hatala and Waldram 2017).

People sometimes view healers from other cultures or societies as more powerful and efficacious than their own healers. For example, the Iraqw of northern Tanzania often seek healers from other ethnic groups, such as the Sukuma, the Ihanzu, the coastal Swahili, and ones from Somalia, even if these healers do not speak the Iraqw language. The Iraqw often state that "we believe the medicine of other's peoples to be stronger than our own" (quoted in Rekdal 1999:460). Cross-cultural healing may serve as a means of fostering social connections that prevent or reduce interethnic conflict and avoiding local healers who may be viewed as capable of malevolent or exploitative acts. Ironically, in developed societies, including the United States and Australia, large numbers of people have come to embrace non-Western medical systems, such as traditional Chinese medicine, Tibetan medicine, and Ayurveda.

Typologies of Healing Systems

In their efforts to summarize the broad diversity and range of health beliefs and practices found in a wide diversity of sociocultural systems around the world, medical anthropologists have created various typologies of conceptions of disease etiology within medical systems. One scheme, for example, offers a matrix of the geographic distribution of various ethnomedical theories of disease causation depicting that three types of causation beliefs are worldwide: biomedical understandings, theories of illness due to loss of internal balance within body systems, and acceptance that illness can be caused by exposure to someone with an "evil eye" (Erickson 2008). We present three alternative ethnomedical typologies here, although these by no means exhaust the range of schemes that have been developed. One of the earliest of these suggested that every medical system embraces both a "disease theory system" and a "health-care system" and, further, that medical systems include both an ideological component and a social structural component (Foster and Anderson 1978). The disease theory system includes ideas about the nature of health and ideas about the causes of disease or illness. This typology makes a distinction between personalistic systems and naturalistic systems. The former view disease as resulting from the action of a "sensate who may be a supernatural being (a deity or god), a nonhuman being (such as a ghost, ancestor, or evil spirit), or a human being (a witch or sorcerer)" (Foster and Anderson 1978:53). Naturalistic systems, by contrast, view disease as emanating from the imbalance of certain inanimate elements in the body, such as the male and female principles of yin and yang in Chinese medicine. Personalistic and naturalistic explanations are not mutually exclusive. The health-care system

refers to the social relationships and interactions between the healers and their patients. The healer may be assisted by various assistants and in the case of complex societies may work in an elaborate bureaucratic structure, such as a clinic, health maintenance organization, or hospital.

Another approach, one based on types of healers within an ethnomedical system, categorizes shamans, mediums, sorcerers, and priests as examples of personalistic practitioners, while they group herbalists, chemists, bodyworkers, and lay midwives into the category of naturalistic practitioners (Loustanunau and Sobo 1997). Herbalists are found in the great majority of societies and prescribe or treat patients using medicinal plants but also minerals, as is the case in the Madagascar village studied by Harper as described in chapter 2. Chemists or pharmacists dispense both over-the-counter and prescription drugs and in developing countries often give injections. Indeed, in developing countries, chemists, without a prescription from a biomedical physician, often dispense both brand name and generic drugs, including what are generally legally defined as controlled substances in developed countries. While surgery is a highly prestigious specialty within biomedicine, various individuals in indigenous and archaic state societies practice or did practice surgery. One common surgical technique of traditional surgery is trephination. The Incas have already been mentioned as practitioners of this surgical art, but many other types of surgery have been described in indigenous societies around the world. Bodyworkers are musculoskeletal specialists who massage or manipulate the body to alleviate muscular tensions or skeletal misalignments. Finally, lay, or "direct-entry," midwifery around the world has been the focus of much medical anthropological research (Davis-Floyd et al. 2001). Midwives have generally delivered infants in indigenous societies and continue to do so, particularly in developing countries but increasingly so in developed societies as well. In reality, specific healers may be a blend of the various types delineated by Loustanunau and Sobo, such as when humoral doctors act as both chemists and herbalists or healers mix naturalistic and personalistic techniques.

Individuals follow various paths in becoming traditional healers. One is by "divine selection," in which the prospective healer has a dream involving the visitation of a spiritual guide who reveals that the dreamer has been chosen to serve the people in this manner. Individuals who themselves have been healed from a serious disease commonly are chosen to become traditional healers, thus the term "wounded healer," which has even been applied as well to many professional psychologists and psychiatrists in modern and postmodern societies. In addition, individuals who desire to become healers may undergo an apprenticeship with an established healer, often a member of one's kinship group.

Allan Young (1976a), a health anthropologist with a strong interest in medical knowledge, has developed two separate models that focus on the way medical systems organize knowledge about diseases. In the first of these, he distinguishes between (1) internalizing systems, which focus on physiological explanations and biophysical signs that the healer can detect, indicating the nature of disease episodes (such is the case in biomedicine), and (2) externalizing systems, which stress events outside the sick body. In the latter, the disease-causing agents are

generally animate and even human or at least humanlike, such as a witch or a sorcerer. It is the healer's job to determine what events prompted the disease-causing agent to attack the patient, as is the case in Navajo medicine where the singer attempts to restore a balance between the patient and social group, natural environment, or the wider cosmos. Young argues that externalizing medical systems tend to develop into internalizing medical systems with growing social complexity.

In his second model, Young (1976b, 1977) posits two types of systems: (1) accumulating medical systems, which consist of accumulated, formalized teachings, generally in written form, that are shared with prospective practitioners in training institutions or with colleagues at conferences or in professional associations (e.g., biomedicine, Ayurveda and Unani in South Asia, and Chinese medicine), and (2) diffusing medical systems, in which practitioners generally do not share medical knowledge (thus making it rather diffuse or unsystematic) with one another and regard it to be secret (e.g., shamans and magical healers).

Finally, using a biocultural or medical ecological perspective, Horacio Fabrega (1997) has developed an elaborate scheme of medical systems on which we draw in this chapter and elaborate on in our discussion of medical pluralism in the next chapter. Fabrega proposes the abbreviation "SH" to refer to a hypothesized biological adaptation for sickness and healing (i.e., like the biological adaptations that have been made among desert-dwelling animals to allow them to live in great heat with limited moisture, species have also adapted to the survival challenge of sickness with healing behaviors). For example, he maintains that chimpanzees exhibit some basic behaviors, such as the use of leaves to wipe themselves and the use of leaf napkins to dab at bleeding wounds, suggesting a prehuman SH adaptation, but also observes that chimps exhibit some non-SH responses as well, such as aversion to and exploitation of sick group members. More recently, it has been discovered that certain species of caterpillars when infected with various diseases will eat plants that have chemicals that are effective against the specific attacking disease agent. Fabrega suggests that many of the SH characteristics of chimpanzees existed in early hominid societies and that SH became more refined during prehominid stages of human evolution, as is implied by the presence of healed fractures in some Neanderthal remains. He maintains that patterns of food sharing and reciprocity between the sexes provided a social context for the SH adaptation because the latter also entails cooperative behavior. Asserting that "SH constitutes the foundational material for the elaboration of medicine as a social institution," Fabrega (1997:70) argues that the providers of SH (i.e., healers) in early foraging societies were highly insightful individuals who possessed an elaborate knowledge of the social organization of their society.

An Evolutionary Model of Disease Theories and Healing Systems

Drawing on the work of several health anthropologists, we propose an evolutionary model of disease theories and healing systems, shown in table 6.2.

TABLE 6.2 An Evolutionary View of Illness Theories and Healing Systems

Type of Society	Purported Causes of Illness	Types of Healers
Nomadic foraging	Self, ancestors, gods, outsiders	Shamans, diviners
Village level—simple horticulturalists	Same as foragers plus group members as possible agents, simple ethnopathologies	Shamans, magico-healers, mediums, herbalists
Nomadic pastoralists	Same as simple horticulturalists and in some cases imbalances between hot and cold foods	Healers, mediums, exorcists, and sometimes religious healers associated with world religions
Chiefdoms—sedentary foragers and intensive horticulturalists	Same as simple horticulturalists	Healers, mediums, herbalists, shamans among sedentary foragers
Early state societies	Same as above	Same as above plus priests
Early empires/civilizations	Individual behaviors, moral failings, elaborate ethnopathologies, imbalances in body humors	Priests, professional physicians, folk and religious healers
Modern industrial societies	Germs, genes, lifestyles, highly elaborate ethnopathologies	Orthodox and heterodox professional physicians, religious and folk healers, diversity of additional healers
Postmodern societies	Same as above plus emotions interacting with body, return to humor theories in some medical subsystems	Same as above plus movement to integrate orthodox and heterodox modalities

Health, Illness, and Medicine in Family-Level Foraging Societies

Family-level foraging societies, such as Australian Aborigines and the Inuit of the Arctic region, tend to have shamanic healers who often employ idiosyncratic techniques that they try to conceal from other practitioners. The juncture of medicine and religion is particularly exemplified in the role of the shaman, who has been depicted by anthropologists, scholars of comparative religion, transpersonal psychologists, and New Agers as the prototypical traditional healer. The word "shaman" is a term associated with the Tungus, a reindeer-herding group in Siberia who use it to label a part-time religious healer who is in direct contact with the supernatural realm. Fascination with shamans harkens back to Russian scientific expeditions into Siberia in the seventeenth and eighteenth

centuries. Other terms used to refer to this practitioner include "magician," "medicine person," and "witch doctor." The shaman has one or more spirit guides at his or her command, may be male or female, and is often "called" by spirit familiars through dreams or visions. Shamans often are considered eccentric and have been described by various ethnographers as moody, highly introverted, highly perceptive and intelligent, and intimately aware of their sociocultural system. Shamans among the Chuckee people of Siberia often are homosexuals, and those in some societies may be transvestites. There are several ways by which individuals become shamans, including inheriting the position from a parent, receiving a "call" from a spirit, recovering from a serious illness, or possessing outstanding abilities, such as being an excellent hunter, having psychic abilities, or being insightful.

In essence, the shaman is a complex figure with a multiplicity of roles. As a religious intermediary, the shaman visits the supernatural realm to petition favors, including the curing of diseases; as a diviner, the shaman locates wild game for hunting or lost objects or diagnoses illness; as a judge, the shaman determines when people have violated rules, such as having eaten taboo food or expressed anger inappropriately; and as an entertainer, the shaman carries out elaborate performances, including dance, sleight of hand, ventriloquism, shaking the ceremonial tent by pulling a series of strings and thus simulating the entrance of the spirits, and wearing colorful costumes. As a healer and psychotherapist, the shaman uses both naturalistic and supernaturalistic techniques. Ritual healing techniques include soul retrieval, removal of disease-causing objects, exorcism, confession, and sacrifice.

Michael Winkelman (2000) argues that shamanistic practices induce extraordinary experiences and healing by producing integrative processes in the limbic system or paleomammalian brain. He further asserts that shamanism evolved during the course of human history as a practice for symbolically and psychophysiologically enhancing well-being, both in the shaman and in patients. In entering into an altered state of consciousness, the shaman—unlike the Haitian *houngan* described in chapter 2—is not possessed by the spirit guides but rather controls them in order to achieve many tasks, including healing, dream interpretation, divination, clairvoyance, handling of fire, communications with spirits of the dead, recovery of lost souls, mediation between the gods and mortals, and protection against spirits and malevolent practitioners. The shaman's curing ceremony functions as an important social event that provides a means of integrating the patient into the immediate social group but also contributes to the social and psychic unity of the larger collectivity.

Winkelman (2000:276) maintains that shamanistic practices serve as a "set of sophisticated traditions for managing self, emotions, and consciousness." He views shamanism primarily as a phenomenon of foraging or hunting-and-gathering societies but maintains that aspects of classical shamanism persist in later or more complex societies. Altered states of consciousness induced by shamanic mimetic rituals, such as chanting, singing, drumming, dancing, and other repetitive behaviors, may activate the human relaxation response that provided early human societies with a survival advantage (McClennon 1997). Shamanism may constitute one of the earliest forms of religion, which, like later

variants of religion, constituted an adaptation to inner anxieties and a culturally constituted defense mechanism for society facing significant environmental challenges or dealing with confusing and crisis-torn times (LaBarre 1980).

Shamans are found in societies such as the !Kung or San of the Kalahari Desert, the Semang of the Malay Peninsula, and the Jivaro of Bolivia. Winkelman argues that shamanistic practitioners and somewhat similar magico-religious healers are found in technologically more complex sociocultural systems. He maintains that sociocultural processes have contributed to the disappearance of shamans and other shamanistic healers. For example, in state societies, these practitioners are in competition with the monopolistic claims of institutional religions over which priests preside. The shaman does not completely disappear in modern state societies but goes underground in the form of spiritualists, faith healers, and New Age psychics and neoshamans.

Shamans among the !Kung

Most adult males and some adult females among the !Kung of the Kalahari Desert of Namibia and Botswana function as shamans and perform community healing rituals about four times a month. Several men, who are sometimes joined by women, dance around a fire and a group of singers, thus inducing a trancelike state of "boiling" energy, or *num*. The healers may also ingest plant substances that contain *num*. While in trance !Kung healers treat sick spectators by imploring the ancestral spirits to restore them to health. Among the !Kung and other foraging societies, shamanic healing techniques tend to be public because they are utilized by numerous individuals in the group. Healing techniques tend to be focused on immediate restoration of well-being or accommodation to death through ritual activities and social practices.

In a classic work on health-related issues among the Yirrakala or Yolngu, who were traditionally foragers, of northeastern Arnhem in the Northern Territories of Australia, two types of healers were identified: the *marrnggitj*, or traditional healer, and the *raglak*, or sorcerer (Reid 1983). Some individuals become a *marrnggitj* in the wake of a frightening supernatural experience in which a spirit confers healing powers on them. Ultimately, of course, the prospective traditional healer must have the ability to counteract sorcery or the work of the *raglak* to consistently attract patients. Reid found only two men, a boy, and a woman who still practice aboriginal medicine in the Yirrakala vicinity.

Not all foraging societies have shamanic healers per se. A case in point is the Penan, a group in Brunei Darussalam on the island of Borneo in Indonesia who were traditionally hunters and gatherers but who are in the process of being transformed into village horticulturalists. Voeks and Sercombe (2000) argue that the rudimentary medical belief system associated with their foraging lifestyle is a consequence of the fact that they did not suffer from many debilitating diseases, as was the case with surrounding village peoples. Nevertheless, the Penan do recognize the existence of numerous unnamed spirits that can inflict harm on humans for violating various taboos and rely on "dream readers" to divine the sources of illness episodes, but they often disregard the advice of these medical practitioners. The Hadza, a foraging people in Tanzania, do not have any medical specialists other than women who perform clitoridectomies. Every adult has

knowledge about medicinal plants and can treat anyone with an illness. In addition to shamans, the Toba, a traditionally foraging society that engaged in some horticulture, have herbalists (some of whom today are called *curanderos* and act as healers in Christian evangelical sects) and other specialists who treat specific illnesses, such as those found among infants and children.

Historically shamans have tensions with state societies and established churches (Oppitz 2013:25). Nordic state societies beginning in the eighteenth century confiscated drums used by Sami shamans. Siberian shamans encountered repression on the part of the Russian empire under Czar Peter the Great, the Russian Orthodox Church, and the Stalinist purges during the Soviet era. Buddhist lamas suppressed shamans in Mongolia and Tibet.

Health, Illness, and Medicine in Village-Level Societies

Village-level societies, such as the Ainu of Hokkaido in northern Japan and the Yanomamo in the Amazon basin of northern Brazil and southern Venezuela, are characterized by the appearance of specialized healers, elaborate healing ceremonies attended by community members, and an expansion of the sick role manifested, for example, in growing attention to the psychosocial needs of sick individuals. The healer role in these societies confers social power on "individuals selected in part because of their psychological, social and political as well as strictly medical talents" and who are in a position to "profit from positive and favorable outcomes of healing" (Fabrega 1997:99).

As is the case in foraging societies, shamans are also often found in village-level societies. For example, among the Yanomamo, shamans call the spirits, who take the form of tiny humanoids, to implore them to release their victims from disease. Conversely, women are the primary repositories of herbal lore that is derived primarily from wild plants but also from some cultivated ones. Yanomamo shamans, called *shapori*, in their capacity as "masters of the spirits" perform both benign and malevolent acts, curing members of their community but attacking and reputedly killing enemies in other villages (Jokic 2014).

The Fore, a New Guinea highlands horticultural people who, like the Yanomamo, had a village-level society at the moment of contact with European expansion, depend on what they call Dream Men or Smoke Men, healers who rely on dreams and hallucinations induced by inhaling tobacco smoke in order to identify offending sorcerers.

The Warao of Venezuela have three types of shamans: priest shamans, light shamans, and dark shamans (Wilpert 2004). Priest shamans induce a trance by consuming tobacco and take spirit journeys that take them past monsters to an ancestral mountain where they will eventually reside after they die. While still alive, however, they obtain assistance from the ancestors in punishing people who threaten harmonious relationships with the ancestors. Light shamans come from the land to the east where the sun rises and are connected with the Tobacco Spirit. While they use a sucking cure when treating patients, they function as assault sorcerers by casting arrow-like projectiles at enemies. Dark shamans are associated with the setting sun and Hoebo, a spirit that takes the form of a red macaw that drinks human blood. They are likened to vampires and blood-sucking insects.

The Dani, a horticultural village people who also inhabit the New Guinea highlands, have two types of magico-religious healers. One type is the *hathale*, who tend to be women who carry out a variety of medico-religious functions, ranging from serving as midwives, carrying out abortions, administering medicinal herbs, and performing simple curing ceremonies or even allegedly casting spells. The other type includes political leaders who seek to appease the ancestral spirits by sacrificing pigs in order to ensure fertility both among women and in garden plots as well as general health and prosperity.

Robert Desjarlais, a medical anthropologist, participated in some twenty healing rituals as a shamanic apprentice to a sixty-five-year-old healer called Meme during his fieldwork among the Yolmo Sherpa, an ethnically Tibetan horticultural people residing in the Helambu region of north-central Nepal. Desjarlais (1992:241) notes that while Yolmo shamans often are not successful in their efforts to cure primarily physical ailments, such as arthritis or tuberculosis, "they are adept at healing illnesses related to personal distress or social conflicts (especially when compared with biomedical treatments)."

Recovered patients in various horticultural village societies became members of curer societies or sodalities. The Ndembu, who reside on a wooded plateau in the Mwinilunga District in the North-Western Province of Zambia, are a case in point. According to Victor Turner, a diviner who diagnoses the source of patients' afflictions resulting from an offended spirit directs his clients to a senior practitioner who propitiates the spirit and assigns a "band of adepts, consisting of persons who have themselves undergone as patients or novices" portions of curative ritual. He observes, "If the patient is deemed to be cured, he or she may then assist at later performances as an adept" (Turner 1968:16). Whereas the term "wounded healer" was initially applied to shamans (Eliade 1964), the concept may also be applied to other types of healers, including diviners or *ngoma* practitioners in Swaziland of southern Africa. In Swazi traditional healing the person possessed by an evil spirit can be healed while undergoing training as a diviner. Furthermore, "suffering certainly is a lived reality for diviners, but as a sign of a calling it is also an ideology" (Reis 2000:73).

Health, Illness, and Medicine in Pastoralist Societies

The Datoga, a pastoral people in Tanzania, make a distinction between an "illness of God" and an "illness of human beings," although many illnesses may be ascribed to both agents. Human-caused illnesses include witchcraft (generally believed to emanate from neighboring indigenous groups), the evil eye, and the administration of poisons. Male healers belong to one of two healing clans and treat patients by spitting on their bodies as a type of blessing ritual that may be combined with the application on patients' bodies of mixtures of herbs and animal fat. Women healers, who are far fewer in number, function as mediums through which patients can petition the spirits for advice or a cure. The Tuareg, a traditional pastoral people found in the Saharan and Sahelian areas of West and North Africa who have been heavily influenced by Islam, believe that disease may be a result of imbalances of hot and cold in the body, malicious gossip, or sorcery. The Tuareg have a wide array of healers to which they can turn, including

herbalists, bonesetters, Islamic religious healers, diviners, and exorcists. Like many pastoral peoples around the world, the Tuareg have been influenced by the medical practices of other peoples, particularly the Hausa, another Islamic group. The Mongols, the most renowned and feared pastoral nomads in human history, blended their naturalistic practices, including massage, bone setting, midwifery, and shamanistic practices that called on spirits to alleviate disease or misfortune, with medical beliefs and practices from first Tibetan Buddhism and later Russian biomedicine.

Health, Illness, and Medicine in Chiefdom Societies

According to Fabrega (1997), chiefdom (or prestate) societies, such as those of Hawaii and the Maori of Polynesia; the sedentary foraging societies of the Pacific Northwest Coast; and early state societies, such as Sumer, a city-state in Mesopotamia, exhibit the beginnings of the "institution," or "system," of medicine, which includes (1) an elaborate corpus of medical knowledge that embraces aspects of cosmology, religion, and morality and (2) the beginnings of medical pluralism (a topic on which we elaborate in chapter 7), manifested by the presence of a wide variety of healers, including general practitioners, priests, diviners, herbalists, bonesetters, and midwives who undergo systematic training or apprenticeships. Traditional Maori recognize two categories of disease: *matetangata* (those caused by human mishaps) and *mate atua* (those caused by supernatural forces). As in other parts of Polynesia, both types of diseases emanated from a violation of a *tapu*, or social norm that invoked the wrath of the gods. *Tohunga*, or traditional healers, either recited chants to induce religious healing or prescribed medicinal plants.

Some male healers, or *fojo*, in Samoa, another Polynesian chiefdom society, are also chiefs, and some women healers are the wives of chiefs. The former tend to restrict their practices to kinfolk whereas the latter do not. Some *fojo* are general practitioners who treat a wide range of ailments and others are specialists who treat either natural illnesses or supernatural illness, or employ certain medical procedures, such as massage or administration of herbs. Healers also vary in their range and depth of knowledge of the disease etiology and diagnostic and treatment techniques (Macpherson and Macpherson 1990: 102, 117–18, 140).

Folk Healers in Modern Societies

A wide variety of folk healers continue to function in modern societies, particularly in developing societies but also in developed societies. These include herbalists, bonesetters, lay or direct-entry midwives, and religious healers. Even shamans persist in modern societies, including in urban areas. Indeed, shamanism has undergone a resurgence in parts of Siberia since the collapse of the Soviet Union, in large part as a form of ethnic identity. Indeed, the Russian government now recognizes shamanism as an official religious tradition.

Konstantinos Zorbas (2017) conducted 18 months of ethnographic research on shamanism in Kyzyl (population approximately 100,000), the capital city of Tuva Republic situated near the Mongolian border. Shamans in Kyzyl operate

either under the umbrella of three shamanic associations or independently. All shamans in Kyzyl both engage in healing rituals and place curses on individuals who have placed a curse on a client. Kyzyl residents often express hostile impulses through supernatural retaliation facilitated by a shaman, thus making shamans a "cultural equivalent of the justice system, by virtue of their 'supernatural' power (sanctioned by this culture) for redressing injustices" (Zorbas 2017:146).

Female Shamans in Central Asia

Many folk healers in modern societies are women. Ironically, while Islam only allows men to be religious leaders, known as *mullahs*, and while Islamic clerics in Central Asia sought to do away with shamanism, they were not successful in doing so and women found a niche as religious leaders within the parameters of shamanism. Razia Sultanova (2011), a scholar of female shamanism and Sufism, reports that in Central Asia shamanism took on a Muslim texture. For example, shamans in their ritual first appeal to Allah, then the Muslim saints, and finally to the helping spirits or *jinn* mentioned in the Quran. Female shamans in Central Asia have also embraced Sufism, a mystical form of Islam popular in Iran, Iraq, Central Asia, and North Africa, which traditionally is famous for induced trance-like behavior through ecstatic dancing. Central Asian women often play musical instruments that historically were associated with shamanism but became incorporated into Sufism.

Arab Women Healers in Israel

Israeli anthropologist Ariela Popper-Giveon (2015) has written a fascinating account of Arab women healers, referred to as *Sheikha*, *Darwisha*, *Haijah*, or *Fataha*, and their patients in Israel, a society in which they are a part of a heavily discriminated ethnic minority. She conducted ethnographic research on Arab women healers in two settings—urban Arab society in central Israel and Bedouin society in the Negev Desert. Arabs in Israel are discriminated against in numerous ways, including in terms of access to education, employment, health care, politics, and living areas. Other than a few urban Arab Christians, most Israeli Arabs are Sunni Muslims, at least culturally if not religiously. While Arabs living on the West Bank and Gaza live in apartheid-like conditions, Arabs living in mixed-population cities in central Israel come into contact with the Jewish population, even residing in the same neighborhoods, working together in factories and offices, and visiting the same shopping malls, restaurants, and cafés. They also listen to Hebrew radio stations and read Hebrew newspapers and their children often attend schools where they study with Jewish students who are numerically and sociologically the majority group. In contrast, the Bedouin Arabs living in the Negev Desert generally experience only indirect contact with Jews, finding themselves living in separate housing, attending separate schools, and participating in separate leisure activities. Bedouin Arabs in the Negev tend to speak primarily Arabic in everyday life and maintain contacts with people in Arab countries, particularly Jordan and Lebanon and even Saudi Arabia.

Popper-Giveon worked with a sample of twenty women, five women healers and five patients in her urban group and five women healers and five patients in her Negev group. Most of her informants were fairly fluent in Hebrew, particularly those living in urban settings. Popper-Giveon (2015:26) describes her position as a Jewish Israeli woman studying Arab Israeli women as an outsider-insider, noting that in some ways she was alien to the women whom she interviewed and in other ways she was a kindred spirit in that she conducted the interviews in Hebrew and in her own country.

Traditional Arab women healers in Israel rely on a wide array of treatment techniques. The amulet (*hajab*), however, is a central device used in healing, particularly in the treatment of problems induced by the evil eye or magic spells: "despair, depression and anxiety; extended bachelorhood; sterility and livelihood problems" (Popper-Giveon 2015:84). Arab women healers also often read verses from the Quran in treating problems emanating from magic spells or the evil eye. Magical potions, consisting of bodily organic matters, such as menstrual blood, secretions, nails, hair, and urine may be used in counter-spells to ward off evil caused by an enemy. They may also be used as love charms for arousing a man's desire or maintaining a woman's fidelity and love.

Popper-Giveon contends that the traditional Arab women healers and their patients in her study experience dual discrimination, as members of the Arab minority and as members of a patriarchal Muslim community. She maintains that traditional Arab healing constitutes a coping mechanism for both the healers and patients in her sample in dealing with the numerous personal and social vagaries in their lives. Women healers achieve personal power, self-esteem, economic benefits, and respect from their families and communities. Women patients also feel personally empowered by the treatments that they receive, not only ritually but in terms of advice and social support.

Conversely, the traditional Arab women's healing complex does not pose a threat to the repressive structure of Israeli society, even if it constitutes the ultimate source of patients' problems. As Popper-Giveon (2015:255) observes, the "healers do not involve themselves in the overt public activity of various organizations and governments agencies, nor do they join forces with one another to consolidate and entrench their power." In reality, the healers often compete with each other for their patients' affection and financial payments. Furthermore, Arab women's healing serves to reinforce traditional patriarchal patterns.

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Case Study: Are the Therapeutic Aspects of Religion Something That Partially Address Refugee Health Problems?

In chapter 2, we touched upon migration and refugee health issues, a topic that organizations such as the UN, WHO, and the World Bank have acknowledged. The United Nations (2017) estimates that 65.6 million people were internally displaced worldwide in 2017 for a wide variety of reasons, such as civil

wars but even in order to move indigenous and peasants in order to protect endangered animal species and environmental degradation purportedly caused by these populations. While environmentalists often support such programs, they often overlook the social and health impacts upon the displaced populations. Anthropologist Jeffrey G. Snodgrass and two of his colleagues in other disciplines investigated in a highly quantitative manner the mental health benefits of religious rituals among indigenous Sahariya conservation refugees who had been displaced from a wildlife sanctuary in central India to make room for Asiatic lions from a neighboring area (Snodgrass et al. 2017). Bronislaw Malinowski (1954), a pioneer anthropologist, discussed how magical and religious rituals serve to alleviate anxiety and instill a sense of personal empowerment in uncertain situations, ranging from hunger, the danger of predation, whether the crops will grow, or the possibility of death in particular. Conversely, other anthropologists have argued that religion can instill anxiety and fear in people in that it often poses the possibility of punishment from ancestral or other spirits for violating social norms. Fundamentalist Christians often argue that unrepented sin can result in eternal damnation in hell.

The Sahariyas are one of some seventy-five proto-horticultural, largely foraging groups, referred to as *Adivasis*, meaning "first inhabitants" or "aboriginal peoples." They are poor, generally illiterate, and live in remote geographical areas, but historically have formed linkages with Hindu communities with whom they exist in patron-client relationships. Sahariya have adopted Hinduism and participate in Hindu festivals, such as Holi, the Hindu Festival of Colors, and Navrati, the Nine Nights of the Goddess, which occurs shortly after Holi.

The research team surveyed 159 Sahariya heads of households, 78 in a relocated village called Maziran and 81 in a buffer zone control village called Behruda. It proposed four hypotheses:

> H1. Health indicators would improve over the course of the festivals, with the greatest improvement occurring at peaks in the celebration;
> H2. Health improvements would be greatest among the more affluent Sahariya;
> H3. Health improvement would be affected by perceived insecurity;
> H4. Health improvement would be the greatest for Sahariya who were suffering from a more serious psychiatric ailment.

The team measured diurnal cortisol, stress, and PANAS (Positive and Negative Affect Schedule) levels for each subject over the course of the festivals. Based upon these measures, they found that ritual participation in both Holi and Navrati "promotes positive over negative psychosomatic experience" (H1; Snodgrass et al. 2017:271). The researchers also found physiological and psychiatric support for hypotheses 2, 3, and 4.

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Biomedicine as the Predominant Ethnomedicine in Modern Societies

Various scholars have debated the time of the emergence of biomedicine and its branching off from the traditional allopathic medicine that was practiced, for example, during the colonial era in the United States. Michel Foucault argued in *The Birth of the Clinic* (1975) that biomedicine emerged around 1800 in Europe as it systematically began to classify diseases into families and species and focused more on the body, thus the term the "clinical gaze," than on the ill person. According to Foucault (1975:169), "Clinical experience sees a new space opening up before it; the tangible space of the body . . . the medicine of organs, sites, causes, a clinic wholly ordered in accordance with pathological anatomy." He went on in subsequent writings to portray medicine as one of a series of forms of social control, along with the state and prisons. Chrystal Jaye and her colleagues apply Foucault's social constructionist analysis of biopower to biomedical education at the University of Otago on the South Island of New Zealand (Jaye et al. 2006). Based upon interviews with preclinical medical students, clinical instructors, and medical educators, they observed that the new crop of biomedical students is expected to be more humane, compassionate, and culturally sensitive than their mentors, in essence internalizing new forms of subjectivity:

> These new subjectivities are subject to scrutiny through surveillance techniques that assess behaviours and attributes such as reflectivity, self-awareness, patient-centredness, humanitarian values, ethics and maintenance of professional boundaries. Mechanisms of formative and summative feedback constitute regimes of normalizing judgment from teachers—but these normalizing judgments are also exercised through subtle positive and negative reinforcements from within the learning environment as successive generations of doctors teach the new generation what it needs to know. (Jaye et al. 2006:150–51)

In contrast to the social constructionist perspective adopted by Jaye and her colleagues in their analysis of biomedical education, Brian McKenna (2010) applies a critical medical anthropological perspective in his analysis of biomedical education at Michigan State University. He examined a six-year project called the Community/University Health Partnership, which aimed to produce more community-oriented primary care practitioners. Because the project challenged biomedicine's orientation toward curative care, professional rivalry, overspecialization, and hospital-based education, it met with stiff resistance and ultimately collapsed, despite support from a dean who ultimately resigned from his position.

Biomedicine is often depicted as an example of a profession par excellence because its physicians generally enjoy prestige, high incomes, relative autonomy over work, and a monopoly over the prescribing of dangerous drugs, declaring people mentally incompetent, signing birth and death certificates, and referring patients to other health practitioners. Freidson (1970) likens the position of biomedicine in modern societies to that of state religions in preindustrial civilizations. But state religions as instruments of social control were always subordinate to the wishes of ruling elites of their societies. Thus, the professional dominance

of biomedical physicians is *delegated* rather than *absolute*. The corporate class and its political allies in advanced developed and many developing societies delegate power to biomedicine in the form of financial support and licensure or statutory registration (Baer 2001:46–47). Biomedical physicians also often are highly represented on the registration boards of other biomedical health practitioners, such as nurses and allied health professionals. They also tend to be overrepresented on hospital boards, government health departments, health advisory boards, the National Institutes of Health, and the World Health Organization. Nurses and allied health workers often embark upon drives for professionalization that emulate the training and organizational structures of their biomedical superiors.

In recent decades, biomedical physicians have encountered challenges to their professional dominance from several quarters, including new models of health focused on prevention, the social determinants of health, and community; the rise of corporate medicine with its strong emphasis on managed care; the changing nature of state involvement and regulation, which includes cost-cutting measures and performance indicators; the lessening public status of biomedicine due to media exposés of medical fraud and negligence; the recognition of patient rights; and the increasing popularity of complementary and alternative medical systems. Beginning in the early 1970s, particularly in the United States, an increasing number of biomedical physicians became salaried employees rather than independent entrepreneurs. Scholars began to see biomedicine as a "big business" manifested in an array of large health-care corporations and proprietary hospitals that were part and parcel of a medical-industrial complex (Wohl 1984). While biomedical physicians tend to enjoy considerable prestige in most societies, in some developing societies their status is relatively low or declining. For example, in Mexico, biomedical physicians constitute the least prestigious profession out of the ten most popular professions and earn less than engineers, lawyers, and architects (Harrison 2000:301–2). In 1993 about a quarter of Mexican biomedical physicians were unemployed or underemployed, although many rural areas lacked biomedical physicians.

Hospitals

The hospital is the primary biomedical health-care setting in the provision of not only acute care but also outpatient services, such as lab work, X-rays, physical therapy, and programs addressing alcohol and drug addiction, weight reduction, and fitness. The hospital started out as a charitable institution that served as a last resort for the critically ill poor. In the nineteenth century, affluent patients received health care at home administered by either a house-calling physician or a full-time private-duty nurse. Hospitals come in different forms, such as voluntary community hospitals, or religiously supported hospitals, proprietary hospitals, and public hospitals. Hospitals may be general in scope or specialized, such as in the case of mental or heart hospitals.

The hospital has evolved into an elaborate social system with subsystems and a wide array of occupational subcultures, including those of administrators, physicians, nurses, medical technologists, and even patients. The hospital is a highly

authoritarian (some would even say quasi-militaristic) organization where orders are to be executed without question. It manifests a dual pattern of authority consisting of (1) administration, which includes a board of trustees, administrators, and paid staff, and (2) medical staff, some of whom are paid employees but others who are visitors, and an elaborate occupational support structure that includes nurses, nurse aides, medical technologists, therapists, social workers, orderlies, and (in teaching hospitals like the one studied by Katz as described in chapter 2) medical students as well as, sometimes, medical anthropologists. Registered nurses function in two worlds in that they are responsible for carrying out administrative policy and regulations and serve as the physicians' representative on the floor by carrying out orders for patients. At the same time, nurses have struggled to define their own arena of expertise, emphasizing the "care" aspects of healing as opposed to the "cure" activities undertaken by physicians. Increasingly, however, with the rise of the nurse-practitioner and related roles (e.g., nurse-anesthesiologists), nurses have penetrated the curer role as well.

Patients often experience *culture shock*, a term used by anthropologists for adjusting to the challenges of life in a culture that is foreign to them, on being admitted to the hospital because they undergo an ordeal of depersonalization, loss of self-identity, exposure to unfamiliar jargon, and a "stripping" process in which they must check their personal belongings and forgo many preferences. They find themselves in a state of total or near-total dependency in which they are fed meals not of their own choosing, may be bathed by strangers, receive medications at all times of the day and night, and must wear standardized, ill-fitting, and, for many people, far too revealing clothing. Poked, prodded, questioned, and awakened for testing at odd hours, they often respond by trying to personalize their immediate space by decorating their room or section of a room with gifts from friends and relatives or pictures from home, sometimes getting cranky or resistant, and, for many people, asking frequently when they can "go home." Hospitalization, in short, is often experienced as a kind of sacrifice that is endured in exchange for the hope of returning from the world of the sick to the land of the well.

An increasing number of health anthropologists conduct both teaching and research activities in hospitals. Some of this work is carried out by physician-anthropologists, but even more of it is carried out by nurse-anthropologists, many of whom belong to the Council of Nursing and Anthropology, a constituent unit of the Society for Medical Anthropology. Within this context, some health anthropologists are involved in what is termed clinical anthropology or clinically applied medical anthropology. Much of this work occurs in hospitals or other clinical settings and often involves the anthropologist serving as a cultural translator who facilitates the physician-patient relationship. In contrast to sociologists Gail Henderson and M. Cohen, who produced a book-length ethnography of the Second Attachment Hospital complex on the outskirts of Wuhan, the fifth-largest city in China, no anthropologists to date have conducted an ethnography of an entire hospital.

Nevertheless, in keeping with a growing interest in hospital ethnography, Anna Harris (2009), a biomedical physician, conducted an in-depth study of how

Nurse-Anthropologists and Physician-Anthropologists

Many nurses have found anthropology and specifically health anthropology to be of great utility in their work. Consequently, numerous undergraduate nursing programs require their students to take a course in cultural anthropology, and many nursing schools now teach courses in health anthropology and related subjects. Indeed, many of the people teaching these courses are nurses who have obtained graduate degrees in health anthropology. Some anthropologists have argued that, in contrast to biomedical physicians, who tend to be "disease oriented," nurses are inclined to be more "person oriented." Conversely, while many biomedical students regard medical anthropology and related fields to be intrusions in their demanding studies, medical anthropology includes some prominent physician-anthropologists, including Arthur Kleinman and Paul Farmer, both of whom are based at Harvard University and have had significant productive influence on health anthropology.

overseas biomedical physicians, particularly ones from developing countries, in two outer suburban hospitals in a large Australian city have adjusted to the larger national health-care system as well as the larger society. In addition to learning about a different hospital system and its biomedical routines and procedures, they spent much of their time "improving their English language fluency, sorting out their visas, making sure their families were settled and they had a decent salary to support them." In addition to occupying the less prestigious positions in urban hospitals, overseas biomedical physicians in Australia increasingly find themselves practicing in "country towns" or rural areas, places that domestic Australian general practitioners and specialists have in many instances abandoned. Beyond the hospital is the clinic or health station, often set up in more isolated, lower population density, and rural areas to serve the health needs of dispersed communities. The health station, like the one studied by Leslie Butt (figure 6.2), is especially important in areas with limited health budgets and comparatively fewer medical professionals.

As health anthropology has evolved, it has taken on new topical interests, including ones intimately related to biomedicine. Two of these new topic interests are pharmaceuticals and biotechnology, both of which touch on the anthropology of the body.

Health Anthropology and the Pharmaceutical Industry

Health anthropologists have become increasingly aware of the importance of studying the enormous pharmaceutical industry; pharmaceutical drugs are dispensed around the globe with tremendous impact on health and health behavior and are used in ways that are culturally meaningful, whether or not they are in line with or reflect medically approved patterns of use. Etkin (1988), for example, has examined the cultural construction of the efficacy of pharmaceutical drugs



FIGURE 6.2 Government health post in the highlands of western Papua New Guinea. Photo by Leslie Butt.

in light of varying cultural understandings of what efficacy means. While pharmaceutical drugs are intended to have specific physical, emotional, or behavioral effects, their effectiveness may be judged by quite different standards in local cultural settings.

Moreover, to understand the impact of the pharmaceutical industry we must recognize its global character: "a pharmaceutical company might have its corporate headquarters in the United States, [test its drugs in Uganda], produce its drugs in Ireland, assemble its capsules in Brazil, and sell the products in Bolivia" (Bodenheimer 1985:190). Furthermore, pharmaceutical/biotechnology companies cannot be pinpointed to one location; they function as any multinational corporation does, worldwide (Mehrabadi 2005). Operations are carried out depending on the price of labor, where raw materials are the least expensive, where taxes can be easily evaded, and where market regulations are the least strict.

As a result of the pharmaceutical industry's global reach, its products appear in societies that lack a developed health-care infrastructure, sometimes with dramatic effect. In the case of El Salvador, for example, the arrival of pharmaceutical drugs pushed out locally produced indigenous medicines and the healers who

produced and dispensed them (Ferguson 1981). Local community structures then began to break down. In a similar case in South India, the traditional use of herbal remedies has been overwhelmed by the widespread sale of pharmaceutical drugs, which were associated by local populations with modernity and progress (Nichter 1989). By using Western medicines, in effect, people are attempting to reap the benefits of development, including the standard of living and concentration of wealth found in Western countries. In this sense, pharmaceuticals sometimes acquire culturally constructed fetishized qualities in the same sense that possessing a lucky charm will bring personal fortune. The popularity of pharmaceuticals internationally has led to the formation of an informal sector for the distribution of commercial drugs, such as folk injectors who give single shots of antibiotics in community markets in various countries. Given the need for receiving a course of antibiotic treatment, not only is this emergent folk practice ineffectual from a medical standpoint, but it can also contribute to the development of strains of bacteria that are drug resistant. In Cameroon, for example, there is evidence not only that self-medication with pharmaceutical drugs can be detrimental to health but also that the available resources of low-income households are wasted on useless medication (van der Geest and Whyte 1988).

Moving from the micro level to the macro level, we can see that because pharmaceutical corporations are profit driven, their actions are not necessarily shaped by a health agenda, sometimes with significant health consequences. For example, in 2005 the pharmaceutical company Eli Lilly agreed to plead guilty to a single misdemeanor count and to pay a fine of $36 million for its violation of the Food, Drug, and Cosmetic Act in its promotional campaign for the anti-osteoporosis drug Evista (Corporate Crime Reporter 2005). According to this law, pharmaceutical companies must specify the medical purpose(s) of new drugs in their applications to receive marketing approval from the Food and Drug Administration (FDA). Once the FDA approves a drug, it can be legally promoted and sold only for the purposes for which it has been approved. Promotion for what is called "off label" uses is outlawed. In the case of Evista, Lilly executives were disappointed with sales levels during the first year that the drug was on the market; it brought in only about a fourth of the projected annual sales of $400 million. To make up for this loss in expected income, Lilly decided to widen the market for its drug by promoting it for several unapproved uses. Although application by Lilly to the FDA to include these new uses was rejected, the company began pushing use of the drug in the prevention of breast cancer and cardiovascular disease. The consequences of such action are not trivial in that doctors, assuming that the drug is effective in cancer and heart disease prevention, may prescribe it instead of drugs that have proven in clinical trials to be effective in treating these diseases. In its expanded promotion efforts, Lilly used a variety of strategies, including training sales representatives how to get physicians to ask about other uses of Evista, sending unsolicited medical letters to physicians promoting unapproved uses of the drug, distributing an "Evista Best Practices" videotape that states that "Evista truly is the best drug for the prevention of all these diseases" (referring to osteoporosis, breast cancer, and cardiovascular disease), and distributing a medical reprint highlighting findings about the various

benefits of Evista that was written by company employees and paid consultants while hiding a line in the reprint that acknowledged that the effectiveness of the drug in reducing cancer risk or cardiovascular disease had not been established, among other legerdemain strategies to promote and sell Evista (M. Singer 2007).

In another case involving the painkiller Vioxx in which it appears that improper procedures were followed in clinical trials, it is estimated that at least 25,000 heart attacks and strokes occurred after taking the drug (Bazerman and Chugh 2006). In the estimation of one pharmaceutical drug expert, David Graham, associate director for science in the Office of Drug Safety at the FDA, Vioxx may have contributed to the deaths of 88,000 to 138,000 people, putting it at the top of the list of known deadly pharmaceuticals to be released to the public (M. Singer and Baer 2007).

In summary, this chapter starts out with a discussion of several domains of ethnomedicine, such as ethnophysiology, ethnopsychiatry, and ethnopharmacology. It also considers various indigenous and folk medicine systems; typologies of healing systems; an evolutionary model of disease theories and healing systems; biomedicine as the predominant ethnomedicine in modern societies with the hospital as the central biomedical institution; and the relationship between health anthropology and the pharmaceutical industry.

Discussion Questions

1. Discuss the concerns of the various domains and sub-domains in Quinlan's overview of the purview of ethnomedicine (table 6.1). What might be some types of culture-bound syndromes other than the ones mentioned in this chapter?
2. Discuss the relationship between medicine and magico-religious rituals in traditional healing systems. Conversely, what types of naturalistic techniques are found in traditional healing systems around the world?
3. Based upon the evolutionary model of disease theories and healing systems elaborated upon in this chapter, discuss the general features of health, illness, and medicine in family-level foraging societies, village-level societies, pastoral societies, and chiefdom societies.
4. Give examples for the persistence of shamans and other folk healers in modern societies.
5. In what ways has biomedicine evolved into the predominant form of ethnomedicine in modern societies? Discuss the centrality of the hospital and the pharmaceutical industry to biomedicine.

7 Plural Medical Systems

Complexity, Complementarity, and Conflict

Throughout this century and even before, there has been a general assumption—even a conviction . . . that folk and popular systems of health beliefs and practices would inevitably decline in modern and industrialized societies. . . . Yet this has not been the case.

—Bonnie Blair O'Connor (1995:1)

Introduction and Overview

In the previous chapter, we examined the concept of ethnomedicine and conceptions of health, illness, and medicine in family-level foraging, village-level pastoral, and chiefdom societies. We move on in this chapter to consider the phenomenon of medical pluralism in complex societies by examining it in two locales in developing societies, namely, a village in the Bolivian Andes and a city in Java, Indonesia, and then in a technologically developed society, namely, Australia. We then examine various typologies of plural medical systems but particularly stress the concept of a dominative medical system, which points to the fact that medical systems do not exist in a vacuum but rather reflect the class, racial, ethnic, and gender relations and inequalities of the wider society. In a dominative medical system, several different healing traditions coexist in the same society, but one tends to be more closely aligned with the dominant social groups in that society and to be the dominant healing tradition as well.

Indigenous or tribal societies exhibit a diversity of healing beliefs and practices as well as an array of different kinds of healers, but these tend to be rather loosely organized and often reflect idiosyncratic patterns resulting in part from the fact that healers commonly are secretive about their medical knowledge. In contrast, state societies manifest the coexistence of a highly elaborate array of medical traditions at both the conceptual and the practice level, a pattern that health anthropologists call medical pluralism. In other words, the medical system of state-level societies tends to be made up of various medical subsystems that coexist in a social environment that ranges between extensive cooperation and open conflict.

We present three case studies of medical pluralism in three distinct sociocultural settings. The first is a small village in the Bolivian high plain (altiplano) based on ethnographic research done by Libbet Crandon-Malamud, the second is an administrative/university city in Java based on ethnographic research done by Steve Ferzacca, and the last is Australia based on social historical and ethnographic research conducted by Hans Baer (2009).

■ ■ ■

A Case Study of Medical Pluralism in a Rural Area in a Developing Society: The Altiplano of Bolivia

A classic case in research in medical pluralism is based upon Libbet Crandon-Malamud's (1991) ethnographic fieldwork conducted between 1976 and 1978 on medical pluralism in Kachitu (pseudonym), a rural town on the Bolivian altiplano. Kachitu is the center for a canton consisting of some sixteen thousand Aymara Indians living in thirty-six *communidades*. The town proper has some one thousand residents consisting of three ethnoreligious groups: Aymara peasants, Methodist Aymara, and Catholic mestizos (those of mixed European and indigenous blood). Before the Revolution of 1952, the mestizos, who constitute about half the population of Kachitu, supervised the labor of the Aymara on behalf of the national elites, who generally resided in La Paz and other cities. Since the revolution, the mestizos have been seeking to avoid poverty and are a socially fragmented ethnic category that includes teachers, small shopkeepers, and poor occasional agricultural laborers. Before 1952, the Aymara, who make up another third of the village, lived in Indian communities that were heavily taxed or on haciendas (plantations).

Some Aymara were miners and some performed personal services, such as domestic work, for mestizos. Many Aymara peasants migrated to Kachitu from the haciendas upon acquiring land through fictive kinship ties with mestizos. Following the revolution and land reform, others moved to town and claimed land on the outskirts that had been confiscated from the mestizos. The Methodist Aymara are converts or descendants of converts to the Methodist church, which established a mission in Kachitu in the early 1930s. The Methodist Aymara attended mission schools and became entrepreneurs who took over local administrative and political offices when the revolutionary government threw the mestizos out of these positions. They function as the economic and political backbone of Kachitu. Crandon-Malamud (1991) argues that the people of Kachitu tend to conflate ethnicity and social class, noting that Aymaraness, mestizoness, and even Methodism constitute "masks for social class."

Themes

Crandon-Malamud identifies four themes that permeate the dialogue about medical etiology and diagnosis in Kachitu—pervasive hunger, subordination, victimization, and exploitation—within the context of the Bolivian political economy. Pervasive hunger is a by-product of race and differentiates the purportedly white elites from mestizos and Indians as a means to justify unequal access to the political process and economic resources. The second theme entails an economic system that exploits Indian labor, the third consists of caudillo (strong authoritarian leadership) political structures that suppress dissent on the part of

rural mestizos and Indians, and the fourth is conflict that exists between mestizos and Indians.

Medical Subsystems

Five different medical subsystems are used by those in Kachitu, depending on the diagnosis: Aymara home care, shamanistic or *yatiri* care, mestizo folk home care, biomedical clinical care, and hospital care in La Paz. Crandon-Malamud examines the intricate ways that the residents of Kachitu utilize the local plural medical system for purposes of establishing their sense of cultural identity and obtaining the few resources available to them. Contrary to the wishes of biomedical practitioners and indigenous healers, decisions concerning illness etiology and diagnosis tend to be made primarily by patients themselves, their families, and other interested parties. Biomedical physicians who practice in Kachitu must abandon many of their preconceptions and adjust themselves to the local belief systems if they expect to establish rapport with their patients. The various medical ideologies in Kachitu function as options that address different types of ailments. As Crandon-Malamud (1991:202–3) observes, "All things being equal, if one has tuberculosis, one goes to the physician in the Methodist clinic; if one suffers from khan achachi, one goes to yatiri; if one has a stomach upset, one resorts to medicinas casera [home care]." (*Khan achachi* refers to a sickness stemming from a phantom, *khan achachi*, who consumes meat, alcohol, and other gifts that it demands from people.)

Kachitunos, regardless of their social standing, tend to be pragmatic when it comes to seeking medical treatment. Medical dialogue serves as an idiom by which a person identifies his or her ethnic identity within the larger context of Bolivian society—one that is characterized by frequent economic crises, unstable governments, and military coups. Mestizos in Kachitu who find themselves downwardly mobile may, in essence, gain access to greater health care by turning to Indian indigenous medicine. Crandon-Malamud argues that they use medical dialogue, namely, debate about the most appropriate explanations of disease etiology and therapies to eradicate disease, as a mechanism of empowerment in the face of overwhelming external oppressive political-economic forces at both the national and the international level. Unfortunately, this medical dialogue has served as a rather limited form of empowerment and in reality serves more as a coping mechanism than an oppositional fulcrum within the larger Bolivian political economy and the global economy.

Status of Health Care in Bolivia in the Pre-Revolutionary Era

Health services in Bolivia underwent a serious decline during the 1980s. The infant mortality rate at the time was the highest in Latin America, with a national average of over 200 per 1,000 live births in 1985, as high as 650 per 1,000 live births in some rural areas, as opposed to 150 per 1,000 live births in 1975

(Morales 1992:135). Furthermore, half of the children between ages one and six were malnourished; 60 percent of school-aged children manifested goiter and 45 percent anemia due to iodine and iron deficiencies, respectively (Morales 1992:135).

Cuts in government spending contributed to a lack of potable water in 65 percent of urban households. Between 1960 and 1990 life expectancy in Bolivia increased by only five years, from the mid-forties to about fifty, some twenty years less than more developed Latin American countries (Klein 1992:280).

To no small degree, Bolivia's health problems and other socioeconomic problems emanate from a national social structure in which "the wealthiest 5 percent control 39 percent of the national income and the poorest 20 percent, only 2 percent" (Morales 1992:203), a situation which in part was corrected by the coming to power of a left-wing populist political party and indigenous president in 2006.

Social and Health Conditions in Bolivia after the Election of Indigenous President Evo Morales

President Evo Morales and his governing Movement Towards Socialism (MAS) party have adopted a policy of "communitarian socialism" and proposed a "Law of Mother Earth," which stipulates living in harmony with nature. Morales called for Socialism of the South or Indigenous Socialism that includes ten "commandments." These call for the eradication of the capitalism in order to "save the planet"; ending wars that have benefited empires, multinational corporations, and selected families; developing "relations of coexistence" among the countries of the world; treating access to water as a human right; developing "clean energies"; respecting Mother Earth; treating "basic services, such as water, electricity, education, health care, communications, and collective transportations" as human rights; consuming only what is necessary and seeking to consume local products; promoting "cultural and economic diversity"; and developing a "communitarian socialism that is in harmony with Mother Earth" (Derber 2011:135). The Bolivian legislature established eleven new rights for nature, including the right to life and for existence, the right to clean water and air, the right to be free from pollution, and the right to "not be affected by mega-infrastructure and development projects that affect the balance of ecosystems" (Bell 2016:81).

The MAS emerged as the political arm of the indigenous-peasant movement based in the department of Cochabamba in the mid-1990s and models itself upon an "assembly-style, rank-and-file democracy of peasant unions in region" that consisted in large part of coca growers who opposed the U.S.-supported "drug war" in the region (Webber 2011:3). Over the course of its ascent, the party has become more moderate as an urban, mestizo middle-class intelligentsia has assumed influence. The MAS has also sought to appease the capitalist class in the eastern section of Bolivia in terms of the assembly's governance structure, in part to stave off a secession movement. Garcia Linera, whom

Morales chose as his vice president in 2005, developed the notion of "Andean-Amazonian state capitalism."

In 2006 Morales signed a law that nationalized Bolivia's natural gas reserves and required all foreign energy companies to renegotiate their contracts so that they would benefit the Bolivian people (Derber 2011:138). His administration embarked upon a program of "nationalization without expropriation," which nationalized oil and natural gas reserves within Bolivia but did not want multinational corporations in the hydrocarbon sector to cease operations, but enter into joint ventures with the government (Kaup 2015). Under this arrangement, foreign direct investment in Bolivia increased from $281 million in 2006 to $1.75 billion in 2013. In addition to providing financial support for social programs, the government's profits from the hydrocarbon sector have substantially increased international monetary reserves in Bolivia.

Bolivia has embarked upon a program of food sovereignty by promoting local and sustainable food production, which seeks to allow peasants to obtain ownership of seeds, thereby breaking them for dependence upon agribusinesses that dominate seed production. Morales withdrew Bolivia from the World Bank and the International Monetary Fund and undertook nationalizing the oil, natural gas, and lithium resources, a move that required altering the constitution (Vattimo and Zabala 2011).

In January 2009, the Bolivian people in a national vote of 61.4 percent approved a new constitution declaring that Bolivia constitutes a "pluri-national state, communitarian state" based on respect and equality for all and a policy of providing communal access to water, work, education, health, and housing for all.

Despite the best of intentions, the movement toward implementing Morales's vision of socialism has been slow and patchy. Bolivia under Morales has preserved private property rights, not undergone serious land reform measures, and experienced fiscal austerity. In 2011 Morales slashed fuel subsidies in order to reassure foreign investors. Also the Morales government provides hundreds of millions of dollars in loans, export subsidies, and tax incentives to wealthy agricultural exporters and has expelled landless indigenous squatters from large agricultural estates (Petras 2011:15). In essence, despite MAS's stated commitment to socialism over the long run, Bolivia between at least 2005 and 2010 followed a "reconstituted neoliberalism" based largely on exports of natural gas, oil, and mineral resources (Webber 2011:232). Furthermore, while Morales initially embraced direct democracy, many coca growers upon whose support he relied in his ascent to the presidency feel that he has "slipped into old fashion patronage politics, rewarding friends and pushing opponents" (Grisaffi 2013:49). Despite Morales's capitulation to neoliberal pressures and retreat from participatory democracy, there still exists a left-indigenous collection of critical thinkers committed to the transformation of Bolivia into a highly equitable and environmentally sustainable society.

On the positive side, poverty in Bolivia has been reduced from 60.6 percent of the population in 2005 to 43.5 percent in 2012 (Fidler 2014:18). Under the Morales government, by 2012 life expectancy increased to 67 years, and infant mortality dropped to 33 per 1,000 live births (UNICEF 2013). Literacy has increased tremendously and there has been expanded rural electrification and access to employment, education, and health care. The Bolivian government has created small state-owned enterprises that allow local producers input into their operation, and has allocated more than thirty-five million hectares of land as communal land or indigenous territories.

Perhaps the most profound contradiction of the "socialism for the twenty-first century" experiment in Bolivia is, on the one hand, ongoing reliance on fossil fuels for generating revenue, much of it for needed social programs, and, on the other hand, the government's defense of Mother Earth. The case of Bolivia is a profound illustration that socialism cannot be achieved in one country, especially one on the periphery of the capitalist world system, and that the construction of socialism for the twenty-first century will have to be a process occurring in numerous countries.

Under the Morales government, indigenous medicine has achieved notable headway, particularly in terms of the institutionalization of indigenous medical associations and the licensing of indigenous healers. Even prior to the Morales government, in 1984 with the creation of the Bolivian Society of Traditional Medicine Bolivia became the first Latin American country to start the process of legalizing indigenous medicine (Babis 2014). A new constitution approved in a referendum in 2009 explicitly acknowledged the rights of indigenous people, who comprise about 60 percent of the population (the highest percentage of indigenous people in all of the Latin American countries), "with a special chapter devoted to indigenous medicine" (Babis 2014:290). In 2012 a National Register for indigenous medicine commenced with a unit for indigenous medicine in the nine provincial health offices. Accreditation under the registry requires that a practitioner show evidence of birth in Bolivia, be over thirty years of age, possess a certificate from their indigenous community or indigenous association, have practiced for at least five years, and submit treatment reports for the practitioner's last ten patients.

Situated between biomedicine and indigenous medicine one finds a relatively new category of health practitioners situated in Bolivian cities called *los naturistas* (LN) who seek to integrate indigenous medicine and natural medicine (Bruun and Elverdam 2006). They generally are mestizos but treat both mestizos and indigenous people. In addition to relying heavily upon herbal medicine, LNs may use biomedical devices such as sphygmomanometers, or refer their patients to a biomedical physician for ultrasound or other tests, with the latter referring the patient back to the LN.

A Case Study of Medical Pluralism in an Urban Setting of a Developing Society: A View from Central Java

Steve Ferzacca, an anthropologist based at Lethbridge University in Canada, adopts a meaning-centered perspective, described in chapter 1, in seeking to unravel "phenomenological meanings of plurality of health care and perception in lives" (2001:5) among residents of Yogyakarta in the Republic of Indonesia. This city is the site of Gadjah Mada University (the country's first postindependence university) as well as many other colleges, institutes, and vocational schools, and it is a mecca of the classical arts, dance, music, and literature. In addition to visiting various biomedical clinics in Yogyakarta, Ferzacca conducted ethnographic research in Rumah Putri, a neighborhood within a subdistrict of the city consisting primarily of working-class people and lower-middle-class clerks, teachers, and public servants. Like various other countries in Southeast Asia, Indonesia, particularly under President Suharto's New Order, has promoted an authoritarian program of modernization, or *pembangunan* (development), and westernization in the form of elaborate Five Year Plans. Ferzacca (2001:210) views medical pluralism in Yogyakarta generally and among the residents of Rumah Putri specifically as a "social practice that produces hybrid forms of medicine," that is, a mixture of traditional and modern medicine, utilized by people who lead "hybrid lives." Medical pluralism in Yogyakarta is manifested largely in two broad forms: biomedicine and *pengobatan tradisional*, or traditional medicine. Biomedicine is practiced in clinics and hospitals that serve as "entry sites into modernity where exposures to the hegemony of scientific medicine takes place" (Ferzacca 2001:68). It is more or less universally available to everyone on the Indonesian island of Java, particularly in urban areas, and is situated in various settings, including private biomedical clinics, public health centers, mobile units, integrated health posts, and numerous government health programs. It is provided not only by biomedical physicians but also by nurse practitioners and health cadres. Many Javanese view biomedicine as a system that is useful in treating diseases (such as adult-onset type 2 diabetes, hypertension, and heart disease) associated with development or modernization, and hence they see it as complementing traditional indigenous and newly emerging medical subsystems.

Medical Subsystems

Traditional medicine itself consists of a wide array of medical subsystems, therapies, and practitioners, including herbalism, massage, the use of amulets and talismans, patent medicines, Chinese herbalists and acupuncturists, Ayurvedic and Unani doctors, and indigenous Javanese healers and psychics. *Dukuns*, or traditional Javanese healers, practice various forms of shamanism, divination, exorcism, magic, midwifery, pediatrics, massage, spinal manipulation or bone setting, herbalism, and counseling and, rather than being valued as culturally familiar, are often regarded with suspicion because they may seek to exploit patients or place curses on unsuspecting victims. Conversely, psychics, or

paranormals, are generally held in high esteem as paragons of modernity and development. These individuals find missing persons, perform astrology, tell fortunes, foretell the future, heal the sick, and perform rituals at various public events and rites of passage. The paranormals in particular have found a devoted clientele among politicians, businesspeople, and professional people, including members of the Javanese intelligentsia, such as professors, film and music stars, and even Catholic priests. In essence, while associated with *pengobatan tradisional*, paranormals practice a hybrid medicine that blends traditional practices with modern ones associated with Western and Eastern psychic traditions. In summary, Ferzacca (2001) argues that medical pluralism in urban Java provides a public sphere where both practitioners and patients can try to cope with everyday hardships of living under the domination of an oppressive regime intent on forcing them into modern life and the global economy.

More recently, an ethnographic research on the plural medical system in eastern Java examines treatment of mental illness, with a focus on three types of healers, namely the *dukun*, the *kyai*, and the psychiatrist (Woods 2007). The dukun is a traditional shamanic-like healer who may go into trance or serve as a spirit medium and often touches the patient, sometimes gently and other times more forcefully. He or she may provide the patient with an amulet, water blessed by a prayer or spell, paper containing a magical formula, or an herbal medicine. The dukun seeks to create balance between the patient and the cosmos. The patient is expected to compensate the dukun for his or her services with a gift or money. Dukuns do not generally belong to professional associations and tend to be viewed with ambivalence in Javanese society.

In contrast to the dukun, the kyai is also a male practitioner who often surrounds himself with Islamic symbols and attire. He may perform either in a public or private setting and interprets symptoms as emanating from transgressions against Allah or people. The kyai prescribes amulets or prayers and admonishes the patient to observe the pillars of Islam. While he is believed to have supernatural powers, the kyai has studied Islamic law, history, and texts and is ultimately a representative of Allah. Unlike the dukun, kyais belong to a number of regional or national professional associations. For his services, the patient provides the kyai with a gift, food, or money, always given "surreptitiously in an offered palm or envelope" (Woods 2007:444).

Finally, the psychiatrist, while a biomedical specialist, does not dismiss the cultural appropriateness of the treatment administered by the dukun or kyai. Psychiatrists, generally male, may see their patients in private clinics or hospitals, the latter which tend to be loud and chaotic settings in Java. They prescribe advice and medications in an authoritarian and often rushed manner and often dismiss many of the symptoms presented by their patients, although high-income patients may be treated in a more solicitous manner. Reportedly, Indonesian psychiatrists have been profoundly influenced by cultural psychiatry, particularly as manifested in the work of psychiatrist-anthropologist Arthur Kleinman.

A Case Study of Medical Pluralism in a Developed Society: The Australian Dominative Medical System

Australia, which has about twenty-four million people in an area slightly smaller than the lower forty-eight U.S. states, is unique in that it has gone further than perhaps any other developed country in providing public support for the training of practitioners of complementary and alternative medicine (CAM), both in terms of degree-granting programs in chiropractic, osteopathy, Chinese medicine, and naturopathy in public universities and in terms of partnerships that have developed between private CAM colleges and public universities that allow students in the former to upgrade an advanced diploma into a bachelor's degree. In contrast, aside from public colleges of osteopathic medicine, which in the United States have evolved into a parallel medical system to biomedicine with an emphasis on family medicine and reliance on manipulative therapy as an adjunct treatment approach, all training programs in chiropractic, Chinese medicine, and naturopathy are embedded in private institutions in both the United States and Canada. As the Australian case indicates, the manner in which medical pluralism is expressed in a particular society is significantly shaped by political, economic, and social structural forces specific to that society, a point that was made in the case of Bolivia described previously. Whereas the Australian medical system with its various medical subsystems was relatively pluralistic in the nineteenth century, early in the twentieth century, with the transformation of regular or allopathic medicine into biomedicine, the latter adopted a scientific guise to distinguish itself from heterodox or alternative competitors. The Australian medical system evolved into a plural system in which biomedicine achieved dominance over other medical subsystems, a configuration that medical anthropologists call a "dominative medical system" (Baer 2001). Table 7.1 depicts a model of the Australian dominative medical system.

Beginning in the late nineteenth century, Australia as a relatively open society conducive to maverick individuals and social experiments served as what one aptly terms a "paradise of quacks"—a veritable haven for homeopaths, herbalists, phrenologists, hydropaths, lay midwives, spiritual healers, and purveyors of home remedies and patent medicines (Martyr 2002). The nature of some of the medical subsystems in Australia is examined next.

Medical Subsystems

Osteopathy

Although initially developed in the United States, osteopathy diffused to various countries, particularly Anglophone countries, such as Canada, the United Kingdom, Australia, and New Zealand, where, unlike in the United States, it has remained primarily a manual medical system. In contrast to various other countries, the development of Australian osteopathy has been intertwined with that of chiropractic. Some Australian practitioners attempted to institutionalize a merger

TABLE 7.1 The Australian Dominative Medical System

Biomedicine

Fully legitimized professionalized heterodox medical systems

Osteopathy
Chinese medicine in Victoria

Semi-legitimized professionalized heterodox medical systems

Chinese medicine outside of Victoria
Naturopathy or natural therapies
Direct-entry midwifery
Homeopathy

Limited or marginal heterodox medical systems

Homeopathy
Massage therapy
Reflexology
Reiki
Kinesiology

Religious healing systems

Spiritualism
Pentecostalism
Liberal Catholic Church
Scientology
New Age healing

Folk and ethnic medical systems

Anglo-Australian folk medicine
European immigrant groups' folk medical systems
Asian folk medical systems
Aboriginal medical systems

of chiropractic and osteopathy as was evidenced by names of various professional bodies and training institutions (e.g., the Chiropractic and Osteopathy College of Australia). Several osteopathic colleges reportedly transformed themselves into chiropractic colleges in the mid and late 1960s.

Chiropractic

There have been at least four periods in the development of Australian chiropractic: (1) the establishment period (1918–1953), in which various groupings of chiropractors emerged in Victoria, including ones drawn from the ranks of British-trained osteopaths and others who obtained chiropractic training in the United States; (2) the period of expansion (1954–1961), which witnessed a considerable increase in the number of chiropractors trained in Australia (as well as many trained overseas); (3) the period of agitation (1961–1973), resulting in 1964 in the passage of the Western Australian Chiropractors' Act, which provided

statutory registration for chiropractors in that state and included them under private health insurance plans; and (4) the period of legitimation, which began with a federal parliamentary committee report in 1977 that recommended registration for both chiropractors and homeopaths although not naturopaths (Willis 1989). Chiropractic and osteopathy constitute fully legitimized professionalized heterodox medical systems in the sense that their training programs are embedded in public universities and one private university, namely, the chiropractic program at Murdoch University.

Chinese Medicine in Australia

Although traditional Chinese medicine was first introduced in Australia in the mid-1850s in places like the gold fields of Victoria, where Chinese migrants first settled, it did not attain widespread popularity until the development of the holistic health movement. Chinese medicine has been and continues to be a fractionated profession as is evidenced by its wide array of associations and private training programs, some of which are situated in Chinese medicine colleges and others in colleges of natural therapies. Four public universities, namely, the Royal Melbourne Institute of Technology, the University of Western Sydney, the University of Technology–Sydney, and Victoria University, offer degree programs in Chinese medicine. Despite the opposition of organized biomedicine, the Victoria Parliament passed the Chinese Registration Act in May 2000, making Victoria the only Australian jurisdiction to formally recognize regular Chinese medicine practitioners. However, Chinese medicine later obtained federal statutory registration under the Allied Health Professions Act. In essence, Chinese medicine finds itself situated on the border between fully legitimized and partially legitimized status as a heterodox medical system.

Naturopathy

The historic development of Australian naturopathy can be divided into three periods: its emergence between the 1920s and 1940s, a holding pattern between the 1950s and early 1970s, and its explosion from the late 1970s to the present under the umbrella of the holistic health movement. While the terms "naturopath" and "naturopathy" continue to be used in a variety of contexts in Australia, including academic programs and professional associations, the terms "natural therapies" and "natural medicine" have become commonplace and are applied to a wide array of therapeutic subsystems, including Western herbalism, acupuncture, massage therapy, homeopathy, reflexology, and aromatherapy. Naturopathy constitutes a semi-legitimized professionalized heterodox medical system in that it has not achieved statutory registration in any political jurisdiction. It for a while achieved some degree of legitimation inasmuch as naturopathic training programs became situated in public institutions of higher education or in partnerships with private colleges and public universities. The fact that both the federal government and certain state governments have

commissioned inquiries into various aspects of CAM also constitutes a de facto form of legitimation.

Lay or Direct-Entry Midwifery

There are a large number of nurse-midwives in Australia, as in other Western societies. The women's liberation movement stimulated development of the natural childbirth movement in Australia. Several public universities, including Flinders University in Adelaide and Monash University in Melbourne, came to offer direct-entry programs in which midwifery is offered as an undergraduate degree. Organizations that promote lay midwifery care in both hospitals and homes include the Australian College of Midwives, the Australian Society of Independent Midwives, a consumer group called Maternity Coalition, and the Community Midwifery Program, which since 1990 has been funded by the government to seek alternative models of childbirth for low-risk women (Sutton 2004:6–7).

Teaching CAM

Other alternative therapeutic systems such as homeopathy, massage therapy, reflexology, aromatherapy, Reiki, polarity therapy, kinesiology, and various other CAM therapies constitute limited or marginal heterodox medical systems in Australia. Many of these systems function as specific treatment modalities that are taught in schools of natural therapy or that are used as adjuncts by naturopaths or natural therapists in their practices. Furthermore, when the tenets of these subsystems are taught at a school of natural therapy as a specialized program of study or at a training institution that focuses on a specific therapy, such as homeopathy or reflexology, they often entail a shorter period of study and the granting of a "diploma" or "certificate" rather than a "degree" or "advanced diploma."

At the next level, one finds various Euro-Australian religious healing systems, such as Christian Science, Spiritualism, Seventh-Day Adventism, the Liberal Catholic Church, Scientology, New Age healing, and Pentecostalism. Finally, as a multiethnic society, Australia has numerous folk medical subsystems. Of these, aboriginal healing systems in particular have received considerable attention, particularly from anthropologists. In contrast, the folk medical systems of various European and Asian immigrant groups have received relatively little attention. A study of the health status of Korean immigrants in Sydney revealed that most of them take *hanbang* (which, in Korean, means Chinese medicine) herbal medicine and health food, such as ginseng and deer antlers, but also rely heavily on biomedicine (Han 2000).

CAM and Biomedicine

In the past, the Australian Medical Association (AMA) had been a virulent opponent of CAM. Despite the policies of the AMA, many biomedical general practitioners have been adopting CAM therapies, especially acupuncture, or are now willing to refer patients to CAM therapists. Discussion of CAM reportedly

has been introduced into the undergraduate curriculum of several biomedical schools. Moreover, in 1992, various biomedical physicians formed the Australian Integrative Medicine Association. Ironically, as a result of the increasing popularity of CAM, the AMA has in recent years relaxed its long-standing dismissal of CAM systems (Easthope et al. 2000).

As in other developed societies, the corporate class and the state in Australia have since the 1970s come to express concern about rising health-care costs. The growing support in various ways for CAM exhibited by the Australian government may constitute a covert strategy for curtailing rising health costs. Given that CAM subsystems often emphasize individual responsibility for health, they are compatible with the strong interest among government health administrators, health policy makers, and academics in preventive medicine and health promotion. Since the 1970s, both Labor and Coalition (Liberal and National parties) governments have encouraged citizens to obtain private health insurance and have sought to make them more self-reliant and responsible (White 2002:95). The government's support for CAM is an integral part of its neoliberal effort to divest itself of as much health-care expenditure as the Australian public will tolerate. While the federal government provides support for complementary medicine programs, which tend to be relatively inexpensive compared to biomedical schools because of their low-tech approach, for the most part it does not provide reimbursement under the guidelines of Medicare—the Australian national health plan—to complementary practitioners for services rendered.

Periodically, the issue of whether the government should reimburse complementary practitioners for their services arises. Several years ago, Medicare created new regulations that all patients with a referral from a biomedical general practitioner could receive free chiropractic or osteopathic treatment. However, the Enhanced Primary Care plan permits reimbursement for only five visits per year. Given that chiropractors in particular are prone to treat patients on a quite-regular basis, the patient will be forced to either pay for additional visits out of pocket or receive a partial rebate (perhaps around 20 percent) from a private health plan.

The future of CAM as well as integrative medicine in Australia is difficult to predict but will be subject to numerous larger political, economic, and social structural forces in the larger society. So long as CAM is not fully incorporated into Medicare, it will tend to serve primarily the more affluent sectors of Australian society. Patients who use CAM "represent a generally more privileged segment of the Australian population" (Gray 2006:245). Furthermore, although chiropractors, osteopaths, naturopaths, Western herbal medicine practitioners, homeopaths, and Chinese medicine practitioners view themselves as primary care practitioners, so long as they cannot practice in hospitals on a regular basis, they will remain subordinate partners in the Australian dominative medical system.

Integrative Medicine as a Biomedical Specialty

Integrative medicine has by and large evolved into a biomedical specialty that incorporates or even, one could argue, co-opts CAM therapies and CAM practitioners. This is highlighted by an ethnographic study with four clinical fellows (Salkeld 2014) of an integrative medicine training program at the University of Arizona. The clinical fellows, all biomedical physicians, drew upon the expertise of selected CAM therapists, namely practitioners of naturopathic medicine, traditional Chinese medicine, energy medicine, botanical medicine, mind-body therapy, manual medicine, and a sleep hygienist, as well as a few spiritually oriented practitioners, including a Christian minister and a practitioner who focused in chakra balancing and past life therapy. In assessing patients' histories, the fellows examined biomedical diagnosis, family relationships, religious orientation, nutrition, and experience with CAM, thus making their approach much more holistic than conventional biomedical physicians. While the fellows sought input from the CAM practitioners in their observations of patients, they never challenged a biomedical diagnosis in the study. Furthermore, as the researcher observed (Salkeld 2014:62), "although fellows in this case conference study felt comfortable incorporating select complementary therapies, they were reluctant to adopt explicitly different explanations for disease etiology," thus preserving biomedical dominance over CAM.

■ ■ ■

Typologies of Plural Medical Systems

A number of social scientists have created typologies to help understand the nature of medical pluralism in complex societies of the sort described in the three cases that were presented here. Drawing on the meaning-centered perspective, one model delineates three overlapping sectors in the health-care system (Chrisman and Kleinman 1983). The popular sector consists of health care performed by patients themselves along with their families, social networks, and communities, or the "therapeutic management group" (Janzen 1978:4–5). It includes a wide range of therapies, such as special rituals, diets, prohibitions on certain behaviors, herbs, teas, exercise, rest, baths, and massage, as well as, with the rise of commercial pharmacies, articles such as over-the-counter drugs, vitamin and nutritional supplements, humidifiers, heating pads, and hot water bottles.

The second level of this typology, the folk sector, encompasses various healers who are self-trained or undergo an apprenticeship and tend to practice independently, often out of their home on a quasi-legal or illegal basis. These include shamans, mediums, magicians, herbalists, bonesetters, lay midwives, psychics, and faith healers, such as the Maprofeita healer from Mozambique shown in figure 7.1. An example of the folk sector was described in chapter 2 in the case of Brodwin's study of *houngans* in Haiti and the ways in which people make decisions about which type of healer to consult when they feel ill.



FIGURE 7.1 Maprofeita faith healer in urban Mozambique. Photo by Ippolytos Andreas Kalofonos.

The professional sector includes the practitioners and bureaucratic structures, such as clinics, hospitals, and associations, that are associated with both biomedicine and professionalized heterodox medical systems, such as Ayurveda and Unani in South Asia; acupuncture and herbalism in China; homeopathy, osteopathy, chiropractic, and naturopathy in Britain; and *Heilpraktikers* (naturopaths) in Germany. The department of surgery studied by Katz that was presented in chapter 2 is an example of one part of the professional sector in biomedicine.

Healing and History

In the previous chapter we introduced Fabrega's biocultural approach to medical systems, in which he maintains that medical pluralism is first manifested in chiefdom societies, prestate societies, and early state societies and then develops an even more elaborate form in more complex societies, starting out with empires and civilizations, such as ancient China, ancient India, ancient Greece, and medieval Islamic societies, followed by European societies and finally postmodern societies.

According to Fabrega, in state societies, sickness and health adaptation (SH) is characterized by two tiers: (1) an official, scholarly, academic medical system oriented to the care of elites and (2) a wide array of less prestigious physicians

and folk healers who treat subordinate segments of the society. The state, with its various government bodies, plays an increasing role in medical care by hiring practitioners for elites and providing free or nominal care for the poor, especially during famines and epidemics. The literate or "great" medical tradition, such as in traditional Chinese medicine or Ayurveda in South Asia, includes the formation of a medical profession, the beginnings of clinical medicine, and increasing commercialization (and technologization) of the healing endeavor.

The first literate or "great medical tradition" commenced in the ancient city-state of Sumer and its hinterland, the first instance of medical pluralism in an early state society. Sumeria, which existed between 4000 and 3000 BCE, possessed three categories of cuneiform texts that included various kinds of medical information: therapeutic or medical texts per se, omen collections or "symptom texts," and miscellaneous texts that included information on ailments and medical practices (Magner 1992:19). Sumerian physicians diagnosed symptoms by taking health histories rather than performing direct physical examination. Conversely, the "conjurer," "diviner," or "priest-healer" in Sumer society conducted a direct physical examination and used the patient's symptoms and life circumstances as omens to help diagnose disease. The Sumerian pharmacopoeia included 250 medicinal plants and some 120 mineral substances as well as alcoholic beverages, fats and oils, animal parts and products, honey, wax, and various kinds of milk.

In the centuries after Sumer, various empires and civilizations developed particularly elaborate corpora of medical knowledge, such as manifested in traditional Chinese medicine and Ayurveda in South Asia, that are incorporated into writings and commentaries that over time both exhibit an internal order and also include numerous inconsistencies. While much has been written about medical systems in Old World ancient empires and civilizations, out of which biomedicine ultimately developed, including those of Egypt, Greece, and Rome, far less has been written about the medical systems of New World civilizations and empires, such as those of the Mayas, Aztecs, and Incas.

The Aztecs, who developed an extensive empire in central Mexico that ended with the Spanish conquest in the early sixteenth century, had a complex medical system that combined naturalistic and supernaturalistic elements and included priests, shamans, and physicians (Ortiz de Montellano 1989). The state-sanctioned presence of both priests and shamans was a unique situation worldwide because of the strong antagonism that generally exists between the two. The Aztecs believed that the human body contains several souls, each with a specific function affecting growth, development, physiology, and fate in the afterlife. The three primary souls were the *tonally*, situated in the head; the *teyolia*, located in the heart; and the *ihiyotl*, found in the liver. Health depended on the relative size of each soul at any point in time and the degree of balance among the three souls. Diet and personal behavior influenced the size of each soul. Offended deities, particularly the creator god Tezcatlipoca (or one of his many manifestations), or certain animals could inflict disease on an individual. Aztec astrology posited that the equilibrium of the universe affected the human body. The Aztecs also believed that disease could be inflicted by a sorcerer through certain rituals.

The Aztec priesthood became more elaborate and specialized as the Aztec civilization and empire expanded. Various sorts of priests officiated at rituals, including those featuring human sacrifice; managed temple treasuries; taught school; listened to confessions and prescribed penance (Wolf 1999:153). Priests conducted worship of the Aztec pantheon that consisted of numerous deities, each of which had one or more temples where its idol was kept. The vast majority of priests were males, but there were some female priests who generally served briefly before marrying.

Aztec shamans were capable of sending their *tonally* on magical flights back to the time of creation or, alternatively, transforming them into the appearance of an animal. This process of spiritual transformation by the shamans was assisted by the use of various hallucinogenic plants. Shamans focused on curing ailments caused by imbalance in the relationships between the gods and humans or by the intrusion of beings from other cosmic realms into the sufferer's body.

Aztec physicians could be either male or female. Sahagun, a Catholic priest who chronicled many aspects of Aztec culture following the Spanish conquest, described the good general practitioner as being one who worked with herbs, stones, and roots; examined, experimented, and alleviated disease; set bones and administered massages, purges, and potions; bled his or her patients; and dressed wounds (Smith 2003:255).

Aztec medicine also had an array of medical specialists who focused on diagnosing specific ailments by identifying the offended deity who had induced the disease and prescribing the appropriate propitiatory ritual, which could be making an offering, confessing a wrongdoing, engaging in acts of penance, or praying. It included, as well, empirical knowledge about hundreds of medicinal plants that were applied by herbal specialists. The Aztec medical system relied on highly sophisticated surgeons, including those who served in the many battles that the Aztecs fought while enlarging their empire, some of whom even practiced plastic surgery. Aztec society also had fortune-tellers who could be either male or female and who divined by casting maize kernels and gazing in water.

Sickness and health adaptation in modern state societies worldwide is characterized by the emergence of biomedicine as a dominant and hegemonic profession. Biomedicine, in turn, is characterized by patterns of secularization, including the weakening of the role of institutional religion in public life; scientific knowledge, with its embrace of biological reductionism, including the tendency to view disease as due primarily to biological factors, such as germs; the emergence of the hospital as the center of healing and research; and the universalization of categories of sickness. For many social scientists, modernity is generally associated with the rise of advanced technology, conventional Western science, the notion of progress, rationalism, efficiency, and secularization. In this scenario, the hospital emerges as the center of healing and the "clinical gaze" in which the body is subjected to close probing and scrutiny, such as in the use of the stethoscope and surgery.

Biomedicine during the modern era became increasingly specialized with the appearance of practitioners who focused on certain parts of the body, such as the heart, the lungs, and the immunological system. Furthermore, in keeping

with a pattern of medical specialization is the "growth of the discipline of psychiatry and the increasing importance given to mental illness in the society and in conceptions of social disability, welfare, and crime" (Fabrega 1997:136). At the same time, medical pluralism in Western societies and Japan manifests itself in the continuing existence of alternative practitioners, such as homeopaths, herbalists, osteopaths, chiropractors, and naturopaths, who are held in contempt by many biomedical physicians.

Despite persisting biomedical dominance, the pursuit of alternative therapies, including ones based on Western spirituality or Eastern mysticism, manifests itself most profoundly in what Fabrega and other social scientists term postmodern societies. The postmodern world is associated with corporate globalization involving the diffusion of corporate control over economic, political, and cultural processes throughout the world; a sense of social fragmentation and uncertainty about social institutions and values; an emphasis on consumerism; a growing faith in intuition; and "hybridity," which is the blending of elements from different cultures and parts of the world. Many individuals in postmodern societies, particularly those belonging to higher socioeconomic categories, regard health as an achieved status obtained through education, prevention, and lifestyle. Further, postmodern societies manifest an "obsessive preoccupation with health and fitness" (Fabrega 1997:137), thus resulting in the phenomenon of the "worried well."

Patients of CAM

These patterns result in "dissatisfaction with physicians, with 'establishment' medicine, with the escalating costs of a more routinized and procedural approach to healing, and with tentacles of medical insurance companies," all leading to an "increasing interest in holistic medical traditions and the growth of health-promoting and life extension industries that compete with orthodox biomedicine" (Fabrega 1997:244).

Numerous patient utilization studies conducted in the United States, Canada, Britain, Australia, and various Western European countries have sought to identify the types of people who tend to turn to complementary and alternative therapies and practitioners. These studies indicate that users of CAM, aside from various folk medical systems, tend to be female, relatively young, employed, highly educated, and often but not necessarily involved in alternative lifestyles. Based upon interviews with seventeen "high performing" individuals, all of whom had tertiary levels of education, in the Canadian province of Alberta, CAM has been found in part to function as a self-care system, entailing practices such as vegetarian and vegan diets, yoga, meditation, and tai chi (Fries 2013).

At the same time, CAM has found some reception among working-class people in developed societies. One study, for example, found that many of the researcher's respondents in Oceanport (pseudonym), a suburb of an Australian city, reported that they and other household members followed a "mixed therapy regimen" in which they used a CAM therapy either prior to or in combination with biomedical treatment (Connor 2004).

CAM often overlaps with New Age spirituality and New Age Orientalism and for affluent Westerners may entail travel to health spas in developing countries, particularly India. For example, Vedic Village in India offers its clients access to five holistic physicians, two of whom have degrees in Ayurvedic medicine and surgery and three of whom have degrees in homeopathic medicine and surgery (Nazrul 2012). One of the holistic physicians has naturopathic training and the other five hold expertise in acupuncture. In addition to the Western visitors to Vedic Village, the site is visited by Indian middle-class professional people or wealthy business people who seek relaxation therapy (Nazrul 2012).

CAM in Denmark

Based upon interviews with forty CAM practitioners and three hundred patients of CAM clinics in Denmark, one study delineates three principal actor-networks: a technocrat network, a social-democratic network, and a neoliberal network (Johannessen 2007). The technocrat network exhibits body praxis that operates with a plumbing model in which the body is viewed as a series of pipes that "reflects technologies of drainage systems, circulation of water, heat and electricity in the house so familiar to contemporary people" (Johannessen 2007:270). In Danish CAM clinics, reflexology in particular, which entails the massage of reflex zones on the feet that are purportedly linked with various organs, follows the plumbing model associated with the technocrat network. The social-democratic network model seeks to achieve an internal balance among body organs reflective of the need for a more macroscopic balance between the body's immune system and external disease agents, such as allergens, pollutants, bacteria, and viruses. CAM therapies focused upon proper nutrition and the technologies of biopathy illustrated, for instance, by use of an "electrical apparatus showing signs of light and sound" exemplify this model, which is characterized by "principles of democracy, consultancy, and uniqueness" (Johannessen 2007:273). Finally, the neoliberal network model views the body as a computer governed by the brain that can be corrected through a process of reprogramming when an autonomous individual begins to malfunction. In order to reprogram the patient's body-mind, kinesiology with its tests of muscle strengths operates on the basis of this model in that its practitioners administer bodywork based upon patients' responses to questions about their experiences in childhood and even previous lives, substance intolerance, malfunctioning organs, and strained social relations. Patients may prefer one or another actor-network, but in reality they move between them, thus following a sort of mixed therapy regimen within the confines of CAM.

Traditional Healers in Mozambique

While CAM systems have, since the 1970s, become increasingly popular in developed societies, the high cost of biomedicine and the fact that it tends to be based in urban areas means that many people in developing societies, particularly in rural areas but also in cities, continue to rely heavily on "traditional" medical systems or on new healers who combine aspects of folk medicine and biomedicine. A study of indigenous and church-based healers in Mozambique revealed that traditional healing has been altered appreciably due to the impacts

of both colonialism and market capitalism (Pfeiffer 2005). Men, in particular, in poverty-stricken Mozambique have responded to structural adjustment policies imposed on their country by the World Bank and the International Monetary Fund in 1987 by turning to traditional healing as a means of making a livelihood. Indeed, the Mozambique Association of Traditional Healers, a government-based body to which many *curandeiros* belong, lists the availability of numerous treatments, including ones to help a client obtain a job, avoid mishaps, contact ancestral spirits, and cope with a wide array of problems of living. Some traditional healers, however, view the association as a government ploy to tax them. In response to the growing pecuniary orientation of many traditional healers, a large number of people in Mozambique have turned to Pentecostal sects and African independent churches, which syncretize or blend elements of traditional African religions and Christianity, to create healing rituals designed to help them cope with their plight.

CAM and Class

In keeping with Navarro's (1986:1) assertion that classes as well as ethnic groups and genders within capitalist societies "have different ideologies which appear in different forms of culture," it may be argued that these social categories also construct different health understandings, behavior patterns, and medical subsystems to coincide with their respective views of reality. Regarding gender differences, for example, various medical anthropologists concerned with reproductive health have argued that men and women do not always share reproductive goals and strategies. As has been observed, "Contested reproductive interests may derive from women's and men's differing experiences of pregnancy and birth, varying domestic and productive responsibilities, or discrepant material interests" (Sargent 2006:32). How men influence women's reproductive behaviors, in fact, remains an understudied issue. One possible reason for this is that earlier anthropologists viewed reproduction as a women's issue, and therefore peripheral to the discipline, a telling comment on how social factors influence the course of social research within and beyond medical anthropology, an issue addressed in chapter 2 (Browner 2001:773).

Class differences in health understandings, behaviors, and medical subsystems are also of considerable importance. Critical medical anthropologists, for example, argue that the patterns of medical pluralism found in state societies tend to reflect hierarchical and unequal social relations based on class, caste, ethnicity, region, religion, and gender distinctions found in the larger society. Thus, critical medical anthropologists argue that national medical systems in the modern world should be described as "plural" rather than "pluralistic" inasmuch as biomedicine enjoys a dominant status over both heterodox (e.g., chiropractic and naturopathy) and folk medical systems (e.g., *curanderismo* and African American spiritualism). In reality, plural medical systems may be best described as "dominative" in that one subsystem within the larger complex of coexistent medical traditions generally enjoys a preeminent status over the others. As noted earlier, within the modern context, biomedicine exerts dominance over all other complementary

and alternative medical subsystems (see chapter 1), although people are quite capable of "dual use" of distinct medical subsystems, either simultaneously or sequentially (Romanucci-Ross 1977). Thus, it is not unusual for a patient to be treated during a period of time for the same or different health problems by a folk healer and a biomedical physician (with or without the knowledge of one or both healers).

In the United States, as a result of financial backing for its research activities and educational institutions, initially from corporate-sponsored foundations and later the federal government as well, biomedicine evolved into the dominant medical system. It then asserted scientific superiority and clearly established authority over professionalized heterodox medical systems, such as homeopathy (a medical system developed by Samuel Hahnemann, a disenchanted German physician), eclecticism (a hybrid of regular medicine and botanical medicine), hydropathy (an elaborate system of water cures), osteopathy, and chiropractic. Homeopathy and eclecticism have been somewhat absorbed by biomedicine at both the organizational and the therapeutic level. Osteopathy initially constituted a manual medical system created by Andrew Taylor Still, a dissatisfied regular physician who viewed the spine as the key to good health. It evolved into osteopathic medicine and surgery and achieved full practice rights in all fifty states and the District of Columbia by the early 1970s. Chiropractic, another manual medical system developed by D. D. Palmer, an American magnetic healer, evolved into the leading professionalized heterodox medical system in the United States (and in a growing number of settings in the United States is being practiced as a junior partner to orthopedics). At the beginning of the twentieth century, hydropathy became part of naturopathy, a highly eclectic medical system that incorporated botanical medicine, exercise, dietetics, and colonic irrigations. Naturopathy declined in the 1930s but underwent a rejuvenation with the emergence of the holistic health movement during the 1970s and 1980s.

Biomedical dominance over rival medical systems has never been absolute. In capitalist societies, the corporate class historically has come to exert greater and greater influence over politicians and government bureaucrats through processes such as lobbying, campaign contributions, and the dissemination of information favorable to its positions. Nevertheless, although the state serves the interests primarily of the wealthiest class in society, it must periodically make concessions to subordinate social groups in the interests of maintaining social order and the capitalist mode of production. As a result, certain complementary and alternative practitioners, with the backing of clients and particularly influential patrons, were able to obtain legitimation in the form of full or limited practice rights. Lower social classes, ethnic minorities, and women have often used CAM therapies as a forum for challenging not only biomedical dominance but also, to a degree, the hegemony of the corporate class or state elites. CAM systems sometimes resist, at least subtly, the elitist, hierarchical, and bureaucratic patterns of biomedicine. In contrast to biomedicine, which is dominated ultimately by the corporate class or state elites, folk healing systems are more generally the domain of common people. Biomedical physicians often have significant representation on the registration boards of other health practitioners, tend to be overrepresented on

hospital and health policy advisory boards and in government health departments, and dominate health research funding.

As was the case in Australia, the growing interest in CAM therapies by corporate and government elites around the world is related to the cost of high-technology biomedicine. Even in countries where explicit financial and/or legal support is absent, governments often prefer to support CAM approaches because they focus on self-limiting diseases, that is, diseases that tend to run their natural course without treatment.

While some anthropologists have commended efforts to integrate aspects of biomedicine and CAM practitioners as constituting a key to more comprehensive and culturally sensitive health care, others have viewed the therapeutic alliance between biomedicine and folk healers as a manifestation of a "new colonialism" or a strategy to co-opt CAM (P. Singer 1977, Baer and Coulter 2008). An emancipatory therapeutic alliance ultimately requires an egalitarian relationship between various medical systems, one that transcends the hierarchical structure of existing dominative systems associated with modern societies.

Societies vary considerably in the degree to which they tolerate medical pluralism. This reality is effectively captured in a scheme depicted in table 7.2, which depicts the "range of subcultures permitted within a national medical culture" (Last 1996:380). In the case of the exclusive system, the national power structure recognizes and tolerates only one medical subculture, namely, biomedicine, as acceptable or at least dominant over alternative systems. Such a system existed in the former Soviet Union and former Soviet-bloc countries where alternative healers were viewed as feudalistic anachronisms associated with superstition and economic backwardness. In the French model, which is followed not only in France but also in francophone Africa and parts of Latin America, the national power structure recognizes only biomedical practitioners and defines alternative medical systems as illegal. In a supposedly exclusive biomedical system, such as found in Portugal, where the Medical Council forbids its members to use CAM therapies, various biomedical physicians have opted to contravene this mandate by incorporating acupuncture and homeopathy into their practices (Almeida 2012).

TABLE 7.2 **Plural Medical Systems in Terms of the Officially Accepted Range of Medical Subcultures**

Exclusive systems
Soviet model
French model
American model
Tolerant systems
British model
German model
Integrated systems
Third World systems

In the American model, which theoretically is based on free market principles, the national power structure defines "hospital medicine" or biomedicine as the dominant medical subculture legally and in terms of funding for medical education and research. Conversely, this variant permits a restricted tolerance of alternative medical systems, some of which, such as chiropractic and acupuncture, enjoy more official acceptance than others, such as herbalism and various forms of bodywork.

While giving preeminence to biomedicine, tolerant systems adopt a more or less laissez-faire policy toward alternative medical systems. The British system, which is followed not only in Britain but also in Australia, New Zealand, and many developing nations, operates on the basis of common law, which essentially states that if the law does not specifically restrict certain procedures, such as prescribing legally defined drugs, performing surgery, and signing death certificates, as exclusively the domain of the biomedical practitioner, individuals may practice an alternative medical system, such as herbalism or naturopathy. While this policy applied for a long time to both osteopaths and chiropractors, both healing groups obtained statutory recognition, in Australia in the early 1980s and in Britain in the early 1990s. Another example of the tolerant system is the German model mentioned earlier. While biomedical physicians may work within the traditions of both *Schulmedizin* and *Naturheilkunde*, Germany has a diverse group of practitioners called *Heilpraktikers* (the counterparts to naturopaths in North America, Britain, Australasia, and India) who vary tremendously in terms of their training but are permitted to practice a wide array of natural therapies as long as they have passed an examination indicating that they understand the laws regulating medical practice.

The integrated system is especially characteristic of India and China. In India, the government permits practitioners of various ancient healing systems, such as Ayurvedic, Siddha, and Unani medicine, to establish their own professional associations, training institutions, hospitals, and pharmaceutical industry and provides these systems with statutory recognition and funding. Ayurveda is based upon Sanskrit texts; Unani, or Greek medicine, on Arabic and Persian texts; and Siddha is a tradition of humoral medicine in South India. India also has two transplanted Western heterodox medical systems, namely homeopathy and nature cure or naturopathy. Nature cure, which has received Indian government recognition, made inroads into India in part due to Mohandas K. Gandhi's advocacy of dietary reforms, hydrotherapy, enemas, and mud baths (Alter 2016). Furthermore, Tibetan medicine has been practiced for many centuries in Himalayan India. Tibetan exiles established the first Tibetan medical college in Dharamsala in 1961, one that has gained world renown (Pordié and Blaikie 2014). The Central Institute of Buddhist Studies in Ladakh established a Tibetan medicine course of study in 1989, thus allowing local people to study close to home.

Indeed, in 2010 the Indian government extended official recognition to Tibetan medicine. Unfortunately, the increasing commodification of indigenous medicinal substances in recent times "threatens to rob the poor sections of Indian society of access to Indian medicine because they cannot afford the relatively expensive Ayurvedic and Unani brands" (Bode 2006:233). The poor

farmer from Bankura district in West Bengal, India, in figure 7.2, is an example of the part of Indian society whose access to Indian medicine is becoming more limited.

The People's Republic of China officially recognizes both biomedicine and traditional Chinese medicine and has integrated both to a certain degree in state-operated institutions, including some clinics, hospitals, and colleges. During the Maoist era, barefoot doctors, who were highly touted as paragons of revolutionary dedication, served as medical auxiliaries trained in both biomedicine and Chinese medicine. China also has made provisions for incorporating various indigenous or folk healers. Despite the preeminence of biomedicine in contemporary China, traditional Chinese medicine continues to be very popular and has become fashionable among young professional people in large cities, such as Beijing, perhaps a parallel development similar to the popularity of CAM among professional people in Western societies (Boeke 2014:176–77). Whereas many Chinese people view biomedicine as "fast" in its results, they view traditional Chinese medicine as "slow" but more thorough, focusing more on the roots or causes of disease than the symptoms.

In contrast to China, which has witnessed the rise of the new middle classes with disposable incomes as a result of tremendous economic growth, although massive poverty persists in particularly the rural areas of the country, many people in Russia since the collapse of the Soviet Union in 1991 have fallen into economically desperate situations. In order to cope with gaps in the national health-care



FIGURE 7.2 Impoverished small farmer dries grains in Bankura District in West Bengal, India. Photo by Anna Marie Nicolaysen.

system, older women, referred to as *babushka*, surviving on meager pensions, have turned to self-care utilizing CAM therapies. In the city of Ulan-Ude, the capital of Buryatia situated on the border of Russia and Mongolia, older women draw upon the services of the East-West Hospital, which offers not only biomedical care but also acupuncture, herbalism, hydrotherapy, massage, and even Tibetan medicine under the rubric of *restorative medicine*. This is a new specialty designed to manage chronic ailments (Chudakova 2016). Some of the women purchase Tibetan medicines, some of them of dubious value and posing a drain on their limited budgets. Furthermore, many for-profit clinics and practitioners offering CAM emerged in Ulan-Ude during the twenty-first century, an issue that also occurred in Russia and led to much debate in a country where once all health care fell under the purview of the state.

Arab Herbal Medicine in Israel

Israel is, at least for its Jewish citizens, a largely developed society with an integrative medical system in which CAM modalities have been incorporated into biomedical clinics and hospitals. In recent times, in an effort to appease its Arab citizens, most of whom live in conditions more reminiscent of a developing country, Israeli biomedicine has made efforts to incorporate Arab herbal medicine into the larger integrative health-care system (Keshet and Popper-Giveon 2013). Biomedical research not only in Israel but in Egypt, Turkey, Jordan, and Morocco increasingly has incorporated Arab herbal medicine into its purview of investigation. Conversely, even among religious Jews, a ritual blessing, referred to as "deep healing," is applied by a rabbi in dealing with personally and socially distressful situations, a practice that by and large remains outside the larger integrative health-care system (Sharabi 2014).

The Third World model is situated in many developing societies, particularly the poorest ones in which biomedicine is primarily urban based, poorly financed, and largely a privilege of elites and a small middle class. While the power structure privileges biomedicine and even makes it available on a restricted basis to the masses, it tolerates a wide array of alternative medical systems consisting of local folk healers, bonesetters, injectionists (folk healers who use syringes to inject vitamins, antibiotics, or other drugs), barber-surgeons, and midwives who serve the majority of people—a fact recognized by the World Health Organization. For example, in Mozambique, which has one biomedical physician for some fifty thousand people, there is a traditional healer available for every two hundred people. Furthermore, pharmacists in developing nations often serve as a major source of medical information and dispense drugs freely that generally require a prescription written by a biomedical physician in developed societies. In many developing societies, folk and indigenous healers have organized themselves into professional associations in their efforts to obtain greater legitimation and recognition (Last 1996).

The Third World Model in Morocco

Morocco is yet another country in which the Third World model prevails and while biomedicine constitutes the dominant medical sub-system, only 15 percent of the country's citizens were covered by the public health system, and

TABLE 7.3 The Moroccan Dominative Medical System

Medical Sub-System	Origins
European medicine/ biomedicine	Initially utilized by Franciscan priests around 1800 Authorized by the reigning sultan around 1860 to attend to the health-care needs of resident Europeans Expansion after independence from Spain in 1956
Classical Arab humoral medicine	Greco-Roman tradition spread by Nestorian Christians in the late fifth century CE throughout the Middle East
Prophetic medicine	Based upon the teachings of the Prophet Muhammad and entailing rituals designed to cure illnesses arising from failure to abide by Islamic norms and customs
Popular medicine	Consists of a great variety of medical specialists, most of whom treat a limited number of illnesses using minor surgery and administration of vegetable, animal, and mineral medical products

Source: Adapted from Dieste (2013:166–203).

still patients often had to pay for medicines and operations (Dieste 2013:209). The Moroccan dominative medical system consists of the layers depicted in table 7.3.

Under the rubric of popular medicine are found herbalists and small stores that sell vegetable, mineral, and animal medicinal products; cauterizers who apply iron and wooden tips to cleanse afflicted body organs, such as the intestines, or the nervous system; "barbers" who engage in bloodletting; bonesetters; tooth pullers; women healers who administer herbs, deliver babies, and exorcise evil spirits; and religious sanctuaries and spas linked to the healing abilities of Islamic saints. Approximately 60 percent of the Moroccan population utilizes both biomedicine and a wide array of traditional medical systems, with slightly less than 40 percent relying exclusively on biomedicine and a small minority relying only on traditional medicine (Dieste 2013).

New Directions in the Study of Medical Pluralism

Whereas the early work on medical pluralism tended to focus on levels in plural medical systems, more recent research on this phenomenon recognizes that, as Stoner (1986:47) asserts, "pluralism can now be examined as a multiplicity of healing techniques, rather than of medical systems." In a similar vein, Pool and Geissler (2005:45) maintain:

> Rather than trying to reveal "[medical] systems" we should focus on studying practice (what people actually do when they are ill or suffer misfortune). Health-seeking behavior is not simply enactment of "beliefs" within the confines of "culture" or a "system," but a creative process in which we must recognize the role of invention, innovation, and disorder.

In reality, practitioners from various specific medical systems borrow therapies or modalities from other systems, such as when folk healers in developing societies prescribe manufactured biomedical drugs or even inject them. In South Kanara, India, healers often mixed Ayurvedic medicine with biomedicine and traditional healing techniques (Nichter 1989:187–215).

In response to the growing popularity of CAM in various developed societies, biomedical physicians have increasingly been incorporating various therapeutic techniques from homeopathy, herbalism, acupuncture, and bodywork into their regimen of treatment in an effort to create an "integrative medicine." In essence, while biomedicine continues to enjoy dominance and preeminence worldwide, it increasingly has come to find itself challenged by CAM practitioners of various sorts, including professionalized heterodox ones (e.g., chiropractors and naturopaths), partially professionalized heterodox ones (e.g., herbalists and bodyworkers), and religious and folk healers. Some biomedical practitioners continue to resist the inroads being made by CAM practitioners, yet an increasing number of them are choosing to collaborate with them, but in a manner in which the CAM practitioners later face the danger of co-optation by being transformed into medical auxiliaries similar to nurses, physicians' assistants, and physical therapists.

Some anthropologists believe that the analysis of medical pluralism has reached a theoretical impasse because categorization efforts produced static functionalist typologies or employed incomparable terms (Brodwin 1996). Health anthropologists turned to concerns such as the political economy of health, biomedical domination, CAM systems in developed societies, reproduction, the mindful body, biotechnology, substance abuse, and AIDS. Aside from the issues of the shortcomings of much of the research on medical pluralism, various critical health anthropologists, as we have seen, developed an interest in how power relations shape plural medical systems. In essence, medical pluralism is a topic that continues to be of central concern in medical anthropology.

Medical Syncretism

In keeping with increasing globalization, medical anthropologists have come to find a growing pattern of *syncretism* or *hybridization*—a process in which two relatively distinct medical systems merge to form an essentially new one or at least elements of different medical systems are blended in the treatment of a specific health problem (Pool and Geissler 2005:44–45). Medical syncretism is illustrated in modern Ayurvedic medicine, which is drastically different from the Hindu medical system based on ancient Sanskrit medical texts. Ayurveda has drawn heavily upon the Galenic (Unani) concepts of Islamic medicine. Both professionalized Ayurvedic and Unani medicine have incorporated aspects of biomedicine. Ayurvedic acupuncture constitutes a fascinating example of medical hybridization, one that entails the blending of Ayurveda and traditional Chinese medicine or Taoist medicine. The popularity of acupressure and acupuncture in modern India has resulted in the development of an entity called Ayurvedic acupuncture (Alter 2005). Both Ayurveda and Chinese medicine contend that the unimpaired flow of bioenergy, *Prana* in the case of the former and *chi* in the case of the latter, are

essential to health maintenance. *Prana* flows through each and every organ of the body much in the same way that *Tying* and *chi* flow through channels or twelve regular meridians and eight subsidiary meridians, respectively.

Another example of medical hybridization entails the simultaneous use of the biomedical drug Viagra and herbal medicine in China in the treatment of male sexual impotence. Viagra initially officially entered the Chinese market in July 2000, yet for ages Chinese medicine has used *zhuangyang* herbal tonics that purportedly nourish kidney yang in order to cure impotence. One study entailed interviews with some two dozen physicians, both biomedical and traditional Chinese medical, and about 350 patients suffering from impotence or other sexually related diseases (Zhang 2007). While very few men relied exclusively on Viagra in treating their impotence, many of them engaged in a pattern of switching between the use of Viagra and herbal tonics in their treatment regimen. This switching pattern is illustrated in the case of Mr. Wang (pseudonym) who took ten herbal *yikanwan* pills (in batches of five twice a day) each day over the course of four months and took only two Viagra pills, apparently to give him an extra sexual boost (Zhang 2007:65). Even some biomedical physicians advise that herbal medicine be taken in conjunction with Viagra. While biomedicine does not support the notion that Viagra directly affects sexual desire, only the ability to have a penile erection, some Chinese patients took Viagra to increase their sexual desire. In essence, medical pluralism now operates not only *between* medical systems but also *within* those systems, in both the consultation room and the lab. Thus, in seeking to achieve and provide sexual potency in China, both male patients and physicians essentially hybridize traditionalism, which has drawn from an ancient civilization, and modernity, which has been heavily shaped by the West. These and other incidences of the blending of biomedicine and traditional Chinese medicine can be referred to as a "mangle of practice" (Pickering 1995, Scheid 2002).

Elsewhere in East Asia, Korean medicine also has undergone a process of hybridization as it has sought scientific validation for its herbal formulas. In efforts to prove the efficacy of Korean medicines, "K[orean] M[edicine] doctors blend traditional medicine with several techniques and theories of biomedicine" (Kim 2009). Korean medicine doctors and South Korean businessmen have joined to transform Korean medicine into a "transcultural medical system" that is marketed abroad, including in China, Taiwan, and the United States.

Medical Diversity

Various anthropologists have proposed the term *medical diversity* in lieu of the established term *medical pluralism* (Krause 2013). Medical diversity entails the mutual borrowing of medical beliefs and practices from various medical traditions, which may be biomedical or indigenous or hybridized forms, on the part of patients who are seeking to cure or alleviate their ailment. For example, in African countries such as Kenya and Tanzania, the healer-patient relationship occurs within in a "triangle of healing sources" consisting of biomedical practitioner, local indigenous healer, and healer-prophet-witchfinder (Parkin 2013:129). If the patient and his or her patient management group do not obtain satisfactory treatment from a local healer, they may turn to distant biomedical practitioners

and/or prophet-healers for treatment. With the increasing involvement of Chinese state companies involved in mining and agricultural activities in African countries, traditional Chinese medicine practitioners have added to the range of medical diversity. In the city of Bishkek in Kyrgyzstan, patients rely on a wide array of health practitioners, ranging from Kyrgyz spiritual healers, bonesetters, phytotherapists, Chinese, Korean, and Tibetan medicine practitioners, alongside of biomedical practitioners (Penkala-Gawecka and Ratjar 2016).

Developed societies are also characterized by a pattern of medical diversity. A case in point are spa therapists in a Norwegian health hotel who cater to relatively affluent women who feel overburdened by the stresses of modern life (Anderssen 2016). The spa therapists provide services such as massages, non-surgical body treatments, manicures and pedicures, workouts, and dieting regimens. The therapists view themselves as health workers, but of a different genre than biomedical practitioners and CAM therapists, in that they focus on the well-being of their clients rather than upon primarily their ailments. In reality, the spa therapy settings are part of a long list of different kinds of cure resorts throughout Central and Eastern Europe. For example, in the Czech Republic cure resorts have become part and parcel of *wellness tourism* in which spa therapists and wellness managers with little or no biomedical or CAM training create wellness packages for visitors (Spier 2011).

Medicoscapes

The concept of *medicoscapes* even further broadens the notion of medical pluralism. It recognizes that in their treatment of various diseases, a wide array of national, transnational, and international organizations and institutions, including entities such as the World Health Organization or the Bill and Melinda Gates Foundation, and a wide array of NGOs, along with biomedical practitioners and complementary and alternative practitioners, both professionalized and indigenous, may play a role in the treatment of specific diseases (Hoerbst and Wolf 2014). Certain individuals and groups who generally accept the legitimacy of biomedicine and complementary medicine may opt for religious or philosophical reasons to refuse certain biomedical treatment modalities. For example, in Germany where both biomedicine and *Naturheilkunde* (the art of natural healing or natural medicine) are highly esteemed, Jehovah's Witnesses routinely refuse blood transfusions, as is the case in other countries (Rajtar 2016).

The Globalization of Traditional Medicine and CAM

Just as biomedicine has become a globalized medical system due to colonialism, the migration of Europeans and North Americans to other parts of the world, many heterodox medical systems and CAM systems have also become globalized. For example, naturopathy, which had historical roots in Europe and crystalized into a distinct eclectic CAM system in the United States, has diffused to many parts of the world, including not only Australia and New Zealand but also India, South Africa, and Guatemala (Baer and Sporn 2009). Traditional Chinese

medicine is yet another example that has been diffused to many countries as a result of a global China diaspora. Yet another example of the globalization of a CAM system is Tibetan medicine, which of course made its way to India as a result of the relocation of Tibetan refugees to that country as a result of China's colonization of Tibet after the Chinese Revolution of 1949. Ironically, Tibetan doctors wearing their traditional garb instead of the white coats they wear in Tibetan clinics and hospitals treat patients in the Lhasa Holiday Inn (Janes 2002). Tibetan doctors who were trained in either Tibet or India now are found in many European and American cities. Furthermore, the Chinese government beginning in the 1970s commenced upon a program of reformulating Tibetan medicine and incorporating it into biomedicine as well as fostering the privatization of Tibetan medicine practice and production of Tibetan medicines for commercial purposes (Pordié and Blaikie 2014).

Discussion Questions

1. Compare and contrast the pattern of medical pluralism in the Bolivian altiplano, the city of Yogyakarta in central Java, and Australia.
2. Discuss the concepts of complementary and alternative medicine (CAM) and integrative medicine. Compare and contrast how biomedical practitioners use these terms in contrast to health anthropologists.
3. In what ways does Australian medical pluralism constitute a case of a dominative medical system?
4. Based upon your own experience, how is the Australian dominative medical system similar to or different from dominative medical systems in North American and European societies?
5. In what ways does the dominative medical system model apply to instances of medical pluralism in societies such as China, India, Indonesia, and Morocco? Are there instances where the model does not effectively describe medical pluralism in these societies?
6. Compare and contrast medical pluralism in early state societies and modern societies.
7. Discuss the types of people who seek out CAM therapies in modern societies, including in countries such as the United States, Denmark, Australia, and India.
8. Discuss some of alternatives to the traditional focus on levels in plural medical systems, such as the concepts of medical syncretism, medical diversity, and medicoscapes. In what ways do these approaches provide new insights into medical pluralism?

8 The Biopolitics of Life

Biotechnology, Biocapital, and Bioethics

We all live with a concrete awareness that we cannot say No to science, technology and medicine. Even if we wanted to, we cannot say No to the medical complex that appropriates our bodies, defines our state of health, and positions us in a continuum of fitness from the temporarily abled to the permanently disabled. . . . How are we to understand our often intense hunger to say Yes?

—Downey and Dumit (1997:5)

Introduction and Overview

We begin this chapter with an outline of the way health anthropologists approach biotechnology and introduce the reader to some of the debates and international examples of impacts of biotechnologies. We end the chapter with specific case studies of history, ramifications, and applications of these ideas. We talk about the way applications of biotechnologies have formed and been formed by conceptualizations of nature and the ways in which biotechnology has been used to define humanity as separate from both animals and machines. We consider ethical implications of the use of these technologies and, through the lens of biocapital, explore how they interact within the contemporary global economic environment.

We will then consider the ways in which biotechnology has allowed us to shape notions of kinship and how it has enabled us to divide the body and package biomatter in line with the demands of the global market system. We examine the gap between anthropology and the cognitive sciences, and see how advances in biotechnology have forced us to productively question scientific disciplinary boundaries. After that, we examine ways in which biotechnology has allowed us to manipulate the foundational building blocks of life and the ways in which science has dealt with the capitalist structures of development pipelines, corporate infrastructures, and the market.

We offer three anthropological case studies of socially transformative biotechnological application: the development of human Growth Hormone (hGH), the invention of the Polymerase Chain Reaction (PCR) process, and the impacts of visual scan technologies. We close the chapter by considering the combined effect of biotechnologies on how we use them to structure nature, society, and family and a case study of ancestry, conceptions of race, culture, and belonging.

Critical Health Anthropology and Biotechnology

Within health anthropology, biotechnology is used interchangeably to refer to both technology based on biology, and as a shorthand for biomedical

technology. For anthropologists, medical technology can be any type of technology used in healing processes, including a thermometer; crystal singing bowls used for Tibetan healing ceremonies; medical visualization equipment such as ultrasound or a CAT scan; complex laboratory equipment used for genetic testing, as well as the science that went into mapping the genome and developing the testing procedures; or a pipe used to smoke Mapacho, a Peruvian tobacco used as a plant medicine. Biotechnology includes the material culture of health, the research and knowledge to understand and interpret knowledge extracted from bodies and their environment, and technological manipulation of the organic world for health purposes.

It is difficult to imagine health anthropology research that doesn't involve biotechnology in some way or another. As has been discussed elsewhere in this book, anthropologists do not see science as separate from culture. Science and scientific developments take place within particular cultural contexts and are informed by cultural beliefs and practices. They are shaped by the cultures in which they are embedded, and shape those cultures in turn. Health anthropologists engage with biotechnology when we explore cultures of science and scientific endeavor, impacts of health technology on understandings and constructions of personhood, and the complex ways in which technology impacts on health equity. Biotechnology has the capacity to both ameliorate and exacerbate health inequities, and anthropologists have long been important contributors to the complex bioethics debates that surround biotechnology.

When viewed through a global lens, there are striking patterns present in many applications of health technologies: wealthy, first world bodies often benefit at the cost of bodies in the developing world. The health of the wealthy frequently benefits from the ill health of the poor. Clinical trials are undertaken in impoverished parts of the world, often in slum areas where the offer of regular medical check-ups can be an enticement for participation. At their worst, some trials have been found to be scientifically flawed and deeply unethical, with side effects not monitored or recorded, and in some cases trial participants have been left with life-threatening health consequences from their participation (Kamat 2014). Even when undertaken ethically and with scientific rigor, conducting trials of pharmaceuticals and other biomedical products in impoverished areas has deep ethical implications. For the most part, these products will be made most easily available, when proven safe, to wealthy consumers, and are often financially out of reach of people in the communities on which they were tested.

Science, Nature, and Culture

People are meaning making creatures. We make sense of the world through the construction of complex cosmologies that assign a categorical order to everything in the world. The applications of biotechnologies to human bodies often challenge these systems of order and offer many productive inroads for anthropological insight. In their landmark study *Laboratory Life*, originally published in 1979, Bruno Latour and Steve Woolgar state that rather than focusing on the "facts" that scientists "discovered" in their laboratory, they aimed to "determine

how a fact came to acquire its character in the first place" (1986:273). Latour and Woolgar suggest that what they discovered was something that lab scientists had known all along, but wasn't as well understood outside of laboratories. Discovery is subjective and cultural beliefs, values, and relationships play a significant role in the construction of scientific categories and knowledge. This insight has informed the work of many anthropologists, including David Schneider's (1984) analysis of the cultural construction of the concept of blood relatedness (more about that later in this chapter), and Emily Martin, whose article "The Egg and The Sperm: How Science Has Constructed a Romance Based on Stereotypical Male-Female Roles" (1991), and has been deeply influential for a generation of scholars working on bioethics and social implications of biomedical interventions. Latour and Woolgar's articulation of the social and cultural construction of "scientific" knowledge continues to inform health anthropologists' research into established and emerging health technologies.

Another important contribution to health anthropologists' understandings of biotechnology emerged from Donna Haraway's work *A Cyborg Manifesto* (1991), where she suggests that the twentieth century had heralded the breaking down of three previously rigid boundaries that defined what it was to be a human being. The first boundary breach she discusses is the boundary between human and animal. With the development of animal to human transplantation, the idea that organs of animals of other species can be used to treat humans has opened fierce ethical, spiritual, and scientific discussions. Disease transmission across species boundaries, such as avian flu and swine flu, have all impacted on scientific and lay concepts of boundaries between humans and other species. Haraway (1997) developed this even further when she articulated feelings of kinship with Oncomouse™, a laboratory mouse that was genetically designed to be susceptible to particular cancers, so that cancer treatments can be tested on it. Critiques of clinical trials as being gendered, and gender discriminatory, have argued that women's bodies, especially the bodies of impoverished women, are often used unethically in medical trials (Rajan 2017). The idea of women being treated as "guinea pigs" in clinical trials resonates through much critical biotechnology literature (Arditti et al. 1984, Corea et al. 1987, Finkler 2000).

The second of Haraway's boundary breaches occurs in the ways in which boundaries between humans and machines are increasingly permeable. With pacemakers, joint replacements, bionic limbs, and other machine-like enhancements, Haraway argued that the cyborg human was no longer science fiction, but very much a reality. Finally, Haraway argued that the physical and non-physical worlds became increasingly less boundaried. A quarter of a century later, in light of developments of the internet and virtual realities, this aspect of Haraway's work appears especially prescient. Haraway's contribution challenged taken-for-granted assumptions about how people articulate nature and humanity through biotechnology. Her work opened a series of crucial ethical and methodological issues, raising questions about the position of humanity in relation to nature and technology.

In addition to Latour and Woolgar's repositioning of science-as-culture, and Haraway's repositioning of personhood by defining human beingness as

being merged with, rather than apart from, technology, there emerged a body of anthropological work focused specifically on the nature of nature. "Nature" had often been posited as the opposite of "culture," with nature being the natural world created by God or evolution, depending on your belief system, and culture being the parts of the world created by human activity. Marilyn Strathern's work *After Nature* (1992) suggests that rather than being in an oppositional relationship with culture, nature itself was a cultural construction. Yanagisako and Delaney suggest that calling something "natural" was a way of investing it with particular sorts of power and legitimation (1995). Many of the core debates around bioethics and emerging biotechnologies center on that point: if a technology can be said to be creating something seen as "unnatural," arguments opposing its development are often strengthened, whereas if discourses of naturalness can be called upon, the ethical argument for a particular technology is often enhanced. For example, when the idea of in vitro fertilization (IVF) was initially introduced as a scientific possibility, opponents argued that the idea of a baby conceived in a test tube was an aberration (against God or Nature or both). Advocates of IVF mobilized the "naturalness" of the desire for people to become parents, especially the "natural" instincts of women to become mothers by whatever means possible, to suggest that fertility interventions such as facilitating the joining of egg and sperm in a test tube rather than a womb were fulfilling a natural desire, and were therefore a morally legitimate use of biotechnology. That nature can be and is taken up by both the opponents and proponents of biotechnologies demonstrates both its malleability and its potential for disruption.

Biocapital: Bodies of Profit

Studies in biocapital explore relationships between forms of life and forms of exchange, including phenomena as diverse as the market value of body organs; attempts to patent genetic material; slavery and human trafficking; economic value of reproductive material and services; and clinical trials of pharmaceuticals (Dumit 2012; Franklin 2003; Rajan 2006, 2017; Rose 2007; Waldby and Mitchell 2006). Biocapital involves "product-making and profit-seeking" by transforming "biotic material and information" into "value, markets, wealth and profit" (Helmreich 2008:464). Biotic material and information can include entire populations or individuals, as well as parts of individuals from organs down to the level of cells, molecules, genomes, and genes.

> New capitalist practices see university and corporate biosciences becoming porous to one another, with the circulation of biomaterials between labs governed by novel regimes of buying and selling—regimes set in place by regulatory transformations, primarily in the United States, that permit the holding of intellectual property in biological matter and knowledge. (Helmreich 2008:464)

In the sections that follow we explore a number of case studies in biotechnology that highlight relationships between science, biology, culture, capital, and health. We present ways in which concepts of kinship and personhood have been constructed, deconstructed, and disrupted through increasing normalization of reproductive technologies, biotechnological imaginaries of race and ancestry, and

through the lens of designer baby debates. We look at ways in which molecular biotechnologies break down what it means to be human and ways in which imaging technology challenges knowledge construction and disciplinary boundaries. We examine ways in which market pressures and the culture of the laboratory influence scientific ideas and challenge classic scientific claims of objectivity. We finish by looking at the ways in which combinations of biotechnologies are key in constructing our ideas of self, family, and society at large.

Biotechnology is applicable to all manner of life. In the context of health anthropology, questions of biotechnology tend to revolve around our constructions of the human-self and society. In popular discourse, biotechnology is often reified, given a kind of superhuman agency. From an anthropological perspective, however, the sophistication of technology does not stop it from being a tool for the expression of human desires and the construction of human designs. As we alter the world around us, we also alter ourselves. Biotechnologies are tools that we use on ourselves. Subsequently, the transformations we generate with these technologies are loaded with reflections of our norms, aspirations, and anxieties, and with reflections of power relations, ethics, and inequities of the sociocultural contexts in which they are embedded.

Reproductive Technologies

In the last half-century, in middle- and upper-class industrialized communities across the globe, there has been a profound separation of sex and reproduction (Bates 2010:93). With the advent of reliable and accessible contraception, it is ordinary to have sex without reproducing. With the advent of IVF and associated technologies, it has become technically possible, socially acceptable, and increasingly normalized to reproduce without having sex. Developments in reproductive technology have not only meant changes in the way babies are conceived, but they have also had a fundamental impact on the ways in which we understand (conceive of!) kinship and personhood. Donor sperm has paved the way for social acceptability of a person's biological father and social father being different people. Surrogate gestation, where an embryo is implanted in a woman who is not necessarily the child's biological mother, means a child may be carried and birthed by a woman who is neither her biological mother nor the mother who will raise her. The section that follows will consider the way that biotech-facilitated procedures like IVF, surrogacy, and biomedical interventions in gender complicate and disrupt Western notions of kinship, as well as enrich our understandings of it.

Early anthropology kinship diagrams did not differentiate between different types of mothering and fathering: each person on a kinship diagram had a linear relationship with a male person and a female person who were assumed to be their social and biological mother and father.

Figure 8.1 shows us Hulya, her mother, father, older sister, and younger brother. Aspects of relatedness that did not fit the acceptable Euro-American family model, such as Latin American comadres and compadres (co-mothers and fathers) or milk kinship (relatedness established through breastfeeding) were

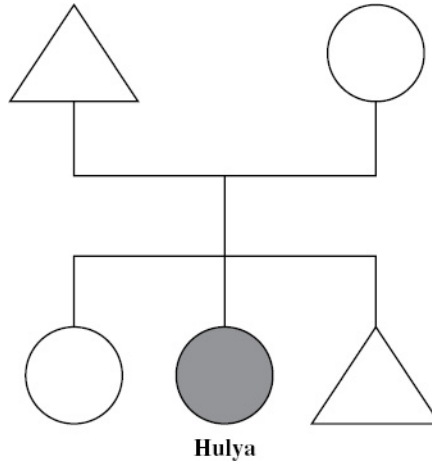


FIGURE 8.1 Huyla's Family.

delegitimized and labeled "fictive" kinship (Altorki 1980; Khatib-Chahidi 1992; Long 1996, 2003; Schneider 1984; Yanagisako and Delaney 1995). So-called fictive relationships were drawn on to kinship diagrams in dotted lines and were differentiated from "real" "blood" relationships. Schneider, in making the point that blood relatedness was a metaphor rather than a scientific fact, drew our attention to ways in which representations of relationships in kinship diagrams conflated the Euro-American concept of blood kinship with a normative social kinship based on Euro-American ideas of what a normal (i.e., good, moral, acceptable) family was (Carsten 2011, Schneider 1984, Strathern 1992, Yanagisako and Delaney 1995).

Looking at the blood groups of family members of one of the authors of this book illustrates this. Figure 8.2 shows the blood types of Debbi, her mother,

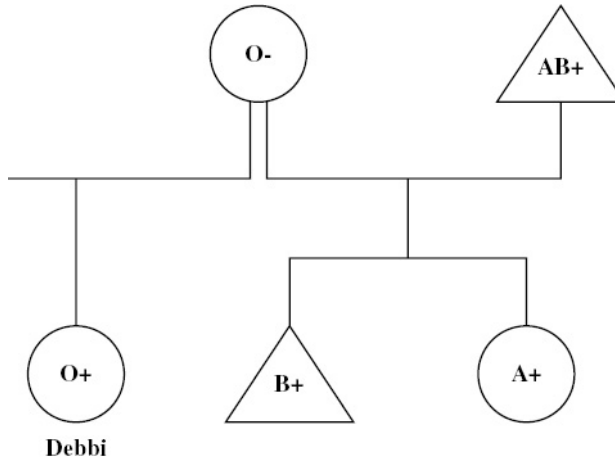


FIGURE 8.2 Blood Types in Debbi's Family of Origin.

her mother's husband, and their two children. Even though blood relatives are understood to "have the same blood" in folk understandings of Euro-American kinship, it is possible for people to be related and have completely different blood types.

In this case, although everyone is related "by blood" to two or more of the others in the diagram, not one of them share the same blood type. Although Debbi's mother could safely donate blood to everyone, Debbi could not safely donate blood to her mother, while she could safely donate blood to her stepfather, to whom she is not related "by blood."

At the same time, people totally unrelated to each other can be the same blood type. Figure 8.3 shows Debbi, her son, and the father of her son. Although Debbi and her son's father are from different continents and are not genetically related, they and their son all share the same blood type and can safely donate blood to each other.

With the advent of what Nelkin and Tancredi (1989) labeled the new reproductive technologies (NRTs), the ways in which the relatedness of a new human coming into the world is understood has shifted dramatically in Euro-American cultures. Prior to the era of the new reproductive technologies, while adopted children could have different biological and social parents, the "normalized" child-parent relationship was for a child's mother to be his or her genetic, birth, and social mother, coupled with the expectation that the social father should also be the genetic father.

In some communities around the globe it is now not only possible, but also not especially uncommon, for children to have five different people as part of their parenting cohort: social mother and father, genetic mother and father, and gestational mother. The role of biological mother was previously indivisible. NRTs provide for the possibility for the genetic mother and the woman who carries the child to not be the same person. Genetic and gestational motherhood are no longer inseparable (see figure 8.4).

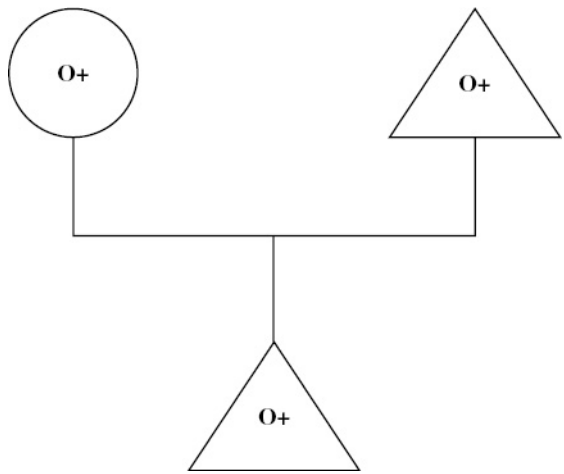


FIGURE 8.3 Blood Types in Debbi's Family of Procreation.

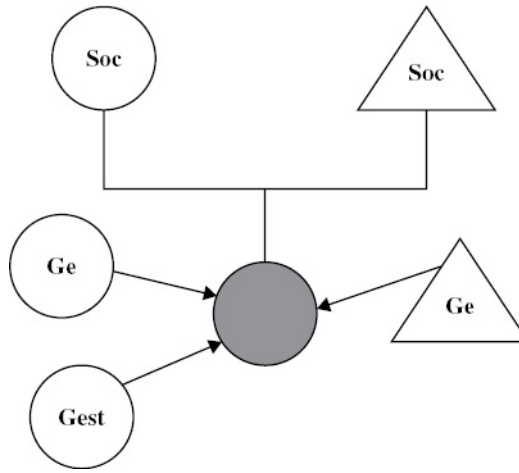


FIGURE 8.4 Social (Soc), Genetic (Ge), and Gestational (Gest) Relatedness.

The increasing divisibility of parenthood is occurring at a time when gender categories are increasingly, and heatedly, contested. Binary gender configurations, where there are just two genders, male and female, and every individual is either one or the other, are being challenged by increasingly fluid gender configurations. While there is a long-standing body of scholarship in anthropology around "third gender" categories (Herdt 1996), binary gender categories have, in recent years, come under much broader public scrutiny. Fluid gender categories are challenging biological determinism, while hormonal and surgical technologies are creating possibilities to change aspects of personhood that were previously seen as inalterable. For many transgendered people, this allows them the possibility to inhabit a body that is in alignment with their internalized sense of gendered self (Herdt 2010, Valentine 2007).

At the end of the twentieth century, there were few countries in the world where same-sex couples could be socially and legally recognized as parents. In less than two decades, the marriage equality movement has meant that in many countries, same-sex couples, or people in gender fluid relationships, can have their kin relationships with children and each other legally recognized. A kinship diagram in an anthropology textbook with parents of the same sex would not have made sense thirty or forty years ago. Now because of technological and social changes in the ways that families are conceptualized, it makes perfect sense for many people (see figure 8.5).

The case of reproductive technology demonstrates how entangled the social and technological are. Carol Delaney suggests that in all cultures, beliefs about creation and procreation are inextricably interlinked (1991). The beliefs we have for how life on earth came into being are consistent with the logic of our beliefs for how new lives come into being. Adoption of new reproductive technologies brings new concepts about creation into the world and shifts the understandings of the communities they impact in their relations with each other and with their environments.

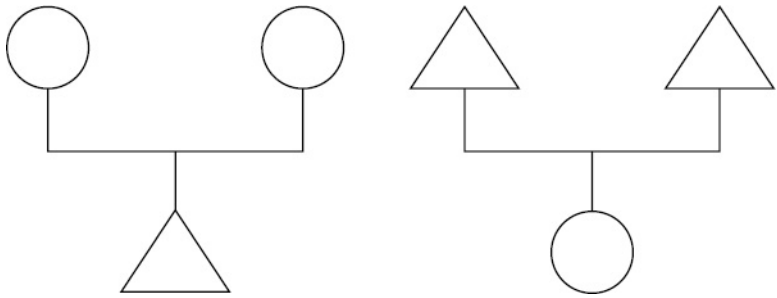


FIGURE 8.5 Kinship Diagrams with Same-Sex Parents.

Reproductive technologies have introduced new ways in which inequities of maternal opportunity are enhanced. There is a long history of women oppressed by status beliefs (including class, color, and/or ethnicity) being forced to relinquish part or all of their own maternal relationships to perform mothering functions to the children of wealthier families. With the advent of NRTs, new inequities have arisen whereby poorer women sell their eggs and/or bodies to contribute toward the production of children for wealthier families (Lock and Nguyen 2010, Pande 2011, Panitch 2013, Whittaker 2014). In an example of biocapital exchange, local beliefs, economics, and laws combine to generate many manifestations of surrogacy relationships: in the United States laws do not prohibit paid surrogacy, and there are many poorer American women who act as paid surrogates for wealthier families, sometimes with their own eggs, sometimes being implanted with a fertilized egg of the social mother and/or father. Famous examples of celebrity children born as a result of surrogacy relationships include Michael Jackson's children and the children of Nicole Kidman and Keith Urban. In Australia, where the law prohibits paid surrogacy, there are established "trade routes" of families using Thai, Indian, and other impoverished women of color as paid surrogates. In Europe, impoverished Eastern European women act as surrogates for wealthier Western European families. In most cases of paid surrogacy, the pregnancy is not initiated by sex, but in a test tube, and transfer into the womb is facilitated with a speculum, pipette, and dry ice storage to freeze and keep the embryos viable until they are ready to be implanted into the surrogate womb.

Divisible Bodies

While surrogacy involves commodification of whole humans, intersections of globalization, wealth inequity, and biotechnologies have created new "industries" in body parts in the form of body organs being made available, through donation, sale, or theft, for transplant. Anthropologist Nancy Scheper-Hughes (2000) has been central in bringing to public attention global trafficking in transplant organs, while other ethnographic accounts have focused on the meanings of the relationships created through donor transplants (Kisch et al. 2018). Organ transplant is one of the areas where the multidisciplinary nature of medical

anthropology is most stark: for anthropologists this job entails holding power accountable when abuse is uncovered; for many clinician-anthropologists the strength of ethnography is in providing insights that can enhance the effectiveness of medical systems, rather than challenge them.

Permeating all of these areas is the enormous variation within and between cultures in etiquettes, norms, and values around corporeal reciprocity. Taking anthropologist Marcel Mauss's concept of reciprocity articulated in *The Gift* (1925), British sociologist Lloyd Titmuss's landmark book *The Gift Relationship: From Human Blood to Social Policy* (1970), which was named by the *New York Times* as one of the ten most important books in the year of its publication, compared the United Kingdom's system of blood donation in which blood was given without economic recompense, and seen as a gift from donor to recipient, with the system in the United States in which blood is framed as a commodity that the donor sells and the recipient buys. Legal academic Kara Swanson's work *Banking on the Body: The Market in Blood, Milk, and Sperm in Modern America* (2014) explores the commodification of what Long (2016:7) refers to as the nutritive and procreative body fluids. Health anthropologists have made significant contributions to discourses on commodification of the body (Scheper-Hughes and Wacquant 2002). We suggest that there is also value in the anthropological approach to understanding these complex reciprocities as globalized exchanges, some of which involve profit making, but many of which do not.

With the exponential growth of genomic and diagnostic technologies, human beingness is being dividuated (Ott 2018) and globalized in increasingly smaller increments. These increments are transformed into market-ready products, ready for participation in the process of biocapital exchange. While in the 1970s and 1980s reproductive technologies focused on the creation of new humans were emerging, and in the 1990s human body parts were enabled by technology to develop into a global market, a much more recent trend is the emergence of global networks of biobanks that are "structured collections of biological samples and associated data, stored for the purposes of current and future research" (Parodi 2015). Ethnographic attention on the high-tech, globally networked, biobanking industry with its associated utilization of emerging big data set analytics has drawn attention to challenges of standardization of sample collection, heterogeneity of sampling, and ethical considerations around confidentiality and consent (Mason et al. 2016), and has shown that even when disembodied, human tissue retains a "persisting personal and cultural identity" (Dive et al. 2017:301).

Many of the implications of recent developments in big data set analytics are yet to be explored, but one that health anthropologists will no doubt be watching is the implications of big data set analysis on population screening. There is an important difference between diagnostic testing and screening that anthropologists have commented on, that goes almost unremarked in clinical literature. Whereas a diagnostic test is undertaken on a person who is showing signs of a disease, a screening test is undertaken on a healthy, or at least asymptomatic, person who is deemed to be in an at-risk category for a disease. This slippage from testing people to diagnose an already present set of symptoms to

screening an asymptomatic group of people is an increasingly ordinary occurrence in biomedicalized health care (Kaufert 2000). As costs of particular tests go down, statistics become more available, or pharmaceutical companies search for increased sales for diagnostic tests, it is not uncommon for an increasingly larger group of people to have screening recommended. More than medical factors form the basis for recommendations of who should be screened: economic and political factors also play a significant role (Sandall et al. 2004). As Nelkin and Tancredi (1989:10) note, Western biomedical reductionism and the faith in simplistic, numerically constituted data "has obscured the uncertainties intrinsic to such diagnostic tests, and they are widely accepted as neutral, necessary and benign." They suggest that "the boundaries of the 'normal' or the 'healthy' are often fuzzy. . . . When interpretive margins are fluid, they can easily be manipulated by diagnostic evaluations cloaked in the neutral garb of science" (1989:19), warning that the use of diagnostic tests for population screening "has always been directly linked to social or medical intervention, through therapy, prevention or exclusion" (1989:23–24).

Bringing the Lab into the Field: Anthropology and the Neurosciences

The development of biotechnologies has significant implications for the way research into cognition and human nature is conducted. In the field of neuroscience and clinical psychology, new biotechnologies are forcing cognitive scientists to deal with questions that have previously been the focus of anthropologists. Recent developments have started to undermine long-standing disciplinary boundaries and are bringing anthropology into closer contact with other scientific and medical traditions. Health anthropology, with its close connections to the biomedical and the anthropological fields, stands to be the connective link between the neuro-cognitive and the anthropological.

Anthropology and the cognitive sciences share a common subject matter—humanity—but have historically approached their subjects in entirely different ways. Cognitive scientists have a strong interest in the internal mechanics of cognition. The individual human subject is the focus of inquiry. Neurology and psychiatry are primarily about the mechanics of the brain and related biological systems. In psychology, the subjects' interpretive schema and cognitive frameworks are the focus. Anthropology takes an interest in these matters but is largely focused on the shaping effect of environments and social and cultural structures. This difference in focus has been key to the separation of these disciplinary domains. It also underpins their methodologies.

Neuroscience, psychiatry, and psychology have traditionally conducted their studies within a clinical setting. In these disciplines the patients and research subjects were expected to come to the scientists, who would apply the laboratory ideals of isolation and controlled observation to deduce principles of brain function and pathology. The clinic, like the laboratory, houses the various biotechnologies that facilitate the study of the mind. The clinical environment is sterile and can be easily controlled, allowing clinicians to regulate stimuli that

patients/research subjects are exposed to. Imaging machines and other medical diagnostic equipment can be bulky and, due to their complexity, require specialist knowledge to operate. Biotechnologies are often tied to a technological infrastructure that allows data to be collected, analyzed, and interpreted with each step, often needing additional equipment and specialized conditions. For these reasons, the clinic makes a logical and convenient site for the cognitive sciences to operate from. Anthropologists, on the other hand, do their work largely in the places that their subjects routinely occupy.

These circumstances mean that cognitive scientists and anthropologists work with different samples of humanity. It is important to note that this trend is not without exception. For example, social psychologists take their subjects from a variety of different locales and physical anthropologists often work in lab settings. However, the trend toward limited sampling and lab-centered studies is well noted in cognitive sciences. Psychologists Joseph Henrich, Steven J. Heine, and Ara Norenzayan (2010) drew on anthropology and other disciplines to point out the dangers of this sampling trend in the behavioral sciences, suggesting that the people who are the most common subjects of research in fields such as psychology, economics, and cognitive science are in fact outliers on the spectrum of humanity. These people, whose behavior and neurology are supposed to be representative of humanity at large were, in fact, "weird":

> Western, Educated, Industrialized, Rich, and Democratic (WEIRD). In particular, it is about the Western, and more specifically American, undergraduates who form the bulk of the database in the experimental branches of psychology, cognitive science, and economics, as well as allied fields. . . . The sample of contemporary Western undergraduates that so overwhelms our database is not just an extraordinarily restricted sample of humanity; it is frequently a distinct outlier vis-a-vis other global samples. (2010:61, 82)

This article has set the stage for disciplinary crossover, with authors suggesting that cognitive scientists need to diversify their subject base and that anthropologists need to get more involved in experimental studies in their various fields (Henrich et al. 2010:29). However, despite best intentions, the practical restrictions of biotechnologies have made looking at an active brain in a natural environment difficult. Isolation in clinical conditions and relative immobility have been mainstay requirements of neuroimaging. Size of machinery and the desire to control stimuli have contributed to the methodological and disciplinary gap between neuroscience and health anthropology.

Technologies are changing, however. The pace of technological change is increasing as our machines become increasingly sophisticated and integrated into daily life. Biotech is no exception. Brain scan technologies are becoming increasingly compact and resilient. Wireless transmission has allowed the subject of study to transmit relevant data as they leave the lab behind. Experiments with a portable electroencephalography (EEG) system have allowed people to monitor their electronic brain function in real time with their smartphones. This process required the subject to wear an unfashionable swim cap, but the device was lightweight and relatively unobtrusive. This allows the subjects to engage in their

everyday activity as they were being scanned (Stopczynski et al. 2014). A more sophisticated positron emission tomography (PET) has also been tested (Bauer et al. 2016, Melroy et al. 2017).

These early applications of portable biotechnology have delivered new data and have posed challenging questions about previous brain-scan findings. Being able to record the EEG information of people cycling through the streets of Alberta, Canada, has allowed researchers to see that "noise" is a part of everyday life function. In other words, divided attention is a part of complex daily activity and reaction tasks in the real world require a different kind of brain function to reaction tasks done in the lab (Scanlon et al. 2017).

For health anthropologists and cognitive scientists, this technological progression opens a wide range of application. The ability to observe brain function in real time has been used by anthropologists to see inside the heads of their participants by cross-referencing observational data with real-time measurements of arousal, relaxation, attention, and stress as they go about their everyday life (Yasui 2009). Evolutionary anthropologists theorized about the brain structure of our ancestors by monitoring the brain function of contemporary humans engaged in making stone tools (Putt et al. 2017). This kind of research allows medical, physical, and evolutionary anthropology to transgress on the territory previously held by cognitive scientists. Simultaneously, the cognitive sciences stand to benefit as these increasingly portable technologies venture more and more outside of the laboratory.

Molecular Biotechnologies: Tiny Pieces, Giant Infrastructures

Over the last several decades, technological and scientific advances have given people the ability to manipulate the most fundamental blocks of life. Human beings can now transform the function of organisms on a cellular and molecular level, allowing us to treat illness and create new material compounds and even life forms. Molecular biotechnology evokes images of pristine laboratories, complex machinery, advanced treatments, large pharmaceutical brands, and armies of lab coat–wearing scientists. This is a type of biotechnology that regularly captures the public imagination. It represents the level of our current scientific ability and achievement. These biotechnologies are drugs, genetic and cellular therapies, hormones, and vaccines. They allow us to alter the behavior of cells, bacteria, and DNA and to direct their function.

There is a mythology that surrounds the large biochemical and pharmaceutical companies that manufacture these products. That mythology rests on two common presumptions. The first presumption is that science research is driven by an objective credo of scientific progress. In this view, discoveries are shared and built upon by an integrated community of researchers all working toward a unified idea of progress. The second presumption is that research direction in medical biotech is based on a hierarchy of need—the desire to improve health and cure disease. These presumptions are the epitome of humanist enlightenment values and are assumed to be encoded in scientific ethics and the medical

Hippocratic Oath. These presumptions work to instill biotechnological users and manufacturers with credibility in the eyes of the public and hope for the patient. This is what anthropologist Mary-Jo DelVecchio Good (2001) refers to as the "medical imaginary" and the "biotechnical embrace." This medical imaginary is actively promoted by biotechnology companies and the medical establishment (see figure 8.6).

Working with DNA, the "essential building blocks of life," is a resource-intensive and labor-intensive process. Laboratories function to isolate biological fragments, and through this isolation lay claim to profits from the production of life-saving medications, therapies, and objective knowledge. These technologies are often thought to be produced through processes of objective science. Those who make them require substantial resources and wield tremendous power. Health anthropologists approach biotechnological products as agents in a human process, intertwining in assemblages with their own mythologies, agency, social context, and culture. Because they are so complex, biotechnologies and the people who produce them are enmeshed in a vast infrastructure and are never outside of economic and cultural pressures (Fukushima 2016). Far from being objective and outside of the influence of subjectivity, biotechnologies are products of a culture of science and capitalism.

The development of molecular biotech rests on a promise and an idea. For those participating in the product-making and profit-seeking, this is a direct investment in the biocapital of our molecular makeup. The development of a new drug or therapy is contingent on an investment in an idea and, in a contemporary capitalist environment, the promise of return. This introduces significant risk for those who are seeking to invest in the expensive process of research and

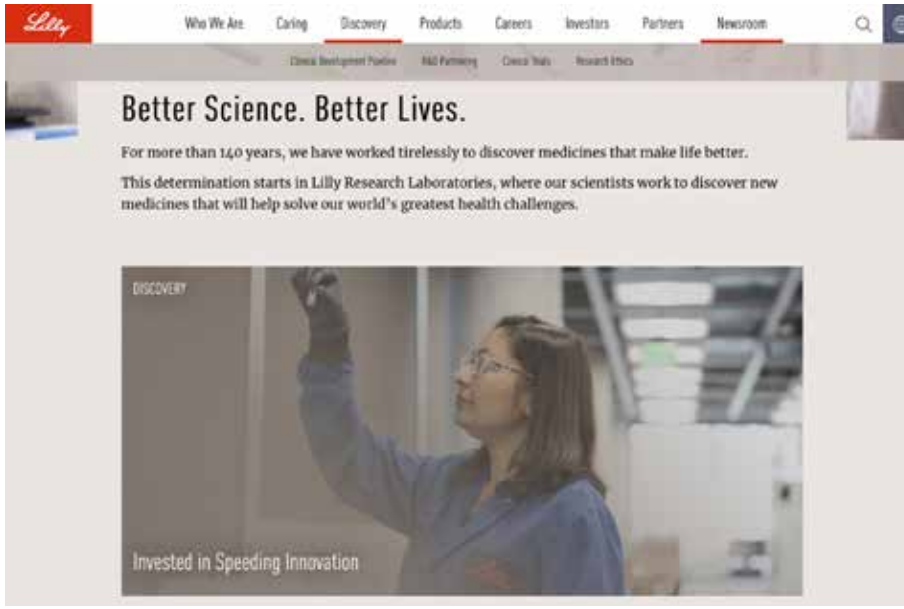


FIGURE 8.6 Biotechnology Ad Promoting Biochemical and Pharmaceutical Mythology.

development and this risk is worked into the infrastructure and culture of biotech production. Biotech developers make partnerships at every stage of production and marketing of a new treatment or drug, creating networks and connections to facilitate the development of each product. Charles Mather (2005:1324) has described this process of development as the "drug pipeline":

> The drug pipeline is not seamless. Movement across stages involves institutions and individuals from different domains or sub-cultures including science, medicine, industry, and government. This makes the drug pipeline different from that of other products and developmental assets. . . . One of the dilemmas that pharmaceutical companies face is making the drug pipeline more predictable and thereby ensuring product development and market share. To make the pipeline more predictable companies aspire to control settings and activities outside the domain of industry. Industry seeks to control the pipeline by funding the research programs of academic physician and physician–scientists, financially supporting conferences, rounds, and journal clubs in hospitals and medical schools, running continuing medical education events (CMEs) for non-academic physicians, and giving physicians a wide range of gifts including free drug samples.

This perspective challenges the objective and clinical medical imaginary of biotech and pharmaceutical production (Singer and Allen 2017). It encourages us to see biotechnologies not as passive products of the pure scientific rationality driven by need and progress, but as complex agents meshed in complex structures. Mather (2006) describes biotech products as "entities" that mature through exposure to different stages of development, grow in complexity, and even experience rites of passage.

The development of retroviral drugs to combat HIV/AIDS offers a number of well-documented case studies of the complex intersections between local, national, and international public health policies and practice; local, national, and international activists and advocates; Big Pharma (the global industrialized pharmaceutical industry) and international patent law. While many of the early retrovirals were tested in impoverished communities, once they were registered for use by the Food and Drug Administration (FDA) in the United States, prices became prohibitive for patients in most developing countries. The 1995 TRIPS Agreement (Trade-Related Aspects of International Property Rights), promoted by wealthy nations, mandated minimum standards for the protection of international copyright, including pharmaceutical patents. Although there were some exemptions, protection of patent rights (argued on the basis of the expense of the pharmaceutical pipeline development costs, as discussed above) put AIDS medication out of reach of many people in the developing world. In 2001 the Doha agreement created wider exemptions for developing countries to produce generic medications that made public provision of HIV medication affordable, although the uptake of these has been uneven (Wise 2006). In South Africa, stigma around HIV prevented effective retroviral accessibility for many years (Fassin 2007b). The successful campaign for retrovirals in Brazil has resulted in what Biehl (2007a:1183) refers to as a "pharmaceutically-centered model of public health" in which "the sustainability of the Brazillian AIDS policy has to be constantly renegotiated in light of global drug markets," while Hickel

(2012:513) suggests that the continuing "high prevalence of HIV in Swaziland [now eSwatini] is less a biomedical condition than a symptom of neoliberal market policy." As the global pharmaceutical industry grows ever more complex, the imperative increases for health anthropologists to continue to pay attention to what Whyte and colleagues (2002) refer to as the "social life of medicines" (Hardon and Sanabria 2017).

The Story of hGH—Growing up Growth Hormone

Human growth hormone (hGH) is a hormone that is naturally secreted in childhood from the pituitary gland to stimulate growth. Individuals who are unable to produce this hormone naturally are a well-established medical treatment group, experiencing a range of conditions, most notably pituitary dwarfism. Prior to 1985, these conditions were treated with hGH extracted from cadavers. These treatments came with significant health risks and supply of extracted hGH was insufficient to meet medical demand (Conrad and Potter 2004). With a need identified, the biotechnology company Genentech established a partnership with pharmaceutical manufacturers Eli Lilly and Novo Nordisk to create a drug pipeline for a synthesized version of hGH created through recombinant DNA processes.

In 1985 use of cadaver-extracted hGH was banned by the American Food and Drug Administration. Genentech introduced a synthesized hGH to the market in time to fill the void. This version of hGH could be manufactured in unlimited qualities and the medical need was met. However, the infrastructure for this pipeline was extensive, with a multitude of interested parties seeking return on their investment that the original targeted patient group could not provide. Mather (2006) presents hGH as an example of a biomedical entity with too much infrastructural momentum to stop at having met medical need. Invested parties sought to expand the application of hGH beyond prescribed use.

Anthropologist Alan Goodman (2013) describes height as a condition of economic and genetic circumstances. Nutrition and genes both influence how tall we grow and, for some, height can be an indicator of a socially favorable childhood or "good breeding." Inversely, shortness has been associated with inadequacy and stigma. Goodman notes that synthetic hGH could now be used to redress these potential "inadequacies." hGH is a recombinant DNA treatment, a supplement to natural biological processes, and was subsequently applied to a population that was previously not a target of biotechnological intervention. Borrowing from literature in the biotech and medical fields, Goodman (2013:365) notes that "hGH injection transforms short stature, a signifier of individual genetics or group-resource access, into a medically treatable disease." In direct contradiction of the biomedical imaginary of need-based intervention, due to the economic and social pressure of infrastructural requirements of the hGH pipeline, new biomedical need was created.

Medical sociologists Peter Conrad and Deborah Potter (2004) describe a situation that went even further. Seeking to further extend the market, the hGH industry sought applications outside of medical pathology, into human

enhancement. In addition to treatment of growth hormone deficiency, Conrad and Potter document the use of hGH in antiaging treatments, for cosmetic purposes, and for athletic performance enhancement. A long-standing federal investigation into misuse of this biotechnological intervention found that Genentech had actively cultivated relationships with researchers, medical institutions, and professionals to promote its product. In 1999 Genentech paid $50 million to settle criminal charges from the FBI and the FDA. The FDA alleged that "Genentech had 'begun marketing Protropin [a strain of synthetic hGH] for use in the treatment of medical conditions for which it did not have FDA approval'" (Conrad and Potter 2004:191).

The Culture of PCR

The case study above challenges our ideas of the biomedical imaginary and demonstrates that the weight of biotech infrastructure and ever-expanding corporate hunt for profit drive the development of biotechnology beyond the requirements of human medical need. The push for the dissemination is the result of biotechnological infrastructures, but this push also shapes the culture of development in the lab and in the field of research. Medical anthropologists, working alongside research scientists and within biotechnology companies, paint a more complicated and holistic picture of this culture.

Paul Rabinow (1996) offers one early case study in his work on the development of polymerase chain reaction techniques. The development of this technology demonstrates the human and cultural side of the biotechnological infrastructure and process. Rabinow begins his account in the 1970s and 1980s in the United States with a series of legislative and judicial changes specifically designed to turn scientific discoveries into patentable products. Then, and to this day, biotechnologies were and are a high risk and return industry. New technologies offer huge financial returns, but, because these technologies take time and money to develop, they are often risky investments. With sparse direct government funding, researchers in both the private and the not-for-profit sector enter into partnerships with the business world to fund their work. In exchange, researchers make promises of returns. This generates pressure to pick projects on the basis of viability, time, and the commercial need of investor clients. For example, Rabinow (1996:51) notes that private biotech developers were likely to take on contracts to produce new patents for variations of existing commodities, rather than developing completely new technologies. Similarly, diagnostic technologies were considered to be less risky than working on complicated cures.

Polymerase chain reaction (PCR) was developed as a diagnostic tool through the work of Cetus Corporation. PCR is a method that detects and replicates a predetermined sequence within a chain of DNA or RNA inside a solution. If a genetic sequence exists within a strand of DNA it will be replicated (this is called amplification); if it doesn't then no amplification will occur. This can be used as a quick method of finding sequences that relate to genetic diseases or conditions, verifying heredity, forensic comparison, or to produce an abundance of testable genetic material.

This is a highly versatile technology that has since been widely applied across a wide swath of industries and disciplines. Even a degraded piece of DNA can be tested with PCR methods. It has been used in cloning and DNA sequencing and was an essential tool in the mapping of the human genome. Outside medicine it has been applied in environmental microbiology to verify sources of biotoxins within an environment.

The story of the development of this prolific biotechnology offers many insights into the culture and practices of the biotech industry. Rabinow describes a tense, deadline-driven climate. There was regular friction between members of the research and marketing teams, with scientists often being forced to scramble to keep up with promises made by marketing executives. Information about the process development was seen as a valuable commodity, with tight restrictions on the dissemination of the research progress and constant worries about the activities of competing scientific teams. This serves as an example of the protection of biocapital investment, in the form of patented genetic material, over the romantic scientific ideal of collaboration for the purpose of progress. Rabinow (1996:10–11) notes that in the story of PCRs, the romantic view of science as a pure, objective, cooperative, and unified effort is not evident:

> Anyone the least bit familiar with the history of science . . . could hardly describe it as "essentially cooperative and communal effort." This view of contemporary bioscience—shared by many—is historically and anthropologically inadequate.

Visualization Technologies

Another area of technological development in health that has come under scrutiny from health anthropologists is the explosion of medical imaging technologies that has allowed information from inside the body to be accessible to the external gaze. X-rays, ultrasound technology, MRIs, CAT scans, and PET scans have shifted ways in which health and illness are understood and communicated (Braidotti 1991, Burri and Dumit 2008, Downey and Dumit 1997, Joyce 2005). Ultrasound images of unborn babies have profoundly shifted maternal-fetal relationships: being able to visualize a baby in utero changes ways in which mothers and other family members bond with their child to be (Nabhan and Faris 2010). Ultrasound produces "both a newly independent fetus and a medically reinscribed mother under the powerful sign of normative maternal-child health" (Rapp 1997:32). The ultrasound photo has become a common "first photo," extending a child's place in their families back to include their time in utero and shifting relationships between families and unborn children.

Joyce (2005) describes ways in which MRI (magnetic resonance imaging) scans move from being a diagnostic tool utilized by health professionals—doctors or radiographers—to help understand enfleshed, corporeal bodies, to the machines being ascribed agency and voice in their own right. Rather than merely "revealing" the inner body, Joyce (2005:438) suggests that MRI scans "produce the body, bringing together aspects of physical bodies and cultural, social, and economic factors in unique and locally specific ways." Joyce (2005:438) shows how representations of MRI scans are authoritative and transparent, and the

privileging of medical imaging technology over other forms of diagnosis, including physical examination of the body of the patient, can lead to a neglect of "the importance of interpretation in the production of health and illness."

When Technologies Combine

In this section, we present examples of unintended consequences of combinations of biotechnologies, and for this we return to the area of reproduction and the impact of biotechnologies on definitions of personhood for yet-to-be-born persons. Previously, we discussed ways in which new reproductive technologies have created profound shifts in understandings of family and relatedness. In this section we will look at how technology is intertwined with discourses around acceptable and desirable categories of personhood.

The average age of women birthing in most Western countries is rising, and alongside that is an increase in anxiety about delivering babies with so-called birth defects and/or genetic anomalies. The diagnosis of fetuses with chromosomal differences that indicate the child could be born with a disability, and the subsequent termination of pregnancies found to be compromised, involves a range of medical technologies, equipment, knowledge, and specialized clinical environments. There are a number of individual technologies that combine for second-trimester terminations, including amniocentesis, ultrasound, and genetic testing. Amniocentesis involves the gathering of an amount of amniotic fluid, the fluid that surrounds the baby in the womb. Cells of the fetus/baby are present in the amniotic fluid, meaning that a baby's genetic information may be obtained prior to its birth. Amniotic fluid is obtained with a needle inserted into the uterus through the mother's abdomen. The accuracy of the placement of the needle into the uterus is guided by high-definition ultrasound. The interpretation of the genetic material is undertaken in a specialist laboratory. Based on the medical interpretation of findings, terminations may be undertaken with active obstetric intervention.

As with new conception technologies, diagnostic and termination technologies have generated deep ethical and moral contestations. Beck-Gernsheim (2000:131) comments on euphemisms used in discourses of disability prevention:

> More is at stake here than dental hygiene. What is actually meant here is avoiding the birth of a handicapped child, either by way of renouncing biological parenthood altogether or (more likely in practice) by way of a "tentative pregnancy" (Katz-Rothman 1986) and induced abortion in the case of a genetic deficiency.

The speed with which prenatal diagnostic screening became routine was remarkable. Rapp (1997:36) commented:

> Prenatal diagnostic technologies have moved out of academic settings into commercial labs with great speed over the last decade, and the most experimental technologies are often available, for a price, long before their accuracy is fully tested.

First- and second-trimester pregnancy terminations are very different procedures. The medically measured length of a human pregnancy is taken as forty weeks.

First-trimester pregnancy terminations, undertaken at up to thirteen weeks, however contested and ethically problematic in some communities, are relatively straightforward procedures from a medical perspective. Second-trimester terminations, undertaken at between thirteen and twenty-six or twenty-seven weeks, when the embryo is well established as a fetus, are more complex medical procedures. They are significantly riskier for the mother's health, and can be more emotionally complex for both families and the health-care workers.

Based on biostatistical information that more children with "birth defects" or "chromosomal anomalies" are born to older women, in most Western countries, women over a certain age are advised to undertake an amniocentesis, and to terminate the pregnancy should the test indicate that the baby has a "genetic abnormality." When the use of this technology first became widespread, there were ethical concerns around "designer babies." The fear was that amniocentesis would be used to facilitate the genetic testing of a fetus, and fetuses who weren't carrying the characteristics of what their parents required as their "perfect" baby, would be aborted. Characteristics such as height, intelligence, hair and eye color, and sexual preference were all seen as things that parents may choose to abort for. These fears did not immediately eventuate: the mapping of the genome made it clear that genomics is much more complicated than being able to read off a genetic sequence to determine your baby's height, hair color, or potential heterosexuality. In other areas, where there is more certainty offered by the results of the amniocentesis, second-trimester terminations have become increasingly incorporated as a normalized aspect of pregnancy and of maternity care. We will discuss two examples: the termination of babies carrying the gene for Down syndrome in the United States and Australia, and sex selection in India and China.

Anthropologist Rayna Rapp initiated long-term fieldwork in a genetic clinic in New York. Her introduction to amniocentesis and second-trimester terminations came through her own experience, movingly documented in *XYLO: A True Story* (Rapp 1984). Ethnographic research points to troubling bioethical dilemmas between families expecting babies and the medical professionals who care for them (Rapp and Ginsburg 2011). Down syndrome, the iconic condition most commonly referred to when discussing genetic testing, is seen by most doctors as an unarguable reason for pregnancy termination. Although for some women and their partners this is a comfortable decision, for many it is not, and there are a number of examples in the ethnographic literature that point to women feeling coerced to terminate pregnancies (Long 2016; Rapp 2000a, 2000b, 2014). Long suggests that doctors are singularly unqualified to advise on Down syndrome terminations: they are in a profession where their identity is defined by intelligence, and their imagination falls short when it comes to envisaging potential quality of life with a lower-than-average IQ, whereas evidence from families with Down syndrome members suggests that it is much less cut and dried than that (Rapp 2014).

Another manifestation of the combination of amniocentesis and second-trimester termination technologies can be seen in the Fifty Million Missing movement in India, a social media campaign that has been set up to draw attention to the impact of widespread and systematic prenatal sex selection. In use now for more than thirty years as a sex selection technique, the use of

amniocentesis to determine the sex of the fetus, and the termination of pregnancies carrying female fetuses, has become so widespread in a number of countries, including India and China, that the practice has been significantly influencing sex ratio demographics for more than two decades (Allahbadia 2002, Sen 2005).

Disability advocates have long been vocal in their critique of genetic testing and termination (Finger 1984, Gammeltoft and Wahlberg 2014, Kaufert 2000, Rapp and Ginsberg 2011, Saxton 1984). Many of the early concerns of this discourse are now part of day-to-day maternity care (selective termination of Down syndrome and Spina bifida pregnancies, for example), while other fears of gene manipulation to create designer babies had not, until recently, been scientifically likely to eventuate. In 1987 Japanese scientists working in bacterial immunology discovered what is known as the CRISPR sequence. In later research conducted in the late 1990s it was discovered that CRISPR was a unique defense system that allowed some families of bacteria to "hack up and dismantle the viruses—memorizing, interrogating and cleaving foreign DNA" (Kirksey 2016). This system has been harnessed to be able to identify and alter specific gene sequences within a complete genome, making a kind of molecular scissor that allows for the direct modification of genes. Writing in *Anthropology Now*, Eben Kirksey (2016) describes CRISPR as a potential realization of the "designer babies" dream. Kirksey describes an interesting dilemma with the concept of human perfection: perfection is limited by our historical, scientific, and cultural imaginations. Intentional genetic alteration is now possible, and with sequencing technologies, there is a belief that we can control for a much more complex and wide range of characteristics. As noted below, health anthropologists working in this area make the point that desirable characteristics are historically, culturally, economically, politically, and philosophically idiosyncratic.

Prior efforts to create a community of perfect human beings on the basis of cultural ideals have demonstrated the authoritarian danger of applying a fixed moral order to a diverse human biology. Both European Colonial and Nazi ideas of racial superiority and of eugenics have cast a long and dark shadow over the history of biomedicine. Increasing global accessibility of reproductive technologies brings together complex intersections of cultural ideas about lineage, progeny, and ideals of parenthood. The question of agency has a special resonance in individualistic cultures like those prevalent across the Anglosphere. Many ethical trepidations about prenatal genetic intervention are reflections of the "Western" desire to preserve agency and are bolstered by a cosmopolitan appreciation for diversity. These are not universal human priorities. In a world where market forces play a central role in the availability of technologies, a user-pays model of genetics determines application. Less individualistic communities who see children in more directly utilitarian terms are increasingly turning to biotech to generate desired types of offspring. The biocapital available for sale here are the genetic components of idealized childhood traits.

Suli Sui and Margaret Sleeboom-Faulkner (2016) document a growing market for prenatal genetic testing in China. Businesses advertise being able to screen for beauty, health, and even talent. Though these claims are unlikely to be substantiated, the marketplace is relatively unregulated. Simultaneously, the cultural appetite is less tuned to implications on individual agency and cultural

ideas of filial piety and social conformity. In 2009, the city of Chongqing carried out the "talented baby project" where gene testing was used to test for fifty "talent gene" candidates, with those to be raised according to their talent attribution. This project articulates the desire to create a perfect generation, using the best of available state and market methods to create an ideal citizenry forged by a combination of directed nature and nurture. Unwanted elements, such as tendencies for violence, depression, or disability, are left by the wayside on the way to reaching an imagined socially constructed ideal.

One of the major appeals of medical biotechnology is the promise of certainty. Genetic testing offers the chance of looking back through a family bloodline to establish ancestral connection of the likelihood of a genetic illness. For the media and those in the business of selling genetic knowledge, ancestry information is sold as a kind of ultimate revelation of self. Ruha Benjamin (2015:134) has found that we are often tempted to talk about scientific projects in progress as if they were ready made—extrapolating possibilities and transforming them into imminent realities. This kind of certainty is highly appealing in light of parenthood. Each iteration of genetic biotech sparks a conversation about the dangers of runaway technology. In the 1990s and 2000s the ability to screen embryos prior to implantation through IVF sparked a series of discussions about our ability to design a perfect human. The argument went that our ability to foresee illness would lead to a new form of eugenics by design. However, ethnographic studies of the discussions of new tech (like IVF and preimplantation genetic diagnosis conducted by Sarah Franklin and Celia Roberts in 2006) reveal classic parental worries rephrased in new terms. They argue that parenting has always been a question of intervention, guiding, and design. Practices of prenatal care and postnatal parenting techniques have been prescribed by medical specialists since the 1800s, and direct medical intervention has been used to keep babies alive since the invention of medicine. Similarly, the practice of child selection has a long and dark history in the practices of eugenics and selective infanticide. As these technologies shift and change, they continue to reflect desires to pass on a particular type of genetic heritage or meet the requirements of societal norms and local kinship traditions.

Ancestry, Families, and Genetics: Biotechnology and Belonging

New understandings do not seamlessly replace old beliefs (Martin 1994:16). Among researchers, paradigms take time to shift. Among the general public, lay understandings of the processes of science can make it easy for new biotechnologies to be appropriated to new and sometimes uncomfortable ends. Scientific understanding can challenge old social structures and dispel prejudices, but it can also be used to create new identity categories and biases, many of which are legitimized with a veneer of assumed scientific objectivity. Recently, genetic testing has offered a contemporary example of the benefits and risks of the proliferation of new biotech. It also offers insight into our understanding of the idea of lineage and ways in which it relates to our conceptions of self, family, and ancestry.

Family has always been at the core of ancestry. One outcome of biotech-facilitated accessibility of genetic testing is how this technology has drawn on and resuscitated concepts of race. Anthropologist Jonathan Marks (2008) argues that the idea of race took root at the cross-section of the "Age of Reason" and the "Age of Colonialism" when people of Europe were trying to rationalize their encounters and relationships with global "Others." Marks talks about three scientific paradigms of understanding the concept of race.

The first, appearing in the 1700s, were descriptions of ideal types of people and had a moral and regulative tone. In familial terms, these were our perfect ancestors—mythologized ideal types—as imagined by an eighteenth-century Swedish scientist with a penchant for bureaucracy. Carl Linnaeus gave us the color coding system that underpins simple conceptualization of race today: red Americans, white Europeans, yellow Asians, and black Africans. This categorization was enthusiastically embraced in Europe but required some updating when the concept of evolution reconfigured European understandings of the animal world in the nineteenth century. To make race work within this new paradigm in which humanity interacted with environment, the focus shifted from the ideal individual to evolution of fitness to geographically specific places. To (mis)apply Darwinian notions of speciation, human difference was now attributed to environmental adaptation and, subsequently, geographical typing. Humanity was now to be sorted into discrete units that were modeled on the idea of species that could be linked back to a common ancestor. This new racial "family" was then linked to the idea of an ancestral home.

All of these conceptualizations of race came with entire armories of biotechnological equipment and method. Though this equipment was informed by scientific principles, the application of this gear was informed by a deeply ethnocentric political ideology. A wide array of surgical, measuring, and categorizing equipment failed to establish cause and effect (see the work of Lundy Braun [2014] for one of many examples) and uncritically served to enforce deeply subjective exclusionary beliefs. Similarly, relatively objective evolutionary ideas were adapted to represent European self-conceptions. Social Darwinism introduced the notion of a hierarchy that mirrored existing colonial power structures and placed Europeans at the top of a freshly reimagined evolutionary ladder.

Benedict Anderson (1991) famously described nations as Imagined Communities. Imagined communities are large groups of otherwise unrelated people who are unified by a common imaginary, set of narratives, constructed cultural structures, a fictive kinship, and a shared and mythologized tale of origins. Developments of the concept of race described above were happening in the midst of a robust period of formation and consolidation of European states and national identities. The race narrative, particularly Social Darwinist hierarchies, set up common identities for the residents of nascent European states and justified their oppressive colonial adventures abroad. This kind of national imagining acts as a powerful tool of identity construction and presently contributes to public understandings of race, ancestry, genetics, and biotechnologies.

World War II exposed the European population to the consequences of the Social Darwinian hierarchy and the dangers of the racial purity narrative.

Technologies of health and administration were inverted to inflict maximum casualties. Ideas that have been used to justify the worst atrocities of colonialism were being used to rationalize mass murder on a scale previously unrecorded in Europe. The notion of pure familial bloodlines, rooted in a place and graded on the basis of superiority, lost its allure in the aftermath of this historical event. Marks (2008:24) points to the 1960s postwar period as the starting point of the next paradigm of race in anthropology and the scientific community, the idea of clines.

Drawing on archaeological evidence, anthropology embraced the "Out of Africa" hypothesis. In this conceptualization of human evolution—supported by contemporary genetic studies—the idea of discrete speciation was entirely discredited. Instead of exclusive units of speciation, clines describe human development as an organic flow of populations going through cycles of isolation and interbreeding. In this view people all stem from the same source and spread out and reconnect on the basis of geographical topography. Like a river, populations sometimes branch out to become isolated from other streams due to lay of the land. Adaptations occur here and they present in phenotypical features such as skin tone or nose and eye shape. With time they reconnect and there are always trickles that unify in between. The notion of race, as an imagination of humans being deeply divided by Darwinian ideas of speciation, was made redundant by this paradigm. We were all the same species, and our differences were the product of relatively superficial environmental adaptation. In a landmark 1962 article titled "On the Non-Existence of Human Races" anthropologists Livingstone and Dobzhansky argued that we should drop the ideas of "races" and distinct units of humanity:

> If a central problem of physical anthropology is the explanation of the genetic variability among human population—and I think it is—then there are other methods of describing and explaining this variability which do not utilize the concept of race. This variability can also be described in terms of the concepts of cline and morphism. . . . The theoretical analysis of clines has barely begun but there seems to be no need for the concept of race in this analysis. (Livingstone and Dobzhansky 1962:279)

Evidence from molecular genetics has strengthened this hypothesis. Genetics can offer insight into ancestry, but this insight is more like a map of travels and genetic interactions than a location of a single place. Biotechnologies demonstrated that the segmentation of human population into distinct units was neither productive nor accurate. However, just like the imagined communities that create a national identity, the identities that were created through the various paradigms of race have a powerful allure to this day. Notions of race, conclusively dismissed through the process of biotechnological gene sequencing in the 1990s, have been given new life with the progression of technologies and the commercial and administrative requirements of the nation-state.

One important application of contemporary medical biotech in genetics is the personalized treatment of disease. Not everyone reacts to medicines in the same way and genetics play a role in the process of metabolizing certain drugs. Also, some genetic diseases are more prevalent among groups with shared regional origins. Genetics can be extremely helpful in the application of medical

treatment and in discovering the epidemiology of illness. Health anthropologists Shim and colleagues (2018) and Catharine Bliss (2015) reveal a history of the imposition of old values on new understanding. In the 1990s, public health services in the United States introduced policies that required genomics researchers to use federal race classifications in their work. This was done largely as an effort of administrative convenience and with the good intention of delivering services to a diverse population. This led to a need to create a sample base for "race" sets that fit administrative race requirements. In the 2000s, these classification standards were expanded to the International HapMap Project. The aim was to create a biobank of human genetic diversity. However, this complex new understanding of human genetic mix and variation utilized the loaded language of race.

Genetic biotech has been taken up as a medically useful tool. Justifications for its use are often framed in giving families with genetic predispositions to life-threatening conditions the possibility of having healthy children free from disability. Genetic scientists working in this area are fully aware of the nuances of race and ancestry. The uptake of this technology by the general population, aided by strong marketing campaigns, however, has demonstrated much less nuance. The continuation of the concept of race into contemporary genetic science has resulted in tropes of outdated national mythologies reclothed in notions of contemporary science. This situation has been complicated by the increasing availability of commercial genetic ancestry testing. On one hand, these tests often challenge ideas of mythologized racial purity and introduce the public to the complexities of clinal mixing. On the other hand, a number of studies by health anthropologists (see Benjamin 2015 and Kent et al. 2014 for examples) demonstrate that people internalize such information in ways that often reinforce outdated race values. This is exacerbated by ways in which the language of race permeates ancestry results. The results of such tests appear in the form of national and regional groupings that reflect ideas of race as tied to discrete national location. The percentage breakup of genetic presence that appears in the results of such tests (x percent West African, x percent British Isles, x percent Native American, etc.) is remnant of ideas of racial purity.

One case study of the complexities of gene testing comes from white-nationalist groups such as neo-Nazis, the KKK, and various white identitarian movements. These groups are known to prescribe to a worldview that idealizes a particular national type. This is strongly reminiscent of the first iteration of the racial concept—European nationalist groups draw on imagery of an idealized nomadic knight defined by particular phenotypical traits and held up as a pure specimen of their imagined ancestry. Among the most racist, the idea of intermarriage with people of a distinctly different ethnic and cultural group is expressed in terms of speciation and treated as impure. Social Darwinist hierarchy often applies and people are sorted into orders of superiority with assumptions of fundamental traits. Purity is held sacred and people are strongly tied to place. These ideas are completely unscientific and ancestry testing can serve as an opportunity to revise harmful identity categories with scientific reality. Indeed, according to sociologists Aaron Panofsky and Joan Donovan (2017), the results of ancestry tests can lead members of such groups to question their ideology and arbitrary notions of purity.

Panofsky and Donovan observed a series of ways in which ancestry data were used to enforce white nationalist identity conceptions or discarded when the results didn't fit their existing narrative. For some, Genetic Ancestry Testing (GAT) tests (which only trace five generations back) served as a pedigree of white purity, with any results from western and northwestern European regions being seen as ideal by group members. For those with more regional variation in results, a variety of justifications were at hand. Technical error was blamed and grand global conspiracies were created to excuse outcomes that challenged European purity. White nationalists fell back on personal family trees and on their self-descriptions in relation to phenotype (blue eyes, narrow nose, pale complexion) to reestablish their race narratives.

This kind of outcome is common. Sociologist Alondra Nelson (2008) found a similar pattern of behavior among an African American community in the United States, with genetic information being accepted or rejected on the basis of existing expectations and conceptions of existing racial identity. Sandra Soo-Jin Lee (2013) describes a situation that is opposite of that of white nationalists, with members of her study looking for a more exotic heritage than Europe in their results. The Californians she interviewed were hoping to find a "home village" in a distant global region that might explain their various character idiosyncrasies. Shim and colleagues (2018) found that among their population sample of American Latina communities genetic data were seen as extra information to construct their own identity narratives. Though people were looking at contemporary genetic tests, echoes of older notions of racial heritage punctuated their worldviews: a desire for a perfect type or a distant home that is causal of their various self-perceived character traits.

An interesting case study comes from Brazil (Kent et al. 2014). With one of the most mixed populations on the world, Brazilian national identity is increasingly and deliberately being constructed with a sense of pride in the genetic diversity of its population. Affirmative action policies were put into place in the 1990s, cultural icons of mixed ethnicity were promoted, and the samba was held up as a cultural artifact of innovation through hybridization. Some Brazilian geneticists—most notably, Sergio Pena—contributed to this effort, arguing against regional and phenotypic ideas of race in public and scientific forums. More recently, a number of projects were started to map Brazilian ancestry. They were conducted primarily for demographic and medical reasons and the outcomes, which were presented in the familiar terminology of race in relation to place of origin, were quickly taken up by the media. The ideology of national unity through pride in genetic diversity was challenged by narratives of segmentation through racialized identity construction. To borrow from Kent, Santos, and Wade (2014:738), "we argue that genomics creates a multiplicity of 'imagined genetic communities' or nations, with race playing a varied role."

Another uptake of ancestry screening technology is increasing use by indigenous and other non-dominant societal groups. Activist members of self-identified black Brazilian communities—Brazilians of African heritage—use medical genetic testing data that were being gathered to measure incidents of sickle cell disease to validate and strengthen communities (Kent et al. 2014:743–44). American First Nations communities have used gene-testing biotech to establish heredity

and land title (TallBear 2013). In Australia, the National Centre for Indigenous Genomics (NCIG) has been gathering samples since the 1960s. These samples have been used to unite families who have been separated through the social Darwinist and eugenicist policies that led to children being removed from their families, known in Australia as the Stolen Generation. They have also been used to establish ancestral lineage, which has been used in legal application for government recognition and land ownership. Working in conjunction with communities, gene testing is understood to offer significant benefit by clarifying ancestry and legitimating cultural, legal, economic, and political rights.

Using genes to legitimate membership of cultural groups is, for some, uncomfortably reminiscent of early conceptions of race. Kim TallBear (2013) cautions against the use of genetics to establish ancestry as legitimation of group membership. She argues that genetics is a Western imposition on indigenous measures of culture, family, and belonging. Kent, Santos, and Wade (2014) draw on Anderson to describe groups who connect on the basis of genetic heritage as a new kind of imagined community. Health anthropologists Shim, Alam, and Aouizerat (2018), Benjamin (2015), Bliss (2015), and others point out that contemporary genetics, particularly the type that is available to the general public, is engaged in a kind of terminology switching whereby ancestry is a functional stand-in for problematic ideas of biological essentialism and race. They argue that this is an application of scientific biotechnologies to support an idea that is thoroughly unscientific.

For First Nations people in Canada and Australia, genetic data can help people find lost family and can establish connection to land that is legitimated and recognized in land rights court cases. For white supremacists, the reification of genetics can legitimate racist ideologies. For millions of people ordering the sequencing of their genes via online ancestry websites, genetics offers a kind of imaginary connection to their pasts. Jonathan Marks (2008) suggests that the race-infused notions of ancestry are a stand-in for a much more basic and essential idea: family. For all the complexity of biotechnology, gene analysis, and history of racial classification, below the surface is a desire to connect and be a part of a kinship group. In this way, biotechnology is brought into service for one of the most fundamental of all human experiences: the need to belong.

For all of the pervasive impacts of biotechnology, the anthropological perspective reminds us to consider biotechnology in its broader sociocultural context. The division and packaging of the body, the breaking of disciplinary boundaries, our capacity to understand and manipulate cells, viruses, and genes, the pull of the market, scans, gene editing, and lineage create an image of biotech as a powerful entity with an agency of its own. Press coverage of many of the topics discussed above often draw a picture of something that will imminently break free of our control with consequences we can barely imagine. But anthropological work reminds us that, despite their sophistication and mystique, biotechnologies are tools. Like all tools of human construction, they are used to construct a world that reflects our fears and desires. At the core of biotechnological application is our desire for safety, stability, health, legacy, connection, belonging, and financial profit. And like some tools, they have the potential for harm as well as good.

Discussion Questions

1. Biocapital refers to processes and outcomes in which bodies and human biological material are commodified. Discuss some of the ways in which biological material is transformed into capital in pharmaceutical research.
2. According to Donna Haraway, which three boundaries in our understandings of personhood have been breached?
3. Why is it a problem that a disproportionate amount of research has been undertaken with WEIRD (Western, Educated, Industrialized, Rich, and Democratic) subjects?
4. What are some of the potential benefits that indigenous communities see in genomic research? What are some of the drawbacks identified by members of indigenous communities?
5. What do we mean when we say that "blood relatedness is a metaphor"?
6. Discuss the idea that nature is a cultural construct.
7. What are some of the ethical issues that surround globalized biotechnologies?
8. In what ways does reproductive technology impact on ideas of family and kinship?
9. Discuss the role that culture plays in producing and reproducing scientific knowledge.

9 Strategies and Visions for a Healthier World

Introduction and Overview

This chapter delineates some of the basic dimensions of capitalism, particularly in its latest form, often termed *neoliberalism*. In light of the contradictions of capitalism, there are serious questions about it being a sustainable system. In that there are many reasons to conclude it is not sustainable, how can we build a healthier post-capitalist future?

Students and others reading this text may feel overwhelmed by the enormity of the issues that we have been discussing, as we ourselves often do. Conversely, as Pirages and colleagues (2000:183) observe, "In many respects, the urgent environmental and health issues confronting humanity on the frontier of the twenty-first century are the same as those it has always faced: epidemic disease, resource limitations, poverty, affluence, and the unintended consequences of new technologies." Over time, however, and with increased corporate globalization, the scale and the complexities of these problems have grown and intensified.

We believe that ultimately overcoming many of the contradictions associated with the capitalist world system will require ultimately replacing it with an alternative world system that we term *democratic eco-socialism*, which we view as a pathway to a healthier world. We briefly list various system-challenging reforms that may serve to make the transition from the present world system to an alternative world system based upon social justice, democratic processes, environmental sustainability, and a safe climate, all necessary to achieving a healthier world. Finally, we discuss our contention that health anthropology is not simply a theoretical exercise but an action-oriented endeavor.

Global Capitalism

While global capitalism has resulted in impressive technological innovations, including ones in biomedicine, it is a system fraught with contradictions, including, as Barone (2004:142) asserts, "economic instability, perpetual unemployment for millions, militarism and imperialism, profound inequality and poverty in the face of plenty, deadening work and oppressive work conditions, perpetuation of racism and sexism, irrational materialism, environmental destruction, lack of democratic control of economic destiny, constraints on political democracy, and barriers to attainment of community, solidarity, and cooperation."

Capitalism

It is important to note that terminology is often tricky. Which terms you use reflects your underlying theoretical or philosophical understanding of the world. Until relatively recently, the term "capitalism" tended to be used by progressive thinkers to refer to an economic system in which capital or productive resources (e.g., a factory, a farm, or a mine) were used to generate privately held profits. Indeed, Karl Marx's most important scholarly work is titled *Das Kapital* (*Capital*). Mainstream or more conservative scholars, by contrast, tended to prefer the term "market economy," which places attention on the distribution of commodities rather than on the social relationships that shape their production. In the wake of the collapse of the Soviet Union, however, an increasing number of conventional economists are now employing the term capitalism. Additionally, progressive scholars often also refer to the contemporary period in the evolution of capitalism as being shaped by neoliberalism, a philosophical perspective (which, to complicate matters, is neither new nor liberal) embraced by conservative economists, policy makers, and corporate officials who urge the minimization of government regulation of the "free market," including the free flow of goods and services across national boundaries. Notably, the free flow of workers across national boundaries, however, is not advocated by neoliberal thinkers, who tend to want strict immigration laws. Moreover, despite opposing government restrictions on markets or the ability to sell their products anywhere and everywhere, corporations strongly demand government support for their own needs, such as in the construction of the transcontinental railroad across North America in the nineteenth century or the development of the military-industrial complex during the twentieth. While complicated, it is important to understand this array of concepts and philosophies because they have such a large impact on health. When tobacco corporations push for the elimination of laws in various developing countries that, for example, restrict the open advertisement and promotion of tobacco products, their success can have a major impact on health. Tobacco-related diseases, after all, are expected to be the primary cause of death in the developing world by 2020.

Capitalism, a globalizing political economic system committed to profit-making and continual economic growth, has created a treadmill of production and consumption, ultimately resulting in social inequality, depletion of natural resources, and environmental degradation, including climate change. Even more than in earlier stages of capitalism, under neoliberalism, transnational corporations and their associated bodies, such as the World Bank, the International Fund, and the World Trade Organization, make or break governments and politicians around the world. Although capitalism has been around for some five hundred years, it manifests so many contradictions that it must be replaced by an alternative world system, one oriented toward social parity and justice, democratic processes, environmental sustainability, and a safe climate.

As we have shown in this book, these contradictions entail numerous consequences for people's health and access to health care.

Democratic Eco-Socialism as a Pathway for a Healthier World

The challenge before us is determining how we can collectively address these monumental problems and participate in a larger effort to create a just, environmentally sustainable, and healthy global community, all of which require a safe climate. The answers are not easy and require collaboration with many individuals and groups outside the relatively small world of anthropology. As active members of an even smaller endeavor within that discipline, health anthropologists cannot create a healthy planet single-handedly but can do so only in "pragmatic solidarity" with a broad coalition of progressive people, ranging from health practitioners to academics in other disciplines (particularly sociology, political science, policy studies, and public health) and including the staffs of research institutes, government agencies, international health organizations, and nongovernmental organizations as well as people involved in peace, social justice, environmental, anti–corporate globalization, health rights, women's rights, and other social and labor movements around the world.

As proponents of the critical perspective in health anthropology, we recognize that other theoretical perspectives, including the biocultural and cultural interpretive ones, have provided and continue to provide important insights that elucidate environmental and health issues. Conversely, we strongly believe that the creation of a healthy planet for both humanity and the ecosystem will, over the long run, require the transcendence of the present global economy and a movement toward a more equitable, just, and ecologically sustainable global order. Such movement, however, is dependent on a vision for an alternative to the present global system. Lifestyle changes, such as biking, walking, or taking public transportation rather than driving cars, recycling, being a vegetarian or vegan, are not enough to protect the environment. In sum, humanity needs to undergo a socio-ecological revolution (J. Foster 2009, Baer 2018).

In the nineteenth century, various revolutionaries and reformers sought to develop alternatives to an increasingly globalizing capitalist world order. Efforts at the national level to create such an alternative started out with the Bolshevik Revolution in Russia in 1917 and included subsequent revolutions in other countries, including China in 1949, Vietnam in 1954, Cuba in 1959, and Nicaragua in 1979. Scholars have spilled much ink trying to determine whether these societies constituted examples of "actually existing socialism," transitions between capitalism and socialism, state capitalism, or new class societies and why many of these societies ultimately became fully incorporated into the capitalist world system, beginning with the collapse of the Soviet Union in 1991. Space does not permit a detailed review of why these experiments to provide an alternative to the present global economy fell short of the mark and in many cases ultimately failed. Suffice it to say that their failure to achieve an authentically democratic socialist world system was ultimately related to both internal forces

specific to each of these societies and external forces that created a hostile environment for equitable development.

Numerous observers have viewed the collapse of Communist regimes in the former Soviet Union and Eastern Europe as an indication that capitalism constitutes the end of history and that socialism was a bankrupt experiment that led to totalitarianism, forced collectivization, gulags, ruthless political purges, and inefficient centralized planned economies. What these commentators often overlook is that efforts to create socialist-oriented societies developed mainly in economically underdeveloped or developing societies, such as Russia, Mongolia, China, Cuba, and Vietnam. The efforts of Lenin, Trotsky, and other Bolsheviks to develop the beginnings of a revolutionary process that they hoped would result in socialism occurred under extremely adverse conditions, including constant external threat. Although the Bolsheviks, particularly under the dictatorial leadership of Stalin, managed to transform the Soviet Union into an industrial powerhouse by the 1930s, a variety of external forces, such as World War II and the Cold War, and internal forces, such as the centralized command economy and a political system of one-party rule, prevented the development of socialist democracy. With some modifications, the model of bureaucratic centralism was adopted by various other post-revolutionary societies after World War II, starting with China. The contradictory nature of Leninist regimes imploded first in Eastern Europe in 1989, particularly highlighted by the opening of the Berlin Wall, and in the Soviet Union in 1991. In the case of China, its Communist Party leaders embraced capitalist structures as a means of rapid economic development to the point that some scholars argue that it now constitutes a state capitalist society, entailing tremendous social inequalities and environmental devastation. The collapse of Communist regimes created a crisis for many leftists throughout the world. Many progressives had hoped that somehow these societies, which were characterized in a variety of ways, would undergo changes that would transform them into democratic and ecologically sensitive societies.

Today, global capitalism is a well-entrenched system with support from many sides but increasingly glaring contradictions. But there are voices bucking this massive global system from many quarters, including the anti–corporate globalization movement, the most progressive segments of the labor and environmental movements, peace movements, and indigenous and ethnic rights movements, as well as socialists, anarchists, and even left-leaning liberals and social democrats. Indeed, over the course of the past decade or so, a distinct climate movement has emerged in many developed and developing countries, one that has built upon warning about the dangers of climate change and global warming emanating over the past two or three decades from climate scientists, environmental groups, and indigenous groups, particularly in the Arctic and the South Pacific. Much of the climate movement in developed countries, including the United States and Australia, has tended to stress climate change mitigation strategies that come under the rubric of *ecological modernization*, that is, an emphasis on renewable sources of energy (i.e., solar, wind, geothermal, and wave), energy efficiency, electric cars, and mass transportation, but has tended to deemphasize social justice issues. In contrast, the climate movement emanating from developing

countries has given much more attention to issues of social parity, such as the fact that the developed countries have historically contributed and continue to contribute much more on a per capita basis to greenhouse gas emissions than have or do the developing countries, even China, which has become the leading emitter in the world, even superseding the United States. The Klimaforum at the UN Copenhagen Conference in December 2009 and the World People's Summit on Climate Change and the Rights to Mother Earth in Cochabamba, Bolivia, in March 2010 posited global capitalism as an economic system of production and consumption that seeks profits without limits and separates humans from nature. The latter assemblage, in particular, called for the development of an alternative world committed to social justice or parity and environmental sustainability.

Numerous proposals for achieving an alternative world system have been proposed and go by labels such as *radical democracy*, *earth democracy*, *economic democracy*, and *democratic eco-socialism* (Baer et al. 2013). It is our sense that the creation of what Boswell and Chase-Dunn (2000) term "global democracy" would entail the following components: (1) an increasing movement toward public ownership of productive forces at local, regional, national, and international levels; (2) the development of an economy oriented toward meeting social needs and environmental sustainability rather than profit-making; (3) a blending of both representative and participatory democratic processes; (4) the eradication of social disparities and redistribution of human resources between developed and developing societies and within societies in general; (5) the curtailment of population growth that in larger part would follow from the previously mentioned conditions; (6) the conservation of finite resources and the development of renewable energy resources, such as wind and solar energy, that counteract the present trend toward global warming; and (7) the reduction of wastes through recycling and transcending the culture of consumption.

Many have argued that socialism has been tried in places like the Soviet Union and China, and even Cuba for that matter, and has proven wanting. While of these three societies Cuba comes the closest to embodying socialist ideals and practices, socialism, let alone global democracy or democratic eco-socialism, remains a vision rather than an existing social system per se. Nevertheless, developments in Latin America, particularly Venezuela, Bolivia, and certainly Cuba, raise the hope of creating a "socialism for the 21st century" (C. Katz 2007). As John Bellamy Foster (2009:276) so aptly argues, "Today, the transition to socialism and the transition to an ecological society are one." Although presently or in the near foreseeable future the notion that democratic eco-socialism may be implemented in any developed society may seem utterly ridiculous and utopian, history tells that social changes can occur very quickly once economic, political, and social structural changes have reached a tipping point.

The vision of global democracy provides people everywhere with an alternative to the existing capitalist world system that continues to self-destruct due to its socially unjust and environmental unsustainable commitments and practices. Ultimately, a shift to global democracy in any one country would have to be part of a global process or a "permanent revolution" that no one can fully envision. The history of the Soviet Union and Stalinism tells us that socialism cannot be created in "one country." The struggle for a safe climate needs to be part and

parcel of a larger struggle for social justice and environmental sustainability, both internationally and within specific nation-states, including the United States and Australia. As Magdoff and Foster (2010:25) aptly argue, "Everywhere radical, essentially anti-capitalist, strategies are emerging, based on other ethics and forms of organization, rather than the profit motive: ecovillages; the new urban environment promoted in Curitiba in Brazil and elsewhere; experiments in permaculture, and community-supported agriculture, farming and industrial cooperatives in Venezuela, etc."

While Cuba remains a poor developing country, it has achieved health statistics in terms of infant mortality and life expectancy at birth that rival those of developed societies, and it has the lowest HIV prevalence in the Americas and one of the lowest in the world. Furthermore, Cuba has been exporting biomedical physicians and nurses to other developing countries, such as Venezuela, Timor-Leste, and Pakistan in the aftermath of its tragic earthquake of 2005.

Proposals such as global democracy and democratic eco-socialism constitute longer-term steps in the creation of a better world for both humanity and the health of its inhabitants and the planet. In the meantime, many medical anthropologists, working on different issues in varied places and with differing perspectives, are actively engaged in seeking to use their knowledge, skills, and resources to address pressing health and health-related social and environmental problems. In a recent book, Baer (2018) delineates that transitional system-challenging reforms need to make a shift from the present capitalist world system to a democratic eco-socialist world system, one that will create the possibilities for a socially just, democratic, environmentally sustainable, and healthier world for everyone. Democratic eco-socialism constitutes what critical sociologist Erik Olin Wright (2010) terms a *real utopia*, a utopian vision that is practically achievable through much theorizing and social experimentation.

Reforms, despite the best of intentions, are often problematic in that they may serve to stabilize capitalism, as has repeatedly been the case around the world. In the United States during the 1960s, welfare programs, including Medicaid designed for the indigent poor and Medicare designed for senior citizens, along with civil rights legislation, functioned to stabilize social unrest from various quarters, particularly on the part of African Americans. In light of this reality, Andre Gorz (1974) differentiated between *reformist reforms* and *non-reformist reforms*. He used the term reformist reform to designate the conscious implementation of minor material improvements that avoid any drastic alteration of the existing social system. Between the poles of reformist reform and complete structural transformation or revolution, Gorz identifies a category of applied work that he labels non-reformist reform or what we prefer to term system-challenging reform, which seeks to address immediate concerns, such as better wages and health care, but is part and parcel of a larger effort to make permanent changes in the social alignment of power. The distinction between the two types of reforms is sometimes difficult to make. But one distinction might be whether they are initiated by the powers-that-be or whether they are initiated by the working class, various other subaltern or marginalized groups, or anti-systemic social movements.

While space does not allow for elaboration of these systemic-challenging reforms needed to bring about democratic eco-socialism, they include (1) the creation of new progressive, anti-capitalist parties designed to capture the state in countries around the world; (2) the implementation of greenhouse gas emissions taxes at sites of production that include measures to protect low-income people; (3) social planning and increasing public ownership, socialization, or nationalization in various means of production, including health care; (4) increasing social equality within nation-states and between nation-states and achieving a sustainable global population: (5) implementation of workers' democracy; (6) the creation of meaningful work and shortening the work week; (7) the creation of a new global steady-state or net zero growth economy, which will entail growth for some populations but de-growth for other populations; (8) the adoption of energy efficiency, renewable energy sources, and green jobs; (9) the expansion of public transportation and massive diminution of a reliance on private motor vehicles and air travel; (10) the implementation of sustainable agriculture and forestry; (11) resistance to the culture of consumption and adoption of sustainable and meaningful consumption; (12) the implementation of sustainable trade; and (13) the implementation of sustainable settlement patterns and communities. These transitional steps constitute loose guidelines for shifting human societies or countries toward democratic eco-socialism and a safe climate, but it is important to note that these developments will require a global effort, including the creation of progressive global climate governance regimes. Unfortunately, the existing United Nations Framework Convention on Climate Change (UNFCCC), which has had twenty-three Congresses of the Parties (COPs) since 1995, despite some incremental steps forward, operates by and large within the dictates of a global capitalist system and the parameters of its growth paradigm, often framed under the term "sustainable development" and generally entails voluntary commitments to reducing greenhouse gas emissions.

While we suggest a litany of possible radical transitional reforms, we are not calling for a definitive fool-proof blueprint paving the way to a socio-ecological revolution. In calling for an alternative to the capitalist world system that is "relatively democratic and relatively egalitarian," Wallerstein (2013) cautions that the latter has "never yet existed" and is "only a possibility." An alternative world system cannot be designed in precise detail, but "will evolve as the new system begins to take life" (Wallerstein 2013:33). The application of our suggested transitional reforms will require adaptation to the political, economic, and sociocultural conditions in both developed and developing societies. Furthermore, our suggested transitional reforms do not exhaust possible transitional reforms necessary for creating a democratic eco-socialist world system.

As humanity enters an era of increasingly dangerous climate change accompanied by tumultuous environmental and social consequences, we need to consider alternatives that hopefully will circumvent dystopian scenarios caused by ongoing socioeconomic, ecological, and climate crises if business continues more or less as usual. We propose the imagining and creating of a democratic eco-socialist world system as a real utopia, not just a vehicle for creating a healthier world, but also a more socially just, democratic, and environmentally sustainable world society, as well.

Democratic eco-socialism rejects the capitalist treadmill of production and consumption, and its associated economic growth paradigm. Instead, it recognizes that humans live on an ecologically fragile planet with limited resources, such as clean fresh water and arable land to grow crops (both of which are essential for maintain healthy populations) that must be sustained and renewed as much as possible for future generations. While at the present time or for the foreseeable future, the notion that democratic eco-socialism may eventually be implemented in any society, developed or developing, or in a number of societies, may appear absurd. However, history tells us that social changes can occur very quickly once social, structural, and environmental conditions have reached a tipping point.

Our own sense is that overall things will get worse before they get better, and there is no guarantee that they will get better. Nevertheless, while the capitalist world system appears to be well entrenched on numerous fronts, including within universities, there are numerous cracks in the system, manifested in phenomena such as Brexit in the UK, the election of Donald Trump as president in the United States, tensions within the European Union, the threat of nuclear war, ongoing conflicts in many parts of the world, growing gaps in global wealth, climate crises of many sorts, the acidification of the oceans, and the extinction of many creatures, ranging from birds and animals to marine life.

Presenting a precise timeline for a transition from the existing capitalist world system to a democratic eco-socialist world system is extremely difficult, probably impossible. However, we believe that the stabilization of the earth's climate system needs to occur within the next two or three decades lest large swatches of land become uninhabitable for human beings as well as nonhuman species. Despite the daunting difficulties that much of humanity currently faces and will continue to face over the course of this century, we believe that it is important that progressive people keep plugging away at challenging the system in their conversations, teachings, and writings while staying involved in anti-systemic movements by struggling to create new left and anti-capitalist parties; pointing out alternative ways of organizing the world at global, regional, national, and local levels; and listening to critical input from other progressive perspectives, including eco-anarchism, eco-feminism, and indigenous voices, to mention only a few. In contrast to capitalism, which is focused on profit-making or the creation of surplus value, as Kallis (2017:15) argues, socialist politics are "about specifics, about concrete use values: employment, a decent wage, dignified conditions of living, a healthy environment, education, public health or clean water for all."

Health Anthropology as an Action-Oriented Endeavor

As an applied field, health anthropology, as we have seen, is action oriented; given the significant problems facing human populations, research-informed action is critically needed in the struggles for health and a healthier world. We believe that anthropologists and health anthropologists, as a process of creating a healthier world, but also a socially just and environmentally sustainable

world, need to become politically engaged and work in solidarity with progressive and anti-systemic movements that in various, often cumbersome and even contradictory ways are seeking to create an alternative to the existing global political economy, one that we call democratic eco-socialism.

Around 1848 or so, social movements and nationalist movements that came into being involved in activity opposing some aspect of the capitalist world system "began to create a new institution: the continuing organization with members, officers, and specific political objectives (both long-term and short-term)" (Arrighi et al. 1989:29). Progressive movements exhibit degrees of radicalism in critiquing capitalism, conceptions of democracy and gender relations, and environmental perspective. Some are not evenly clearly anti-systemic but wish to make global capitalism more socially just, transparent, and environmentally sustainable. An example of this would be the human rights movement, which has no clearly delineated critique of corporations, believing that they can either be reformed or shamed into adopting human rights principles. Even large segments of the labor, ethnic minorities rights, women's, peace, and environmental and climate movements can be characterized in this way. As scholar-activists who have over the years been involved in various social movements, including the labor, peace, anti-apartheid, environmental, climate, and socialist movements, we know firsthand that social movements, even those with some semblance of anti-capitalist tendencies, can be quite disparate, contradictory, and factionalized. However, in today's world order they are the site where the greatest potential to transcending global capitalism lies.

In terms of climate change, anthropologists can work in solidarity with indigenous organizations such as the International Forum of Indigenous Peoples and Local Communities on Climate Change, the Inuit Circumpolar Conference, and the Alliance of Small Island States, which have been challenging the limited actions of developed countries in responding to a monumental environmental problem that threatens human settlement patterns, subsistence, lifeways, and health. They can also work in solidarity within the climate and climate justice, anti-coal, anti-coal seam gas, anti-fracking campaigns that emerged around the world, including the United States and Australia.

Various health social scientists engage in health activism on multiple fronts in collaboration with health social movements (HSMs), which challenge political-economic power and scientific and professional authority. In the United States, these movements include "struggles to improve the quality of and access to health care (e.g., health insurance reform and advocacy of a single-payer health system); to eliminate persistent health inequalities based on race, ethnicity, gender, class or sexual (e.g., the HIV/AIDS and breast cancer movement), and to push public health and medical institutions to address fundamental causes of disease and disability by reshaping scientific inquiry on etiology, diagnosis, treatment, and prevention (e.g., the HIV/AIDS and breast cancer movements)" (Morello-Frosch et al. 2012:5). Health social movements include (1) health access movements that "seek equitable access to health care and improved provision of health care services"; (2) embodied health movements that "address disease, disability or illness experience by challenging science on aetiology, diagnosis, treatment, and prevention"; (3) constituency-based health movements that

"address health inequality and health inequity based on race, ethnicity, gender, class and/or sexuality differences"; (4) environmental health social movements that address environmental factors, such as air and water pollution, toxic waste dumps, and climate change, which adversely impact upon health; and (5) complementary and alternative medicine movements that promote various CAM therapies and seek to obtain greater legal and social recognition for them (Brown and Zavestoski 2004:685–88). In reality, HSMs may overlap with each other. Brown and Zavestoski (2004:691) maintain that "Studying health social movements offers insight into an innovative and powerful form of political action aimed at transforming the health care system, modifying people's experience of illness and addressing broader social determinants of health and disease in diverse communities."

The Contested Illness Research Group (CIRG) established in 1999 at Brown University has collaborated with HSMs, not only examining "citizen-science alliances," but also participating in them, thus functioning as scholar-activists (Senier et al. 2012:269).

Action-oriented health anthropologists can consider working in solidarity with the People's Health Movement, which has chapters in some seventy countries, particularly developing ones. While not a radical organization per se, the People's Health Movement definitely is a progressive one that challenges the impact of neoliberal globalization on the global health crisis. Its vision states:

> Equity, ecologically sustainable development and peace are at the heart of our vision of a better world—a world in which a healthy life for all is a reality: a world that respects, appreciates and celebrates all life and diversity; a world that enables the flowering of people's talents and abilities to enrich each other; a world in which people's voices guide the decisions that shape our lives. (People's Health Movement n.d.)

Its objectives include promotion of health for all through equitable, participatory, and inter-sectoral movements; to advocate for governments and health agencies to ensure universal access to quality health care, education, and social services dictated by their needs and not incomes; to promote the involvement of people and organizations in the formulation and implementation of health and social policies; and to urge people to develop solutions to their health needs based upon local conditions.

The People's Health Movement has published to date four editions of the *Global Health Watch* in collaboration with various other organizations. Like the 2005, 2008, and 2011 editions, volume 4 examines five issues that impact upon health: (1) the global political and economic architecture that has created a global health crisis; (2) current issues and debates within national health-care systems, particularly ones calling for universal health-care coverage in countries around the world; (3) strategies seeking to address the "multiple social and structural developments of health," (4) scrutinizing global institutions, such as the World Health Organization and the World Bank, which impact upon global health; and (5) campaigns and actions seeking to improve health and health care (People's Health Movement 2014:1).

As this chapter attempts to show, creating a healthier world will ultimately require transcending global capitalism with an alternative world system that we

term democratic eco-socialism. In terms of health, it is essential that everyone on the planet has access to adequate and nutritious food, clean water, basic housing, comprehensive health care, satisfying and unalienated work, a sense of community, health education, a clean environment, and a safe climate. These will require a drastic redistribution of resources around the world, which the capitalist class and its political allies will resist with incredible and violent force if it so deems necessary. However, progressive people need to strategize how to unify their efforts in the endeavor to create a better world. We hope as critical health anthropologists that this book has enlightened you about the complexities of the human condition in what some call the Anthropocene and others perhaps more appropriately the Capitalocene, particularly in terms of health and health care around the world, and some possible avenues for creating a healthier world for all.

Discussion Questions

1. Define capitalism and its latest incarnation, often termed neoliberalism.
2. What are some of the ways in which transnational corporations and their associated bodies make or break governments and politicians around the world? How is this process operating in your own country?
3. Discuss why earlier efforts to create socialist societies, in places such as Russia, China, and elsewhere, failed. What lessons can be drawn from the successes and failures of these societies in seeking to reinvent the concept of socialism?
4. What are the basic components of democratic eco-socialism and how does it vary from ways that socialism has been defined in the past?
5. Discuss some examples of reformist reforms and non-reformist or system-challenging reforms, especially in terms of health care. Into which category would you place Obamacare, which the Trump administration has been seeking to dismantle?
6. Envision a possible single-payer health-care system for the United States, drawing upon examples of other single-payer health-care system around the world, perhaps particularly the Canadian model.
7. Discuss possibilities for making health anthropology an action-oriented endeavor. To what social movements, nongovernmental organizations (NGOs), or political parties in your own community or country could action-oriented health anthropologists contribute and what are some actions that they might take?

Source Material for Students

Health Anthropology and Related Journals

- *Anthropologies of African Biosciences* (https://africanbiosciences.wordpress.com/)
- *Anthropology and Medicine* (https://www.tandfonline.com/loi/canm20)
- *Anthropology in Action: Journal for Applied Anthropology in Policy and Action* (journals.berghahnbooks.com/aia/index.php)
- *Culture, Medicine and Psychiatry* (web.mit.edu/dumit/www/cmp.html)
- *Global Change and Human Health* (www.springerlink.com/content/106598/)
- *International Journal of Bio-Anthropological Practice* (sites.google.com/site/ijbapractice/current-issue)
- *Journal of Complementary and Alternative Medicine* (www.liebertpub.com/products/product.aspx?pid=26)
- *Journal of Ethnobiology and Ethnomedicine* (www.ethnobiomed.com/)
- *Journal of Human Ecology* (www.krepublishers.com/02-Journals/JHE/JHE-00-0-000-000-1990-Web/JHE-00-0-000-000-1990-1-Cover.htm)
- *Medical Anthropology and Bioethics* (en.tuva.asia/82-medical.html)
- *Medical Anthropology Quarterly* (https://anthrosource.onlinelibrary.wiley.com/journal/15481387)
- *Medical Anthropology: Cross-Cultural Studies in Health and Illness* (https://www.tandfonline.com/loi/gmea20)
- *Social Science and Medicine* (https://www.journals.elsevier.com/social-science-and-medicine)

Health Anthropology Organizations and Institutions

- Centre for Health and Society, University of Melbourne (Australia) (www.chs.unimelb.edu.au/)
- Durham University's Medical Anthropology Research Group (UK) (www.dur.ac.uk/anthropology/research/marg/)
- Medanthro Study Group (www.h-net.org/~medanthro/)
- Medical Anthropology Network of the European Association of Social Anthropologists (https://www.easaonline.org/networks/medical/)
- The Medical Anthropology Working Group (Germany) (www.medicalanthropology.de/)
- National Academy of Medicine (https://nam.edu/)
- Society for Applied Anthropology (www.sfaa.net/)
- Society for Ethnomedicine (Germany) (www.agem-ethnomedizin.de/)
- Society for Medical Anthropology (http://www.medanthro.net/)

Health Anthropology Blogs

- AccessDenied: A Conversation on Unauthorized Im/migration and Health! (https://accessdeniedblog.wordpress.com/)
- AnthroGenetics Blog (anthrogenetics.wordpress.com/category/medical-anthropology/)
- Living Anthropologically (https://www.livinganthropologically.com/anthropology-blogs-2017/)
- Somatosphere: Science, Medicine, and Anthropology (http://somatosphere.net/)
- Voices of Medical Anthropology (socmedanthro.wordpress.com/2010/01/07/who-are-we-in-the-public-imagination-carolyn-sargent/)

Health Anthropology Films

A list of health anthropology and related films is available at http://h-net.msu.edu/cgi-bin/logbrowse.pl?trx=vx&list=h-medanthro&month=1008&week=b&msg=XqNYA9Wn8Cvf4YGBth214w&user=&pw=.

Universities in North America with Anthropology Graduate Programs

Programs at these schools and what they have to offer the prospective student can be searched individually.

- Boston University School of Medicine
- Brandeis University
- Brown University
- Case Western Reserve University
- City College of New York (CUNY)
- Columbia University
- Cornell University
- East Carolina University
- Emory University
- Harvard University
- McGill University
- Michigan State University
- New School for Social Research
- Northwestern University
- Ohio State University
- Pennsylvania State University
- Portland State University
- Rensselaer Polytechnic Institute
- San Diego State University
- Southern Methodist University
- Stanford University

- State University of New York (SUNY)–Binghamton
- University of Alabama
- University of Alberta
- University of Arizona
- University of Buffalo
- University of California–Berkeley
- University of California–Los Angeles
- University of California–San Diego
- University of California–San Francisco
- University of California–Santa Barbara
- University of California-Santa Cruz
- University of Chicago
- University of Colorado–Denver
- University of Colorado–Boulder
- University of Connecticut
- University of Denver
- University of Hawaii
- University of Illinois
- University of Iowa
- University of Kansas
- University of Kentucky
- University of Manitoba
- University of Maryland
- University of Massachusetts
- University of Memphis
- University of Michigan
- University of Missouri–Columbia
- University of Montana
- University of Montréal
- University of North Carolina–Chapel Hill
- University of Oregon
- University of Pennsylvania
- University of Pittsburgh
- University of South Florida
- University of Texas–Austin
- University of Toronto
- University of Virginia
- University of Washington
- University of Wisconsin–Madison
- Washington State University
- Wayne State University
- Yale University
- York University

Encyclopedias

- Encyclopedia of Medical Anthropology (https://www.springer.com/us/book/9780306477546)
- Ethnic Diseases Sourcebook (https://www.amazon.com/Ethnic-Diseases-Sourcebook-Information-Unnumbered/dp/0780803361)
- The Gale Encyclopedia of Alternative Medicine (https://www.cengage.com/search/productOverview.do?N=197+4294917621+4294892091+4294888430+4294916915&Ntk=P_EPI&Ntt=154348299167640833623286228813593842 65&Ntx=mode%2Bmatchallpartial)

Online Medical Anthropology Tutorials and Resources

- AnthroSource (https://anthrosource.onlinelibrary.wiley.com/)
- Medical Anthropology: How Illness Is Traditionally Perceived and Cured around the World (anthro.palomar.edu/medical/default.htm)
- Resources (http://www.medanthro.net/academic-resources/)

Glossary

adaptation practice that human societies are implementing to adjust to climate change, such as increasing surface temperatures, rising sea levels, storm surges, droughts, heavy rains, and floods

allopathic medicine medical system that evolved into biomedicine; term commonly used today to differentiate biomedicine from homeopathic and osteopathic medicine

anthropogenic usually refers to changes in the environment, including the climate, that are a product of human activity

Ayurvedic medicine global system of traditional medicine that originated in India as far back as three to four thousand years ago, based on healthful living throughout the life cycle and maintaining and reestablishing balance among five elements of life: earth, water, fire, air, and ether

bioethics ethical issues in medical practice

biomedicine globally dominant medical system often called Western, allopathic, or cosmopolitan medicine

biopathy emergent perspective on health care that seeks to restore the body to a fundamental harmonious state, including restoration of the body's natural immune and cell repair systems, and that emphasizes avoiding unnatural products (including biomedicines, processed foods)

biosocial interaction of biological, social structural, and social relationship factors in the creation of health status

biotechnology broad range of technological devices and procedures, such as reproductive technologies, organ transplants, mechanical ventilators, magnetic resonance imaging devices, X-ray machines, prosthetic limbs, and even Band-Aids and cotton balls

blaming the victim blaming the behavior of socially subordinated persons or groups for their adverse health and social status

body theory explanatory approach that focuses on the role of the body in societal processes and the role of society in bodily attitudes, experiences, and practices

capitalism political economic system based on private property ownership of the means of production, profit-making, and continual economic expansion

Chinese medicine ancient medical system based on concepts of energy flow that has diffused to many regions of the world

chiropractic form of manual medicine developed in the 1890s in Iowa by D. D. Palmer, a magnetic healer

climate change change in the climate over either a short or long period that may be related to natural events or human-created (i.e., anthropogenic) activities or even both

commodified bodies body organs and fluids that are transformed into entities that can be bought and sold, such that they come to be thought of as commodities with a commercial value

complementary and alternative medicine healing systems other than biomedicine; this expression locates biomedicine as the core of medicine and views other healing systems as potentially complementary to this dominant system

consensus analysis quantitative research procedure that is used to determine the modal (i.e., most frequent) answers provided by a group to a set of questions about a particular topic (e.g., "what is disease?")

contextual approach seeing attitudes and behaviors within the influencing

natural and social settings in which they emerge and function

critical medical anthropology theoretical perspective in medical anthropology that emphasizes the importance of the interface between the political economic/ecological system and local cultural patterns in shaping health

cultural broker role sometimes played by medical anthropologists as intercessors between local and dominant social and/or medical systems

cultural competence ability of health-care providers to understand and work successfully with the cultural traditions of their patients

cultural emotions culturally shaped feelings that help shape behavior within a society

cultural epidemiology integration of the anthropological concept of culture into the study of disease outbreaks and epidemics

culture of medicine shared set of beliefs and practices acquired by medical students and residents that shape day-to-day behavior and understandings among physicians

cyborg human being who functions with an artificial part or device or even an organ transplant from another person

dengue mosquito-borne viral disease, known as the "bone crusher"

disease diagnosis of a sickness made by a biomedical physician or some type of professionalized heterodox medical practitioner

dominative medical system system under which biomedicine politically, economically, and culturally dominates other medical subsystems, both professionalized and folk

ecological modernization shift to technological innovations, such as increasing energy efficiency, renewable sources of energy, and public transport, which are being proposed as strategies to create a more sustainable environment and mitigate climate change

eco-socialism proposed alternative to global capitalism, based upon a commitment to social parity and environmental sustainability that some theorists view as an important form of climate change mitigation

ecosyndemic syndemic disease interactions caused by human restructuring of the environment (e.g., anthropogenic climate change enables vector-borne diseases to spread to new areas and interact with other diseases found in the new locations)

epidemiology study of population health patterns and the emergence and spread of disease within and across populations

ethnocultural communities historic ethnic cultural traditions of subgroups within a larger society, synonymous with "subcultures"

ethnomedicine culturally constituted medical systems, including biomedicine

ethnosemantic elicitation type of interview in which the researcher seeks to learn the cognitive categories study participants use to organize their experience and understanding of the world

explanatory model group's understanding of the nature, causes, and symptoms of a disease

folk healing systems indigenous and local health-care systems; also called complementary and alternative health-care systems

folk injectors folk healers who use syringe injection of medicines that contain pharmaceutical and folk medicine mixtures

free listing research method in which researchers ask study participants to name all of the kinds of items that exist within a cultural domain (e.g., all of the causes of disease)

global warming important manifestation of climate change in which the earth or portions of the earth experience increases in temperature due to various greenhouse gases, particularly carbon dioxide, methane, and nitrous oxide

health anthropology study of human health and disease, health-related behavior, and healing and health care from the perspective of anthropology

health disparities differences in health and disease patterns and health-care access across populations or segments of populations

health inequalities differences in health and disease patterns and health-care access across populations or segments of populations that are the consequence of social inequalities

health transition improvements in life expectancy and changes in the relative importance of various causes of mortality that have occurred worldwide since the end of World War II

holistic core anthropological concept that refers to viewing specific components of the human condition (e.g., health behavior, religious behavior) within a broader social, environmental, and cultural context

homeopathy form of alternative professionalized medicine in which practitioners administer diluted forms of substances believed to be the cause of adverse symptoms following the principle that "like is cured by like" not, as in biomedicine, by opposites (e.g., antibiotics).

hospital ethnography relatively new type of ethnographic research in which medical anthropologists conduct observations in various clinics and other care units in a hospital

humoral medical system set of beliefs and healing practices based on the premise that the organs, body systems, and fluids inside the body parallel elements and structures of the natural world outside the body and that balance must be maintained across elements to remain healthy

illness label for the patient's experience of sickness

illness behavior cultural patterning of behavior among the sick

illness narratives personal, storylike accounts of illness experience

integrative medicine medicine that theoretically blends together the best elements of biomedicine and complementary and alternative medical systems

life course stages of life that find expression in differing health issues and configurations

life history interview method used to elicit detailed narratives of participants' life stories

meaning-centered medical anthropology theoretical perspective that prioritizes symbolism and meaning in the interpretation of health-related behavior

medical anthropology. See **health anthropology**

medical ecology theoretical perspective in medical anthropology that emphasizes the interface between populations and their environments in the shaping of health

medical hybridization blending together of elements of various medical subsystems or the borrowing of elements on the part of one medical system from other medical subsystems

medical pluralism conflation of medical subsystems in complex or state societies that entails competition and even collaboration among the subsystems

medical-industrial complex profit-making or pecuniary relationship between biomedicine and various businesses, including health insurance, health care, and pharmaceutical companies

meme small cultural unit (e.g., bit of knowledge and emotion) transmitted through social learning across generations within a society

mindful bodies conception of the body developed within medical anthropology that sees the body as a nexus of individual experience, social relationship, and a set of representations of the body as a natural symbol with which to think about nature, society, and culture

mitigation wide set of practices that are being proposed to reduce the impacts of climate change or global warming

multi- or mixed-method research integration of a suite of methods, qualitative and quantitative, in a research initiative

naturopathy highly eclectic medical subsystem that relies on a wide array of therapeutic modalities, including herbalism, homeopathy, reflexology, nutritious eating, and exercise

neoliberal reforms within the health arena, changes in national health-care policies that are favored by big development lending institutions that stress privatization of disease treatment and the use of market mechanisms to set the price of health care

nurse-anthropologist nurse who holds either an MA or PhD in anthropology

osteopathy manual medical system developed in the mid-1800s in Kansas by Andrew Taylor Still, a disenchanted American regular physician

paleopathology study of diseases in the past, especially during prehistoric times

panopticon term introduced by Michel Foucault that refers to an ever-growing ability to observe, internally and externally, and control individuals in society at all times and places

physician-anthropologist biomedical physician who holds either an MA or PhD in anthropology

pile sort research method in which study participants are asked to sort into like piles (as seen by participants) a set of behaviors, ideas, objects, and so forth that have been inscribed on a set of cards (e.g., sort disease symptoms into like piles)

psychogenesis in psychology, the development of a physical disorder resulting from psychological, rather than physiological, sources

Q-sorts research method in which participants are asked to rank (on some criteria) a set of behaviors, ideas, objects, and so forth that have been inscribed on a set of cards

racism discrimination in health care (or elsewhere) based on an individual's presumed membership in a biologically defined racial category. Because gene distributions are no more uniform within "identified races" than across culturally imposed racial boundaries, anthropologists discount the existence of distinct races among humans

scatter plot analytic strategy in which a computer program represents the overall picture or pattern in a set of sorted cards, based on an assessment of which cards people tended to put in the same pile or keep apart in different piles

self-limiting diseases short-term diseases that the body overcomes without treatment

semistructured interview flexible interview format in which new questions are inserted by the researcher in response to participant responses to previous questions

shaman part-time religious healer who contacts the supernatural realm so as to determine the cause of illness or petition the spirits to release patients from their distress

social stigmatization situation involving the creation of what are called socially "discredited selves" in which disease sufferers are blamed for their illness and discriminated against for being a sufferer

social suffering immediate personal and shared experience of group members of broader human problems caused by the unjust exercise of political and economic power

syndemic adverse interactions between two or more diseases or other health-related conditions (e.g., stress, malnutrition), often promoted by social inequality and subordination, that increase the total disease burden of a population

typology categorization of items in a cultural domain into two or more tiers of inclusivity based on the degree of relationship of items (e.g., in U.S. society the cultural domain of "pets" consists of dogs, cats, fish,

and the like; the subcategory of dogs consists of Labradors, terriers, poodles, and so on; the subcategory of poodles includes standards and miniatures, and so forth)

wounded healers shamans and other healers, including modern-day psychologists, who initially underwent treatment for some type of illness

writing up final act of research by the anthropologist that involves the preparation of a report, article, chapter, or book to present a research project and its findings

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